# Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type and to Flexible m × n Nets” — Helper

Tested on: Mathematica 14.0
Revised with the assistance of the Claude models Opus 5.5, Fable 5, and Fable 5.1 (Anthropic)
All computations were checked by the authors

## Section 1

In[•]:=
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(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[α, β, γ, δ, σ, a, b, c, d, r, s, f, M, assumptions, isTrueQ, idL, idR, okId,
  idxRules, dispL, dispR, statusChip, smallChip, s1colTest, s1colHdr];
(*The computation concerns one vertex:the internal symbols carry no index,
and the panels display the vertex index i*)
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α];
b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ];
d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
idL[1] = HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])];
idR[1] = HoldForm[r (r - M) / (M (r - 1))];
idL[2] = HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])];
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idR[2] = HoldForm[(r - M) / (r (r - 1))];
idL[3] = HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])];
idR[3] = HoldForm[s (s - M) / (M (s - 1))];
idL[4] = HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])];
idR[4] = HoldForm[(s - M) / (s (s - 1))];
idL[5] = HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])];
idR[5] = HoldForm[f (f - M) / (M (f - 1))];
idL[6] = HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])];
idR[6] = HoldForm[(f - M) / (f (f - 1))];
okId[n_] := okId[n] = isTrueQ[ReleaseHold@idL[n], ReleaseHold@idR[n]];
(*Display form of the identities:the same held expressions,
with the vertex index i shown on every symbol*)
idxRules = {HoldPattern[α] → HoldForm[Subscript[α, i]],
   HoldPattern[β] → HoldForm[Subscript[β, i]],
   HoldPattern[γ] → HoldForm[Subscript[γ, i]], HoldPattern[δ] →
    HoldForm[Subscript[δ, i]], HoldPattern[σ] → HoldForm[Subscript[σ, i]],
   HoldPattern[r] → HoldForm[Subscript[r, i]], HoldPattern[s] →
    HoldForm[Subscript[s, i]], HoldPattern[f] → HoldForm[Subscript[f, i]],
   HoldPattern[M] → HoldForm[Subscript[M, i]]};
dispL[n_] := idL[n] /. idxRules;
dispR[n_] := idR[n] /. idxRules;
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s1colTest = RGBColor[0.10, 0.35, 0.75];
s1colHdr = RGBColor[0.15, 0.35, 0.45];

(* ============================================================*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* ============================================================*)
Module[{okAll, starLine, chkLine}, okAll = And @@ Table[okId[n], {n, 6}];
  starLine[n1_, n2_] :=
   Style[Row[{Style[TraditionalForm[HoldForm[#1 == #2] &[dispL[n1], dispR[n1]]],
       Bold], ",      ", Style[TraditionalForm[
        HoldForm[#1 == #2] &[dispL[n2], dispR[n2]]], Bold]}], 14, s1colTest];
  chkLine[n1_, n2_] :=
   Style[Row[{TraditionalForm[HoldForm[#1 == #2] &[dispL[n1], dispR[n1]]], "   ",
      smallChip[okId[n1]], "         ", TraditionalForm[HoldForm[#1 == #2] &[
        dispL[n2], dispR[n2]]], "   ", smallChip[okId[n2]]}], 13];
  Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
      Style[Row[{"For each  ", Style["i", Italic], " = 1, 2, 3, 4 :"}], 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[α, i]]],
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        ", ", TraditionalForm[HoldForm[Subscript[β, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[γ, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[δ, i]]], " ∈ (0, π)"}], 13],
     Style[Row[{TraditionalForm[HoldForm[Subscript[σ, i] == (Subscript[
              α, i] + Subscript[β, i] + Subscript[γ, i] + Subscript[δ, i]) / 2]],
        " ,   ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[α, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[β, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[γ, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[δ, i]]],
        " ∉ πℤ"}], 13], Style[Row[{TraditionalForm[HoldForm[Subscript[a, i] ==
           Sin[Subscript[α, i]] / Sin[Subscript[σ, i] - Subscript[α, i]]]],
        ",   ", TraditionalForm[HoldForm[Subscript[b, i] ==
           Sin[Subscript[β, i]] / Sin[Subscript[σ, i] - Subscript[β, i]]]],
        ",   ", TraditionalForm[HoldForm[Subscript[c, i] ==
           Sin[Subscript[γ, i]] / Sin[Subscript[σ, i] - Subscript[γ, i]]]],
        ",   ", TraditionalForm[HoldForm[Subscript[d, i] ==
           Sin[Subscript[δ, i]] / Sin[Subscript[σ, i] - Subscript[δ, i]]]]}],
      13], Style[Row[{TraditionalForm[HoldForm[Subscript[r, i] ==
           Subscript[a, i] Subscript[d, i]]], ",   ", TraditionalForm[
         HoldForm[Subscript[s, i] == Subscript[c, i] Subscript[d, i]]],
        ",   ", TraditionalForm[HoldForm[Subscript[f, i] ==
           Subscript[a, i] Subscript[c, i]]], ",   ",
        TraditionalForm[HoldForm[Subscript[M, i] == Subscript[a, i]
            Subscript[b, i] Subscript[c, i] Subscript[d, i]]]}], 13],
     Style[Row[{TraditionalForm[HoldForm[Subscript[r, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[s, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[f, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[M, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[M, i] / Subscript[r, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[M, i] / Subscript[s, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[M, i] / Subscript[f, i]]],
        "  ≠ 0, 1 ;  hence both sides of the identities  ",
        Style["⋆", 13, s1colTest], "  below are well-defined"}], 13],
     "", Style[Row[{"Each  ", smallChip[True],
        "  below certifies that the statement displayed beside it has been
          verified by a symbolic computation at that point;  a  ",
        smallChip[False], "  would mean that this verification failed."}],
      12, GrayLevel[0.35]], "", Style[
      Row[{Style["We want to test the following identities  ", Bold, 14],
        Style["⋆", 14, s1colTest], Style[" :", Bold, 14]}]],
     starLine[1, 2], starLine[3, 4], starLine[5, 6]}, Spacings → 1],
   Background → RGBColor[0.99, 0.96, 0.88],
   FrameMargins →
    12]];
Print[
 Panel[Column[{Row[{Style["The six ratio identities", Bold, 15, s1colHdr],
      Spacer[20], statusChip[okAll]}], Style["Verification:", Bold, 13],
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    Style[Row[{"The identities  ", Style["★", 13, s1colTest],
       "  hold under the given hypotheses:"}], 13],
    chkLine[1, 2], chkLine[3, 4], chkLine[5, 6]}, Spacings → 1],
  Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
  FrameMargins → 12]];
Print[Framed[Style[If[okAll, "Section 1: ALL CLAIMS VERIFIED",
    "Section 1: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[okAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses:

For each $i = 1, 2, 3, 4$ :

$\alpha_i, \beta_i, \gamma_i, \delta_i \in (0, \pi)$

$\sigma_i = \frac{1}{2}(\alpha_i + \beta_i + \gamma_i + \delta_i)$ , $\sigma_i - \alpha_i, \sigma_i - \beta_i, \sigma_i - \gamma_i, \sigma_i - \delta_i \notin \pi\mathbb{Z}$

$a_i = \frac{\sin(\alpha_i)}{\sin(\sigma_i - \alpha_i)}$, $b_i = \frac{\sin(\beta_i)}{\sin(\sigma_i - \beta_i)}$, $c_i = \frac{\sin(\gamma_i)}{\sin(\sigma_i - \gamma_i)}$, $d_i = \frac{\sin(\delta_i)}{\sin(\sigma_i - \delta_i)}$

$r_i = a_i d_i$, $s_i = c_i d_i$, $f_i = a_i c_i$, $M_i = a_i b_i c_i d_i$

$r_i, s_i, f_i, M_i, \frac{M_i}{r_i}, \frac{M_i}{s_i}, \frac{M_i}{f_i} \neq 0, 1$ ; hence both sides of the identities ★ below are well −defined

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following identities ★ :

$$\frac{\sin(\alpha_i)\sin(\delta_i)}{\sin(\beta_i)\sin(\gamma_i)} = \frac{r_i(r_i - M_i)}{M_i(r_i - 1)}, \quad \frac{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \delta_i)}{\sin(\sigma_i - \beta_i)\sin(\sigma_i - \gamma_i)} = \frac{r_i - M_i}{r_i(r_i - 1)}$$

$$\frac{\sin(\gamma_i)\sin(\delta_i)}{\sin(\alpha_i)\sin(\beta_i)} = \frac{s_i(s_i - M_i)}{M_i(s_i - 1)}, \quad \frac{\sin(\sigma_i - \gamma_i)\sin(\sigma_i - \delta_i)}{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \beta_i)} = \frac{s_i - M_i}{s_i(s_i - 1)}$$

$$\frac{\sin(\alpha_i)\sin(\gamma_i)}{\sin(\beta_i)\sin(\delta_i)} = \frac{f_i(f_i - M_i)}{M_i(f_i - 1)}, \quad \frac{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \gamma_i)}{\sin(\sigma_i - \beta_i)\sin(\sigma_i - \delta_i)} = \frac{f_i - M_i}{f_i(f_i - 1)}$$

### The six ratio identities — Verified

Verification:

The identities ★ hold under the given hypotheses:

$$\frac{\sin(\alpha_i)\sin(\delta_i)}{\sin(\beta_i)\sin(\gamma_i)} = \frac{r_i(r_i - M_i)}{M_i(r_i - 1)}$$ ☑ $$\frac{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \delta_i)}{\sin(\sigma_i - \beta_i)\sin(\sigma_i - \gamma_i)} = \frac{r_i - M_i}{r_i(r_i - 1)}$$ ☑

$$\frac{\sin(\gamma_i)\sin(\delta_i)}{\sin(\alpha_i)\sin(\beta_i)} = \frac{s_i(s_i - M_i)}{M_i(s_i - 1)}$$ ☑ $$\frac{\sin(\sigma_i - \gamma_i)\sin(\sigma_i - \delta_i)}{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \beta_i)} = \frac{s_i - M_i}{s_i(s_i - 1)}$$ ☑

$$\frac{\sin(\alpha_i)\sin(\gamma_i)}{\sin(\beta_i)\sin(\delta_i)} = \frac{f_i(f_i - M_i)}{M_i(f_i - 1)}$$ ☑ $$\frac{\sin(\sigma_i - \alpha_i)\sin(\sigma_i - \gamma_i)}{\sin(\sigma_i - \beta_i)\sin(\sigma_i - \delta_i)} = \frac{f_i - M_i}{f_i(f_i - 1)}$$ ☑

**Section 1: ALL CLAIMS VERIFIED**

# Section 2

In[ ]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
```

```
(* ============================================================*)
ClearAll[f, l, m, M, isTrueQ, idL, idR,
  okId, statusChip, smallChip, s2colTest, s2colHdr];
(*The identities below are identities of
 rational functions in Subscript[f,l],Subscript[f,m],M;
the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
idL[1] =
  HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
    Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))];
idR[1] = HoldForm[(Subscript[f, l] - Subscript[f, m])
    (Subscript[f, l] Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
     (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[2] =
  HoldForm[(Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
     (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1];
idR[2] = HoldForm[(Subscript[f, l] Subscript[f, m] - M)
    (Subscript[f, l] Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
     (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[3] = HoldForm[(Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1)) -
    (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))];
idR[3] = HoldForm[(Subscript[f, m] - Subscript[f, l])
    (Subscript[f, l] Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
     (Subscript[f, l] Subscript[f, m]
       (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
idL[4] = HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
     ((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1];
idR[4] = HoldForm[(Subscript[f, l] Subscript[f, m] - M)
    (Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] - M) /
     (Subscript[f, l] Subscript[f, m]
       (Subscript[f, l] - 1) (Subscript[f, m] - 1))];
okId[n_] := okId[n] = isTrueQ[ReleaseHold@idL[n], ReleaseHold@idR[n]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s2colTest = RGBColor[0.10, 0.35, 0.75];
s2colHdr = RGBColor[0.15, 0.35, 0.45];
(* ============================================================*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* ============================================================*)
Module[{okAll, starLine, chkLine}, okAll = And @@ Table[okId[n], {n, 4}];
  starLine[n_] :=
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```
 Style[Style[TraditionalForm[HoldForm[#1 == #2] &[idL[n], idR[n]]], Bold],
  14, s2colTest];
chkLine[n_] :=
 Style[Row[{TraditionalForm[HoldForm[#1 == #2] &[idL[n], idR[n]]],
    "   ", smallChip[okId[n]]}], 13];
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
    Style[Row[{"For  ", TraditionalForm[HoldForm[l]], ", ",
       TraditionalForm[HoldForm[m]], " ∈ {1, …, 4},   ",
       TraditionalForm[HoldForm[l ≠ m]], " :"}], Bold, 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[f, l]]], ", ",
       TraditionalForm[HoldForm[Subscript[f, m]]], ", ", TraditionalForm[
        HoldForm[M]], ", ", TraditionalForm[HoldForm[M / Subscript[f, l]]],
       ", ", TraditionalForm[HoldForm[M / Subscript[f, m]]],
       "  ≠ 0, 1 ;  hence both sides of the identities  ",
       Style["⋆", 13, s2colTest], "  below are well-defined"}], 13],
    "", Style[Row[{"Each  ", smallChip[True],
       "  below certifies that the statement displayed beside it has been
         verified by a symbolic computation at that point;  a  ",
       smallChip[False], "  would mean that this verification failed."}],
     12, GrayLevel[0.35]], "", Style[
     Row[{Style["We want to test the following identities  ", Bold, 14],
       Style["⋆", 14, s2colTest], Style[" :", Bold, 14]}]],
    starLine[1], starLine[2], starLine[3], starLine[4]}, Spacings → 1],
  Background → RGBColor[0.99, 0.96, 0.88],
  FrameMargins → 12]];
Print[Panel[Column[
   {Row[{Style["The four factorization identities", Bold, 15, s2colHdr],
      Spacer[20], statusChip[okAll]}], Style["Verification:", Bold, 13],
    Style[Row[{"The identities  ", Style["⋆", 13, s2colTest],
       "  hold under the given hypotheses:"}], 13],
    chkLine[1], chkLine[2], chkLine[3], chkLine[4]}, Spacings → 1],
  Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
  FrameMargins → 12]];
Print[Framed[Style[If[okAll, "Section 2: ALL CLAIMS VERIFIED",
    "Section 2: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[okAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];];
```

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Given hypotheses:

For $I, m \in \{1, \ldots, 4\}$, $I \neq m$ :

$f_I, f_m, M, \frac{M}{f_I}, \frac{M}{f_m} \neq 0, 1$ ; hence both sides of the identities ★ below are well −defined

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following identities ★ :

$$\frac{f_I\,(f_I-M)}{M\,(f_I-1)} - \frac{f_m\,(f_m-M)}{M\,(f_m-1)} = \frac{(f_I-f_m)\,(f_I\,f_m-(f_I+f_m)+M)}{M\,(f_I-1)\,(f_m-1)}$$

$$\frac{(f_I\,(f_I-M))\,(f_m\,(f_m-M))}{(M\,(f_I-1))\,(M\,(f_m-1))} - 1 = \frac{(f_I\,f_m-M)\,(f_I\,f_m-M\,(f_I+f_m-1))}{M^2\,(f_I-1)\,(f_m-1)}$$

$$\frac{f_I-M}{f_I\,(f_I-1)} - \frac{f_m-M}{f_m\,(f_m-1)} = \frac{(f_m-f_I)\,(f_I\,f_m-M\,(f_I+f_m-1))}{f_I\,f_m\,(f_I-1)\,(f_m-1)}$$

$$\frac{(f_I-M)\,(f_m-M)}{(f_I\,(f_I-1))\,(f_m\,(f_m-1))} - 1 = \frac{(f_I\,f_m-M)\,(f_I+f_m-f_I\,f_m-M)}{f_I\,f_m\,(f_I-1)\,(f_m-1)}$$

The four factorization identities — Verified

Verification:

The identities ★ hold under the given hypotheses:

$$\frac{f_I\,(f_I-M)}{M\,(f_I-1)} - \frac{f_m\,(f_m-M)}{M\,(f_m-1)} = \frac{(f_I-f_m)\,(f_I\,f_m-(f_I+f_m)+M)}{M\,(f_I-1)\,(f_m-1)}$$ ☑

$$\frac{(f_I\,(f_I-M))\,(f_m\,(f_m-M))}{(M\,(f_I-1))\,(M\,(f_m-1))} - 1 = \frac{(f_I\,f_m-M)\,(f_I\,f_m-M\,(f_I+f_m-1))}{M^2\,(f_I-1)\,(f_m-1)}$$ ☑

$$\frac{f_I-M}{f_I\,(f_I-1)} - \frac{f_m-M}{f_m\,(f_m-1)} = \frac{(f_m-f_I)\,(f_I\,f_m-M\,(f_I+f_m-1))}{f_I\,f_m\,(f_I-1)\,(f_m-1)}$$ ☑

$$\frac{(f_I-M)\,(f_m-M)}{(f_I\,(f_I-1))\,(f_m\,(f_m-1))} - 1 = \frac{(f_I\,f_m-M)\,(f_I+f_m-f_I\,f_m-M)}{f_I\,f_m\,(f_I-1)\,(f_m-1)}$$ ☑

**Section 2: ALL CLAIMS VERIFIED**

# Section 3

In[ ]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, rad, sq, assumptions,
  okSq, okEps, okRadLow, okRadHigh, okRadical, okMerge, s3chk, cellChk,
  clSym, intvStr, intIdx, snVal, cnVal, lblCell, statusChip, smallChip,
  s3colGiven, s3colTest, s3colHdr, dn, t, p, q, k, tI, pqI, sgI];
(*The computation concerns one vertex:the internal symbols carry no index,
and the panels display the vertex index i*)
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α];
b = Sin[β] / Sin[σ - β];
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```
c = Sin[γ] / Sin[σ - γ];
d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
 assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
these hold under the given hypotheses*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi,
   0 < δ < Pi, 0 < σ - α < Pi, 0 < σ - β < Pi, 0 < σ - γ < Pi, 0 < σ - δ < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[σ] / Sin[σ - δ]) ^ 2;
(*Two-stage verification of the sigma-
 identity:STAGE 1 (okSq) identifies rad with the perfect square sq.STAGE
   2 (okEps) uses Stage 1:the quotient Sin[sigma]/Sin[sigma-delta]
   has the sign eps (u/v checks,because Sin[sigma-delta]>0),
so Sqrt[rad] equals eps times this quotient (abstract radical law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
 eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
   over trigonometric arguments reaches the kernel.*)
okSq = TrueQ[FullSimplify[rad == sq, Assumptions → assumptions]];
okEps =
  Module[{tt, u, v}, okSq && TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
    TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
    TrueQ[Assuming[tt > 0, Simplify[Sqrt[tt^2] == tt]]] &&
    TrueQ[Assuming[tt < 0, Simplify[Sqrt[tt^2] == -tt]]] &&
    TrueQ[Simplify[(Sin[σ] / Sin[σ - δ]) Sin[σ - δ] == Sin[σ]]]];
s3chk[e1_, e2_] := Together[e1 - e2] === 0 || Simplify[e1 - e2] === 0;
(*Radicand identities per regime,rational in fresh symbols*)
okRadLow = Module[{rv, sv, fv, mv}, s3chk[
    fv (fv - mv) (rv - 1) (sv - 1) / (fv (1 - mv)), (fv - mv) (rv - 1) (sv - 1) / (1 - mv)]];
okRadHigh = Module[{rv, sv, fv, mv}, s3chk[(fv - mv) (rv - 1) (sv - 1) / (fv (1 - mv)),
    (1 / fv - 1 / mv) (rv - 1) (sv - 1) / (1 - 1 / mv)]];
(*Abstract radical laws:k-extraction and the dn-fold*)
okRadical = Module[{x, kk, dd},
   TrueQ[Assuming[kk > 0 && x ≥ 0, Simplify[Sqrt[x / kk^2] == Sqrt[x] / kk]]] &&
    TrueQ[Assuming[dd > 0 && x ≥ 0, Simplify[dd Sqrt[x] == Sqrt[dd^2 x]]]]];
(*Class-wise radical merges,as displayed:atoms m,w,v,
A>0 parametrize|p q|,Sqrt[dn^2-1],Sqrt[dn^2-k'^2],Sqrt[k'^2-dn^2];
in class 2,dn^2-1=-(A^2+k^2) and dn^2-k'^2=-A^2*)
okMerge[1] = Module[{m, w, v},
   TrueQ[Assuming[m > 0 && w > 0, Simplify[Sqrt[w^2 m^2] == m Sqrt[w^2]]]] &&
    TrueQ[Assuming[m > 0 && v > 0, Simplify[Sqrt[v^2 m^2] == m Sqrt[v^2]]]]];
okMerge[2] = Module[{m, A, kk},
   TrueQ[Assuming[m > 0 && A > 0 && kk > 0, Simplify[Sqrt[(-(A^2 + kk^2)) (-m^2)] ==
        -I (I m) Sqrt[A^2 + kk^2]]]] && TrueQ[Assuming[m > 0 && A > 0,
      Simplify[Sqrt[(-A^2) (-m^2)] == -I (I m) Sqrt[A^2]]]]];
okMerge[3] = Module[{m, w, v},
   TrueQ[Assuming[m > 0 && w > 0, Simplify[Sqrt[w^2 m^2] == -(-m) Sqrt[w^2]]]] &&
```

```
    TrueQ[Assuming[m > 0 && v > 0, Simplify[Sqrt[v^2 m^2] == -(-m) Sqrt[v^2]]]]];
(*Cell checks:class cl of p q (1:pos real,2:pos imaginary,3:neg real) and eps=
 sgn(sin sigma);
same atoms;
radicals merged per the given sign identity;
returns the pair of verdicts {sn agrees,cn agrees}*)
cellChk[cl_, eps_] :=
  Module[{kk, m, w, A, pq, W, V, snS, cnS, snT, cnT}, Which[cl == 1, pq = m;
    W = w m;
    V = m Sqrt[w^2 + kk^2];
    snT = If[eps == 1, I w / kk, -I w / kk];
    cnT = If[eps == 1, Sqrt[w^2 + kk^2] / kk, -Sqrt[w^2 + kk^2] / kk],
    cl == 2, pq = I m;
    W = m Sqrt[kk^2 + A^2];
    V = A m;
    snT = If[eps == 1, Sqrt[kk^2 + A^2] / kk, -Sqrt[kk^2 + A^2] / kk];
    cnT = If[eps == 1, -I A / kk, I A / kk], cl == 3, pq = -m;
    W = w m;
    V = m Sqrt[w^2 + kk^2];
    snT = If[eps == 1, -I w / kk, I w / kk];
    cnT = If[eps == 1, -Sqrt[w^2 + kk^2] / kk, Sqrt[w^2 + kk^2] / kk]];
   snS = I eps W / (kk pq); cnS = eps V / (kk pq);
   {s3chk[snS, snT], s3chk[cnS, cnT]}];
(*Display helpers for the indexed symbols*)
tI = TraditionalForm[HoldForm[Subscript[t, i]]];
pqI = TraditionalForm[HoldForm[Subscript[p, i] Subscript[q, i]]];
sgI = TraditionalForm[HoldForm[Subscript[σ, i]]];
clSym[1] = Subscript["ℝ", ">0"];
clSym[2] = Row[{"i ", Subscript["ℝ", ">0"]}];
clSym[3] = Subscript["ℝ", "<0"];
intvStr[1, 1] = "(0, iK′)";
intvStr[1, -1] = "(2K, 2K + iK′)";
intvStr[2, 1] = "(K, K + iK′)";
intvStr[2, -1] = "(3K, 3K + iK′)";
intvStr[3, 1] = "(2K, 2K + iK′)";
intvStr[3, -1] = "(0, iK′)";
intIdx[1, 1] = 1; intIdx[1, -1] = 3; intIdx[2, 1] = 2;
intIdx[2, -1] = 4; intIdx[3, 1] = 3; intIdx[3, -1] = 1;
snVal[1] =
  TraditionalForm[HoldForm[I Sqrt[Superscript[dn, 2] Subscript[t, i] - 1] / k]];
snVal[2] =
  TraditionalForm[HoldForm[Sqrt[1 - Superscript[dn, 2] Subscript[t, i]] / k]];
snVal[3] =
  TraditionalForm[HoldForm[-I Sqrt[Superscript[dn, 2] Subscript[t, i] - 1] / k]];
snVal[4] =
  TraditionalForm[HoldForm[-Sqrt[1 - Superscript[dn, 2] Subscript[t, i]] / k]];
cnVal[1] = TraditionalForm[
```

```
    HoldForm[Sqrt[Superscript[dn, 2] Subscript[t, i] - 1 + k^2] / k]];
cnVal[2] = TraditionalForm[
    HoldForm[-I Sqrt[1 - k^2 - Superscript[dn, 2] Subscript[t, i]] / k]];
cnVal[3] = TraditionalForm[
    HoldForm[-Sqrt[Superscript[dn, 2] Subscript[t, i] - 1 + k^2] / k]];
cnVal[4] = TraditionalForm[
    HoldForm[I Sqrt[1 - k^2 - Superscript[dn, 2] Subscript[t, i]] / k]];
lblCell[cl_, eps_] :=
  Row[{pqI, " ∈ ", clSym[cl], ",   ", sgI, If[eps == 1, " < π", " > π"],
    "   →   ", TraditionalForm[HoldForm[Subscript[ε, i]]],
    If[eps == 1, " = 1", " = -1"], ",   ", tI, " ∈ ", intvStr[cl, eps]}];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
    Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
    Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
s3colGiven = RGBColor[0.85, 0.45, 0.00];
s3colTest = RGBColor[0.10, 0.35, 0.75];
s3colHdr = RGBColor[0.15, 0.35, 0.45];

(* ============================================================*)
(*PART 2:GIVEN PANEL AND VERIFICATION PANEL*)
(* ============================================================*)
Module[{cells, gridPairs, gridSn, gridCn, okAll, hdrCell, tab1, tab2, mergeRow},
  cells = {{1, 1}, {1, -1}, {2, 1}, {2, -1}, {3, 1}, {3, -1}};
  gridPairs = cellChk @@@ cells;
  gridSn = gridPairs[[All, 1]]; gridCn = gridPairs[[All, 2]];
  okAll = okSq && okEps && okRadLow && okRadHigh && okRadical && okMerge[1] &&
    okMerge[2] && okMerge[3] && (And @@ gridSn) && (And @@ gridCn);
  hdrCell[cl_] := Row[{pqI, " ∈ ", clSym[cl]}];
  tab1 = Grid[{{"", hdrCell[1], hdrCell[2], hdrCell[3]},
     {Row[{sgI, " < π"}], intvStr[1, 1], intvStr[2, 1], intvStr[3, 1]},
     {Row[{sgI, " > π"}], intvStr[1, -1], intvStr[2, -1], intvStr[3, -1]}},
    Frame → All, FrameStyle → GrayLevel[0.75], Spacings → {2, 0.8}];
  tab2 = Grid[{{tI, intvStr[1, 1], intvStr[2, 1], intvStr[3, 1], intvStr[2, -1]},
     {Row[{"sn ", tI}], snVal[1], snVal[2], snVal[3], snVal[4]},
     {Row[{"cn ", tI}], cnVal[1], cnVal[2], cnVal[3], cnVal[4]}},
    Frame → All, FrameStyle → GrayLevel[0.75], Spacings → {2, 0.8}];
  mergeRow[1] = Row[{pqI, " ∈ ", clSym[1], " :   ",
     TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - 1)
          Subscript[p, i]^2 Subscript[q, i]^2] == Subscript[p, i]
         Subscript[q, i] Sqrt[Superscript[dn, 2] Subscript[t, i] - 1]]], ",   ",
     TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - k'^2)
          Subscript[p, i]^2 Subscript[q, i]^2] == Subscript[p, i]
```

```
        Subscript[q, i] Sqrt[Superscript[dn, 2] Subscript[t, i] - k'^2]]],
   "      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]], "   ",
   smallChip[okMerge[1]]}];
mergeRow[2] = Row[{pqI, " ∈ ", clSym[2], " :   ",
   TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - 1)
        Subscript[p, i]^2 Subscript[q, i]^2] == -I Subscript[p, i]
       Subscript[q, i] Sqrt[1 - Superscript[dn, 2] Subscript[t, i]]]], ",   ",
   TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - k'^2)
        Subscript[p, i]^2 Subscript[q, i]^2] == -I Subscript[p, i]
       Subscript[q, i] Sqrt[k'^2 - Superscript[dn, 2] Subscript[t, i]]]],
   "      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]], "   ",
   smallChip[okMerge[2]]}];
mergeRow[3] = Row[{pqI, " ∈ ", clSym[3], " :   ",
   TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - 1)
        Subscript[p, i]^2 Subscript[q, i]^2] == -Subscript[p, i]
       Subscript[q, i] Sqrt[Superscript[dn, 2] Subscript[t, i] - 1]]], ",   ",
   TraditionalForm[HoldForm[Sqrt[(Superscript[dn, 2] Subscript[t, i] - k'^2)
        Subscript[p, i]^2 Subscript[q, i]^2] == -Subscript[p, i]
       Subscript[q, i] Sqrt[Superscript[dn, 2] Subscript[t, i] - k'^2]]],
   "      ", Style["●", 14, RGBColor[0.40, 0.15, 0.60]], "   ",
   smallChip[okMerge[3]]}];
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
    Style[Row[{"For each  ", Style["i", Italic], " = 1, 2, 3, 4 :"}], 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[α, i]]],
       ", ", TraditionalForm[HoldForm[Subscript[β, i]]],
       ", ", TraditionalForm[HoldForm[Subscript[γ, i]]], ", ",
       TraditionalForm[HoldForm[Subscript[δ, i]]], " ∈ (0, π)"}], 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[σ, i] == (Subscript[
              α, i] + Subscript[β, i] + Subscript[γ, i] + Subscript[δ, i]) / 2]],
       " ,   ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[α, i]]],
       ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[β, i]]],
       ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[γ, i]]],
       ", ", TraditionalForm[HoldForm[Subscript[σ, i] - Subscript[δ, i]]],
       " ∈ (0, π),   ", TraditionalForm[HoldForm[Sin[Subscript[σ, i]] ≠ 0]],
       " ,   ", TraditionalForm[HoldForm[
         Subscript[ε, i] == Sign[Sin[Subscript[σ, i]]]]], " ∈ {±1}"}], 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[a, i] ==
          Sin[Subscript[α, i]] / Sin[Subscript[σ, i] - Subscript[α, i]]]],
       ",   ", TraditionalForm[HoldForm[Subscript[b, i] ==
          Sin[Subscript[β, i]] / Sin[Subscript[σ, i] - Subscript[β, i]]]],
       ",   ", TraditionalForm[HoldForm[Subscript[c, i] ==
          Sin[Subscript[γ, i]] / Sin[Subscript[σ, i] - Subscript[γ, i]]]],
       ",   ", TraditionalForm[HoldForm[Subscript[d, i] ==
          Sin[Subscript[δ, i]] / Sin[Subscript[σ, i] - Subscript[δ, i]]]]}],
     13], Style[Row[{TraditionalForm[HoldForm[Subscript[r, i] ==
          Subscript[a, i] Subscript[d, i]]], ",   ", TraditionalForm[
       HoldForm[Subscript[s, i] == Subscript[c, i] Subscript[d, i]]],
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   ",   ", TraditionalForm[HoldForm[Subscript[f, i] ==
      Subscript[a, i] Subscript[c, i]]], ",   ",
   TraditionalForm[HoldForm[M == Subscript[a, i] Subscript[b, i]
       Subscript[c, i] Subscript[d, i]]]}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[r, i]]], ", ",
   TraditionalForm[HoldForm[Subscript[s, i]]],
   ", ", TraditionalForm[HoldForm[Subscript[f, i]]],
   ", ", TraditionalForm[HoldForm[M]], ", ",
   TraditionalForm[HoldForm[M / Subscript[f, i]]], "  ≠ 0, 1,   ",
   TraditionalForm[HoldForm[Subscript[f, i] > 0]],
   " ,   ", TraditionalForm[HoldForm[M > 0]]}], 13],
Style[Row[{TraditionalForm[HoldForm[k == Sqrt[1 - M]]], ",   ",
   TraditionalForm[HoldForm[k' == Sqrt[M]]], ",   ", "dn ", tI, "  =  ",
   TraditionalForm[HoldForm[Sqrt[Subscript[f, i]]]], "   if  M < 1"}],
 13], Style[Row[{TraditionalForm[HoldForm[k == Sqrt[1 - 1 / M]]],
   ",   ", TraditionalForm[HoldForm[k' == Sqrt[1 / M]]], ",   ", "dn ",
   tI, "  =  ", TraditionalForm[HoldForm[1 / Sqrt[Subscript[f, i]]]],
   "   if  M > 1"}], 13], Style[Row[{TraditionalForm[
    HoldForm[Subscript[p, i] == Sqrt[Subscript[r, i] - 1]]], ",   ",
   TraditionalForm[HoldForm[Subscript[q, i] == Sqrt[Subscript[s, i] - 1]]],
   " ∈ ", Subscript["ℝ", ">0"], " ⋃ ", "i ", Subscript["ℝ", ">0"]}], 13],
Style[Row[{TraditionalForm[HoldForm[k]], ", ",
   TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 13], Style[Row[
  {TraditionalForm[HoldForm[Sign[Superscript[dn, 2] Subscript[t, i] - 1] ==
      Sign[Superscript[dn, 2] Subscript[t, i] - k'^2] ==
      Sign[Subscript[p, i]^2 Subscript[q, i]^2]]], "  ≠ 0"}], 13],
"", Style["Given formulas:", Bold, 14], Style[
 Row[{Style[TraditionalForm[HoldForm[cn Subscript[t, i] ==
       dn Subscript[t, i] Sin[Subscript[σ, i]] / (Subscript[p, i]
           Subscript[q, i] Sin[Subscript[σ, i] - Subscript[δ, i]])]],
    Bold], "   if  M < 1 ;      ", Style[TraditionalForm[
     HoldForm[cn Subscript[t, i] == Sin[Subscript[σ, i]] / (Subscript[p, i]
          Subscript[q, i] Sin[Subscript[σ, i] - Subscript[δ, i]])]],
    Bold], "   if  M > 1"}], 14, s3colGiven], "",
Style[Row[{"Table 1 (the interval of  ", tI, " ):"}], Bold, 14],
tab1, "", Style[Row[{"Table 2 (the values of  sn ", tI,
   "  and  cn ", tI, "  on each interval):"}], Bold, 14],
tab2, "", Style[Row[{"Each  ", smallChip[True],
   "  below certifies that the statement displayed beside it has been
     verified by a symbolic computation at that point;  a  ",
   smallChip[False], "  would mean that this verification failed."}],
 12, GrayLevel[0.35]], "", Style[
 Row[{Style["We want to test the following formulas  ", Bold, 14],
   Style["⋆", 14, s3colTest], Style[" :", Bold, 14]}]], Style[Row[
  {Style[TraditionalForm[HoldForm[sn Subscript[t, i] == I Subscript[ε, i]
       Sqrt[(Superscript[dn, 2] Subscript[t, i] - 1) Subscript[p, i]^2
          Subscript[q, i]^2] / (k Subscript[p, i] Subscript[q, i])]],
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        Bold], ",      ", Style[TraditionalForm[HoldForm[cn Subscript[t, i] ==
           Subscript[ε, i] Sqrt[(Superscript[dn, 2] Subscript[t, i] - k'^2)
               Subscript[p, i]^2 Subscript[q, i]^2] / (k Subscript[p, i]
               Subscript[q, i])]], Bold]}], 14, s3colTest]},
   Spacings → 1], Background → RGBColor[0.99, 0.96, 0.88],
  FrameMargins →
   12]];
Print[Panel[Column[
   {Row[{Style["The derivation:  the formulas for  ", Bold, 15, s3colHdr],
      Style[Row[{"sn ", tI}], Bold, 15, s3colHdr], Style["  and  ",
       Bold, 15, s3colHdr], Style[Row[{"cn ", tI}], Bold, 15, s3colHdr],
      Spacer[20], statusChip[okAll]}], Style["Verification:", Bold, 13],
    Style["The following identities hold:", 13], Style[
     Row[{TraditionalForm[HoldForm[(Subscript[f, i] - M) (Subscript[r, i] - 1)
           (Subscript[s, i] - 1) / (Subscript[f, i] (1 - M)) ==
          (Sin[Subscript[σ, i]] / Sin[Subscript[σ, i] - Subscript[δ, i]])^2]],
       "   ", smallChip[okSq]}], 13], Style[
     Row[{TraditionalForm[HoldForm[Sin[Subscript[σ, i]] ==
          Subscript[ε, i] Sqrt[(Subscript[f, i] - M) (Subscript[r, i] - 1)
             (Subscript[s, i] - 1) / (Subscript[f, i] (1 - M))]
           Sin[Subscript[σ, i] - Subscript[δ, i]]]], "      ",
       Style["●", 14, RGBColor[0.55, 0.33, 0.10]], "   ", smallChip[okEps]}],
     13], Style[Row[{"Hence, substituting the identity  ",
       Style["●", 13, RGBColor[0.55, 0.33, 0.10]], "  into ",
       Style[Row[{"the given formulas for  cn ", tI}],
        Italic, 13, s3colGiven], " ,  and using  ",
       TraditionalForm[HoldForm[Subscript[p, i]^2 Subscript[q, i]^2 ==
          (Subscript[r, i] - 1) (Subscript[s, i] - 1)]],
       "  and the values of  ", TraditionalForm[HoldForm[k]], ", ",
       TraditionalForm[HoldForm[k']], ",  dn ", tI, "  above, we get"}], 13],
    Style[Row[{TraditionalForm[HoldForm[cn Subscript[t, i] == Subscript[ε, i]
           Sqrt[(Superscript[dn, 2] Subscript[t, i] - k'^2) Subscript[p, i]^2
              Subscript[q, i]^2 / k^2] / (Subscript[p, i] Subscript[q, i])]],
       "      ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]], "   ",
       Style["( M < 1 :", 12, GrayLevel[0.3]],
       "  ", smallChip[okSq && okRadLow && okRadical],
       Style[" ;   M > 1 :", 12, GrayLevel[0.3]],
       "  ", smallChip[okSq && okRadHigh && okRadical],
       Style[" )", 12, GrayLevel[0.3]]}], 13],
    Style[Row[{"and, because  ", TraditionalForm[HoldForm[k > 0]],
       " ,  the identity  ", Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
       "  gives the formula  ", Style["⋆", 13, s3colTest],
       "  for  cn ", tI, " ."}], 13], "", Style[Row[
      {"By the given sign identity, the radicals merge on each class of  ",
       pqI, "  as follows:"}], 13],
    mergeRow[1], mergeRow[2], mergeRow[3], "",
    Style[Row[{"Finally, on each cell below, the range of  ", sgI,
```

```
        "  determines  ", TraditionalForm[HoldForm[Subscript[ε, i] ==
           Sign[Sin[Subscript[σ, i]]]]], " ,  the interval of  ", tI,
        "  is read from Table 1 and the values of  sn ", tI, "  and  cn ",
        tI, "  from Table 2, and, substituting the identities  ",
        Style["●", 13, RGBColor[0.40, 0.15, 0.60]],
        " ,  the formulas  ", Style["⋆", 13, s3colTest],
        "  for  sn ", tI, "  and  cn ", tI, "  (for  cn ", tI,
        " ,  this is a second, independent verification of the formula
          derived above) agree with these values:"}], 13],
     Grid[Table[{Style[lblCell @@ cells[[ic]], 12, GrayLevel[0.3]],
        Column[{Style[Row[{"sn ", tI, " = ", TraditionalForm[HoldForm[
                I Subscript[ε, i] Sqrt[(Superscript[dn, 2] Subscript[t, i] - 1)
                     Subscript[p, i]^2 Subscript[q, i]^2] / (k Subscript[p, i]
                     Subscript[q, i])]]], " = ", snVal[intIdx @@ cells[[ic]]]}],
           12, GrayLevel[0.3]], Style[Row[{"cn ", tI, " = ", TraditionalForm[
              HoldForm[Subscript[ε, i] Sqrt[(Superscript[dn, 2] Subscript[t,
                       i] - k'^2) Subscript[p, i]^2 Subscript[q, i]^2] /
                  (k Subscript[p, i] Subscript[q, i])]], " = ",
             cnVal[intIdx @@ cells[[ic]]]}], 12, GrayLevel[0.3]]}],
        Column[{smallChip[gridSn[[ic]]], smallChip[gridCn[[ic]]]},
         Spacings → 0.3]}, {ic, Length[cells]}],
      Alignment → {Left, Center}, Spacings → {2, 1}], "",
     Style[Row[{"Together, these give the formulas  ",
        Style["⋆", 13, s3colTest], " ."}], 13]}, Spacings → 1],
   Background → Lighter[RGBColor[0.15, 0.35, 0.45], 0.93],
   FrameMargins →
    12]];
 Print[Framed[Style[If[okAll, "Section 3: ALL CLAIMS VERIFIED",
     "Section 3: SOME CLAIMS FAILED"], Bold, White, 16], Background →
    If[okAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
   FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];];
```

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Given hypotheses:

For each $i = 1, 2, 3, 4$ :

$\alpha_i,\ \beta_i,\ \gamma_i,\ \delta_i \in (0,\ \pi)$

$\sigma_i = \frac{1}{2}\,(\alpha_i + \beta_i + \gamma_i + \delta_i)\ ,\quad \sigma_i - \alpha_i\ ,\ \sigma_i - \beta_i\ ,\ \sigma_i - \gamma_i$

$,\ \sigma_i - \delta_i\ \in (0,\ \pi),\quad \sin(\sigma_i) \neq 0\ ,\quad \varepsilon_i = \text{sgn}(\sin(\sigma_i)) \in \{\pm 1\}$

$a_i = \frac{\sin(\alpha_i)}{\sin(\sigma_i - \alpha_i)},\quad b_i = \frac{\sin(\beta_i)}{\sin(\sigma_i - \beta_i)},\quad c_i = \frac{\sin(\gamma_i)}{\sin(\sigma_i - \gamma_i)},\quad d_i = \frac{\sin(\delta_i)}{\sin(\sigma_i - \delta_i)}$

$r_i = a_i\,d_i,\quad s_i = c_i\,d_i,\quad f_i = a_i\,c_i,\quad M = a_i\,b_i\,c_i\,d_i$

$r_i,\ s_i,\ f_i,\ M,\ \frac{M}{f_i} \neq 0,\ 1,\quad f_i > 0\ ,\quad M > 0$

$k = \sqrt{1 - M},\quad k' = \sqrt{M},\quad \text{dn}\,t_i = \sqrt{f_i}\quad \text{if}\ \ M < 1$

$k = \sqrt{1 - \frac{1}{M}},\quad k' = \sqrt{\frac{1}{M}},\quad \text{dn}\,t_i = \frac{1}{\sqrt{f_i}}\quad \text{if}\ \ M > 1$

$p_i = \sqrt{r_i - 1},\quad q_i = \sqrt{s_i - 1} \in \mathbb{R}_{>0} \bigcup i\ \mathbb{R}_{>0}$

$k,\ k' \in (0,\ 1)$

$\text{sgn}(\text{dn}^2\,t_i - 1) = \text{sgn}(\text{dn}^2\,t_i - (k')^2) = \text{sgn}(p_i^2\,q_i^2)\ \neq 0$

Given formulas:

$\text{cn}\,t_i = \frac{\text{dn}\,t_i\,\sin(\sigma_i)}{p_i\,q_i\,\sin(\sigma_i - \delta_i)}\quad \text{if}\ \ M < 1\ ;\qquad \text{cn}\,t_i = \frac{\sin(\sigma_i)}{p_i\,q_i\,\sin(\sigma_i - \delta_i)}\quad \text{if}\ \ M > 1$

Table 1 (the interval of $t_i$):

| | $p_i\,q_i \in \mathbb{R}_{>0}$ | $p_i\,q_i \in i\ \mathbb{R}_{>0}$ | $p_i\,q_i \in \mathbb{R}_{<0}$ |
|---|---|---|---|
| $\sigma_i < \pi$ | $(0,\ iK')$ | $(K,\ K + iK')$ | $(2K,\ 2K + iK')$ |
| $\sigma_i > \pi$ | $(2K,\ 2K + iK')$ | $(3K,\ 3K + iK')$ | $(0,\ iK')$ |

Table 2 (the values of $\text{sn}\,t_i$ and $\text{cn}\,t_i$ on each interval ):

| $t_i$ | $(0,\ iK')$ | $(K,\ K + iK')$ | $(2K,\ 2K + iK')$ | $(3K,\ 3K + iK')$ |
|---|---|---|---|---|
| $\text{sn}\,t_i$ | $\frac{i\,\sqrt{\text{dn}^2\,t_i - 1}}{k}$ | $\frac{\sqrt{1 - \text{dn}^2\,t_i}}{k}$ | $-\frac{i\,\sqrt{\text{dn}^2\,t_i - 1}}{k}$ | $-\frac{\sqrt{1 - \text{dn}^2\,t_i}}{k}$ |
| $\text{cn}\,t_i$ | $\frac{\sqrt{\text{dn}^2\,t_i - 1 + k^2}}{k}$ | $-\frac{i\,\sqrt{1 - k^2 - \text{dn}^2\,t_i}}{k}$ | $-\frac{\sqrt{\text{dn}^2\,t_i - 1 + k^2}}{k}$ | $\frac{i\,\sqrt{1 - k^2 - \text{dn}^2\,t_i}}{k}$ |

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following formulas ★ :

$$\text{sn}\,t_i = \frac{i\,\varepsilon_i\,\sqrt{(\text{dn}^2\,t_i - 1)\,p_i^2\,q_i^2}}{k\,p_i\,q_i},\qquad \text{cn}\,t_i = \frac{\varepsilon_i\,\sqrt{(\text{dn}^2\,t_i - (k')^2)\,p_i^2\,q_i^2}}{k\,p_i\,q_i}$$

## The derivation: the formulas for $\text{sn}\,t_i$ and $\text{cn}\,t_i$ — Verified

Verification:

The following identities hold:

$\frac{(f_i - M)\,(r_i - 1)\,(s_i - 1)}{f_i\,(1 - M)} = \left(\frac{\sin(\sigma_i)}{\sin(\sigma_i - \delta_i)}\right)^2$ ☑

$$\sin(\sigma_i) = \varepsilon_i \sqrt{\frac{(f_i - M)(r_i - 1)(s_i - 1)}{f_i (1 - M)}} \sin(\sigma_i - \delta_i)$$ ● ☑

Hence, substituting the identity ● into *the given formulas for* $\mathrm{cn}\, t_i$, and using $p_i^2 q_i^2 = (r_i - 1)(s_i - 1)$ and the values of $k$, $k'$, $\mathrm{dn}\, t_i$ above, we get

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{\frac{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}{k^2}}}{p_i q_i}$$ ● ( M < 1 : ☑ ; M > 1 : ☑ )

and, because $k > 0$, the identity ● gives the formula ★ for $\mathrm{cn}\, t_i$.

By the given sign identity, the radicals merge on each class of $p_i q_i$ as follows:

$p_i q_i \in \mathbb{R}_{>0}$ : $\sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2} = p_i q_i \sqrt{\mathrm{dn}^2 t_i - 1}$, $\sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2} = p_i q_i \sqrt{\mathrm{dn}^2 t_i - (k')^2}$ ● ☑

$p_i q_i \in i\,\mathbb{R}_{>0}$ : $\sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2} = -i\, p_i q_i \sqrt{1 - \mathrm{dn}^2 t_i}$, $\sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2} = -i\, p_i q_i \sqrt{(k')^2 - \mathrm{dn}^2 t_i}$ ● ☑

$p_i q_i \in \mathbb{R}_{<0}$ : $\sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2} = -p_i q_i \sqrt{\mathrm{dn}^2 t_i - 1}$, $\sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2} = -p_i q_i \sqrt{\mathrm{dn}^2 t_i - (k')^2}$ ● ☑

Finally, on each cell below, the range of $\sigma_i$ determines $\varepsilon_i = \mathrm{sgn}(\sin(\sigma_i))$, the interval of $t_i$ is read from Table 1 and the values of $\mathrm{sn}\, t_i$ and $\mathrm{cn}\, t_i$ from Table 2, and, substituting the identities ●, the formulas ★ for $\mathrm{sn}\, t_i$ and $\mathrm{cn}\, t_i$ (for $\mathrm{cn}\, t_i$, this is a second, independent verification of the formula derived above) agree with these values:

$p_i q_i \in \mathbb{R}_{>0}$, $\sigma_i < \pi$ → $\varepsilon_i = 1$, $t_i \in (0,\ iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{i \sqrt{\mathrm{dn}^2 t_i - 1}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{\sqrt{\mathrm{dn}^2 t_i - 1 + k^2}}{k}$$ ☑

$p_i q_i \in \mathbb{R}_{>0}$, $\sigma_i > \pi$ → $\varepsilon_i = -1$, $t_i \in (2K,\ 2K + iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{i \sqrt{\mathrm{dn}^2 t_i - 1}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{\sqrt{\mathrm{dn}^2 t_i - 1 + k^2}}{k}$$ ☑

$p_i q_i \in i\,\mathbb{R}_{>0}$, $\sigma_i < \pi$ → $\varepsilon_i = 1$, $t_i \in (K,\ K + iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{\sqrt{1 - \mathrm{dn}^2 t_i}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{i \sqrt{1 - k^2 - \mathrm{dn}^2 t_i}}{k}$$ ☑

$p_i q_i \in i\,\mathbb{R}_{>0}$, $\sigma_i > \pi$ → $\varepsilon_i = -1$, $t_i \in (3K,\ 3K + iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{\sqrt{1 - \mathrm{dn}^2 t_i}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{i \sqrt{1 - k^2 - \mathrm{dn}^2 t_i}}{k}$$ ☑

$p_i q_i \in \mathbb{R}_{<0}$, $\sigma_i < \pi$ → $\varepsilon_i = 1$, $t_i \in (2K,\ 2K + iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{i \sqrt{\mathrm{dn}^2 t_i - 1}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = -\frac{\sqrt{\mathrm{dn}^2 t_i - 1 + k^2}}{k}$$ ☑

$p_i q_i \in \mathbb{R}_{<0}$, $\sigma_i > \pi$ → $\varepsilon_i = -1$, $t_i \in (0,\ iK')$

$$\mathrm{sn}\, t_i = \frac{i\,\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - 1) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{i \sqrt{\mathrm{dn}^2 t_i - 1}}{k}$$ ☑

$$\mathrm{cn}\, t_i = \frac{\varepsilon_i \sqrt{(\mathrm{dn}^2 t_i - (k')^2) p_i^2 q_i^2}}{k\, p_i q_i} = \frac{\sqrt{\mathrm{dn}^2 t_i - 1 + k^2}}{k}$$ ☑

Together, these give the formulas ★.

**Section 3: ALL CLAIMS VERIFIED**

# Section 4

In[ ]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[f, i, j, l, m, c, sgnM, sgnI, sgnL, fi, fj, fl, fm, M, chi, χ, display,
  propEq, standingOf, assumptions, isTrueQ, solvesTo, statusChip, smallChip,
  caseCheck, results, starRef, s4colTest, fj3, fm3, fj4, fm4, okS2, okS3, okS4];
Format[fi, TraditionalForm] = Subscript[f, i];
Format[fl, TraditionalForm] = Subscript[f, l];
Format[chi, TraditionalForm] = χ;
(*Standing hypotheses:M and all four f's are positive,
distinct from 1 and from M,and each pair-product (f-1)(f-M) is positive;
these hold under the given hypotheses.The verification below is branch-wise:
  sign data entering through chi and the regime of M is enumerated explicitly,
so every radical reaching the kernel has a sign-determined
  radicand.Parametrization:the first term of the identity carries zeta1 alone;
in every case the verifying choice is zeta1=chi (the value of chi
      on the branch).All display-only rows are wrapped in HoldForm,
so they cannot evaluate through symbols defined in other sections.*)
standingOf[fjE_, fmE_] := M > 0 && M ≠ 1 && And @@ (# > 0 & /@ {fi, fjE, fl, fmE}) &&
   And @@ (# ≠ 1 & /@ {fi, fjE, fl, fmE}) && And @@ (# ≠ M & /@ {fi, fjE, fl, fmE}) &&
   And @@ ((# - 1) (# - M) > 0 & /@ {fi, fjE, fl, fmE});
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(*Solving a case condition for one unknown:True if Solve
   returns exactly one solution and it equals the expected value*)
solvesTo[eq_, var_, expected_] := Module[{sol = Quiet[Solve[eq, var]]},
   Length[sol] == 1 && Simplify[(var /. First[sol]) - expected] === 0];
display[zeta1_, zeta2_, chiv_, fjE_, fmE_] :=
  zeta1 M Sqrt[(fi - 1) (fjE - 1) (fl - 1) (fmE - 1)] + zeta2 (1 - M) Sqrt[fi fjE fl fmE] -
   zeta2 chiv Sqrt[(fi - M) (fjE - M) (fl - M) (fmE - M)];
propEq[zeta1_, zeta2_, chiv_, fjE_, fmE_] := Style[TraditionalForm[
    HoldForm[#] &@(display[zeta1, zeta2, chiv, fjE, fmE] == M (1 - M))], 16];
s4colTest = RGBColor[0.10, 0.35, 0.75];
starRef = Style["★", Bold, 14, s4colTest];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 15, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 15, Bold, RGBColor[0.82, 0.10, 0.10]];
(*Verification per case:in the cases with a "that is" row,
```

```
the case condition is solved for f_j and f_m
  (check mark beside that row).STAGE 1 (the collapse)
  identifies the instantiated left-hand side with its form
  after resolving the radicals under the branch assumptions;
STAGE 2 verifies that the collapsed form equals M(1-M).The check
 mark beside the instantiated identity certifies both stages;
the check mark beside the resolving row certifies stage 2;
the chip in the title row is Verified only if all checks of the case hold.*)
caseCheck[title_, col_, condRow_, solvedRow_, okSolved_, zeta1_, zeta2_,
    branchRow_, chiv_, fjE_, fmE_, extraAssum_, extraRow_, collapsedHeld_] :=
  Module[{okCollapse, okZero, okStages, ok},
    assumptions = standingOf[fjE, fmE] && extraAssum;
    okCollapse =
     isTrueQ[display[zeta1, zeta2, chiv, fjE, fmE], ReleaseHold[collapsedHeld]];
    okZero = isTrueQ[ReleaseHold[collapsedHeld], M (1 - M)];
    okStages = okCollapse && okZero;
    ok = okSolved && okStages;
    Print[
     Panel[Column[{Row[{Style[title, Bold, 15, col], Spacer[16], Style[condRow,
           13, col], Spacer[26], statusChip[ok]}], If[solvedRow === None,
        Nothing, Row[{Style["that is:   ", 13], Style[solvedRow, 13], "   ",
          smallChip[okSolved], Style[Row[{"   (the condition solved for  ",
             TraditionalForm[HoldForm[Subscript[f, j]]], "  and  ",
             TraditionalForm[HoldForm[Subscript[f, m]]], ",  using  ",
             TraditionalForm[HoldForm[Subscript[f, i]]], ", ",
             TraditionalForm[HoldForm[Subscript[f, l]]], " ≠ 0, 1, M)"}],
           GrayLevel[0.35], 12]}]], Row[{Style["testing  ", 13],
         starRef, Style[Row[{"  with   ", Subscript["ζ", 1], " = ",
            TraditionalForm[zeta1], ",   ", Subscript["ζ", 2], " = ",
            TraditionalForm[zeta2]}], Bold, 13], Spacer[24], branchRow}],
       Row[{propEq[zeta1, zeta2, chiv, fjE, fmE], "   ", smallChip[okStages]}],
       Row[{Style["resolving the radicals:   ", GrayLevel[0.35], 12],
         Style[TraditionalForm[HoldForm[#] &@(collapsedHeld == M (1 - M))], 13],
         "   ", smallChip[okZero]}],
       Style[Row[{"additionally used:   ", extraRow}],
        GrayLevel[0.35], 12]}, Spacings → 1],
     Background → Lighter[col, 0.93], FrameMargins → 12]];
    ok];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
     Style[Row[{TraditionalForm[HoldForm[Subscript[f, i]]], ", ",
        TraditionalForm[HoldForm[M]], " > 0,   ", TraditionalForm[HoldForm[
          Subscript[f, i]]], ", ", TraditionalForm[HoldForm[M]], " ≠ 1,   ",
        TraditionalForm[HoldForm[Subscript[f, i] ≠ M]], ",   ", TraditionalForm[
```

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      HoldForm[(Subscript[f, i] - 1) (Subscript[f, i] - M) > 0]],
     "   for each  ", TraditionalForm[HoldForm[i]],
     " = 1, …, 4,   ", TraditionalForm[HoldForm[χ ∈ {±1}]]}],
   13], "", Style["Given cases:", Bold, 14],
  Style[Row[{"for some distinct  ", TraditionalForm[HoldForm[i]], ", ",
     TraditionalForm[HoldForm[j]], ", ", TraditionalForm[HoldForm[l]],
     ", ", TraditionalForm[HoldForm[m]], " ∈ {1, …, 4},  that is,  ",
     TraditionalForm[HoldForm[{i, j, l, m} == {1, 2, 3, 4}]],
     ",  one of the following conditions holds:"}], 13],
  Style[Row[{"(i)      ", TraditionalForm[HoldForm[
       Subscript[f, i] == Subscript[f, j]]], ",    ", TraditionalForm[
      HoldForm[Subscript[f, l] == Subscript[f, m]]], ",    ", TraditionalForm[
      HoldForm[χ (Subscript[f, i] - 1) (Subscript[f, l] - 1) > 0]]}], 13],
  Style[Row[{"(ii)     ", TraditionalForm[HoldForm[Subscript[f, i]
         Subscript[f, j] == Subscript[f, l] Subscript[f, m] == M]]}], 13],
  Style[Row[{"(iii)    ", TraditionalForm[HoldForm[Subscript[f, i] +
         Subscript[f, j] - Subscript[f, i] Subscript[f, j] == Subscript[f, l] +
         Subscript[f, m] - Subscript[f, l] Subscript[f, m] == M]],
     ",    ", TraditionalForm[HoldForm[χ (1 - M) > 0]]}], 13],
  Style[Row[{"(iv)     ", TraditionalForm[HoldForm[
       (Subscript[f, i] + Subscript[f, j] - 1) / (Subscript[f, i]
           Subscript[f, j]) == (Subscript[f, l] + Subscript[f, m] - 1) /
         (Subscript[f, l] Subscript[f, m]) == 1 / M]]}], 13],
  "", Style[Row[{"Each  ", smallChip[True],
     "  below certifies that the statement displayed beside it has been
       verified by a symbolic computation at that point;  a  ",
     smallChip[False], "  would mean that this verification failed."}],
   12, GrayLevel[0.35]], "", Style[
   Row[{Style["We want to test the following identity  ", Bold, 14],
     Style["⋆", 14, s4colTest],
     Style[Row[{" ,  for some  ", Subscript["ς", 1], ", ", Subscript["ς", 2],
        " ∈ {±1},  in each of the given cases:"}], Bold, 14]}]],
  Style[RawBoxes[FormBox[RowBox[{SubscriptBox["ς", "1"], "", "M", SqrtBox[
        RowBox[{UnderoverscriptBox["∏", RowBox[{"i", "=", "1"}], "4",
           LimitsPositioning → False], RowBox[{"(", RowBox[{SubscriptBox[
               "f", "i"], "-", "1"}], ")"}]}]]], "+", SubscriptBox["ς", "2"],
      "", RowBox[{"(", RowBox[{"1", "-", "M"}], ")"}], SqrtBox[
       RowBox[{UnderoverscriptBox["∏", RowBox[{"i", "=", "1"}], "4",
           LimitsPositioning → False], SubscriptBox["f", "i"]}]], "-",
      SubscriptBox["ς", "2"], "", "χ", SqrtBox[RowBox[{UnderoverscriptBox[
           "∏", RowBox[{"i", "=", "1"}], "4", LimitsPositioning → False],
          RowBox[{"(", RowBox[{SubscriptBox["f", "i"], "-", "M"}], ")"}]}]]],
      "=", RowBox[{"M", RowBox[{"(", RowBox[{"1", "-", "M"}], ")"}]}]}]],
     TraditionalForm]], 16, s4colTest]}, Spacings → 1.1],
Background → RGBColor[0.99, 0.96, 0.88], FrameMargins →
 14]];
```

```
(* ============================================================*)
(*PART 3:CASE-BY-CASE VERIFICATION*)
(* ============================================================*)
results = {};
(*Condition (i):fj=fi,fm=fl;
chi (fi-1)(fl-1)>0;
zeta1=chi=sgnI*sgnL per branch,
zeta2=1. The four branches enumerate the signs of fi-1 and fl-1;
by the standing pair-products,fi-M and fi-1 have the same sign,
and similarly at l,so chi=sgnI*sgnL.*)
Do[AppendTo[results, caseCheck["Condition (i):", RGBColor[0.10, 0.30, 0.70],
    Row[{TraditionalForm[HoldForm[Subscript[f, i] == Subscript[f, j]]],
      ",   ", TraditionalForm[HoldForm[Subscript[f, l] == Subscript[f, m]]],
      ",   ", TraditionalForm[
       HoldForm[χ (Subscript[f, i] - 1) (Subscript[f, l] - 1) > 0]]}],
    None, True, sgnI sgnL, 1, Style[Row[{"branch:  sign(",
       TraditionalForm[HoldForm[fi - 1]], ") = ", sgnI, ",  sign(",
       TraditionalForm[HoldForm[fl - 1]], ") = ", sgnL, "  ⟶  χ = ",
       sgnI sgnL, ",  so  \!\(\*SubscriptBox[\(ℊ\), \(1\)]\) = χ"}],
     13, GrayLevel[0.3]], sgnI sgnL, fi, fl, sgnI (fi - 1) > 0 &&
     sgnI (fi - M) > 0 && sgnL (fl - 1) > 0 && sgnL (fl - M) > 0,
    Row[{TraditionalForm[HoldForm[#] &@(sgnI (fi - 1) > 0)], ",  ",
      TraditionalForm[HoldForm[#] &@(sgnI (fi - M) > 0)],
      ",  ", TraditionalForm[HoldForm[#] &@(sgnL (fl - 1) > 0)],
      ",  ", TraditionalForm[HoldForm[#] &@(sgnL (fl - M) > 0)],
      "   (same-sign facts by the given hypotheses)"}],
    HoldForm[M (fi - 1) (fl - 1) + (1 - M) fi fl - (fi - M) (fl - M)]]],
  {sgnI, {1, -1}}, {sgnL, {1, -1}}];

(*Condition (ii):fi*fj=fl*fm=M,that is,fj=M/fi,fm=M/fl;
zeta1=chi (symbolic),zeta2=1;
no sign condition is consumed and the two chi-
 dressed terms cancel syntactically.*)
okS2 = Module[{xj, xm},
   solvesTo[fi xj == M, xj, M / fi] && solvesTo[fl xm == M, xm, M / fl]];
AppendTo[results, caseCheck["Condition (ii):",
   RGBColor[0.55, 0.25, 0.70], TraditionalForm[HoldForm[
     Subscript[f, i] Subscript[f, j] == Subscript[f, l] Subscript[f, m] == M]],
   Row[{TraditionalForm[HoldForm[Subscript[f, j] == M / Subscript[f, i]]],
     ",   ", TraditionalForm[
      HoldForm[Subscript[f, m] == M / Subscript[f, l]]]}], okS2, chi, 1, Style[
    Row[{"single symbolic branch,  \!\(\*SubscriptBox[\(ℊ\), \(1\)]\) = χ"}],
    13, GrayLevel[0.3]], chi, M / fi, M / fl, True,
   "none  (χ remains symbolic; no sign condition is consumed)",
   HoldForm[chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)] +
     (1 - M) M - chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)]]]];
```

```
(*Condition (iii):fi+fj-fi*fj=fl+fm-fl*fm=M,
that is,fj=(M-fi)/(1-fi),fm=(M-fl)/(1-fl);
chi (1-M)>0;
zeta1=chi=c per branch,zeta2=1. Two branches:chi=+1 with M<1,
and chi=-1 with M>1.*)
fj3 = (M - fi) / (1 - fi); fm3 = (M - fl) / (1 - fl);
okS3 = Module[{xj, xm},
   solvesTo[fi + xj - fi xj == M, xj, fj3] && solvesTo[fl + xm - fl xm == M, xm, fm3]];
Do[AppendTo[results, caseCheck["Condition (iii):",
    RGBColor[0.00, 0.50, 0.50], Row[{TraditionalForm[HoldForm[
        Subscript[f, i] + Subscript[f, j] - Subscript[f, i] Subscript[f, j] ==
         Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] == M]],
      ",   ", TraditionalForm[HoldForm[χ (1 - M) > 0]]}], Row[{TraditionalForm[
       HoldForm[Subscript[f, j] == (M - Subscript[f, i]) / (1 - Subscript[f, i])]],
      ",   ", TraditionalForm[HoldForm[Subscript[f, m] ==
          (M - Subscript[f, l]) / (1 - Subscript[f, l])]]}], okS3, c, 1,
    Style[Row[{"branch:  χ = ", c, ",  ", If[c == 1, "M < 1", "M > 1"],
       ",  so  \!\(\*SubscriptBox[\(ℱ\), \(1\)]\) = χ"}],
     13, GrayLevel[0.3]], c, fj3, fm3, If[c == 1, M < 1, M > 1],
    Row[{TraditionalForm[HoldForm[#] &@If[c == 1, M < 1, M > 1]],
      "   (this branch of  ", TraditionalForm[HoldForm[χ (1 - M) > 0]], ")"}],
    HoldForm[M (1 - M) + (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))] -
      (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))]]]], {c, {1, -1}}];

(*Condition (iv):(fi+fj-1)/(fi*fj)=(fl+fm-1)/(fl*fm)=1/M,
that is,fj=M(fi-1)/(fi-M),fm=M(fl-1)/(fl-M);
zeta1=chi=c per branch,and zeta2=-1 if chi(1-M)>0,
zeta2=+1 if chi(1-M)<0,encoded as zeta2=
 -c*sgnM with sgnM=sign(1-M).Four branches over chi and the regime of M;
the collapse row uses per-branch literal templates
  so the cancelling pair of radicals is shown.*)
fj4 = M (fi - 1) / (fi - M); fm4 = M (fl - 1) / (fl - M);
okS4 = Module[{xj, xm}, solvesTo[(fi + xj - 1) / (fi xj) == 1 / M, xj, fj4] &&
    solvesTo[(fl + xm - 1) / (fl xm) == 1 / M, xm, fm4]];
Do[AppendTo[results, caseCheck["Condition (iv):",
    RGBColor[0.75, 0.35, 0.05], TraditionalForm[HoldForm[
      (Subscript[f, i] + Subscript[f, j] - 1) / (Subscript[f, i] Subscript[f, j]) ==
       (Subscript[f, l] + Subscript[f, m] - 1) / (Subscript[f, l] Subscript[f, m]) ==
       1 / M]], Row[{TraditionalForm[
       HoldForm[Subscript[f, j] == M (Subscript[f, i] - 1) / (Subscript[f, i] - M)]],
      ",   ", TraditionalForm[HoldForm[Subscript[f, m] ==
         M (Subscript[f, l] - 1) / (Subscript[f, l] - M)]]}], okS4, c, -c sgnM,
    Style[Row[{"branch:  χ = ", c, ",  ", If[sgnM == 1, "M < 1", "M > 1"],
       "  ⟶  ", If[c sgnM == 1, "χ(1-M) > 0", "χ(1-M) < 0"],
       ",  \!\(\*SubscriptBox[\(ℱ\), \(1\)]\) = χ"}],
     13, GrayLevel[0.3]], c, fj4, fm4, If[sgnM == 1, M < 1, M > 1],
    Row[{TraditionalForm[HoldForm[#] &@If[sgnM == 1, M < 1, M > 1]],
```

```
      "   (this branch of the rule:  ", Subscript["ζ", 2], " = -1 if ",
       TraditionalForm[HoldForm[χ (1 - M) > 0]], ",    ", Subscript["ζ", 2],
       " = +1 if ", TraditionalForm[HoldForm[χ (1 - M) < 0]], ")"}], If[c sgnM == 1,
      HoldForm[(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] -
        (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] + M (1 - M)],
      HoldForm[- (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
        (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
        M (1 - M)]]], {c, {1, -1}}, {sgnM, {1, -1}}];

(* ============================================================*)
(*PART 4:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ results, "Section 4: ALL CLAIMS VERIFIED",
     "Section 4: SOME CLAIMS FAILED"], Bold, White, 16], Background →
    If[And @@ results, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
   FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses:

$f_i, M > 0, \quad f_i, M \neq 1, \quad f_i \neq M, \quad (f_i - 1)(f_i - M) > 0$ for each $i = 1, \ldots, 4, \quad \chi \in \{\pm 1\}$

Given cases:

for some distinct $i, j, l, m \in \{1, \ldots, 4\}$, that is,
$\{i, j, l, m\} = \{1, 2, 3, 4\}$, one of the following conditions holds:

(i) $f_i = f_j, \quad f_l = f_m, \quad \chi (f_i - 1)(f_l - 1) > 0$

(ii) $f_i f_j = f_l f_m = M$

(iii) $f_i + f_j - f_i f_j = f_l + f_m - f_l f_m = M, \quad \chi (1 - M) > 0$

(iv) $\frac{f_i + f_j - 1}{f_i f_j} = \frac{f_l + f_m - 1}{f_l f_m} = \frac{1}{M}$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following identity ★, for some $\zeta_1, \zeta_2 \in \{\pm 1\}$, in each of the given cases:

$$\zeta_1 M \sqrt{\prod_{i=1}^{4} (f_i - 1)} + \zeta_2 (1 - M) \sqrt{\prod_{i=1}^{4} f_i} - \zeta_2 \chi \sqrt{\prod_{i=1}^{4} (f_i - M)} = M (1 - M)$$

Condition (i): $f_i = f_j,\ f_l = f_m,\ \chi\,(f_i-1)\,(f_l-1) > 0$ Verified

testing ★ with $\zeta_1 = 1,\ \zeta_2 = 1$
branch: sign $(f_i-1) = 1$, sign $(f_l-1) = 1 \longrightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2\,f_l^2}\,(1-M) - \sqrt{(f_i-M)^2\,(f_l-M)^2} + \sqrt{(-1+f_i)^2\,(-1+f_l)^2}\,M = (1-M)\,M \quad ☑$$

resolving the radicals: $M\,(f_i-1)\,(f_l-1) + (1-M)\,f_i\,f_l - (f_i-M)\,(f_l-M) = (1-M)\,M$ ☑

additionally used:
$-1+f_i > 0,\ f_i - M > 0,\ -1+f_l > 0,\ f_l - M > 0$ (same −sign facts by the given hypotheses )

Condition (i): $f_i = f_j,\ f_l = f_m,\ \chi\,(f_i-1)\,(f_l-1) > 0$ Verified

testing ★ with $\zeta_1 = -1,\ \zeta_2 = 1$
branch: sign $(f_i-1) = 1$, sign $(f_l-1) = -1 \longrightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2\,f_l^2}\,(1-M) + \sqrt{(f_i-M)^2\,(f_l-M)^2} - \sqrt{(-1+f_i)^2\,(-1+f_l)^2}\,M = (1-M)\,M \quad ☑$$

resolving the radicals: $M\,(f_i-1)\,(f_l-1) + (1-M)\,f_i\,f_l - (f_i-M)\,(f_l-M) = (1-M)\,M$ ☑

additionally used:
$-1+f_i > 0,\ f_i - M > 0,\ 1-f_l > 0,\ -f_l + M > 0$ (same −sign facts by the given hypotheses )

Condition (i): $f_i = f_j,\ f_l = f_m,\ \chi\,(f_i-1)\,(f_l-1) > 0$ Verified

testing ★ with $\zeta_1 = -1,\ \zeta_2 = 1$
branch: sign $(f_i-1) = -1$, sign $(f_l-1) = 1 \longrightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2\,f_l^2}\,(1-M) + \sqrt{(f_i-M)^2\,(f_l-M)^2} - \sqrt{(-1+f_i)^2\,(-1+f_l)^2}\,M = (1-M)\,M \quad ☑$$

resolving the radicals: $M\,(f_i-1)\,(f_l-1) + (1-M)\,f_i\,f_l - (f_i-M)\,(f_l-M) = (1-M)\,M$ ☑

additionally used:
$1-f_i > 0,\ -f_i + M > 0,\ -1+f_l > 0,\ f_l - M > 0$ (same −sign facts by the given hypotheses )

Condition (i): $f_i = f_j,\ f_l = f_m,\ \chi\,(f_i-1)\,(f_l-1) > 0$ Verified

testing ★ with $\zeta_1 = 1,\ \zeta_2 = 1$
branch: sign $(f_i-1) = -1$, sign $(f_l-1) = -1 \longrightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2\,f_l^2}\,(1-M) - \sqrt{(f_i-M)^2\,(f_l-M)^2} + \sqrt{(-1+f_i)^2\,(-1+f_l)^2}\,M = (1-M)\,M \quad ☑$$

resolving the radicals: $M\,(f_i-1)\,(f_l-1) + (1-M)\,f_i\,f_l - (f_i-M)\,(f_l-M) = (1-M)\,M$ ☑

additionally used:
$1-f_i > 0,\ -f_i + M > 0,\ 1-f_l > 0,\ -f_l + M > 0$ (same −sign facts by the given hypotheses )

### Condition (ii): $f_i f_j = f_l f_m = M$ — Verified

that is: $f_j = \frac{M}{f_i}$, $f_m = \frac{M}{f_l}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = \chi$, $\zeta_2 = 1$ single symbolic branch, $\zeta_1 = \chi$

$$(1-M)\sqrt{M^2} + \chi M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{M}{f_i}\right)\left(-1+\frac{M}{f_l}\right)} - \chi \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{M}{f_i}\right)\left(-M+\frac{M}{f_l}\right)} = (1-M)M$$ ☑

resolving the radicals: $\chi M \sqrt{(f_i-1)\left(\frac{M}{f_i}-1\right)(f_l-1)\left(\frac{M}{f_l}-1\right)} + (1-M)M - \chi M \sqrt{(f_i-1)\left(\frac{M}{f_i}-1\right)(f_l-1)\left(\frac{M}{f_l}-1\right)} = (1-M)M$ ☑

additionally used: none ($\chi$ remains symbolic; no sign condition is consumed)

### Condition (iii): $f_i + f_j - f_i f_j = f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$ — Verified

that is: $f_j = \frac{M-f_i}{1-f_i}$, $f_m = \frac{M-f_l}{1-f_l}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M < 1$, so $\zeta_1 = \chi$

$$(1-M)\sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} + M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} - \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$ ☑

resolving the radicals: $M(1-M) + (1-M)\sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M)\sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$ ☑

additionally used: $M < 1$ (this branch of $\chi(1-M) > 0$)

### Condition (iii): $f_i + f_j - f_i f_j = f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$ — Verified

that is: $f_j = \frac{M-f_i}{1-f_i}$, $f_m = \frac{M-f_l}{1-f_l}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M > 1$, so $\zeta_1 = \chi$

$$(1-M)\sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} - M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$ ☑

resolving the radicals: $M(1-M) + (1-M)\sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M)\sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$ ☑

additionally used: $M > 1$ (this branch of $\chi(1-M) > 0$)

Condition (iv): $\frac{f_i+f_j-1}{f_i f_j} = \frac{f_l+f_m-1}{f_l f_m} = \frac{1}{M}$ Verified

that is: $f_j = \frac{M(f_i-1)}{f_i-M}$, $f_m = \frac{M(f_l-1)}{f_l-M}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = 1$, $M < 1 \longrightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M)\sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}}\right) + M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$ ☑

resolving the radicals:

$$(1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$ ☑

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $\frac{f_i+f_j-1}{f_i f_j} = \frac{f_l+f_m-1}{f_l f_m} = \frac{1}{M}$ Verified

that is: $f_j = \frac{M(f_i-1)}{f_i-M}$, $f_m = \frac{M(f_l-1)}{f_l-M}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M > 1 \longrightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M)\sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}} + M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} - \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$ ☑

resolving the radicals:

$$-(1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$ ☑

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $\frac{f_i+f_j-1}{f_i f_j} = \frac{f_l+f_m-1}{f_l f_m} = \frac{1}{M}$ Verified

that is: $f_j = \frac{M(f_i-1)}{f_i-M}$, $f_m = \frac{M(f_l-1)}{f_l-M}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M < 1 \longrightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M)\sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}} - M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$ ☑

resolving the radicals:

$$-(1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$ ☑

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $\frac{f_i+f_j-1}{f_i f_j} = \frac{f_l+f_m-1}{f_l f_m} = \frac{1}{M}$ Verified

that is: $f_j = \frac{M(f_i-1)}{f_i-M}$, $f_m = \frac{M(f_l-1)}{f_l-M}$ ☑ (the condition solved for $f_j$ and $f_m$, using $f_i, f_l \neq 0, 1, M$)

testing ★ with $\zeta_1 = -1$, $\zeta_2 = -1$ branch: $\chi = -1$, $M > 1 \longrightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M)\sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}}\right) - M\sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} - \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$ ☑

resolving the radicals:

$$(1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M)\sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$ ☑

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

**Section 4: ALL CLAIMS VERIFIED**

# Section 5

In[ ]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[gi, gj, kk, kq, kk2, kp2, SS, DD, sig, em, ev, snD, cnD, dnD, S2, C2, D2, dd,
  chkR, statusChip, smallChip, s9colTest, s9colHdr, shiftIK, shift2mK, starOK,
  dupOK1, dupOK2, dupOK3, relOK1, relOK2, finOK1, finOK2, finOK3, okAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
```

```
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
    Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s9colTest = RGBColor[0.10, 0.35, 0.75];
s9colHdr = RGBColor[0.15, 0.35, 0.45];
(*Every check below is a rational identity in a fixed set of atoms,
decided by Together;
no simplification search is involved.Atoms:gi and gj are free;
sig stands for sn t_i and appears
 only through even powers in the final checks;
SS is the solved value of sn^2 t_i;
kk2 and kp2 are the resulting values of k^2 and k'^2;
kq is a generic symbol for k^2 in the duplication checks;
S2,C2,D2 stand for the values of sn,cn,dn at 2 t_i in the shift step;
em stands for (-1)^m and enters every final identity linearly,
so a real symbol suffices;
in the shift step,
(-1)^(-m)=(-1)^m is checked for both values ev=1 and ev=-1 of (-1)^m;
kk stands for k itself and appears only to the first power in denominators,
where it cancels.Divisions occurring in the displayed identities are by k,
g_i,g_j,sn(2 t_i),and 1-k^2 sn^4 t_i,all nonzero by the given
  hypotheses.All display-only rows are wrapped in HoldForm.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
(*shift step:first v→v-iK',then a shift by 2(-m)K*)
shiftIK[{snV_, cnV_, dnV_}] := {1 / (kk snV), I dnV / (kk snV), I cnV / snV};
shift2mK[{snV_, cnV_, dnV_}, e_] := {e snV, e cnV, dnV};
starOK =
  And @@ Table[And @@ MapThread[chkR, {shift2mK[shiftIK[{S2, C2, D2}], 1 / ev],
       {ev / (kk S2), ev I D2 / (kk S2), I C2 / S2}}], {ev, {1, -1}}];
(*duplication step*)
dd = 1 - kq sig^4;
dupOK1 = chkR[2 sig (-I gj sig) gi / dd, -2 I gi gj sig^2 / dd];
dupOK2 = chkR[((-I gj sig)^2 - sig^2 gi^2) / dd, -(gi^2 + gj^2) sig^2 / dd];
dupOK3 = chkR[(gi^2 - kq sig^2 (-I gj sig)^2) / dd, (gi^2 + kq gj^2 sig^4) / dd];
(*relations step*)
SS = 1 / (1 - gj^2);
kk2 = (1 - gi^2) (1 - gj^2);
kp2 = 1 - kk2;
relOK1 = chkR[SS + (-I gj)^2 SS, 1];
relOK2 = chkR[1 - kk2 SS, gi^2];
(*final step*)
DD = 1 - kk2 SS^2;
snD = -2 I gi gj SS / DD;
cnD = -(gi^2 + gj^2) SS / DD;
```

```
dnD = (gi^2 + kk2 gj^2 SS^2) / DD;
finOK1 = chkR[em / (kk snD), I em (gi^2 - gj^2) / (2 kk gi gj)];
finOK2 = chkR[em I dnD / (kk snD), em (gi^2 gj^2 - kp2) / (2 kk gi gj)];
finOK3 = chkR[I cnD / snD, (gi^2 + gj^2) / (2 gi gj)];
okAll = starOK && dupOK1 && dupOK2 &&
    dupOK3 && relOK1 && relOK2 && finOK1 && finOK2 && finOK3;

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
      Style[Row[{"k ∈ (0, 1),   ", TraditionalForm[HoldForm[k'^2 == 1 - k^2]]}],
       14], Style[Row[{"For  ", TraditionalForm[HoldForm[i]], ", ",
          TraditionalForm[HoldForm[j]], " ∈ {1, …, 4} :"}], Bold, 14],
      Style[Row[{TraditionalForm[HoldForm[m ∈ ℤ]], ",   ", TraditionalForm[
           HoldForm[Subscript[t, i] + Subscript[t, j] == 2 m K + I K']]}], 14], Style[
       Row[{TraditionalForm[HoldForm[Subscript[g, i] == dn[Subscript[t, i]]]],
          ", ", TraditionalForm[HoldForm[Subscript[g, j] == dn[Subscript[t, j]]]],
          ",   ", TraditionalForm[HoldForm[Subscript[g, i]]],
          ", ", TraditionalForm[HoldForm[Subscript[g, j]]],
          " ≠ 0,   ", TraditionalForm[HoldForm[
            cn[Subscript[t, i]] == -I Subscript[g, j] sn[Subscript[t, i]]]]}], 14],
      Style[Row[{TraditionalForm[HoldForm[Subscript[t, i]]], "  and  ",
          TraditionalForm[HoldForm[2 Subscript[t, i]]], "  are not poles,   ",
          TraditionalForm[HoldForm[sn[2 Subscript[t, i]] ≠ 0]], ",   ",
          TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, i]]^4 ≠ 0]]}], 14],
      "", Style["Given formulas:", Bold, 14], Style[Row[{TraditionalForm[
           HoldForm[{sn, cn, dn}[v + 2 m K] == {(-1)^m sn[v], (-1)^m cn[v], dn[v]}]],
          ",   ", TraditionalForm[HoldForm[{sn, cn, dn}[v - I K'] ==
             {1 / (k sn[v]), I dn[v] / (k sn[v]), I cn[v] / sn[v]}]], "    ",
          Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],
       14], Style[Row[{TraditionalForm[HoldForm[sn[2 v] ==
            2 sn[v] cn[v] dn[v] / (1 - k^2 sn[v]^4)]], ",   ", TraditionalForm[
           HoldForm[cn[2 v] == (cn[v]^2 - sn[v]^2 dn[v]^2) / (1 - k^2 sn[v]^4)]],
          ",   ", TraditionalForm[HoldForm[
            dn[2 v] == (dn[v]^2 - k^2 sn[v]^2 cn[v]^2) / (1 - k^2 sn[v]^4)]],
          "    ", Style["the duplication formulas", Italic, 13,
           RGBColor[0.10, 0.30, 0.70]]}], 14],
      Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 == 1]],
          ",   ", TraditionalForm[HoldForm[dn[v]^2 == 1 - k^2 sn[v]^2]],
          "    ", Style["the Pythagorean relations",
           Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
       14], Style[Row[{"Each  ", smallChip[True],
          "  below certifies that the statement displayed beside it has been
            verified by a symbolic computation at that point;  a  ",
          smallChip[False], "  would mean that this verification failed."}],
       12, GrayLevel[0.35]], "", Style[
```

```
      Row[{Style["We want to test the following formula  ", Bold, 14],
        Style["⋆", 14, s9colTest], Style[" :", Bold, 14]}]], Style[
      TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[t, i] - Subscript[t, j]] ==
          {I (-1)^m (Subsuperscript[g, i, 2] - Subsuperscript[g, j, 2]) /
             (2 k Subscript[g, i] Subscript[g, j]),
           (-1)^m (Subsuperscript[g, i, 2] Subsuperscript[g, j, 2] - k'^2) /
             (2 k Subscript[g, i] Subscript[g, j]),
           (Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2]) /
            (2 Subscript[g, i] Subscript[g, j])}]], 15, s9colTest]},
    Spacings → 1.1], Background → RGBColor[0.99, 0.96, 0.88],
   FrameMargins →
    14]];

(* ============================================================*)
(*PART 3:VERIFICATION*)
(* ============================================================*)
Print[Panel[Column[
    {Row[{Style["The derivation:  the formula for  ", Bold, 15, s9colHdr],
       Style[TraditionalForm[HoldForm[{sn, cn, dn}[
           Subscript[t, i] - Subscript[t, j]]]], Bold, 15, s9colHdr], Spacer[20],
       statusChip[okAll]}], Style["Verification:", Bold, 13], Style[
      Row[{"By  ", TraditionalForm[HoldForm[Subscript[t, i] - Subscript[t, j] ==
           2 Subscript[t, i] - (Subscript[t, i] + Subscript[t, j]) ==
           -2 m K + (2 Subscript[t, i] - I K')]], "  and  ",
        Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]],
        ",  at  ", TraditionalForm[HoldForm[v == 2 Subscript[t, i]]], " :"}],
      13], Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Subscript[t, i] -
            Subscript[t, j]] == {(-1)^m / (k sn[2 Subscript[t, i]]),
           (-1)^m I dn[2 Subscript[t, i]] / (k sn[2 Subscript[t, i]]),
           I cn[2 Subscript[t, i]] / sn[2 Subscript[t, i]]}]],
        "      ", Style["★", 15, RGBColor[0.10, 0.30, 0.70]],
        "   ", smallChip[starOK]}], 14], Style[
      Row[{"Substituting  ", TraditionalForm[HoldForm[cn[Subscript[t, i]] ==
           -I Subscript[g, j] sn[Subscript[t, i]]]], "  and  ",
        TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
        "  into  ", Style["the duplication formulas",
         Italic, 13, RGBColor[0.10, 0.30, 0.70]], ",  at  ",
        TraditionalForm[HoldForm[v == Subscript[t, i]]], ",  we get"}], 13],
     Style[Row[{TraditionalForm[HoldForm[sn[2 Subscript[t, i]] ==
           -2 I Subscript[g, i] Subscript[g, j]
            sn[Subscript[t, i]]^2 / (1 - k^2 sn[Subscript[t, i]]^4)]],
        "      ", Style["●", 14, RGBColor[0.55, 0.33, 0.10]],
        "   ", smallChip[dupOK1]}], 13],
     Style[Row[{TraditionalForm[HoldForm[cn[2 Subscript[t, i]] ==
           -(Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2])
            sn[Subscript[t, i]]^2 / (1 - k^2 sn[Subscript[t, i]]^4)]],
        "      ", Style["●", 14, RGBColor[0.55, 0.33, 0.10]],
```

```
        "   ", smallChip[dupOK2]}], 13],
      Style[Row[{TraditionalForm[HoldForm[dn[2 Subscript[t, i]] ==
          (Subsuperscript[g, i, 2] + k^2 Subsuperscript[g, j, 2]
              sn[Subscript[t, i]]^4) / (1 - k^2 sn[Subscript[t, i]]^4)]],
        "      ", Style["●", 14, RGBColor[0.55, 0.33, 0.10]], "   ",
        smallChip[dupOK3]}], 13],
      Style[Row[{"By  ", TraditionalForm[HoldForm[cn[Subscript[t, i]] ==
          -I Subscript[g, j] sn[Subscript[t, i]]]], ",   ",
        TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
        ",  and  ", Style["the Pythagorean relations",
         Italic, 13, RGBColor[0.75, 0.35, 0.05]], " :"}], 13],
      Style[Row[{TraditionalForm[HoldForm[sn[Subscript[t, i]]^2 ==
          1 / (1 - Subsuperscript[g, j, 2])]], "      ",
        Style["●", 14, RGBColor[0.00, 0.45, 0.25]], "   ", smallChip[relOK1]}],
       13], Style[Row[{TraditionalForm[HoldForm[
          k^2 == (1 - Subsuperscript[g, i, 2]) (1 - Subsuperscript[g, j, 2])]],
        "      ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]],
        "   ", smallChip[relOK2]}], 13],
      Style[Row[{"Substituting the identities  ",
        Style["●", 13, RGBColor[0.55, 0.33, 0.10]], "  and  ",
        Style["●", 13, RGBColor[0.00, 0.45, 0.25]], "  and the equality  ",
        TraditionalForm[HoldForm[k'^2 == 1 - k^2]], "  into  ",
        Style["★", 13, RGBColor[0.10, 0.30, 0.70]], " ,  we get"}], 13],
      Style[Row[{TraditionalForm[HoldForm[(-1)^m / (k sn[2 Subscript[t, i]]) ==
          I (-1)^m (Subsuperscript[g, i, 2] - Subsuperscript[g, j, 2]) /
            (2 k Subscript[g, i] Subscript[g, j])]],
        "   ", smallChip[finOK1]}], 14], Style[Row[{TraditionalForm[
         HoldForm[(-1)^m I dn[2 Subscript[t, i]] / (k sn[2 Subscript[t, i]]) ==
          (-1)^m (Subsuperscript[g, i, 2] Subsuperscript[g, j, 2] - k'^2) /
            (2 k Subscript[g, i] Subscript[g, j])]],
        "   ", smallChip[finOK2]}], 14], Style[Row[{TraditionalForm[
         HoldForm[I cn[2 Subscript[t, i]] / sn[2 Subscript[t, i]] ==
          (Subsuperscript[g, i, 2] + Subsuperscript[g, j, 2]) /
           (2 Subscript[g, i] Subscript[g, j])]], "   ", smallChip[finOK3]}],
       14], Style[Row[{"Together, these give the formula  ",
        Style["★", 13, s9colTest], " ."}], 13]},
     Spacings → 1], Background → Lighter[s9colHdr, 0.93],
    FrameMargins →
     12]];
Print[Framed[Style[If[okAll, "Section 5: ALL CLAIMS VERIFIED",
     "Section 5: SOME CLAIMS FAILED"], Bold, White, 16], Background →
    If[okAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
   FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

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Given hypotheses:

$k \in (0, 1), \quad (k')^2 = 1 - k^2$

For $i, j \in \{1, \ldots, 4\}$:

$m \in \mathbb{Z}, \quad t_i + t_j = 2\,m\,K + i\,K'$

$g_i = \mathrm{dn}(t_i),\ g_j = \mathrm{dn}(t_j), \quad g_i, g_j \neq 0, \quad \mathrm{cn}(t_i) = -i\,g_j\,\mathrm{sn}(t_i)$

$t_i$ and $2\,t_i$ are not poles, $\mathrm{sn}(2\,t_i) \neq 0, \quad 1 - k^2\,\mathrm{sn}(t_i)^4 \neq 0$

Given formulas:

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(v + 2\,m\,K) = \{(-1)^m\,\mathrm{sn}(v),\ (-1)^m\,\mathrm{cn}(v),\ \mathrm{dn}(v)\}$

$, \quad \{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(v - i\,K') = \left\{\frac{1}{k\,\mathrm{sn}(v)},\ \frac{i\,\mathrm{dn}(v)}{k\,\mathrm{sn}(v)},\ \frac{i\,\mathrm{cn}(v)}{\mathrm{sn}(v)}\right\}$ *the shift formulas*

$\mathrm{sn}(2\,v) = \frac{2\,\mathrm{sn}(v)\,\mathrm{cn}(v)\,\mathrm{dn}(v)}{1 - k^2\,\mathrm{sn}(v)^4}, \quad \mathrm{cn}(2\,v) = \frac{\mathrm{cn}(v)^2 - \mathrm{sn}(v)^2\,\mathrm{dn}(v)^2}{1 - k^2\,\mathrm{sn}(v)^4}$

$, \quad \mathrm{dn}(2\,v) = \frac{\mathrm{dn}(v)^2 - k^2\,\mathrm{sn}(v)^2\,\mathrm{cn}(v)^2}{1 - k^2\,\mathrm{sn}(v)^4}$ *the duplication formulas*

$\mathrm{sn}(v)^2 + \mathrm{cn}(v)^2 = 1, \quad \mathrm{dn}(v)^2 = 1 - k^2\,\mathrm{sn}(v)^2$ *the Pythagorean relations*

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following formula ★ :

$$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(t_i - t_j) = \left\{\frac{i\,(-1)^m\,\left(g_i^2 - g_j^2\right)}{2\,k\,g_i\,g_j},\ \frac{(-1)^m\,\left(g_i^2\,g_j^2 - (k')^2\right)}{2\,k\,g_i\,g_j},\ \frac{g_i^2 + g_j^2}{2\,g_i\,g_j}\right\}$$

The derivation: the formula for $\{\text{sn, cn, dn}\}(t_i - t_j)$ Verified

Verification:

By $t_i - t_j = 2\,t_i - (t_i + t_j) = -2\,m\,K + (2\,t_i - i\,K')$ and *the shift formulas* , at $v = 2\,t_i$ :

$$\{\text{sn, cn, dn}\}(t_i - t_j) = \left\{\frac{(-1)^m}{k\,\text{sn}(2\,t_i)},\ \frac{(-1)^m\,i\,\text{dn}(2\,t_i)}{k\,\text{sn}(2\,t_i)},\ \frac{i\,\text{cn}(2\,t_i)}{\text{sn}(2\,t_i)}\right\}$$ ★ ☑

Substituting $\text{cn}(t_i) = -i\,g_j\,\text{sn}(t_i)$ and $\text{dn}(t_i) = g_i$ into *the duplication formulas* , at $v = t_i$, we get

$$\text{sn}(2\,t_i) = -\frac{2\,i\,g_i\,g_j\,\text{sn}(t_i)^2}{1 - k^2\,\text{sn}(t_i)^4}$$ ● ☑

$$\text{cn}(2\,t_i) = -\frac{\left(g_i^2 + g_j^2\right)\text{sn}(t_i)^2}{1 - k^2\,\text{sn}(t_i)^4}$$ ● ☑

$$\text{dn}(2\,t_i) = \frac{g_i^2 + k^2\,g_j^2\,\text{sn}(t_i)^4}{1 - k^2\,\text{sn}(t_i)^4}$$ ● ☑

By $\text{cn}(t_i) = -i\,g_j\,\text{sn}(t_i)$, $\text{dn}(t_i) = g_i$, and *the Pythagorean relations* :

$$\text{sn}(t_i)^2 = \frac{1}{1 - g_j^2}$$ ● ☑

$$k^2 = (1 - g_i^2)\left(1 - g_j^2\right)$$ ● ☑

Substituting the identities ● and ● and the equality $(k')^2 = 1 - k^2$ into ★ , we get

$$\frac{(-1)^m}{k\,\text{sn}(2\,t_i)} = \frac{i\,(-1)^m\left(g_i^2 - g_j^2\right)}{2\,k\,g_i\,g_j}$$ ☑

$$\frac{(-1)^m\,i\,\text{dn}(2\,t_i)}{k\,\text{sn}(2\,t_i)} = \frac{(-1)^m\left(g_i^2\,g_j^2 - (k')^2\right)}{2\,k\,g_i\,g_j}$$ ☑

$$\frac{i\,\text{cn}(2\,t_i)}{\text{sn}(2\,t_i)} = \frac{g_i^2 + g_j^2}{2\,g_i\,g_j}$$ ☑

Together, these give the formula ★ .

**Section 5: ALL CLAIMS VERIFIED**

# Section 6

In[•]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[i, j, gi, gj, kk, kp2, rhoI, rhoJ, eI, eJ, snTi, cnTi, snTj,
  cnTj, pqI, pqJ, PiT, SPt, r1t, r1pt, r2t, r3t, r4t, xiT, epsT, kkT, chkR,
  statusChip, smallChip, s10colTest, clSym, clHdr, lblPair, braceGrp,
  mkL, pairs3, pairsLoc, addSn, addCn, addDn, expSnOK, expCnOK, expDnOK,
  DG, Dden, denOK, dispRA, dispRB, dispSn, dispCn, resTests, resGrid,
  resDisp, valsOf, prodTests, prodGrid, prodDisp, cmpOf, cmpGrid, snList,
  cnList, dnList, locResTests, locResGrid, locResDisp, locEqList,
  asmSn, asmCn, asmDn, asmLoc, chipTable, ok3, okLoc, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
```

```
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
    Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s10colTest = RGBColor[0.10, 0.35, 0.75];
(*Every check below is a rational identity in a fixed set of radical atoms,
decided by Together;
no simplification search is involved.Class
 regions:each of p_i q_i and p_j q_j lies in one of the three classes R_{>0},
i R_{>0},R_{<0},written p q=rho,i rho,-rho with an atom rho>0;
the sign sgn of p^2 q^2 is then+1,-1,+1,and,by the given sign identities,
sgn is also the sign of g^2-1 and of g^2-k'^2. On each class the atoms A=
 Sqrt[sgn (g^2-1)] and B=Sqrt[sgn (g^2-k'^2)] are real principal
    radicals of positive radicands,and every radical occurring in the
 given and tested formulas resolves into a product of these atoms;
each resolution is certified by squaring,
which is a rational identity,and both sides are nonnegative
 by construction.Every displayed identity has its own check mark,
one for each class region where it depends on the region.The expansions
 by the addition formulas are checked for all four sign choices of e_i,e_j;
elsewhere e_i and e_j enter every identity in degree at most one,so they
 stay symbolic reals.eps_i and eps_j enter only through the product eps_i
 eps_j and its square,so evaluating at eps_i=eps with eps in {1,-1} and eps_j=
  1 covers all four sign choices.k'^2 enters only through the atoms,
so it stays a free symbol kp2.The first equality of the
 second tested chain is analytic and enters as a given;
its remaining equalities are verified here.Divisions occurring in the
 identities are by k,p_l q_l,and k^2-(g_i^2-1)(g_j^2-1),all nonzero by
  the given hypotheses.All display-only rows are wrapped in HoldForm.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
clSym[1] = Subscript["ℝ", "> 0"];
clSym[2] = Row[{"i ", Subscript["ℝ", "> 0"]}];
clSym[3] = Subscript["ℝ", "< 0"];
clHdr[{c1_, c2_}] :=
  Column[{Style[clSym[c1], 11], Style[clSym[c2], 11]}, Alignment → Center];
lblPair[{c1_, c2_}] :=
  Row[{TraditionalForm[HoldForm[Subscript[p, i] Subscript[q, i]]],
    " ∈ ", clSym[c1], ",   ", TraditionalForm[
     HoldForm[Subscript[p, j] Subscript[q, j]]], " ∈ ", clSym[c2]}];
braceGrp[items_List, sz_ : 13, col_ : GrayLevel[0]] := DisplayForm[
   RowBox[{GridBox[({ToBoxes[Style[#, sz, col]]} & /@ items), GridBoxAlignment →
       {"Columns" → {{Left}}}, GridBoxSpacings → {"Rows" → {{1.8}}}],
     StyleBox["}", FontColor → GrayLevel[0.55], SpanMaxSize → Infinity]}]];
mkL[c_, g_, rho_] := <|"pq" → Switch[c, 1, rho, 2, I rho, 3, -rho],
   "sgn" → If[c == 2, -1, 1], "A" → If[c == 2, Sqrt[1 - g^2], Sqrt[g^2 - 1]],
```

```
    "B" → If[c == 2, Sqrt[kp2 - g^2], Sqrt[g^2 - kp2]]|>;
pairs3 = Tuples[{1, 2, 3}, 2];
pairsLoc =
  Select[pairs3, mkL[#[[1]], gi, rhoI]["sgn"] mkL[#[[2]], gj, rhoJ]["sgn"] == 1 &];
(*table of check marks:one row per displayed identity,
one column per class region*)
chipTable[disps_List, grid_, prs_List] :=
  Grid[Prepend[Table[Join[{Style[disps[[n]], 13]}, smallChip /@ grid[[n]],
     {n, Length[disps]}], Join[{Style[Row[{"class regions of  ( ",
         TraditionalForm[HoldForm[Subscript[p, i] Subscript[q, i]]], ", ",
         TraditionalForm[HoldForm[Subscript[p, j] Subscript[q, j]]], " ) :"}],
       11, GrayLevel[0.3]]}, clHdr /@ prs]], Alignment → {Left, Center},
   Spacings → {1.2, 0.9}, Dividers → {None, {2 → GrayLevel[0.8]}}];
(*---expansions by the addition formulas,parity,and dn=g---*)
addSn[sv_, cv_, dv_, sw_, cw_, dw_] := (sv cw dw + sw cv dv) / (1 - kk^2 sv^2 sw^2);
addCn[sv_, cv_, dv_, sw_, cw_, dw_] := (cv cw - sv sw dv dw) / (1 - kk^2 sv^2 sw^2);
addDn[sv_, cv_, dv_, sw_, cw_, dw_] :=
  (dv dw - kk^2 sv sw cv cw) / (1 - kk^2 sv^2 sw^2);
expSnOK = And @@ Flatten[Table[chkR[addSn[a snTi, cnTi, gi, b snTj, cnTj, gj],
      (a snTi cnTj gj + b snTj cnTi gi) / (1 - kk^2 snTi^2 snTj^2)],
     {a, {1, -1}}, {b, {1, -1}}]];
expCnOK = And @@ Flatten[Table[chkR[addCn[a snTi, cnTi, gi, b snTj, cnTj, gj],
      (cnTi cnTj - a b snTi snTj gi gj) / (1 - kk^2 snTi^2 snTj^2)],
     {a, {1, -1}}, {b, {1, -1}}]];
expDnOK = And @@ Flatten[Table[chkR[addDn[a snTi, cnTi, gi, b snTj, cnTj, gj],
      (gi gj - kk^2 a b snTi snTj cnTi cnTj) / (1 - kk^2 snTi^2 snTj^2)],
     {a, {1, -1}}, {b, {1, -1}}]];
(*---the common denominator---*)
DG = (gi^2 - 1) (gj^2 - 1);
Dden = (kk^2 - DG) / kk^2;
denOK =
  And @@ Flatten[Table[chkR[1 - kk^2 (I a Sqrt[(gi^2 - 1) pqI^2] / (kk pqI))^2
        (I b Sqrt[(gj^2 - 1) pqJ^2] / (kk pqJ))^2,
      Dden], {a, {1, -1}}, {b, {1, -1}}]];
(*---radical resolutions,one check per identity and class region---*)
resTests = {Function[{Li, Lj, xi}, chkR[(rhoI Li["A"])^2, (gi^2 - 1) Li["pq"]^2]],
   Function[{Li, Lj, xi}, chkR[(rhoI Li["B"])^2, (gi^2 - kp2) Li["pq"]^2]],
   Function[{Li, Lj, xi}, chkR[(rhoJ Lj["A"])^2, (gj^2 - 1) Lj["pq"]^2]],
   Function[{Li, Lj, xi}, chkR[(rhoJ Lj["B"])^2, (gj^2 - kp2) Lj["pq"]^2]],
   Function[{Li, Lj, xi}, chkR[(rhoI rhoJ)^2, Li["pq"]^2 Lj["pq"]^2 xi]],
   Function[{Li, Lj, xi}, chkR[(Li["A"] Lj["B"])^2, (gi^2 - 1) (gj^2 - kp2) xi]],
   Function[{Li, Lj, xi}, chkR[(Lj["A"] Li["B"])^2, (gj^2 - 1) (gi^2 - kp2) xi]],
   Function[{Li, Lj, xi}, chkR[(Li["B"] Lj["B"])^2, (gi^2 - kp2) (gj^2 - kp2) xi]],
   Function[{Li, Lj, xi}, chkR[(Li["A"] Lj["A"])^2, (gi^2 - 1) (gj^2 - 1) xi]],
   Function[{Li, Lj, xi}, chkR[(Li["A"] Li["B"] Lj["A"] Lj["B"])^2,
     (gi^2 - 1) (gi^2 - kp2) (gj^2 - 1) (gj^2 - kp2)]]};
resGrid = Table[Module[{Li = mkL[pr[[1]], gi, rhoI], Lj = mkL[pr[[2]], gj, rhoJ]},
```

```
     resTests[[n]][Li, Lj, Li["sgn"] Lj["sgn"]]],
    {n, Length[resTests]}, {pr, pairs3}];
(*---product identities,
one check per identity and class region,both eps---*)
valsOf[Li_, Lj_, ep_] :=
  {I ep rhoI Li["A"] / (kk Li["pq"]), ep rhoI Li["B"] / (kk Li["pq"]),
   I rhoJ Lj["A"] / (kk Lj["pq"]), rhoJ Lj["B"] / (kk Lj["pq"])};
prodTests = {Function[{Li, Lj, xi, ep}, Module[{v = valsOf[Li, Lj, ep]},
      chkR[v[[1]] v[[4]], I ep rhoI rhoJ Li["A"] Lj["B"] / (kk^2 Li["pq"] Lj["pq"])]]],
    Function[{Li, Lj, xi, ep}, Module[{v = valsOf[Li, Lj, ep]},
      chkR[v[[3]] v[[2]], I ep rhoI rhoJ Lj["A"] Li["B"] / (kk^2 Li["pq"] Lj["pq"])]]],
    Function[{Li, Lj, xi, ep}, Module[{v = valsOf[Li, Lj, ep]},
      chkR[v[[2]] v[[4]], ep rhoI rhoJ Li["B"] Lj["B"] / (kk^2 Li["pq"] Lj["pq"])]]],
    Function[{Li, Lj, xi, ep}, Module[{v = valsOf[Li, Lj, ep]},
      chkR[v[[1]] v[[3]], -ep rhoI rhoJ Li["A"] Lj["A"] / (kk^2 Li["pq"] Lj["pq"])]]],
    Function[{Li, Lj, xi, ep}, Module[{v = valsOf[Li, Lj, ep]},
      chkR[v[[1]] v[[3]] v[[2]] v[[4]], -xi Li["A"] Li["B"] Lj["A"] Lj["B"] / kk^4]]]};
prodGrid = Table[Module[{Li = mkL[pr[[1]], gi, rhoI], Lj = mkL[pr[[2]], gj, rhoJ]},
     And @@ Table[prodTests[[n]][Li, Lj, Li["sgn"] Lj["sgn"], ep], {ep, {1, -1}}]],
    {n, Length[prodTests]}, {pr, pairs3}];
(*---the tested formulas end-to-end on each class region---*)
cmpOf[{c1_, c2_}, ep_] := Module[{Li = mkL[c1, gi, rhoI], Lj = mkL[c2, gj, rhoJ],
     xi, v, den2, SV, dTar, h1V, h2V, h3V}, xi = Li["sgn"] Lj["sgn"];
    v = valsOf[Li, Lj, ep];
    den2 = 1 - kk^2 v[[1]]^2 v[[3]]^2;
    SV = ep rhoI rhoJ / (Li["pq"] Lj["pq"]);
    dTar = kk^2 - (gi^2 - 1) (gj^2 - 1);
    h1V = (eI gj Li["A"] Lj["B"] + eJ gi Lj["A"] Li["B"]) / dTar;
    h2V = (Li["B"] Lj["B"] + eI eJ gi gj Li["A"] Lj["A"]) / dTar;
    h3V = (kk^2 gi gj + eI eJ xi Li["A"] Li["B"] Lj["A"] Lj["B"]) / dTar;
    {chkR[(eI v[[1]] v[[4]] gj + eJ v[[3]] v[[2]] gi) / den2, I SV h1V],
     chkR[(v[[2]] v[[4]] - eI eJ v[[1]] v[[3]] gi gj) / den2, SV h2V],
     chkR[(gi gj - kk^2 eI eJ v[[1]] v[[3]] v[[2]] v[[4]]) / den2, h3V]}];
cmpGrid = Table[cmpOf[pr, ep], {pr, pairs3}, {ep, {1, -1}}];
snList = Table[And @@ cmpGrid[[ip, All, 1]], {ip, Length[pairs3]}];
cnList = Table[And @@ cmpGrid[[ip, All, 2]], {ip, Length[pairs3]}];
dnList = Table[And @@ cmpGrid[[ip, All, 3]], {ip, Length[pairs3]}];
(*---assembly checks with the radicals as free symbols---*)
asmSn = chkR[
    (eI (I epsT SPt r1t / (kk^2 PiT)) gj + eJ (I epsT SPt r1pt / (kk^2 PiT)) gi) / Dden,
    I (epsT SPt / PiT) (eI gj r1t + eJ gi r1pt) / (kk^2 - DG)];
asmCn = chkR[
    ((epsT SPt r2t / (kk^2 PiT)) - eI eJ (-(epsT SPt r3t / (kk^2 PiT))) gi gj) / Dden,
    (epsT SPt / PiT) (r2t + eI eJ gi gj r3t) / (kk^2 - DG)];
asmDn = chkR[(gi gj - kk^2 eI eJ (-(xiT r4t / kk^4))) / Dden,
    (kk^2 gi gj + eI eJ xiT r4t) / (kk^2 - DG)];
(*---the case k^2=(g_i^2-1)(g_j^2-1)---*)
```

```
locResTests =
  {Function[{Li, Lj}, chkR[(Li["A"] Lj["A"])^2, (gi^2 - 1) (gj^2 - 1)]],
   Function[{Li, Lj}, chkR[(rhoI rhoJ)^2, Li["pq"]^2 Lj["pq"]^2]]};
locResGrid = Table[Module[{Li = mkL[pr[[1]], gi, rhoI], Lj = mkL[pr[[2]], gj, rhoJ]},
    locResTests[[n]][Li, Lj]], {n, Length[locResTests]}, {pr, pairsLoc}];
locEqList = Table[Module[{Li = mkL[pr[[1]], gi, rhoI],
     Lj = mkL[pr[[2]], gj, rhoJ], kkL}, kkL = Li["A"] Lj["A"];
    And @@ Table[chkR[
       -kkL (I ep rhoI Li["A"] / (kkL Li["pq"])) (I rhoJ Lj["A"] / (kkL Lj["pq"])),
       ep rhoI rhoJ / (Li["pq"] Lj["pq"])], {ep, {1, -1}}]], {pr, pairsLoc}];
asmLoc = chkR[-kkT (-(epsT SPt kkT / (kkT^2 PiT))), epsT SPt / PiT];
ok3 = expSnOK && expCnOK && expDnOK && denOK &&
   (And @@ Flatten[resGrid]) && (And @@ Flatten[prodGrid]) && asmSn &&
   asmCn && asmDn && (And @@ snList) && (And @@ cnList) && (And @@ dnList);
okLoc = asmLoc && (And @@ Flatten[locResGrid]) && (And @@ locEqList);
resultsAll = {ok3, okLoc};
(*---displays---*)
dispSn[l_] := TraditionalForm[HoldForm[sn[Subscript[t, l]] ==
     I Subscript[ε, l] Sqrt[(Subsuperscript[g, l, 2] - 1) Subsuperscript[p, l, 2]
         Subsuperscript[q, l, 2]] / (k Subscript[p, l] Subscript[q, l])]];
dispCn[l_] := TraditionalForm[HoldForm[cn[Subscript[t, l]] == Subscript[ε, l]
      Sqrt[(Subsuperscript[g, l, 2] - Superscript[k', 2]) Subsuperscript[p, l, 2]
         Subsuperscript[q, l, 2]] / (k Subscript[p, l] Subscript[q, l])]];
dispRA[l_] := TraditionalForm[
   HoldForm[Sqrt[(Subsuperscript[g, l, 2] - 1) Subsuperscript[p, l, 2]
       Subsuperscript[q, l, 2]] == Sqrt[Abs[Subsuperscript[g, l, 2] - 1]]
      Sqrt[Abs[Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]]]]];
dispRB[l_] :=
  TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, l, 2] - Superscript[k', 2])
       Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]] ==
     Sqrt[Abs[Subsuperscript[g, l, 2] - Superscript[k', 2]]]
      Sqrt[Abs[Subsuperscript[p, l, 2] Subsuperscript[q, l, 2]]]]];
resDisp = {dispRA[i], dispRB[i], dispRA[j], dispRB[j],
   TraditionalForm[HoldForm[Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
        Subsuperscript[p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]] ==
      Sqrt[Abs[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]]
       Sqrt[Abs[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]]],
   TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1)
        (Subsuperscript[g, j, 2] - Superscript[k', 2]) Subscript[ξ, ij]] ==
      Sqrt[Abs[Subsuperscript[g, i, 2] - 1]]
       Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]],
   TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, j, 2] - 1)
        (Subsuperscript[g, i, 2] - Superscript[k', 2]) Subscript[ξ, ij]] ==
      Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]
       Sqrt[Abs[Subsuperscript[g, i, 2] - Superscript[k', 2]]]]],
   TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - Superscript[k', 2])
        (Subsuperscript[g, j, 2] - Superscript[k', 2]) Subscript[ξ, ij]] ==
```

```
      Sqrt[Abs[Subsuperscript[g, i, 2] - Superscript[k', 2]]]
       Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]],
   TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1)
        (Subsuperscript[g, j, 2] - 1) Subscript[ξ, ij]] == Sqrt[Abs[
         Subsuperscript[g, i, 2] - 1]] Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]]],
   TraditionalForm[HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1)
        (Subsuperscript[g, i, 2] - Superscript[k', 2]) (Subsuperscript[g, j, 2] -
          1) (Subsuperscript[g, j, 2] - Superscript[k', 2])] ==
      Sqrt[Abs[Subsuperscript[g, i, 2] - 1]] Sqrt[Abs[Subsuperscript[g, i, 2] -
          Superscript[k', 2]]] Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]
       Sqrt[Abs[Subsuperscript[g, j, 2] - Superscript[k', 2]]]]]};
prodDisp = {TraditionalForm[
    HoldForm[sn[Subscript[t, i]] cn[Subscript[t, j]] == I Subscript[ε, i]
      Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
        Subsuperscript[p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]]
      Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] -
          Superscript[k', 2]) Subscript[ξ, ij]] /
       (k^2 Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
   TraditionalForm[HoldForm[sn[Subscript[t, j]] cn[Subscript[t, i]] ==
      I Subscript[ε, i] Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2]
        Subsuperscript[q, i, 2] Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]
        Subscript[ξ, ij]] Sqrt[(Subsuperscript[g, j, 2] - 1)
        (Subsuperscript[g, i, 2] - Superscript[k', 2]) Subscript[ξ, ij]] /
       (k^2 Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
   TraditionalForm[HoldForm[cn[Subscript[t, i]] cn[Subscript[t, j]] ==
      Subscript[ε, i] Subscript[ε, j]
       Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
        Subsuperscript[p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]]
       Sqrt[(Subsuperscript[g, i, 2] - Superscript[k', 2])
        (Subsuperscript[g, j, 2] - Superscript[k', 2]) Subscript[ξ, ij]] /
       (k^2 Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
   TraditionalForm[HoldForm[sn[Subscript[t, i]] sn[Subscript[t, j]] ==
      -Subscript[ε, i] Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2]
        Subsuperscript[q, i, 2] Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]
        Subscript[ξ, ij]] Sqrt[(Subsuperscript[g, i, 2] - 1)
        (Subsuperscript[g, j, 2] - 1) Subscript[ξ, ij]] /
       (k^2 Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
   TraditionalForm[HoldForm[sn[Subscript[t, i]] sn[Subscript[t, j]]
      cn[Subscript[t, i]] cn[Subscript[t, j]] == -Subscript[ξ, ij]
      Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, i, 2] -
          Superscript[k', 2]) (Subsuperscript[g, j, 2] - 1)
        (Subsuperscript[g, j, 2] - Superscript[k', 2])] / k^4]]};
locResDisp = {TraditionalForm[
    HoldForm[Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)] ==
      Sqrt[Abs[Subsuperscript[g, i, 2] - 1]]
       Sqrt[Abs[Subsuperscript[g, j, 2] - 1]]]],
   TraditionalForm[HoldForm[Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
```

```
        Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]] ==
       Sqrt[Abs[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]]
        Sqrt[Abs[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[
   Column[{Style["Given hypotheses:", Bold, 14], Style["k, k' ∈ (0, 1)", 14],
     Style[Row[{"For  ", TraditionalForm[HoldForm[i]], ", ",
        TraditionalForm[HoldForm[j]], " ∈ {1, …, 4},   ", TraditionalForm[
         HoldForm[i ≠ j]], " :"}], Bold, 14], Style[Row[{TraditionalForm[
         HoldForm[Subscript[g, i] == dn[Subscript[t, i]]]], ",   ",
        TraditionalForm[HoldForm[Subscript[g, j] == dn[Subscript[t, j]]]],
        " ∈ (0, k') ⋃ (1, ∞),   ", TraditionalForm[HoldForm[Subscript[p, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[q, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[p, j]]], ", ",
        TraditionalForm[HoldForm[Subscript[q, j]]], " ∈ ",
        Subscript["ℝ", "> 0"], " ⋃ i ", Subscript["ℝ", "> 0"]}], 14],
     Style[Row[{TraditionalForm[HoldForm[Sign[Subsuperscript[g, i, 2] - 1] ==
           Sign[Subsuperscript[g, i, 2] - Superscript[k', 2]] ==
           Sign[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]]]], " ≠ 0,   ",
        TraditionalForm[HoldForm[Sign[Subsuperscript[g, j, 2] - 1] ==
           Sign[Subsuperscript[g, j, 2] - Superscript[k', 2]] ==
           Sign[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]],
        " ≠ 0", "    ", Style["the sign identities", Italic,
         13, RGBColor[0.00, 0.50, 0.50]]}], 14],
     Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, ij] ==
           Sign[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
             Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]]],
        " ,   ", TraditionalForm[HoldForm[Subscript[e, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[e, j]]], ", ",
        TraditionalForm[HoldForm[Subscript[ε, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[ε, j]]], ", ",
        TraditionalForm[HoldForm[Subscript[ξ, ij]]], " ∈ {±1}"}], 14],
     "", Style["Given formulas and expressions:", Bold, 14],
     Style[dispSn[i], 14],
     Style[dispCn[i], 14],
     Style[dispSn[j], 14], Style[dispCn[j], 14],
     Style[TraditionalForm[HoldForm[Subscript[S, ij] == Subscript[ε, i]
          Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
             Subsuperscript[p, j, 2] Subsuperscript[q, j, 2] Subscript[ξ, ij]] /
           (Subscript[p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]],
      14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] ==
         (Subscript[e, i] Subscript[g, j] Sqrt[(Subsuperscript[g, i, 2] - 1)
               (Subsuperscript[g, j, 2] - Superscript[k', 2])
               Subscript[ξ, ij]] + Subscript[e, j] Subscript[g, i]
```

```
        Sqrt[(Subsuperscript[g, j, 2] - 1) (Subsuperscript[g, i, 2] -
            Superscript[k', 2]) Subscript[ξ, ij]]) /
     (k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
 14], Style[TraditionalForm[HoldForm[Subscript[h, ij2] ==
    (Sqrt[(Subsuperscript[g, i, 2] - Superscript[k', 2])
         (Subsuperscript[g, j, 2] - Superscript[k', 2]) Subscript[ξ, ij]] +
       Subscript[e, i] Subscript[e, j] Subscript[g, i]
        Subscript[g, j] Sqrt[(Subsuperscript[g, i, 2] - 1)
          (Subsuperscript[g, j, 2] - 1) Subscript[ξ, ij]]) /
     (k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
 14], Style[TraditionalForm[HoldForm[Subscript[h, ij3] ==
    (k^2 Subscript[g, i] Subscript[g, j] + Subscript[e, i] Subscript[e, j]
        Subscript[ξ, ij] Sqrt[(Subsuperscript[g, i, 2] - 1)
          (Subsuperscript[g, i, 2] - Superscript[k', 2]) (Subsuperscript[g,
             j, 2] - 1) (Subsuperscript[g, j, 2] - Superscript[k', 2])]) /
     (k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1))]],
 14], Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] ==
      {-sn[v], cn[v], dn[v]}]], "    ",
   Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
 14], Style[Row[{TraditionalForm[HoldForm[sn[v + w] ==
      (sn[v] cn[w] dn[w] + sn[w] cn[v] dn[v]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
   "    ", Style["the first addition formula", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]]}], 14],
Style[Row[{TraditionalForm[HoldForm[cn[v + w] ==
      (cn[v] cn[w] - sn[v] sn[w] dn[v] dn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
   "    ", Style["the second addition formula", Italic,
    13, RGBColor[0.10, 0.30, 0.70]]}], 14],
Style[Row[{TraditionalForm[HoldForm[dn[v + w] == (dn[v] dn[w] -
         k^2 sn[v] sn[w] cn[v] cn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
   "    ", Style["the third addition formula", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]]}],
 14], Style[Row[{"Each  ", smallChip[True],
   "  below certifies that the statement displayed beside it has been
     verified by a symbolic computation at that point;  a  ",
   smallChip[False], "  would mean that this verification failed."}],
 12, GrayLevel[0.35]], "", Style[
 Row[{Style["We want to test these formulas  ", Bold, 14],
   Style["⋆", 14, s10colTest], Style[" :", Bold, 14]}]],
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[
       Subscript[e, i] Subscript[t, i] + Subscript[e, j] Subscript[t, j]] ==
      {I Subscript[S, ij] Subscript[h, ij1], Subscript[S, ij]
        Subscript[h, ij2], Subscript[h, ij3]}]], "    ",
   Style[Row[{"in the case  ", TraditionalForm[HoldForm[k^2 ≠
         (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]]}],
    Italic, 13, GrayLevel[0.35]]}], 15, s10colTest],
Style[Row[{TraditionalForm[HoldForm[(-1)^m == -k sn[Subscript[t, i]]
       sn[Subscript[t, j]] == Subscript[ε, i] Subscript[ε, j]
```

```
            Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2] Subsuperscript[
                p, j, 2] Subsuperscript[q, j, 2]] / (Subscript[p, i] Subscript[
                q, i] Subscript[p, j] Subscript[q, j]) == Subscript[S, ij]]],
        "    ", Style[Row[{"in the case  ", TraditionalForm[HoldForm[k^2 ==
              (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]]}],
         Italic, 13, GrayLevel[0.35]]}], 15, s10colTest]},
    Spacings → 1.1], Background → RGBColor[0.99,
     0.96,
     0.88], FrameMargins → 14]];

(* ============================================================*)
(*PART 3:CASE k^2≠(g_i^2-1)(g_j^2-1)*)
(* ============================================================*)
Print[Panel[
   Column[{Row[{Style["The derivation:  the formula for  ", Bold, 15, RGBColor[
         0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[{sn, cn, dn}[
           Subscript[e, i] Subscript[t, i] + Subscript[e, j] Subscript[t, j]]]],
        Bold, 15, RGBColor[0.10, 0.30, 0.70]], Style["  in the case  ",
        Bold, 15, RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[
          k^2 ≠ (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]],
        Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok3]}],
     Style["Given hypotheses, in addition to those above:", Bold, 13],
     Style[Row[{TraditionalForm[HoldForm[±Subscript[t, i]]],
        ", ", TraditionalForm[HoldForm[±Subscript[t, j]]],
        "  are not poles,   ", TraditionalForm[HoldForm[
          k^2 ≠ (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]]}],
      13], "", Style["Verification:", Bold, 13], Style[
      Row[{"By the given formulas for  ",
        TraditionalForm[HoldForm[sn[Subscript[t, i]]]], "  and  ",
        TraditionalForm[HoldForm[sn[Subscript[t, j]]]], " :"}],
      13], Style[Row[{TraditionalForm[HoldForm[
          1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2 ==
           (k^2 - (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)) /
            k^2]], "   ", smallChip[denOK], "   ",
        Style["which is nonzero in this case", 12, GrayLevel[0.35]]}], 13],
     Style[Row[{"Because  ", TraditionalForm[HoldForm[±Subscript[t, i]]],
        "  and  ", TraditionalForm[HoldForm[±Subscript[t, j]]],
        "  are not poles and  ", TraditionalForm[
         HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2 ≠ 0]],
        ",  ", Style["the addition formulas", Italic, 13,
         RGBColor[0.10, 0.30, 0.70]], ",  at  ",
        TraditionalForm[HoldForm[v == Subscript[e, i] Subscript[t, i]]],
        ",  ", TraditionalForm[HoldForm[w == Subscript[e, j] Subscript[t, j]]],
        ",  hold pointwise and, together with  ", Style[
         "the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]], ",  ",
        TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
        ",  and  ", TraditionalForm[
```

```
    HoldForm[dn[Subscript[t, j]] == Subscript[g, j]]], " ,  give"}], 13],
Grid[{{braceGrp[{Row[{TraditionalForm[HoldForm[sn[Subscript[e, i]
              Subscript[t, i] + Subscript[e, j] Subscript[t, j]] ==
           (Subscript[e, i] sn[Subscript[t, i]] cn[Subscript[t, j]]
               Subscript[g, j] + Subscript[e, j] sn[Subscript[t, j]]
               cn[Subscript[t, i]] Subscript[g, i]) /
            (1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2)]],
       "   ", smallChip[expSnOK]}], Row[{TraditionalForm[
        HoldForm[cn[Subscript[e, i] Subscript[t, i] + Subscript[e, j]
              Subscript[t, j]] == (cn[Subscript[t, i]] cn[Subscript[t, j]] -
              Subscript[e, i] Subscript[e, j] sn[Subscript[t, i]]
               sn[Subscript[t, j]] Subscript[g, i] Subscript[g, j]) /
            (1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2)]],
       "   ", smallChip[expCnOK]}], Row[{TraditionalForm[
        HoldForm[dn[Subscript[e, i] Subscript[t, i] + Subscript[e, j]
              Subscript[t, j]] == (Subscript[g, i] Subscript[g, j] -
              k^2 Subscript[e, i] Subscript[e, j] sn[Subscript[t, i]] sn[
                Subscript[t, j]] cn[Subscript[t, i]] cn[Subscript[t, j]]) /
            (1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2)]],
       "   ", smallChip[expDnOK]}]}, 14],
   Style["✱", 15, RGBColor[0.10, 0.30, 0.70]]}},
 Alignment → {Left, Center}, Spacings → {1.5, 0}],
Style[Row[{"By  ", Style["the given sign identities",
    Italic, 13, RGBColor[0.00, 0.50, 0.50]],
   " ,  every radicand below is positive and the radicals resolve
     as follows  ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]],
   " ;  each identity is verified on each class region:"}],
 13], chipTable[resDisp, resGrid, pairs3],
Style[Row[{"Multiplying the given formulas for  ",
   TraditionalForm[HoldForm[sn[Subscript[t, i]]]],
   ",  ", TraditionalForm[HoldForm[cn[Subscript[t, i]]]],
   ",  ", TraditionalForm[HoldForm[sn[Subscript[t, j]]]],
   ",  ", TraditionalForm[HoldForm[cn[Subscript[t, j]]]],
   " ,  and using the identities  ",
   Style["●", 13, RGBColor[0.00, 0.45, 0.25]],
   " ,  we get the product identities  ",
   Style["●", 14, RGBColor[0.55, 0.33, 0.10]],
   " ;  each is verified on each class region, for both values of  ",
   TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]], " :"}],
 13], chipTable[prodDisp, prodGrid, pairs3], "",
Style[Row[{"Thus, substituting the product identities  ",
   Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
   "  and the verified equality for  ", TraditionalForm[
    HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[Subscript[t, j]]^2]],
   "  into the expansions  ", Style["✱", 13,
    RGBColor[0.10, 0.30, 0.70]], " ,  with  ",
   TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
```

```
        " ,  ", TraditionalForm[HoldForm[Subscript[e, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[e, j]]],
        " ,  and the radicals of the identities  ",
        Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
        "  symbolic, that is, for all their real values
          at once, we get the first formula of  ",
        Style["⋆", 13, s10colTest], " :    sn  ", smallChip[asmSn],
        "    cn  ", smallChip[asmCn], "    dn  ", smallChip[asmDn]}], 13], "",
     Style[Row[{"The first formula of  ", Style["⋆", 13, s10colTest],
        "  is also verified end-to-end separately on each class region,
          with  ", TraditionalForm[HoldForm[Subscript[e, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[e, j]]],
        "  symbolic and for both values of  ", TraditionalForm[
         HoldForm[Subscript[ε, i] Subscript[ε, j]]], " :"}], 13],
     Grid[Join[{{Style["", 11], Style["sn", 11, Bold, GrayLevel[0.3]],
         Style["cn", 11, Bold, GrayLevel[0.3]],
         Style["dn", 11, Bold, GrayLevel[0.3]]}},
       Table[{Style[lblPair[pairs3[[ip]]], 12, GrayLevel[0.3]],
         smallChip[snList[[ip]]], smallChip[cnList[[ip]]],
         smallChip[dnList[[ip]]]}, {ip, Length[pairs3]}]],
      Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
   Background → Lighter[RGBColor[0.10, 0.30, 0.70],
     0.93], FrameMargins → 12]];

(* ============================================================*)
(*PART 4:CASE k^2==(g_i^2-1)(g_j^2-1)*)
(* ============================================================*)
Print[
  Panel[Column[{Row[{Style["The derivation:  the sign identity in the case  ",
        Bold, 15, RGBColor[0.45, 0.15, 0.60]], Style[TraditionalForm[HoldForm[
          k^2 == (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]],
        Bold, 15, RGBColor[0.45, 0.15, 0.60]], Spacer[20], statusChip[okLoc]}],
     Style["Given hypotheses, in addition to those above:", Bold, 13],
     Style[Row[{TraditionalForm[
         HoldForm[Subscript[t, i] + Subscript[t, j] == 2 m K + I K']], ",   ",
        TraditionalForm[HoldForm[m ∈ ℤ]], ",   ", TraditionalForm[
         HoldForm[(-1)^m == -k sn[Subscript[t, i]] sn[Subscript[t, j]]]]}],
      13], Style[Row[{TraditionalForm[HoldForm[
          k^2 == (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]],
        " ,   hence   ", TraditionalForm[HoldForm[
          k == Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)]]],
        "   and   ", TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]]}], 13],
     Style[Row[{"Here  ", TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
        " ,  because  ", TraditionalForm[HoldForm[
          k^2 == (Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1) > 0]],
        "  and, by  ", Style["the given sign identities", Italic, 12,
         RGBColor[0.00, 0.50, 0.50]], " ,  the factors have the signs of  ",
```

```
        TraditionalForm[HoldForm[Subsuperscript[p, i, 2]
           Subsuperscript[q, i, 2]]], "  and  ", TraditionalForm[
         HoldForm[Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]]],
        " ;  hence these signs coincide, and exactly the
          five class regions below occur."}],
      12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
    Style[Row[{"The first equality of the second formula of  ",
       Style["⋆", 13, s10colTest], "  is given above."}], 13],
    Style[Row[{"By  ", Style["the given sign identities",
        Italic, 13, RGBColor[0.00, 0.50, 0.50]],
       " ,  every radicand below is positive and the radicals resolve
         as follows  ", Style["●", 14, RGBColor[0.45, 0.15, 0.60]],
       " ;  each identity is verified on each class region:"}], 13],
    chipTable[locResDisp, locResGrid, pairsLoc],
    Style[Row[{"Substituting  ", TraditionalForm[HoldForm[
         Sqrt[(Subsuperscript[g, i, 2] - 1) (Subsuperscript[g, j, 2] - 1)] == k]],
       "  and the identities  ", Style["●", 13, RGBColor[0.45, 0.15, 0.60]],
       "  into the fourth product identity  ",
       Style["●", 13, RGBColor[0.55, 0.33, 0.10]], " ,  at  ",
       TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
       " ,  and using the given expression for  ",
       TraditionalForm[HoldForm[Subscript[S, ij]]], "  at  ",
       TraditionalForm[HoldForm[Subscript[ξ, ij] == 1]],
       " ,  we get the second and third equalities of the second formula
         of  ", Style["⋆", 13, s10colTest], " ,  with  ",
       TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
       "  and the radicals of the identities  ",
       Style["●", 13, RGBColor[0.45, 0.15, 0.60]],
       "  symbolic, that is, for all their real values at once  ",
       smallChip[asmLoc]}], 13], "",
    Style[Row[{"The second and third equalities of the second formula of  ",
       Style["⋆", 13, s10colTest], "  are also verified end-to-end
         on each class region, for both values of  ",
       TraditionalForm[HoldForm[Subscript[ε, i] Subscript[ε, j]]],
       " :"}], 13], chipTable[{TraditionalForm[
       HoldForm[-k sn[Subscript[t, i]] sn[Subscript[t, j]] == Subscript[ε, i]
          Subscript[ε, j] Sqrt[Subsuperscript[p, i, 2] Subsuperscript[q, i, 2]
            Subsuperscript[p, j, 2] Subsuperscript[q, j, 2]] / (Subscript[
             p, i] Subscript[q, i] Subscript[p, j] Subscript[q, j])]]},
     {locEqList}, pairsLoc]}, Spacings → 1], Background →
   Lighter[RGBColor[0.45, 0.15, 0.60], 0.93],
  FrameMargins →
   12]];

(* ============================================================*)
(*PART 5:SUMMARY*)
(* ============================================================*)
```

```
Print[Framed[Style[If[And @@ resultsAll, "Section 6: ALL CLAIMS VERIFIED",
      "Section 6: SOME CLAIMS FAILED"], Bold, White, 16], Background →
    If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
   FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses:

$k, k' \in (0, 1)$

For $i, j \in \{1, \ldots, 4\}$, $i \neq j$:

$g_i = \mathrm{dn}(t_i)$, $g_j = \mathrm{dn}(t_j) \in (0, k') \bigcup (1, \infty)$, $p_i, q_i, p_j, q_j \in \mathbb{R}_{>0} \bigcup i\,\mathbb{R}_{>0}$

$\mathrm{sgn}(g_i^2 - 1) = \mathrm{sgn}(g_i^2 - k'^2) = \mathrm{sgn}(p_i^2\ q_i^2) \neq 0$,
$\mathrm{sgn}\left(g_j^2 - 1\right) = \mathrm{sgn}\left(g_j^2 - k'^2\right) = \mathrm{sgn}\left(p_j^2\ q_j^2\right) \neq 0$ *the sign identities*

$\xi_{ij} = \mathrm{sgn}\left(p_i^2\ q_i^2\ p_j^2\ q_j^2\right)$, $e_i, e_j, \varepsilon_i, \varepsilon_j, \xi_{ij} \in \{\pm 1\}$

Given formulas and expressions:

$$\mathrm{sn}(t_i) = \frac{i\,\varepsilon_i\,\sqrt{(g_i^2-1)\,p_i^2\,q_i^2}}{k\,p_i\,q_i}$$

$$\mathrm{cn}(t_i) = \frac{\varepsilon_i\,\sqrt{(g_i^2-k'^2)\,p_i^2\,q_i^2}}{k\,p_i\,q_i}$$

$$\mathrm{sn}(t_j) = \frac{i\,\varepsilon_j\,\sqrt{(g_j^2-1)\,p_j^2\,q_j^2}}{k\,p_j\,q_j}$$

$$\mathrm{cn}(t_j) = \frac{\varepsilon_j\,\sqrt{(g_j^2-k'^2)\,p_j^2\,q_j^2}}{k\,p_j\,q_j}$$

$$S_{ij} = \frac{\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}}}{p_i\,q_i\,p_j\,q_j}$$

$$h_{ij1} = \frac{e_i\,g_j\,\sqrt{(g_i^2-1)\,(g_j^2-k'^2)\,\xi_{ij}} + e_j\,g_i\,\sqrt{(g_j^2-1)\,(g_i^2-k'^2)\,\xi_{ij}}}{k^2-(g_i^2-1)(g_j^2-1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2-k'^2)\,(g_j^2-k'^2)\,\xi_{ij}} + e_i\,e_j\,g_i\,g_j\,\sqrt{(g_i^2-1)(g_j^2-1)\,\xi_{ij}}}{k^2-(g_i^2-1)(g_j^2-1)}$$

$$h_{ij3} = \frac{k^2\,g_i\,g_j + e_i\,e_j\,\xi_{ij}\,\sqrt{(g_i^2-1)\,(g_i^2-k'^2)\,(g_j^2-1)\,(g_j^2-k'^2)}}{k^2-(g_i^2-1)(g_j^2-1)}$$

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(-v) = \{-\mathrm{sn}(v), \mathrm{cn}(v), \mathrm{dn}(v)\}$ *the parity formulas*

$$\mathrm{sn}(v+w) = \frac{\mathrm{sn}(v)\,\mathrm{cn}(w)\,\mathrm{dn}(w) + \mathrm{sn}(w)\,\mathrm{cn}(v)\,\mathrm{dn}(v)}{1-k^2\,\mathrm{sn}(v)^2\,\mathrm{sn}(w)^2}$$ *the first addition formula*

$$\mathrm{cn}(v+w) = \frac{\mathrm{cn}(v)\,\mathrm{cn}(w) - \mathrm{sn}(v)\,\mathrm{sn}(w)\,\mathrm{dn}(v)\,\mathrm{dn}(w)}{1-k^2\,\mathrm{sn}(v)^2\,\mathrm{sn}(w)^2}$$ *the second addition formula*

$$\mathrm{dn}(v+w) = \frac{\mathrm{dn}(v)\,\mathrm{dn}(w) - k^2\,\mathrm{sn}(v)\,\mathrm{sn}(w)\,\mathrm{cn}(v)\,\mathrm{cn}(w)}{1-k^2\,\mathrm{sn}(v)^2\,\mathrm{sn}(w)^2}$$ *the third addition formula*

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test these formulas ★:

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(e_i\,t_i + e_j\,t_j) = \{i\,S_{ij}\,h_{ij1}, S_{ij}\,h_{ij2}, h_{ij3}\}$ *in the case* $k^2 \neq (g_i^2 - 1)\left(g_j^2 - 1\right)$

$(-1)^m = -k\,\mathrm{sn}(t_i)\,\mathrm{sn}(t_j) = \frac{\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2}}{p_i\,q_i\,p_j\,q_j} = S_{ij}$ *in the case* $k^2 = (g_i^2 - 1)\left(g_j^2 - 1\right)$

The derivation: the formula for $\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(e_i\,t_i + e_j\,t_j)$

in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$ Verified

Given hypotheses, in addition to those above:

$\pm t_i$, $\pm t_j$ are not poles, $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$

Verification:

By the given formulas for sn $(t_i)$ and sn $(t_j)$ :

$1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2 = \frac{k^2 - (g_i^2-1)(g_j^2-1)}{k^2}$ ☑ which is nonzero in this case

Because $\pm t_i$ and $\pm t_j$ are not poles and $1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2 \neq 0$,

*the addition formulas* , at $v = e_i t_i$, $w = e_j t_j$, hold pointwise and, together with

*the parity formulas* , dn $(t_i) = g_i$, and dn $(t_j) = g_j$ , give

$$\operatorname{sn}(e_i t_i + e_j t_j) = \frac{e_i \operatorname{sn}(t_i) \operatorname{cn}(t_j) g_j + e_j \operatorname{sn}(t_j) \operatorname{cn}(t_i) g_i}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2}$$ ☑

$$\operatorname{cn}(e_i t_i + e_j t_j) = \frac{\operatorname{cn}(t_i) \operatorname{cn}(t_j) - e_i e_j \operatorname{sn}(t_i) \operatorname{sn}(t_j) g_i g_j}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2}$$ ☑

$$\operatorname{dn}(e_i t_i + e_j t_j) = \frac{g_i g_j - k^2 e_i e_j \operatorname{sn}(t_i) \operatorname{sn}(t_j) \operatorname{cn}(t_i) \operatorname{cn}(t_j)}{1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(t_j)^2}$$ ☑

✸

By *the given sign identities*

, every radicand below is positive and the radicals resolve as follows

● ; each identity is verified on each class region:

| class regions of $(p_i q_i, p_j q_j)$ : | $\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $i\,\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $i\,\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $i\,\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{<0}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\sqrt{(g_i^2-1)\,p_i^2\,q_i^2} = \sqrt{\lvert g_i^2-1\rvert}\,\sqrt{\lvert p_i^2\,q_i^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_i^2-k'^2)\,p_i^2\,q_i^2} = \sqrt{\lvert g_i^2-k'^2\rvert}\,\sqrt{\lvert p_i^2\,q_i^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_j^2-1)\,p_j^2\,q_j^2} = \sqrt{\lvert g_j^2-1\rvert}\,\sqrt{\lvert p_j^2\,q_j^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_j^2-k'^2)\,p_j^2\,q_j^2} = \sqrt{\lvert g_j^2-k'^2\rvert}\,\sqrt{\lvert p_j^2\,q_j^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}} = \sqrt{\lvert p_i^2\,q_i^2\rvert}\,\sqrt{\lvert p_j^2\,q_j^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_i^2-1)\,(g_j^2-k'^2)\,\xi_{ij}} = \sqrt{\lvert g_i^2-1\rvert}\,\sqrt{\lvert g_j^2-k'^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_j^2-1)\,(g_i^2-k'^2)\,\xi_{ij}} = \sqrt{\lvert g_j^2-1\rvert}\,\sqrt{\lvert g_i^2-k'^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{(g_i^2-k'^2)\,(g_j^2-k'^2)\,\xi_{ij}} = \sqrt{\lvert g_i^2-k'^2\rvert}\,\sqrt{\lvert g_j^2-k'^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |

$\sqrt{(g_i^2-1)\,(g_j^2-1)\,\xi_{ij}} = \sqrt{|g_i^2-1|}\;\sqrt{|g_j^2-1|}$ ☑ ☑ ☑ ☑ ☑ ☑ ☑ ☑ ☑

$\sqrt{((g_i^2-1)\,(g_i^2-k'^2)\,(g_j^2-1)\,(g_j^2-k'^2))} = \sqrt{|g_i^2-1|}\;\sqrt{|g_i^2-k'^2|}\;\sqrt{|g_j^2-1|}\;\sqrt{|g_j^2-k'^2|}$ ☑ ☑ ☑ ☑ ☑ ☑ ☑ ☑ ☑

Multiplying the given formulas for sn $(t_i)$, cn $(t_i)$, sn $(t_j)$, cn $(t_j)$, and using the identities ●, we get the product identities ●; each is verified on each class region, for both values of $\varepsilon_i\,\varepsilon_j$:

| class regions of $(p_i\,q_i,\,p_j\,q_j)$: | $\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $i\,\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $i\,\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $i\,\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{<0}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\mathrm{sn}(t_i)\,\mathrm{cn}(t_j) = \frac{i\,\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}}\;\sqrt{(g_i^2-1)\,(g_j^2-k'^2)\,\xi_{ij}}}{k^2\,p_i\,q_i\,p_j\,q_j}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\mathrm{sn}(t_j)\,\mathrm{cn}(t_i) = \frac{i\,\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}}\;\sqrt{(g_j^2-1)\,(g_i^2-k'^2)\,\xi_{ij}}}{k^2\,p_i\,q_i\,p_j\,q_j}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\mathrm{cn}(t_i)\,\mathrm{cn}(t_j) = \frac{\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}}\;\sqrt{(g_i^2-k'^2)\,(g_j^2-k'^2)\,\xi_{ij}}}{k^2\,p_i\,q_i\,p_j\,q_j}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\mathrm{sn}(t_i)\,\mathrm{sn}(t_j) = -\frac{\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\,q_i^2\,p_j^2\,q_j^2\,\xi_{ij}}\;\sqrt{(g_i^2-1)\,(g_j^2-1)\,\xi_{ij}}}{k^2\,p_i\,q_i\,p_j\,q_j}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\mathrm{sn}(t_i)\,\mathrm{sn}(t_j)\,\mathrm{cn}(t_i)\,\mathrm{cn}(t_j) = -\frac{\xi_{ij}\;\sqrt{(g_i^2-1)\,(g_i^2-k'^2)\,(g_j^2-1)\,(g_j^2-k'^2)}}{k^4}$ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ | ☑ |

Thus, substituting the product identities ● and the verified equality for $1 - k^2\,\mathrm{sn}(t_i)^2\,\mathrm{sn}(t_j)^2$ into the expansions ★, with $\varepsilon_i\,\varepsilon_j$, $e_i$, $e_j$, and the radicals of the identities ● symbolic, that is, for all their real values at once, we get the first formula of ★: sn ☑ cn ☑ dn ☑

The first formula of ★ is also verified end −to −end separately on each class region, with $e_i$, $e_j$ symbolic and for both values of $\varepsilon_i\,\varepsilon_j$:

| | sn | cn | dn |
|---|---|---|---|
| $p_i\,q_i \in \mathbb{R}_{>0},\;\; p_j\,q_j \in \mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in \mathbb{R}_{>0},\;\; p_j\,q_j \in i\,\mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in \mathbb{R}_{>0},\;\; p_j\,q_j \in \mathbb{R}_{<0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in i\,\mathbb{R}_{>0},\;\; p_j\,q_j \in \mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in i\,\mathbb{R}_{>0},\;\; p_j\,q_j \in i\,\mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in i\,\mathbb{R}_{>0},\;\; p_j\,q_j \in \mathbb{R}_{<0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in \mathbb{R}_{<0},\;\; p_j\,q_j \in \mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in \mathbb{R}_{<0},\;\; p_j\,q_j \in i\,\mathbb{R}_{>0}$ | ☑ | ☑ | ☑ |
| $p_i\,q_i \in \mathbb{R}_{<0},\;\; p_j\,q_j \in \mathbb{R}_{<0}$ | ☑ | ☑ | ☑ |

### The derivation: the sign identity in the case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$ Verified

Given hypotheses, in addition to those above:

$t_i + t_j = 2mK + iK'$, $m \in \mathbb{Z}$, $(-1)^m = -k\,\mathrm{sn}(t_i)\,\mathrm{sn}(t_j)$

$k^2 = (g_i^2 - 1)(g_j^2 - 1)$, hence $k = \sqrt{(g_i^2 - 1)(g_j^2 - 1)}$ and $\xi_{ij} = 1$

Here $\xi_{ij} = 1$, because $k^2 = (g_i^2 - 1)(g_j^2 - 1) > 0$ and, by *the given sign identities*, the factors have the signs of $p_i^2\, q_i^2$ and $p_j^2\, q_j^2$; hence these signs coincide, and exactly the five class regions below occur.

Verification:

The first equality of the second formula of ★ is given above.

By *the given sign identities*, every radicand below is positive and the radicals resolve as follows ●; each identity is verified on each class region:

| class regions of $(p_i q_i, p_j q_j)$: | $\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $i\,\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{<0}$ |
|---|---|---|---|---|---|
| $\sqrt{(g_i^2 - 1)(g_j^2 - 1)} = \sqrt{\lvert g_i^2 - 1\rvert}\,\sqrt{\lvert g_j^2 - 1\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ |
| $\sqrt{p_i^2\, q_i^2\, p_j^2\, q_j^2} = \sqrt{\lvert p_i^2\, q_i^2\rvert}\,\sqrt{\lvert p_j^2\, q_j^2\rvert}$ | ☑ | ☑ | ☑ | ☑ | ☑ |

Substituting $\sqrt{(g_i^2 - 1)(g_j^2 - 1)} = k$ and the identities ● into the fourth product identity ●, at $\xi_{ij} = 1$, and using the given expression for $S_{ij}$ at $\xi_{ij} = 1$, we get the second and third equalities of the second formula of ★, with $\varepsilon_i\,\varepsilon_j$ and the radicals of the identities ● symbolic, that is, for all their real values at once ☑

The second and third equalities of the second formula of ★ are also verified end–to–end on each class region, for both values of $\varepsilon_i\,\varepsilon_j$:

| class regions of $(p_i q_i, p_j q_j)$: | $\mathbb{R}_{>0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{>0}$ $\mathbb{R}_{<0}$ | $i\,\mathbb{R}_{>0}$ $i\,\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{>0}$ | $\mathbb{R}_{<0}$ $\mathbb{R}_{<0}$ |
|---|---|---|---|---|---|
| $-k\,\mathrm{sn}(t_i)\,\mathrm{sn}(t_j) = \frac{\varepsilon_i\,\varepsilon_j\,\sqrt{p_i^2\, q_i^2\, p_j^2\, q_j^2}}{p_i q_i p_j q_j}$ | ☑ | ☑ | ☑ | ☑ | ☑ |

**Section 6: ALL CLAIMS VERIFIED**

# Section 7

```
In[•]:= (* ============================================================*)
        (*PART 1:SETUP*)
        (* ============================================================*)
        ClearAll[gi, gj, k, kp, ei, ej, xi, h1of, h2of, h3of, baseAssum,
          wConfigs, ePairs, dirForward, dirBackward, branchOK, smallChip,
```

```
  statusChip, panelH, resultsAll, condOf, vars, hsym, stmtOf, s5colTest];
Format[gi, TraditionalForm] = Subscript[g, i];
Format[gj, TraditionalForm] = Subscript[g, j];
Format[kp, TraditionalForm] = k';
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s5colTest = RGBColor[0.10, 0.35, 0.75];
vars = {gi, gj, k, kp};
(*The three h's in the case k^2≠(gi^2-1)(gj^2-1),
with xi a numeric sign per branch.*)
h1of[e1_, e2_, x_] :=
  (e1 gj Sqrt[(gi^2 - 1) (gj^2 - kp^2) x] + e2 gi Sqrt[(gj^2 - 1) (gi^2 - kp^2) x]) /
   (k^2 - (gi^2 - 1) (gj^2 - 1));
h2of[e1_, e2_, x_] :=
  (Sqrt[(gi^2 - kp^2) (gj^2 - kp^2) x] + e1 e2 gi gj Sqrt[(gi^2 - 1) (gj^2 - 1) x]) /
   (k^2 - (gi^2 - 1) (gj^2 - 1));
h3of[e1_, e2_, x_] :=
  (k^2 gi gj + e1 e2 x Sqrt[(gi^2 - 1) (gi^2 - kp^2) (gj^2 - 1) (gj^2 - kp^2)]) /
   (k^2 - (gi^2 - 1) (gj^2 - 1));
(*Standing assumptions:k,k' with k'^2=1-k^2;
the case guard;
window assumptions added per branch.*)
baseAssum = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 ≠ (gi^2 - 1) (gj^2 - 1);
wConfigs =
  {{"\!\(\*SubscriptBox[\(g\), \(i\)]\), \!\(\*SubscriptBox[\(g\), \(j\)]\)
      ∈ (1, ∞)", gi > 1 && gj > 1, 1},
   {"\!\(\*SubscriptBox[\(g\), \(i\)]\) ∈ (1, ∞), \!\(\*SubscriptBox[\(g\),
      \(j\)]\) ∈ (0, k')", gi > 1 && 0 < gj < kp, -1},
   {"\!\(\*SubscriptBox[\(g\), \(i\)]\) ∈ (0, k'), \!\(\*SubscriptBox[\(g\),
      \(j\)]\) ∈ (1, ∞)", 0 < gi < kp && gj > 1, -1},
   {"\!\(\*SubscriptBox[\(g\), \(i\)]\), \!\(\*SubscriptBox[\(g\),
      \(j\)]\) ∈ (0, k')", 0 < gi < kp && 0 < gj < kp, 1}};
ePairs = {{1, 1}, {1, -1}, {-1, 1}, {-1, -1}};
(*Two-directional branch check:no zero off the condition locus,
and no nonzero value on it.A condition False
 means the claim on this branch is pure nonvanishing.*)
dirForward[h_, cond_, assum_] :=
  Reduce[assum && h == 0 && ! cond, vars, Reals] === False;
dirBackward[h_, cond_, assum_] :=
  If[cond === False, True, Reduce[assum && cond && h ≠ 0, vars, Reals] === False];
branchOK[h_, cond_, assum_] :=
```

```
  dirForward[h, cond, assum] && dirBackward[h, cond, assum];
(*Per-branch condition of each claim;
False encodes "never vanishes".*)
condOf[1, e1_, e2_] := If[e1 == -e2, gi == gj, False];
condOf[2, e1_, e2_] := If[e1 == -e2, gi gj == kp, False];
condOf[3, e1_, e2_] :=
  If[e1 == e2, k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1), False];
hsym[n_] := Subscript["h", "ij" <> ToString[n]];
stmtOf[n_, e1_, e2_] :=
  Module[{c = condOf[n, e1, e2]}, If[c === False, Row[{hsym[n], " ≠ 0"}],
    Row[{hsym[n], " = 0  if and only if  ", TraditionalForm[c]}]]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style["Given hypotheses:", Bold, 14],
     Style[Row[{"k ∈ (0, 1),   ", TraditionalForm[HoldForm[k'^2 == 1 - k^2]]}],
      13], Style[Row[{"For  ", TraditionalForm[HoldForm[i]],
        ", ", TraditionalForm[HoldForm[j]], " ∈ {1, …, 4},   ",
        TraditionalForm[HoldForm[i ≠ j]], " :"}], Bold, 13],
     Style[Row[{TraditionalForm[HoldForm[Subscript[g, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[g, j]]],
        " ∈ (0, k') ⋃ (1, ∞),   ", TraditionalForm[HoldForm[Subscript[ξ, ij] ==
           Sign[(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)]]],
        " ,   ", TraditionalForm[HoldForm[Subscript[e, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[e, j]]], ", ",
        TraditionalForm[HoldForm[Subscript[ξ, ij]]], " ∈ {±1}"}], 13], "",
     Style["Given formulas, in the case \!\(\*SuperscriptBox[\(k\),
        \(2\)]\) ≠ (\!\(\*SubsuperscriptBox[\(g\), \(i\), \(2\)]\)
        - 1)(\!\(\*SubsuperscriptBox[\(g\), \(j\), \(2\)]\) - 1):",
      Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] ==
       (Subscript[e, i] Subscript[g, j] Sqrt[(Subscript[g, i]^2 - 1)
            (Subscript[g, j]^2 - k'^2) Subscript[ξ, ij]] +
         Subscript[e, j] Subscript[g, i] Sqrt[(Subscript[g, j]^2 - 1)
            (Subscript[g, i]^2 - k'^2) Subscript[ξ, ij]]) /
        (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
     Style[TraditionalForm[HoldForm[Subscript[h, ij2] ==
       (Sqrt[(Subscript[g, i]^2 - k'^2) (Subscript[g, j]^2 - k'^2)
            Subscript[ξ, ij]] + Subscript[e, i]
          Subscript[e, j] Subscript[g, i] Subscript[g, j] Sqrt[
           (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1) Subscript[ξ, ij]]) /
        (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
     Style[TraditionalForm[HoldForm[Subscript[h, ij3] ==
       (k^2 Subscript[g, i] Subscript[g, j] + Subscript[e, i] Subscript[e, j]
          Subscript[ξ, ij] Sqrt[(Subscript[g, i]^2 - 1) (Subscript[g, i]^2 -
              k'^2) (Subscript[g, j]^2 - 1) (Subscript[g, j]^2 - k'^2)]) /
        (k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))]], 14],
```

```
      Style["and, in the case \!\(\*SuperscriptBox[\(k\), \(2\)]\)
         = (\!\(\*SubsuperscriptBox[\(g\), \(i\), \(2\)]\)
         - 1)(\!\(\*SubsuperscriptBox[\(g\), \(j\), \(2\)]\)
         - 1) with \!\(\*SubscriptBox[\(e\), \(i\)]\) =
         -\!\(\*SubscriptBox[\(e\), \(j\)]\):", Bold, 14],
      Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
            Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
              (2 k Subscript[g, i] Subscript[g, j])]],
         "    ", TraditionalForm[HoldForm[Subscript[h, ij2] ==
            (Subscript[g, i]^2 Subscript[g, j]^2 - k'^2) /
             (2 k Subscript[g, i] Subscript[g, j])]], "    ", TraditionalForm[
          HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 + Subscript[g, j]^2) /
             (2 Subscript[g, i] Subscript[g, j])]]}],
       14], Style[Row[{"Each  ", smallChip[True],
         "  below certifies that the statement displayed beside it has been
           verified by a symbolic computation at that point;  a  ",
         smallChip[False], "  would mean that this verification failed."}],
       12, GrayLevel[0.35]], "", Style[
       Row[{Style["We want to test the following claims  ", Bold, 14],
         Style["⋆", 14, s5colTest], Style[
          " ,  over all window configurations of  \!\(\*SubscriptBox[\(g\),
            \(i\)]\), \!\(\*SubscriptBox[\(g\),
            \(j\)]\)  and all  \!\(\*SubscriptBox[\(e\),
            \(i\)]\), \!\(\*SubscriptBox[\(e\), \(j\)]\)
            ,  in both directions:", Bold, 14]}]],
      Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] == 0]],
         "   if and only if   ",
         TraditionalForm[HoldForm[Subscript[g, i] == Subscript[g, j]]],
         "  and  ", TraditionalForm[
          HoldForm[Subscript[e, i] == -Subscript[e, j]]]}], Bold, 14, s5colTest],
      Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] == 0]],
         "   if and only if   ",
         TraditionalForm[HoldForm[Subscript[g, i] Subscript[g, j] == k']],
         "  and  ", TraditionalForm[
          HoldForm[Subscript[e, i] == -Subscript[e, j]]]}], Bold, 14, s5colTest],
      Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij3] == 0]],
         "   if and only if   ", TraditionalForm[
          HoldForm[k^2 (Subscript[g, i]^2 + Subscript[g, j]^2) ==
            k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)]], "  and  ",
         TraditionalForm[HoldForm[Subscript[e, i] == Subscript[e, j]]]}],
       Bold, 14, s5colTest]}, Spacings → 1.1],
    Background → RGBColor[0.99, 0.96, 0.88], FrameMargins →
     14]];

(* ============================================================*)
(*PART 3:CASE k^2≠(gi^2-1)(gj^2-1)*)
(* ============================================================*)
```

```
resultsAll = {};
panelH[n_, hfun_, col_, title_] := Module[{grid, ok, headRow, stmtRow},
   grid = Table[branchOK[hfun[ep[[1]], ep[[2]], w[[3]]], condOf[n, ep[[1]], ep[[2]]],
      baseAssum && w[[2]]], {w, wConfigs}, {ep, ePairs}];
   ok = And @@ Flatten[grid];
   AppendTo[resultsAll, ok];
   headRow = Prepend[
     Table[Style[Row[{"(", ep[[1]], ", ", ep[[2]], ")"}], 12, Bold], {ep, ePairs}],
     Style["(\!\(\*SubscriptBox[\(e\), \(i\)]\), \!\(\*SubscriptBox[\(e\),
        \(j\)]\)) →", 12, GrayLevel[0.3]]];
   stmtRow = Prepend[Table[Style[stmtOf[n, ep[[1]], ep[[2]]], 11, GrayLevel[0.25]],
      {ep, ePairs}], Style["tested →", 11, GrayLevel[0.3]]];
   Print[Panel[
     Column[{Row[{Style[title, Bold, 15, col], Spacer[20], statusChip[ok]}],
       Grid[Join[{headRow}, {stmtRow}, Table[Prepend[smallChip /@ grid[[iw]],
           Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]]], {iw, 4}]],
        Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
     Background → Lighter[col, 0.93], FrameMargins → 12]];];
panelH[1, h1of, RGBColor[0.10, 0.30, 0.70],
  "First claim  (case \!\(\*SuperscriptBox[\(k\),
    \(2\)]\) ≠ (\!\(\*SubsuperscriptBox[\(g\), \(i\),
    \(2\)]\)-1)(\!\(\*SubsuperscriptBox[\(g\), \(j\), \(2\)]\)-1)):"];
panelH[2, h2of, RGBColor[0.55, 0.25, 0.70],
  "Second claim  (case \!\(\*SuperscriptBox[\(k\),
    \(2\)]\) ≠ (\!\(\*SubsuperscriptBox[\(g\), \(i\),
    \(2\)]\)-1)(\!\(\*SubsuperscriptBox[\(g\), \(j\), \(2\)]\)-1)):"];
panelH[3, h3of, RGBColor[0.00, 0.50, 0.50],
  "Third claim  (case \!\(\*SuperscriptBox[\(k\),
    \(2\)]\) ≠ (\!\(\*SubsuperscriptBox[\(g\), \(i\),
    \(2\)]\)-1)(\!\(\*SubsuperscriptBox[\(g\), \(j\), \(2\)]\)-1)):"];

(* ============================================================*)
(*PART 4:CASE k^2==(gi^2-1)(gj^2-1) WITH ei=-ej*)
(* ============================================================*)
Module[{assum7, ok1, ok2, ok3, ok4, ok},
  assum7 = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 == (gi^2 - 1) (gj^2 - 1) &&
    ((gi > 1 && gj > 1) || (0 < gi < kp && 0 < gj < kp));
  ok1 = And @@ Table[
     branchOK[e1 (gi^2 - gj^2) / (2 k gi gj), gi == gj, assum7], {e1, {1, -1}}];
  ok2 = branchOK[(gi^2 gj^2 - kp^2) / (2 k gi gj), gi gj == kp, assum7];
  (*ok3 certifies the displayed bound h_ij3≥1,hence h_ij3≠0*)
  ok3 = Reduce[assum7 && (gi^2 + gj^2) / (2 gi gj) < 1, vars, Reals] === False;
  ok4 = Reduce[assum7 && k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1),
     vars, Reals] === False;
  ok = ok1 && ok2 && ok3 && ok4;
  AppendTo[resultsAll, ok];
  Print[
```

```
  Panel[Column[{Row[{Style["Case \!\(\*SuperscriptBox[\(k\), \(2\)]\) =
          (\!\(\*SubsuperscriptBox[\(g\), \(i\),
          \(2\)]\)-1)(\!\(\*SubsuperscriptBox[\(g\), \(j\),
          \(2\)]\)-1), with \!\(\*SubscriptBox[\(e\), \(i\)]\)
          = -\!\(\*SubscriptBox[\(e\), \(j\)]\):", Bold, 15,
        RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]}],
     Column[{Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
           Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
            (2 k Subscript[g, i] Subscript[g, j])]], ":   ", hsym[1],
        " = 0  if and only if  ", TraditionalForm[HoldForm[
          Subscript[g, i] == Subscript[g, j]]], "   ", smallChip[ok1]}], 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] ==
           (Subscript[g, i]^2 Subscript[g, j]^2 - k'^2) /
            (2 k Subscript[g, i] Subscript[g, j])]], ":   ",
        hsym[2], " = 0  if and only if  ", TraditionalForm[
         HoldForm[Subscript[g, i] Subscript[g, j] == k']],
        "   ", smallChip[ok2]}], 13], Style[
       Row[{TraditionalForm[HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 +
            Subscript[g, j]^2) / (2 Subscript[g, i] Subscript[g, j])]],
        " ≥ 1   ", smallChip[ok3], "   ,  so  ", hsym[3], " ≠ 0"}], 13],
      Style[Row[{"condition of the third claim unsatisfiable here   ",
        smallChip[ok4]}], 13]}, Spacings → 0.8]},
    Spacings → 1], Background → Lighter[RGBColor[0.75, 0.35,
      0.05],
     0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 5:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 7: ALL CLAIMS VERIFIED",
     "Section 7: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

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Given hypotheses:

$k \in (0, 1), \quad (k')^2 = 1 - k^2$

For $i, j \in \{1, \ldots, 4\}, \quad i \neq j$:

$g_i, g_j \in (0, k') \cup (1, \infty), \quad \xi_{ij} = \operatorname{sgn}\left((g_i^2 - 1)(g_j^2 - 1)\right), \quad e_i, e_j, \xi_{ij} \in \{\pm 1\}$

Given formulas, in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$:

$$h_{ij1} = \frac{e_i g_j \sqrt{(g_i^2-1)(g_j^2-(k')^2)}\,\xi_{ij} + e_j g_i \sqrt{(g_j^2-1)(g_i^2-(k')^2)}\,\xi_{ij}}{k^2-(g_i^2-1)(g_j^2-1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2-(k')^2)(g_j^2-(k')^2)}\,\xi_{ij} + e_i e_j g_i g_j \sqrt{(g_i^2-1)(g_j^2-1)}\,\xi_{ij}}{k^2-(g_i^2-1)(g_j^2-1)}$$

$$h_{ij3} = \frac{k^2 g_i g_j + e_i e_j \xi_{ij} \sqrt{(g_i^2-1)(g_i^2-(k')^2)(g_j^2-1)(g_j^2-(k')^2)}}{k^2-(g_i^2-1)(g_j^2-1)}$$

and, in the case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$ with $e_i = -e_j$:

$$h_{ij1} = \frac{e_i (g_i^2 - g_j^2)}{2 k g_i g_j} \qquad h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2 k g_i g_j} \qquad h_{ij3} = \frac{g_i^2 + g_j^2}{2 g_i g_j}$$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following claims ★, over all window configurations of $g_i, g_j$ and all $e_i, e_j$, in both directions:

$h_{ij1} = 0$ if and only if $g_i = g_j$ and $e_i = -e_j$

$h_{ij2} = 0$ if and only if $g_i g_j = k'$ and $e_i = -e_j$

$h_{ij3} = 0$ if and only if $k^2 \left(g_i^2 + g_j^2\right) = k^2 - (g_i^2 - 1)\left(g_j^2 - 1\right)$ and $e_i = e_j$

First claim (case $k^2 \neq (g_i^2-1)(g_j^2-1)$): Verified

| $(e_i, e_j) \to$ | $(1, 1)$ | $(1, -1)$ | $(-1, 1)$ | $(-1, -1)$ |
|---|---|---|---|---|
| tested → | $h_{ij1} \neq 0$ | $h_{ij1} = 0$ if and only if $g_i = g_j$ | $h_{ij1} = 0$ if and only if $g_i = g_j$ | $h_{ij1} \neq 0$ |
| $g_i, g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (1, \infty), g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (0, k'), g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i, g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |

**Second claim (case $k^2 \neq (g_i^2-1)(g_j^2-1)$):** Verified

| $(e_i, e_j) \to$ | (1, 1) | (1, −1) | (−1, 1) | (−1, −1) |
|---|---|---|---|---|
| tested → | $h_{ij2} \neq 0$ | $h_{ij2} = 0$ if and only if $g_i g_j = k'$ | $h_{ij2} = 0$ if and only if $g_i g_j = k'$ | $h_{ij2} \neq 0$ |
| $g_i, g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (1, \infty), g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (0, k'), g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i, g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |

**Third claim (case $k^2 \neq (g_i^2-1)(g_j^2-1)$):** Verified

| $(e_i, e_j) \to$ | (1, 1) | (1, −1) | (−1, 1) | (−1, −1) |
|---|---|---|---|---|
| tested → | $h_{ij3} = 0$ if and only if $k^2(g_i^2+g_j^2) = k^2-(g_i^2-1)(g_j^2-1)$ | $h_{ij3} \neq 0$ | $h_{ij3} \neq 0$ | $h_{ij3} = 0$ if and only if $k^2(g_i^2+g_j^2) = k^2-(g_i^2-1)(g_j^2-1)$ |
| $g_i, g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (1, \infty), g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |
| $g_i \in (0, k'), g_j \in (1, \infty)$ | ☑ | ☑ | ☑ | ☑ |
| $g_i, g_j \in (0, k')$ | ☑ | ☑ | ☑ | ☑ |

**Case $k^2 = (g_i^2-1)(g_j^2-1)$, with $e_i = -e_j$:** Verified

$h_{ij1} = \frac{e_i(g_i^2-g_j^2)}{2 k g_i g_j}$: $h_{ij1} = 0$ if and only if $g_i = g_j$ ☑

$h_{ij2} = \frac{g_i^2 g_j^2-(k')^2}{2 k g_i g_j}$: $h_{ij2} = 0$ if and only if $g_i g_j = k'$ ☑

$h_{ij3} = \frac{g_i^2+g_j^2}{2 g_i g_j} \geq 1$ ☑ , so $h_{ij3} \neq 0$

condition of the third claim unsatisfiable here ☑

**Section 7: ALL CLAIMS VERIFIED**

# Section 8

In[•]:=
```
(* ============================================================*)
(*PART 1:INITIALIZATION*)
(* ============================================================*)
ClearAll[i, UU, XX, YY, ZZ, X3, tt, RR, h1, h2, h3, ee0, ep2, ep3,
  kk, kq, wConfigs, sTriples, dataOf, lhsOf, derOf, assumOf, derCheck,
  resultsAll, statusChip, smallChip, lblZ, lblY, lblX, lblX3, lblY2,
  lblXi, lblUZ, s6colTest, chkR, addSn, Sv, pS, qS, p3S, zS, sS, cS, dS,
  cT, gS, sW, cW, dW, snT1OK, snT2OK, denWOK, expWOK, sSqOK, snWSqOK,
  cnDnWOK, denUOK, expUOK, shiftCOK, w3Configs, thCheck, cCheck];
```

```
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s6colTest = RGBColor[0.10, 0.35, 0.75];
(*Sign regions for i in {1,2}:
 two z_i-windows (from g_i in (0,k') or (1,infinity)) and the sign of y_i;
the sign of x_1 is then forced by u x_1 y_i z_i>0. The window z_i<
 -1 carries 1+u z_i>0. On each region,p_1 and q_i take their
 explicit radical values and the nested radicals collapse;
the identity is checked there for all 8 sign combinations of e',
e_0,eps_i.The displayed expansions and the shift chain are checked
 symbolically:the expansions from the first addition formula with the parity
   formulas and the given denominators,for both signs,and the chain for
 both values of (-1)^m and of e_2,using (-1)^(-e_2 m)=(-1)^m.In these
  symbolic checks x_1 enters as 1/p_1^2,by the given x_1 p_1^2=1.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
addSn[sv_, cv_, dv_, sw_, cw_, dw_, k2_] := (sv cw dw + sw cv dv) / (1 - k2 sv^2 sw^2);
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, i] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, i] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, i] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, i] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, 1] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, 1] < 0]];
wConfigs = {{Row[{lblZ[1], ",  ", lblY[1], ",  hence  ", lblX[1]}], ZZ, 1, 1},
   {Row[{lblZ[1], ",  ", lblY[-1], ",  hence  ", lblX[-1]}], ZZ, -1, -1},
   {Row[{lblZ[2], ",  ", lblY[1], ",  hence  ", lblX[-1]}], -1 - ZZ, 1, -1},
   {Row[{lblZ[2], ",  ", lblY[-1], ",  hence  ", lblX[1]}], -1 - ZZ, -1, 1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, zl_, sy_, sx_}] := Module[{yl = sy YY, x1 = sx XX,
    kkv = Sqrt[UU], kpv = Sqrt[1 - UU], gl, p1, ql, Dd}, gl = Sqrt[1 + 1 / zl];
   p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   Dd = -(1 - x1 tt^2) (1 - UU x1 tt^2);
   {zl, yl, x1, kkv, kpv, gl, p1, ql, Dd}];
lhsOf[{zl_, yl_, x1_, kkv_, kpv_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
  Module[{snl, cnl, RRv = Sqrt[x1 Dd]},
   snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kkv p1 ql);
   cnl = eps Sqrt[(gl^2 - kpv^2) p1^2 ql^2] / (kkv p1 ql);
   (zl ql cnl gl (tt / p1) + ep zl ql snl (e0 I RRv / (XX p1))) / (zl + x1 tt^2)];
derOf[{zl_, yl_, x1_, kkv_, kpv_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
  eps (tt Sqrt[UU x1 yl (1 + zl) (1 + UU zl)] - ep e0 Sqrt[UU yl zl Dd]) /
    (yl UU (zl + x1 tt^2));
```

```
assumOf[{lab_, zl_, sy_, sx_}] := 0 < UU < 1 && XX > 0 && YY > 0 && ZZ > 0 &&
   Element[tt, Reals] && sx XX (-(1 - sx XX tt^2) (1 - UU sx XX tt^2)) ≥ 0 &&
   1 + UU zl > 0 && zl + sx XX tt^2 ≠ 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},
   TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
      assumOf[w]], 180, False] === True];
(*symbolic checks of the intermediate relations*)
snT1OK = chkR[gS^2 + kq ((1 - gS^2) / kq), 1];
snT2OK = chkR[(1 - Sqrt[1 + 1 / zS]^2) / Sqrt[UU]^2, -1 / (UU zS)];
denWOK = chkR[1 - UU (-1 / (UU zS)) (tt / pS)^2, (zS + tt^2 / pS^2) / zS];
sSqOK =
  And @@ Flatten[Table[chkR[(ev Sqrt[pS^2 p3S^2 qS^4 xv] / (pS p3S qS^2))^2, xv],
     {ev, {1, -1}}, {xv, {1, -1}}]];
snWSqOK = And @@ Table[chkR[(I Sv h1)^2 /. Sv^2 → xv, -xv h1^2], {xv, {1, -1}}];
cnDnWOK = chkR[(Sv h2) h3, Sv h2 h3];
denUOK = And @@ Table[chkR[1 - kq (tt^2 / pS^2) (-xv h1^2),
     1 + xv kq h1^2 (tt^2 / pS^2)], {xv, {1, -1}}];
(*symbolic checks of the displayed expansions and of the shift chain*)
expWOK = And @@
   Table[chkR[(qS addSn[tt / pS, cS, dS, ep sS, cT, gS, UU]) /. sS^2 → -1 / (UU zS),
     (zS qS cT gS (tt / pS) + ep zS qS sS cS dS) / (zS + tt^2 / pS^2)], {ep, {1, -1}}];
expUOK = And @@
   Table[chkR[(p3S addSn[tt / pS, cS, dS, -sW, cW, dW, kq]) /. sW^2 → -xv h1^2,
     p3S (cW dW (tt / pS) - sW cS dS) / (1 + xv kq h1^2 tt^2 / pS^2)], {xv, {1, -1}}];
shiftCOK =
  And @@ Flatten[Table[chkR[p3S ev^(-e2) / (kk (tt / pS)), ev pS p3S / (kk tt)],
     {ev, {1, -1}}, {e2, {1, -1}}]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 1  (",
        TraditionalForm[HoldForm[u > 0]], ") :"}], Bold, 14],
     Style[Row[{TraditionalForm[HoldForm[0 < u < 1]], ",   ",
        TraditionalForm[HoldForm[k == Sqrt[u]]], ",   ", TraditionalForm[
         HoldForm[k' == Sqrt[1 - u]]], ",   ", TraditionalForm[HoldForm[k]],
        ", ", TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 14],
     Style[Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
      Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[y, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
        ",   ", TraditionalForm[HoldForm[Subscript[x, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[y, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[z, i]]],
        " ≠ -1, 0,   ", TraditionalForm[HoldForm[Subscript[g, i] ==
           dn[Subscript[t, i]] == Sqrt[1 + 1 / Subscript[z, i]]]]}], 14],
     Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]]], ", ",
```

```
   TraditionalForm[HoldForm[Subscript[q, i]]], " ∈ ",
   Subscript["ℝ", "> 0"], " ⋃ i ", Subscript["ℝ", "> 0"],
   ",   ", TraditionalForm[HoldForm[Subscript[e, 0]]],
   ", ", TraditionalForm[HoldForm[Derivative[1][e]]],
   ", ", TraditionalForm[HoldForm[Subscript[e, i]]],
   ", ", TraditionalForm[HoldForm[Subscript[ε, i]]], ", ",
   TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}],
 14], Style[Row[{TraditionalForm[HoldForm[
     u Subscript[x, i] Subscript[y, i] Subscript[z, i] > 0]], ",   ",
   TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
   ",   ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
Style[Row[{TraditionalForm[HoldForm[𝒟[Subscript[x, 1], t] ==
      -(1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
   ",   ", TraditionalForm[HoldForm[
     R == Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]]]], ",   ",
   TraditionalForm[HoldForm[Subscript[x, 1] 𝒟[Subscript[x, 1], t] > 0]]}],
 14], "", Style["Given formulas:", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[sn[v + w] ==
      (sn[v] cn[w] dn[w] + sn[w] cn[v] dn[v]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
   "    ", Style["the first addition formula", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]]}], 14], Style[
 Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] == {-sn[v], cn[v], dn[v]}]],
   "    ", Style["the parity formulas", Italic,
    13, RGBColor[0.75, 0.35, 0.05]]}], 14],
Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 == 1]],
   ",   ", TraditionalForm[HoldForm[dn[v]^2 + k^2 sn[v]^2 == 1]],
   "    ", Style["the Pythagorean relations",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
 14], Style[Row[{"Each  ", smallChip[True],
   "  below certifies that the statement displayed beside it has been
     verified by a symbolic computation at that point;  a  ",
   smallChip[False], "  would mean that this verification failed."}],
 12, GrayLevel[0.35]], "", Style[
 Row[{Style["We want to test these formulas  ", Bold, 14],
   Style["⋆", 14, s6colTest], Style[" :", Bold, 14]}]],
Style[TraditionalForm[HoldForm[Subscript[W, 2] == Subscript[ε, 2]
     (t Sqrt[u Subscript[x, 1] Subscript[y, 2] (1 + Subscript[z, 2])
           (1 + u Subscript[z, 2])] + Subscript[e, 0] Subscript[e, 2]
         Sqrt[u Subscript[y, 2] Subscript[z, 2] 𝒟[Subscript[x, 1], t]]) /
      (Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]], 15,
 s6colTest], Style[TraditionalForm[HoldForm[Subscript[W, 1] == Subscript[
     ε, 1] (t Sqrt[u Subscript[x, 1] Subscript[y, 1] (1 + Subscript[z, 1])
           (1 + u Subscript[z, 1])] - Subscript[e, 0] Subscript[e, 1]
         Sqrt[u Subscript[y, 1] Subscript[z, 1] 𝒟[Subscript[x, 1], t]]) /
      (Subscript[y, 1] u (Subscript[z, 1] + Subscript[x, 1] t^2))]], 15,
 s6colTest], Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2]
      Subscript[ε, 3] Sign[u Subscript[z, 2]] (Abs[Subscript[x, 1]]
```

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                Subscript[h, 232] Subscript[h, 233] t + Subscript[e, 0]
                Subscript[h, 231] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]]) /
              (Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] (1 + Subscript[ξ, 23] k^
                    2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]], "    ",
        Style["in cases (a) and (b)", Italic, 13, GrayLevel[0.35]]}],
      15, s6colTest], Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2]
          Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
           (k Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] t)]], "    ",
        Style["in case (c)", Italic, 13, GrayLevel[0.35]]}], 15, s6colTest]},
    Spacings → 1.1], Background → RGBColor[0.99, 0.96, 0.88],
   FrameMargins →
    14]];

(* ============================================================*)
(*PART 3:THE W-FORMULAS*)
(* ============================================================*)
resultsAll = {};
Module[{grid, regionOK, ok},
  grid = Table[derCheck[w, sgnTriple], {w, wConfigs}, {sgnTriple, sTriples}];
  regionOK = And @@@ grid;
  ok = snT1OK && snT2OK && denWOK && expWOK && And @@ regionOK;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The derivation:  the formulas for  ", Bold, 15,
         RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[
          HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
        Style["  and  ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
        Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
         15, RGBColor[0.10, 0.30, 0.70]], Style["  for  ", Bold, 15,
         RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u > 0]],
         Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]}],
      Style["Given hypotheses and formulas, in addition to those above:",
       Bold, 13], Style[
       Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2 :"}], Bold, 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[x, 1]
            Subsuperscript[p, 1, 2] == 1]], ",   ", TraditionalForm[
          HoldForm[Subscript[y, i] Subsuperscript[q, i, 2] == 1]], ",   ",
         TraditionalForm[HoldForm[Subscript[p, i] == Subscript[p, 1]]]}], 13],
      Style[TraditionalForm[HoldForm[sn[Subscript[t, i]] == I Subscript[ε, i]
          Sqrt[(Subsuperscript[g, i, 2] - 1) Subsuperscript[p, i, 2]
            Subsuperscript[q, i, 2]] / (k Subscript[p, i] Subscript[q, i])]],
       14], Style[TraditionalForm[HoldForm[cn[Subscript[t, i]] == Subscript[
          ε, i] Sqrt[(Subsuperscript[g, i, 2] - k'^2) Subsuperscript[p, i, 2]
            Subsuperscript[q, i, 2]] / (k Subscript[p, i] Subscript[q, i])]],
       14], Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
         ",   ", TraditionalForm[HoldForm[Subscript[p, 1] cn[ς] dn[ς] ==
          Subscript[e, 0] I R / Abs[Subscript[x, 1]]]]}], 14],
```

```
Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e]
         Subscript[t, i]] == {Derivative[1][e] sn[Subscript[t, i]],
        cn[Subscript[t, i]], dn[Subscript[t, i]]}]], "    ", Style[
    "by the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
 14], "", Style["Verification:", Bold, 13],
Style[Row[{"By  ", Style["the Pythagorean relations",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]], "  and  ",
   TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
   " :   ", TraditionalForm[HoldForm[sn[Subscript[t, i]]^2 ==
      (1 - Subsuperscript[g, i, 2]) / k^2]], "   ", smallChip[snT1OK]}],
 13], Style[Row[{"By  ", TraditionalForm[HoldForm[Subscript[g, i] ==
      Sqrt[1 + 1 / Subscript[z, i]]]], "  and  ", TraditionalForm[
    HoldForm[k == Sqrt[u]]], " :   ", TraditionalForm[HoldForm[
     (1 - Subsuperscript[g, i, 2]) / k^2 == - (1 / (u Subscript[z, i]))]],
   "   ", smallChip[snT2OK]}], 13], Style[Row[{"Hence  ",
   TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[ς]^2 ==
      (Subscript[z, i] + Subscript[x, 1] t^2) / Subscript[z, i]]], "   ",
   smallChip[denWOK], "   ,  which is nonzero by the choice of  ",
   TraditionalForm[HoldForm[t]]}], 13],
Style[Row[{"Because  ", TraditionalForm[HoldForm[±Subscript[t, i]]],
   "  and  ", TraditionalForm[HoldForm[ς]], "  are not poles and  ",
   TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[ς]^2 ≠ 0]],
   ",  ", Style["the first addition formula", Italic, 13, RGBColor[0.10,
     0.30, 0.70]], ",  at  ", TraditionalForm[HoldForm[v == ς]], ",  ",
   TraditionalForm[HoldForm[w == Derivative[1][e] Subscript[t, i]]],
   ",  holds pointwise and, together with  ",
   Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
   ",  gives"}], 13], Style[Row[{TraditionalForm[
    HoldForm[Subscript[q, i] sn[ς + Derivative[1][e] Subscript[t, i]] ==
      (Subscript[z, i] Subscript[q, i] cn[Subscript[t, i]]
          dn[Subscript[t, i]] sn[ς] + Derivative[1][e] Subscript[z, i]
          Subscript[q, i] sn[Subscript[t, i]] cn[ς] dn[ς]) / (Subscript[z,
          i] + Subscript[x, 1] t^2)]], "   ", smallChip[expWOK]}], 14],
Style["Using the given formulas, we get", 13], Style[
 Row[{TraditionalForm[HoldForm[Subscript[q, i] sn[ς + Derivative[1][e]
          Subscript[t, i]] == Subscript[ε, i] (t Sqrt[u Subscript[x, 1]
             Subscript[y, i] (1 + Subscript[z, i]) (1 + u Subscript[z, i])] -
          Derivative[1][e] Subscript[e, 0] Sqrt[u Subscript[y, i]
             Subscript[z, i] 𝒟[Subscript[x, 1], t]]) /
        (Subscript[y, i] u (Subscript[z, i] + Subscript[x, 1] t^2))]],
   "      ", Style["✱", 15, RGBColor[0.10, 0.30, 0.70]]}], 14],
Style[Row[{"Setting  ", TraditionalForm[HoldForm[i == 2]], ",  ",
   TraditionalForm[HoldForm[Derivative[1][e] == - Subscript[e, 2]]],
   "  in  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
   "  gives the first formula of  ",
   Style["⋆", 13, s6colTest], " ."}], 13],
Style[Row[{"Setting  ", TraditionalForm[HoldForm[i == 1]], ",  ",
```

```
        TraditionalForm[HoldForm[Derivative[1][e] == Subscript[e, 1]]],
        "  in  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
        "  gives the second formula of  ",
        Style["⋆", 13, s6colTest], " ."}], 13], "",
     Style[Row[{"The identity  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
        "  is verified separately on each sign region:"}], 13],
     Grid[Table[{Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]],
        Style[Row[{"for all 8 sign combinations of  ",
           TraditionalForm[HoldForm[Derivative[1][e]]], ", ",
           TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
           TraditionalForm[HoldForm[Subscript[ε, i]]], " ∈ {±1}"}],
         12, GrayLevel[0.3]], smallChip[regionOK[[iw]]]}, {iw, 4}],
      Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
    Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
    FrameMargins →
     12]];];

(* ============================================================*)
(*PART 4:U-FORMULA IN CASES (a) AND (b)*)
(* ============================================================*)
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == -1]];
w3Configs = Flatten[
   Table[{Row[{lblX[sx], ",  ", lblX3[s3], ",  ", lblY2[sy], ",  hence  ",
       lblXi[sx s3], ",  ", lblUZ[sx sy]}], sx, s3, sy},
    {sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}], 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{x1 = sx XX, x3 = s3 X3, xi = sx s3, suz = sx sy, p1, p3, q2, S23,
    denom, lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
   q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
   denom = 1 + xi kk^2 x1 h1^2 tt^2;
   lhs = p3 (S23 h2 h3 (tt / p1) - I S23 h1 (ee0 I RR / (XX p1))) / denom;
   rhs = ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (Sqrt[XX X3] denom);
   assum = XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && RR ≥ 0 &&
     Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
   TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{grid, ok}, grid = thCheck /@ w3Configs;
  ok = sSqOK && snWSqOK && cnDnWOK && denUOK && expUOK && And @@ grid;
  AppendTo[resultsAll, ok];
```

```
Print[
 Panel[Column[{Row[{Style["The derivation:  the formula for  ", Bold, 15,
       RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
       Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
       "  in cases (a) and (b) for  ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
      Style[TraditionalForm[HoldForm[u > 0]], Bold, 15,
       RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]}],
    Style["Given hypotheses and formulas, in addition to those above:",
     Bold, 13], Style[Row[{TraditionalForm[
        HoldForm[Subscript[x, 3] Subsuperscript[p, 3, 2] == 1]], ",   ",
       TraditionalForm[HoldForm[Subscript[y, 2] Subsuperscript[q, 2, 2] == 1]],
       ",   ", TraditionalForm[HoldForm[Subscript[p, 2] == Subscript[p, 1]]],
       ",   ", TraditionalForm[HoldForm[Subscript[q, 3] == Subscript[q, 2]]],
       ",   ", TraditionalForm[HoldForm[Subscript[h, 231]]],
       ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
       TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] ==
          Sign[Subscript[x, 1] Subscript[x, 3]]]],
       ",   ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
          Sign[Subscript[x, 1] Subscript[y, 2]]]]}], 13],
    Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
       ",   ", TraditionalForm[HoldForm[Subscript[p, 1] cn[ς] dn[ς] ==
          Subscript[e, 0] I R / Abs[Subscript[x, 1]]]]}], 13],
    Style[TraditionalForm[HoldForm[Subscript[S, 23] ==
        Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2]
           Subsuperscript[p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
         (Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]],
     13], Style[TraditionalForm[HoldForm[{sn, cn, dn}[
        Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
       {I Subscript[S, 23] Subscript[h, 231],
        Subscript[S, 23] Subscript[h, 232], Subscript[h, 233]}]], 13],
    Style[TraditionalForm[HoldForm[sn[ς]^2 == Subscript[x, 1] t^2]], 13],
    "", Style["Verification:", Bold, 13],
    Style[Row[{"By the given expression for  ",
       TraditionalForm[HoldForm[Subscript[S, 23]]], " :   ", TraditionalForm[
        HoldForm[Subsuperscript[S, 23, 2] == Subscript[ξ, 23]]],
       "   ", smallChip[sSqOK]}], 13],
    Style[Row[{"Hence  ", TraditionalForm[HoldForm[
         sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^
           2 == -Subscript[ξ, 23] Subsuperscript[h, 231, 2]]], "   ",
       smallChip[snWSqOK], "   and   ", TraditionalForm[HoldForm[
         cn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]
           dn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t,
               3]] == Subscript[S, 23] Subscript[h, 232] Subscript[h, 233]]],
       "   ", smallChip[cnDnWOK]}], 13], Style[Row[{"Hence  ",
       TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2] Subscript[
                 t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 == 1 + Subscript[ξ,
```

```
          23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2]], "   ",
   smallChip[denUOK], "   ,  which is nonzero by the choice of  ",
   TraditionalForm[HoldForm[t]]}], 13],
Style[Row[{"Because  ", TraditionalForm[HoldForm[± (Subscript[e, 2]
          Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3])]], "  and  ",
   TraditionalForm[HoldForm[ς]], "  are not poles and  ",
   TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2]
             Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 ≠ 0]],
   ",  ", Style["the first addition formula", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]], ",  at  ",
   TraditionalForm[HoldForm[v == ς]], ",  ", TraditionalForm[HoldForm[w ==
      - Subscript[e, 2] Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]]],
   ",  holds pointwise and, together with  ",
   Style["the parity formulas", Italic, 13,
    RGBColor[0.75, 0.35, 0.05]], ",  gives"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e, 2]
           Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
      (Subscript[p, 3] (cn[Subscript[e, 2] Subscript[t, 2] + Subscript[e,
                 3] Subscript[t, 3]] dn[Subscript[e, 2] Subscript[t, 2] +
               Subscript[e, 3] Subscript[t, 3]] sn[ς] - sn[Subscript[e, 2]
                Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] cn[ς]
             dn[ς])) / (1 + Subscript[ξ, 23] k^2 Subscript[x, 1]
            Subsuperscript[h, 231, 2] t^2)]], "   ", smallChip[expUOK]}],
 14], Style["Using the given hypotheses and formulas, we get", 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e, 2]
           Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
      Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
       (Abs[Subscript[x, 1]] Subscript[h, 232] Subscript[h, 233] t +
          Subscript[e, 0] Subscript[h, 231]
           Sqrt[Subscript[x, 1]] 𝒟[Subscript[x, 1], t]]) /
        (Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] (1 + Subscript[ξ, 23]
             k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]],
   "      ", Style["◆", 14, RGBColor[0.55, 0.25, 0.70]]}], 14],
Style[Row[{"which is the third formula of  ",
   Style["⋆", 13, s6colTest], " ."}], 13], "",
Style[Row[{"The identity  ", Style["◆", 13, RGBColor[0.55, 0.25, 0.70]],
   "  is verified separately on each sign region, with  ",
   TraditionalForm[HoldForm[Subscript[h, 231]]],
   ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
   ", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
   ", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
   ", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
   ", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
   "  symbolic, that is, for all their real values at once,
     which covers all their sign choices:"}], 13],
Style[Row[{"In case (a), both values of  ", TraditionalForm[
    HoldForm[Subscript[ξ, 23]]], "  occur, because the sign of  ",
```

```
          TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3]]],
          "  is not constrained;  in case (b),  ",
          TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]], ",  because  ",
          TraditionalForm[HoldForm[k^2 == (Subsuperscript[g, 2, 2] - 1)
              (Subsuperscript[g, 3, 2] - 1) > 0]], "  and  ", TraditionalForm[
           HoldForm[Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]]], ",  so  ",
          TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
          " .  The eight sign regions cover both cases."}],
        12, GrayLevel[0.35]], Grid[Table[{Style[w3Configs[[iw, 1]],
           12, GrayLevel[0.3]], smallChip[grid[[iw]]]}, {iw, 8}],
        Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
     Background → Lighter[RGBColor[0.55, 0.25, 0.70],
       0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 5:U-FORMULA IN CASE (c) FOR u>0*)
(* ============================================================*)
cCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{p1, p3, q2, xi = sx s3, suz = sx sy, S23, lhs, rhs, assum},
   p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
   q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
   lhs = S23 p1 p3 / (kk tt);
   rhs = suz ep2 ep3 sx / (kk Sqrt[XX X3] tt);
   assum =
    XX > 0 && X3 > 0 && YY > 0 && 0 < kk < 1 && Element[tt | ep2 | ep3, Reals] && tt ≠ 0;
   TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] #[[3]] == 1 &];
  grid = cCheck /@ cConfigs;
  ok = shiftCOK && And @@ grid;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The derivation:  the formula for  ", Bold, 15,
         RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
         15, RGBColor[0.00, 0.50, 0.50]], Style["  in case (c) for  ", Bold, 15,
         RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u > 0]],
         Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
      Style["Given hypotheses and formulas, in addition to those above:",
       Bold, 13], Style[Row[{TraditionalForm[
          HoldForm[Subscript[ξ, 23] == Sign[Subscript[x, 1] Subscript[x, 3]]]],
        ",   ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
           Sign[Subscript[x, 1] Subscript[y, 2]]]]}], 13],
      Style[TraditionalForm[HoldForm[Subscript[S, 23] ==
          Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2]
            Subsuperscript[p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
          (Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]], 13],
```

```
  Style[Row[{TraditionalForm[HoldForm[Subscript[t, 2] + Subscript[t, 3] ==
        2 m K + I K']], ",   ", TraditionalForm[HoldForm[m ∈ ℤ]], ",   ",
     TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ",   ",
     TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]}], 13],
  Style[Row[{TraditionalForm[HoldForm[sn[v + 2 m K] == (-1)^m sn[v]]],
     ",   ", TraditionalForm[HoldForm[sn[v ± I K'] == 1 / (k sn[v])]], "    ",
     Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],
   13], Style[Row[{TraditionalForm[HoldForm[sn[ς] == t / Subscript[p, 1]]],
     ",   ", TraditionalForm[HoldForm[t ≠ 0]],
     ",   by the choice of  ", TraditionalForm[HoldForm[t]]}],
   13], "", Style["Verification:", Bold, 13],
  Style[Row[{"Because  ", TraditionalForm[
      HoldForm[Subscript[e, 2] == Subscript[e, 3]]], "  and  ",
     TraditionalForm[HoldForm[(-1)^(-Subscript[e, 2] m) == (-1)^m]],
     ",  ", Style["the shift formulas", Italic, 13,
      RGBColor[0.00, 0.50, 0.50]], "  give"}], 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3]
         sn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
        (-1)^m Subscript[p, 3] / (k sn[ς]) == Subscript[S, 23] Subscript[p, 1]
         Subscript[p, 3] / (k t)]], "   ", smallChip[shiftCOK]}], 14],
  Style["Using the given hypotheses and formulas, we get", 13],
  Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] sn[ς - Subscript[e, 2]
            (Subscript[t, 2] + Subscript[t, 3])] == Subscript[ε, 2]
         Subscript[ε, 3] Sign[u Subscript[z, 2]] Sign[Subscript[x, 1]] /
          (k Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]] t)]],
     "      ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]}], 14],
  Style[Row[{"which is the fourth formula of  ",
     Style["★", 13, s6colTest], " ."}], 13], "",
  Style[Row[{"The identity  ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
     "  is verified separately on each of the four sign
       regions realizable in case (c), with  ",
     TraditionalForm[HoldForm[Subscript[ε, 2]]], ", ",
     TraditionalForm[HoldForm[Subscript[ε, 3]]],
     "  symbolic, that is, for all their real values at once,
       which covers all their sign choices:"}], 13],
  Style[Row[{"Here  ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
     ",  because  ", TraditionalForm[HoldForm[
       k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
     "  and  ", TraditionalForm[HoldForm[
       Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]]], ",  so  ",
     TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
     " ."}], 12, GrayLevel[0.35]],
  Grid[Table[{Style[cConfigs[[iw, 1]], 12, GrayLevel[0.3]],
     smallChip[grid[[iw]]]}, {iw, 4}],
    Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
Background → Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
FrameMargins →
```

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```
      12]];];

(* ============================================================*)
(*PART 6:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 8: ALL CLAIMS VERIFIED",
      "Section 8: SOME CLAIMS FAILED"], Bold, White, 16], Background →
    If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
   FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses, Case 1 $(u > 0)$ :

$0 < u < 1, \quad k = \sqrt{u}, \quad k' = \sqrt{1-u}, \quad k, k' \in (0, 1)$

For $i = 1, 2, 3, 4$ :

$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \mathrm{dn}(t_i) = \sqrt{1 + \frac{1}{z_i}}$

$p_i, q_i \in \mathbb{R}_{>0} \bigcup i\,\mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$

$u\,x_i\,y_i\,z_i > 0, \quad z_i\,(1+z_i) > 0, \quad 1 + u\,z_i > 0$

$\mathcal{D}(x_1, t) = -(1 - x_1\,t^2)\,(1 - u\,x_1\,t^2), \quad R = \sqrt{x_1\,\mathcal{D}(x_1, t)}, \quad x_1\,\mathcal{D}(x_1, t) > 0$

Given formulas:

$\mathrm{sn}(v+w) = \frac{\mathrm{sn}(v)\,\mathrm{cn}(w)\,\mathrm{dn}(w) + \mathrm{sn}(w)\,\mathrm{cn}(v)\,\mathrm{dn}(v)}{1 - k^2\,\mathrm{sn}(v)^2\,\mathrm{sn}(w)^2}$ *the first addition formula*

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(-v) = \{-\mathrm{sn}(v), \mathrm{cn}(v), \mathrm{dn}(v)\}$ *the parity formulas*

$\mathrm{sn}(v)^2 + \mathrm{cn}(v)^2 = 1, \quad \mathrm{dn}(v)^2 + k^2\,\mathrm{sn}(v)^2 = 1$ *the Pythagorean relations*

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test these formulas ★ :

$W_2 = \frac{\varepsilon_2\,(t\,\sqrt{u\,x_1\,y_2\,(1+z_2)\,(1+u\,z_2)} + e_0\,e_2\,\sqrt{u\,y_2\,z_2\,\mathcal{D}(x_1,t)})}{y_2\,u\,(z_2 + x_1\,t^2)}$

$W_1 = \frac{\varepsilon_1\,(t\,\sqrt{u\,x_1\,y_1\,(1+z_1)\,(1+u\,z_1)} - e_0\,e_1\,\sqrt{u\,y_1\,z_1\,\mathcal{D}(x_1,t)})}{y_1\,u\,(z_1 + x_1\,t^2)}$

$U = \frac{\varepsilon_2\,\varepsilon_3\,\mathrm{sgn}(u\,z_2)\,(|x_1|\,h_{232}\,h_{233}\,t + e_0\,h_{231}\,\sqrt{x_1\,\mathcal{D}(x_1,t)})}{\sqrt{|x_1\,x_3|}\,(1 + \xi_{23}\,k^2\,x_1\,h_{231}^2\,t^2)}$ *in cases (a) and (b)*

$U = \frac{\varepsilon_2\,\varepsilon_3\,\mathrm{sgn}(u\,z_2)\,\mathrm{sgn}(x_1)}{k\,\sqrt{|x_1\,x_3|}\,t}$ *in case (c)*

## The derivation: the formulas for $W_2$ and $W_1$ for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $i = 1, 2$ :

$x_1\ p_1^2 = 1, \quad y_i\ q_i^2 = 1, \quad p_i = p_1$

$\mathrm{sn}(t_i) = \dfrac{i\ \varepsilon_i\ \sqrt{(g_i^2-1)\ p_i^2\ q_i^2}}{k\ p_i\ q_i}$

$\mathrm{cn}(t_i) = \dfrac{\varepsilon_i\ \sqrt{(g_i^2-(k')^2)\ p_i^2\ q_i^2}}{k\ p_i\ q_i}$

$\mathrm{sn}(\varsigma) = \dfrac{t}{p_1}, \quad p_1\ \mathrm{cn}(\varsigma)\ \mathrm{dn}(\varsigma) = \dfrac{e_0\ i\ R}{|x_1|}$

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\ \}(e'\ t_i) = \{e'\ \mathrm{sn}(t_i), \mathrm{cn}(t_i), \mathrm{dn}(t_i)\}$ *by the parity formulas*

Verification:

By *the Pythagorean relations* and $\mathrm{dn}(t_i) = g_i$ : $\mathrm{sn}(t_i)^2 = \dfrac{1-g_i^2}{k^2}$ ☑

By $g_i = \sqrt{1 + \dfrac{1}{z_i}}$ and $k = \sqrt{u}$ : $\dfrac{1-g_i^2}{k^2} = -\dfrac{1}{u\ z_i}$ ☑

Hence $1 - k^2\ \mathrm{sn}(t_i)^2\ \mathrm{sn}(\varsigma)^2 = \dfrac{z_i + x_1\ t^2}{z_i}$ ☑ , which is nonzero by the choice of $t$

Because $\pm t_i$ and $\varsigma$ are not poles and $1 - k^2\ \mathrm{sn}(t_i)^2\ \mathrm{sn}(\varsigma)^2 \neq 0$, *the first addition formula* , at $v = \varsigma$, $w = e'\ t_i$, holds pointwise and, together with *the parity formulas* , gives

$q_i\ \mathrm{sn}(\varsigma + e'\ t_i) = \dfrac{z_i\ q_i\ \mathrm{cn}(t_i)\ \mathrm{dn}(t_i)\ \mathrm{sn}(\varsigma) + e'\ z_i\ q_i\ \mathrm{sn}(t_i)\ \mathrm{cn}(\varsigma)\ \mathrm{dn}(\varsigma)}{z_i + x_1\ t^2}$ ☑

Using the given formulas, we get

$q_i\ \mathrm{sn}(\varsigma + e'\ t_i) = \dfrac{\varepsilon_i \left(t\ \sqrt{u\ x_1\ y_i\ (1+z_i)\ (1+u\ z_i)} - e'\ e_0\ \sqrt{u\ y_i\ z_i\ \mathcal{D}(x_1, t)}\right)}{y_i\ u\ \left(z_i + x_1\ t^2\right)}$ ✸

Setting $i = 2$, $e' = -e_2$ in ✸ gives the first formula of ★ .

Setting $i = 1$, $e' = e_1$ in ✸ gives the second formula of ★ .

The identity ✸ is verified separately on each sign region:

| | | |
|---|---|---|
| $z_i > 0,\ y_i > 0,$ hence $x_1 > 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i > 0,\ y_i < 0,$ hence $x_1 < 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i < -1,\ y_i > 0,$ hence $x_1 < 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i < -1,\ y_i < 0,$ hence $x_1 > 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |

## The derivation: the formula for $U$ in cases (a) and (b) for $u > 0$

Verified

Given hypotheses and formulas, in addition to those above:

$x_3\ p_3^2 = 1, \quad y_2\ q_2^2 = 1, \quad p_2 = p_1, \quad q_3 = q_2, \quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$

$\xi_{23} = \text{sgn}(x_1\, x_3), \quad \text{sgn}(u\, z_2) = \text{sgn}(x_1\, y_2)$

$\text{sn}(\varsigma) = \frac{t}{p_1}, \quad p_1\, \text{cn}(\varsigma)\, \text{dn}(\varsigma) = \frac{e_0\, i\, R}{|x_1|}$

$S_{23} = \frac{\varepsilon_2\ \varepsilon_3\ \sqrt{p_1^2\ p_3^2\ q_2^4\ \xi_{23}}}{p_1\ p_3\ q_2^2}$

$\{\text{sn, cn, dn}\}(e_2\, t_2 + e_3\, t_3) = \{i\, S_{23}\, h_{231},\ S_{23}\, h_{232},\ h_{233}\}$

$\text{sn}(\varsigma)^2 = x_1\, t^2$

Verification:

By the given expression for $S_{23}$: $S_{23}^2 = \xi_{23}$ ☑

Hence $\text{sn}(e_2\, t_2 + e_3\, t_3)^2 = -\xi_{23}\ h_{231}^2$ ☑

and $\text{cn}(e_2\, t_2 + e_3\, t_3)\, \text{dn}(e_2\, t_2 + e_3\, t_3) = S_{23}\, h_{232}\, h_{233}$ ☑

Hence $1 - k^2\, \text{sn}(\varsigma)^2\, \text{sn}(e_2\, t_2 + e_3\, t_3)^2 = 1 + \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2$ ☑, which is nonzero by the choice of $t$

Because $\pm(e_2\, t_2 + e_3\, t_3)$ and $\varsigma$ are not poles and $1 - k^2\, \text{sn}(\varsigma)^2\, \text{sn}(e_2\, t_2 + e_3\, t_3)^2 \neq 0$, *the first addition formula*, at $v = \varsigma$, $w = -e_2\, t_2 - e_3\, t_3$, holds pointwise and, together with *the parity formulas*, gives

$$p_3\, \text{sn}(\varsigma - e_2\, t_2 - e_3\, t_3) = \frac{p_3\, (\text{cn}(e_2\, t_2 + e_3\, t_3)\, \text{dn}(e_2\, t_2 + e_3\, t_3)\, \text{sn}(\varsigma) - \text{sn}(e_2\, t_2 + e_3\, t_3)\, \text{cn}(\varsigma)\, \text{dn}(\varsigma))}{1 + \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2} \quad ☑$$

Using the given hypotheses and formulas, we get

$$p_3\, \text{sn}(\varsigma - e_2\, t_2 - e_3\, t_3) = \frac{\varepsilon_2\ \varepsilon_3\ \text{sgn}(u\, z_2)\ \left(|x_1|\, h_{232}\, h_{233}\, t + e_0\, h_{231}\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{\sqrt{|x_1\, x_3|}\ \left(1 + \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2\right)} \quad ◆$$

which is the third formula of ★.

The identity ◆ is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of $\xi_{23}$ occur, because the sign of $z_2\, z_3$ is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = \left(g_2^2 - 1\right)\left(g_3^2 - 1\right) > 0$ and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2\, z_3 > 0$. The eight sign regions cover both cases.

$x_1 > 0,\ x_3 > 0,\ y_2 > 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 > 0,\ x_3 > 0,\ y_2 < 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 > 0,\ x_3 < 0,\ y_2 > 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 > 0,\ x_3 < 0,\ y_2 < 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 > 0,\ y_2 > 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 > 0,\ y_2 < 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 > 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 < 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = 1$ ☑

The derivation: the formula for $U$ in case (c) for $u > 0$ Verified

Given hypotheses and formulas, in addition to those above:

$\xi_{23} = \text{sgn}(x_1 x_3)$, $\text{sgn}(u z_2) = \text{sgn}(x_1 y_2)$

$S_{23} = \frac{\varepsilon_2 \varepsilon_3 \sqrt{p_1^2 p_3^2 q_2^4 \xi_{23}}}{p_1 p_3 q_2^2}$

$t_2 + t_3 = 2mK + iK'$, $m \in \mathbb{Z}$, $(-1)^m = S_{23}$, $e_2 = e_3$

$\text{sn}(v + 2mK) = (-1)^m \text{sn}(v)$, $\text{sn}(v \pm iK') = \frac{1}{k\,\text{sn}(v)}$ *the shift formulas*

$\text{sn}(\varsigma) = \frac{t}{p_1}$, $t \neq 0$, by the choice of $t$

Verification:

Because $e_2 = e_3$ and $(-1)^{-e_2 m} = (-1)^m$, *the shift formulas* give

$p_3 \text{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{(-1)^m p_3}{k\,\text{sn}(\varsigma)} = \frac{S_{23} p_1 p_3}{kt}$ ☑

Using the given hypotheses and formulas, we get

$p_3 \text{sn}(\varsigma - e_2(t_2 + t_3)) = \frac{\varepsilon_2 \varepsilon_3 \text{sgn}(u z_2) \text{sgn}(x_1)}{k \sqrt{|x_1 x_3|}\, t}$ ▲

which is the fourth formula of ★.

The identity ▲

is verified separately on each of the four sign regions realizable in case (c), with $\varepsilon_2$, $\varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = \frac{1}{z_i}$, so $z_2 z_3 > 0$.

$x_1 > 0$, $x_3 > 0$, $y_2 > 0$, hence $\xi_{23} = 1$, $\text{sgn}(u z_2) = 1$ ☑

$x_1 > 0$, $x_3 > 0$, $y_2 < 0$, hence $\xi_{23} = 1$, $\text{sgn}(u z_2) = -1$ ☑

$x_1 < 0$, $x_3 < 0$, $y_2 > 0$, hence $\xi_{23} = 1$, $\text{sgn}(u z_2) = -1$ ☑

$x_1 < 0$, $x_3 < 0$, $y_2 < 0$, hence $\xi_{23} = 1$, $\text{sgn}(u z_2) = 1$ ☑

**Section 8: ALL CLAIMS VERIFIED**

# Section 9

In[•]:=
```
(* ============================================================*)
(*PART 1:INITIALIZATION*)
(* ============================================================*)
ClearAll[i, UU, XX, YY, ZZ, X3, tt, RR, h1, h2, h3, ee0, ep2, ep3, kk, kp, kq, kp2,
  kpS, kS, DdS, Sv, pS, qS, p3S, zS, sS, cS, dS, sT, cT, gS, gT, sW, cW, dW, wConfigs,
  sTriples, dataOf, lhsOf, derOf, assumOf, derCheck, resultsAll, statusChip,
  smallChip, lblZ, lblY, lblX, lblX3, lblY2, lblXi, lblUZ, s7colTest, chkR,
  addCn, nCheck, w3Configs, thCheck, cCheck, e0s, e2s, kRelOK, cnSqOK, dnFormOK,
  normFirstOK, normLastOK, parityOK, snT1OK, snT2OK, snSigOK, denWOK, expWOK,
  sSqOK, snWSqOK, snDnWOK, h3RelOK, denUOK, expUOK, shiftC1OK, shiftC2OK];
```

```
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
s7colTest = RGBColor[0.10, 0.35, 0.75];
(*Case 2:u<0. We write u=-UU with UU>0,so M=1+UU,k'=1/Sqrt[1+UU],
k=k' Sqrt[UU],g_i=1/Sqrt[1+1/z_i].Sign regions for i in {1,2}:
  two z_i-windows and the sign of y_i;
the sign of x_1 is forced by u x_1 y_i z_i>0,
which for u<0 means x_1 y_i z_i<0. The window z_i>0 carries 1+u z_i>0,that is,
1-UU z_i>0. On each region,p_1 and q_i take their explicit radical values;
the final identities are checked there for all 8 sign combinations of e',
e_0,eps_i.Every intermediate relation of the
 derivation is checked symbolically from the given inputs:
  the expansions from the second addition formula with the parity formulas,
for all relevant signs,and the shift chain for both values of (-1)^m and of e_
 2,using (-1)^(-e_2 m)=(-1)^m.In these symbolic checks x_1 enters as 1/p_1^2,
by the given x_1 p_1^2=1,and sn(t_i),cn(t_i),sn,cn,dn at sigma and at e_2 t_2+
 e_3 t_3 are free symbols subject only to the given relations;
in the check of S_23^2=xi_23,p_1,p_3,q_2 are free symbols.*)
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
addCn[cv_, sv_, dv_, cw_, sw_, dw_, k2_] := (cv cw - sv sw dv dw) / (1 - k2 sv^2 sw^2);
lblZ[1] = TraditionalForm[HoldForm[Subscript[z, i] > 0]];
lblZ[2] = TraditionalForm[HoldForm[Subscript[z, i] < -1]];
lblY[1] = TraditionalForm[HoldForm[Subscript[y, i] > 0]];
lblY[-1] = TraditionalForm[HoldForm[Subscript[y, i] < 0]];
lblX[1] = TraditionalForm[HoldForm[Subscript[x, 1] > 0]];
lblX[-1] = TraditionalForm[HoldForm[Subscript[x, 1] < 0]];
wConfigs = {{Row[{lblZ[1], ",  ", lblY[1], ",  hence  ", lblX[-1]}], ZZ, 1, -1},
   {Row[{lblZ[1], ",  ", lblY[-1], ",  hence  ", lblX[1]}], ZZ, -1, 1},
   {Row[{lblZ[2], ",  ", lblY[1], ",  hence  ", lblX[1]}], -1 - ZZ, 1, 1},
   {Row[{lblZ[2], ",  ", lblY[-1], ",  hence  ", lblX[-1]}], -1 - ZZ, -1, -1}};
sTriples = Tuples[{1, -1}, 3];
dataOf[{lab_, zl_, sy_, sx_}] :=
  Module[{yl = sy YY, x1 = sx XX, kkv = Sqrt[UU] / Sqrt[1 + UU],
    kpv = 1 / Sqrt[1 + UU], gl, p1, ql, Dd}, gl = 1 / Sqrt[1 + 1 / zl];
   p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   ql = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
   {zl, yl, x1, kkv, kpv, gl, p1, ql, Dd}];
lhsOf[{zl_, yl_, x1_, kkv_, kpv_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] :=
  Module[{snl, cnl, RRv = Sqrt[x1 Dd]},
   snl = I eps Sqrt[(gl^2 - 1) p1^2 ql^2] / (kkv p1 ql);
```

```
    cnl = eps Sqrt[(gl^2 - kpv^2) p1^2 ql^2] / (kkv p1 ql);
    ((1 + zl) ql cnl (tt / p1) + ep (1 + zl) ql snl gl (e0 I kpv RRv / (XX p1))) /
     (zl + x1 tt^2)];
derOf[{zl_, yl_, x1_, kkv_, kpv_, gl_, p1_, ql_, Dd_}, ep_, e0_, eps_] := Module[
    {uu = -UU}, eps (tt Sqrt[uu x1 yl (1 + zl) (1 + uu zl)] - ep e0 Sqrt[uu yl zl Dd]) /
      (yl uu (zl + x1 tt^2))];
assumOf[{lab_, zl_, sy_, sx_}] := UU > 0 && XX > 0 && YY > 0 && ZZ > 0 &&
    Element[tt, Reals] && sx XX (-(1 - sx XX tt^2) (1 + UU sx XX tt^2)) ≥ 0 &&
    1 - UU zl > 0 && zl + sx XX tt^2 ≠ 0;
derCheck[w_, {ep_, e0_, eps_}] := Module[{d = dataOf[w]},
    TimeConstrained[FullSimplify[lhsOf[d, ep, e0, eps] - derOf[d, ep, e0, eps] == 0,
       assumOf[w]], 180, False] === True];
(*---symbolic checks of the intermediate relations---*)
kRelOK = chkR[(Sqrt[UU] / Sqrt[1 + UU])^2, UU (1 / Sqrt[1 + UU])^2] &&
    chkR[(1 / Sqrt[1 + UU])^2, 1 - (Sqrt[UU] / Sqrt[1 + UU])^2];
cnSqOK = chkR[(tt / pS)^2, tt^2 / pS^2];
dnFormOK = chkR[1 - kq (1 - cS^2), (1 - kq) + kq cS^2];
normFirstOK = chkR[(pS sS dS)^2 /. {sS^2 → 1 - cS^2, dS^2 → kp2 + kq cS^2},
    pS^2 (1 - cS^2) (kp2 + kq cS^2)];
normLastOK = And @@ Table[chkR[(I kpS RR / XX)^2 /. RR^2 → sx XX DdS,
     -kpS^2 sx XX DdS / (sx XX)^2], {sx, {1, -1}}];
parityOK = chkR[(-sS) dS, -(sS dS)];
snT1OK = chkR[gS^2 + kq ((1 - gS^2) / kq), 1];
snT2OK = chkR[(1 - 1 / (1 + 1 / zS)) / (UU / (1 + UU)), (1 + UU) / (UU (1 + zS))];
snSigOK = chkR[1 - (tt / pS)^2, 1 - tt^2 / pS^2];
denWOK = chkR[1 - (UU / (1 + UU)) ((1 + UU) / (UU (1 + zS))) (1 - tt^2 / pS^2),
    (zS + tt^2 / pS^2) / (1 + zS)];
expWOK = And @@
    Table[chkR[(qS addCn[cS, sS, dS, cT, ep sT, gT, kq]) /. {sS^2 → 1 - tt^2 / pS^2,
        sT^2 → 1 / (kq (1 + zS))}, ((1 + zS) qS cT cS - ep (1 + zS) qS sT gT sS dS) /
       (zS + tt^2 / pS^2)], {ep, {1, -1}}];
sSqOK =
  And @@ Flatten[Table[chkR[(ev Sqrt[pS^2 p3S^2 qS^4 xv] / (pS p3S qS^2))^2, xv],
     {ev, {1, -1}}, {xv, {1, -1}}]];
snWSqOK = And @@ Table[chkR[(I Sv h1)^2 /. Sv^2 → xv, -xv h1^2], {xv, {1, -1}}];
snDnWOK = chkR[(I Sv h1) h3, I Sv h1 h3];
h3RelOK = And @@ Table[
    chkR[h3^2 + kq ((I Sv h1)^2 /. Sv^2 → xv), h3^2 - xv kq h1^2], {xv, {1, -1}}];
denUOK = And @@ Table[chkR[1 - kq (1 - tt^2 / pS^2) (-xv h1^2),
     (h3^2 - xv kq h1^2 tt^2 / pS^2) /. h3^2 → 1 + xv kq h1^2], {xv, {1, -1}}];
expUOK = And @@ Table[chkR[(p3S addCn[cS, sS, dS, cW, -sW, dW, kq]) /.
       {sS^2 → 1 - tt^2 / pS^2, sW^2 → -xv h1^2},
      (p3S (cW cS + sW dW sS dS) / (h3^2 - xv kq h1^2 tt^2 / pS^2)) /.
       h3^2 → 1 + xv kq h1^2], {xv, {1, -1}}];
shiftC1OK = And @@
    Flatten[Table[chkR[p3S ev^(-e2) (e2 I dS / (kS sS)), ev e2 I p3S dS / (kS sS)],
      {ev, {1, -1}}, {e2, {1, -1}}]];
```

```
shiftC2OK = And @@ Flatten[Table[chkR[ev e2 I p3S dS / (kS sS),
      (Sv e2 I p3S sS dS / (kS sS^2)) /. Sv → ev], {ev, {1, -1}}, {e2, {1, -1}}]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style[Row[{"Given hypotheses, Case 2  (",
        TraditionalForm[HoldForm[u < 0]], ") :"}], Bold, 14],
     Style[Row[{TraditionalForm[HoldForm[u < 0]], ",   ",
        TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]], ",   ", TraditionalForm[
         HoldForm[k == k' Sqrt[-u]]], ",   ", TraditionalForm[HoldForm[k]],
        ", ", TraditionalForm[HoldForm[k']], " ∈ (0, 1)"}], 14],
     Style[Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2, 3, 4 :"}],
      Bold, 14], Style[Row[{TraditionalForm[HoldForm[Subscript[x, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[y, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[z, i]]], " ∈ ℝ",
        ",   ", TraditionalForm[HoldForm[Subscript[x, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[y, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[z, i]]],
        " ≠ -1, 0,   ", TraditionalForm[HoldForm[Subscript[g, i] ==
           dn[Subscript[t, i]] == 1 / Sqrt[1 + 1 / Subscript[z, i]]]]}], 14],
     Style[Row[{TraditionalForm[HoldForm[Subscript[p, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[q, i]]], " ∈ ",
        Subscript["ℝ", "> 0"], " ⋃ i ", Subscript["ℝ", "> 0"],
        ",   ", TraditionalForm[HoldForm[Subscript[e, 0]]],
        ", ", TraditionalForm[HoldForm[Derivative[1][e]]],
        ", ", TraditionalForm[HoldForm[Subscript[e, i]]],
        ", ", TraditionalForm[HoldForm[Subscript[ε, i]]], ", ",
        TraditionalForm[HoldForm[Subscript[ξ, 23]]], " ∈ {±1}"}],
      14], Style[Row[{TraditionalForm[HoldForm[
          u Subscript[x, i] Subscript[y, i] Subscript[z, i] > 0]], ",   ",
        TraditionalForm[HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]],
        ",   ", TraditionalForm[HoldForm[1 + u Subscript[z, i] > 0]]}], 14],
     Style[Row[{TraditionalForm[HoldForm[𝒟[Subscript[x, 1], t] ==
           -(1 - Subscript[x, 1] t^2) (1 - u Subscript[x, 1] t^2)]],
        ",   ", TraditionalForm[HoldForm[
          R == Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]]]], ",   ",
        TraditionalForm[HoldForm[Subscript[x, 1] 𝒟[Subscript[x, 1], t] > 0]]}],
      14], "", Style["Given formulas:", Bold, 14],
     Style[Row[{TraditionalForm[HoldForm[cn[v + w] ==
           (cn[v] cn[w] - sn[v] sn[w] dn[v] dn[w]) / (1 - k^2 sn[v]^2 sn[w]^2)]],
        "    ", Style["the second addition formula", Italic, 13,
         RGBColor[0.10, 0.30, 0.70]]}], 14], Style[
      Row[{TraditionalForm[HoldForm[{sn, cn, dn}[-v] == {-sn[v], cn[v], dn[v]}]],
        "    ", Style["the parity formulas", Italic,
         13, RGBColor[0.75, 0.35, 0.05]]}], 14],
     Style[Row[{TraditionalForm[HoldForm[sn[v]^2 + cn[v]^2 == 1]],
```

```
        ",   ", TraditionalForm[HoldForm[dn[v]^2 + k^2 sn[v]^2 == 1]],
        "    ", Style["the Pythagorean relations",
         Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
      14], Style[Row[{"Each  ", smallChip[True],
        "  below certifies that the statement displayed beside it has been
          verified by a symbolic computation at that point;  a  ",
        smallChip[False], "  would mean that this verification failed."}],
      12, GrayLevel[0.35]], "", Style[
      Row[{Style["We want to test these formulas  ", Bold, 14],
        Style["⋆", 14, s7colTest], Style[" :", Bold, 14]}]],
     Style[TraditionalForm[HoldForm[Subscript[W, 2] == Subscript[ε, 2]
          (t Sqrt[u Subscript[x, 1] Subscript[y, 2] (1 + Subscript[z, 2])
                (1 + u Subscript[z, 2])] + Subscript[e, 0] Subscript[e, 2]
             Sqrt[u Subscript[y, 2] Subscript[z, 2] 𝒟[Subscript[x, 1], t]]) /
           (Subscript[y, 2] u (Subscript[z, 2] + Subscript[x, 1] t^2))]], 15,
      s7colTest], Style[TraditionalForm[HoldForm[Subscript[W, 1] == Subscript[
          ε, 1] (t Sqrt[u Subscript[x, 1] Subscript[y, 1] (1 + Subscript[z, 1])
                (1 + u Subscript[z, 1])] - Subscript[e, 0] Subscript[e, 1]
             Sqrt[u Subscript[y, 1] Subscript[z, 1] 𝒟[Subscript[x, 1], t]]) /
           (Subscript[y, 1] u (Subscript[z, 1] + Subscript[x, 1] t^2))]],
      15, s7colTest], Style[Row[{TraditionalForm[HoldForm[
          U == Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
            (Abs[Subscript[x, 1]] Subscript[h, 232] t + Subscript[e, 0] k'
               Subscript[h, 231] Subscript[h, 233] Sqrt[Subscript[x, 1]
                 𝒟[Subscript[x, 1], t]]) / (Sqrt[Abs[Subscript[x, 1]
                 Subscript[x, 3]]] (Subsuperscript[h, 233, 2] - Subscript[ξ,
                  23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]],
        "    ", Style["in cases (a) and (b)", Italic, 13,
         GrayLevel[0.35]]}], 15, s7colTest],
     Style[Row[{TraditionalForm[HoldForm[U == Subscript[ε, 2] Subscript[ε, 3]
           Sign[u Subscript[z, 2]] Subscript[e, 0] Subscript[e, 2] k'
           Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] / (k Sqrt[Abs[
                Subscript[x, 1] Subscript[x, 3]]] (1 - Subscript[x, 1] t^2))]],
        "    ", Style["in case (c)", Italic, 13, GrayLevel[0.35]]}],
      15, s7colTest]}, Spacings → 1.1],
   Background → RGBColor[0.99, 0.96, 0.88],
   FrameMargins →
    14]];

(* ============================================================*)
(*PART 3:THE PARAMETER NORMALIZATION*)
(* ============================================================*)
resultsAll = {};
nCheck[sx_] :=
  Module[{x1 = sx XX, p1, Dd, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   Dd = -(1 - x1 tt^2) (1 + UU x1 tt^2);
   assum = UU > 0 && XX > 0 && Element[tt, Reals];
```

```
    TimeConstrained[
      FullSimplify[p1^2 (1 - x1 tt^2) (1 / (1 + UU) + UU / (1 + UU) x1 tt^2) -
          (- (1 / (1 + UU)) x1 Dd / x1^2) == 0, assum], 180, False] === True];
Module[{grid, ok}, grid = nCheck /@ {1, -1};
  ok = kRelOK && cnSqOK && dnFormOK &&
    normFirstOK && normLastOK && parityOK && And @@ grid;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The parameter normalization for  ", Bold, 15,
         RGBColor[0.75, 0.35, 0.05]], Style[TraditionalForm[HoldForm[u < 0]],
         Bold, 15, RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]}],
     Style["Given hypotheses and formulas, in addition to those above:",
      Bold, 13], Style[Row[
      {TraditionalForm[HoldForm[Subscript[x, 1] Subsuperscript[p, 1, 2] == 1]],
       ",   ", TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]]}],
      13], "", Style["Verification:", Bold, 13],
     Style[Row[{"By the given  ", TraditionalForm[HoldForm[k']],
        "  and  ", TraditionalForm[HoldForm[k]], " :   ",
        TraditionalForm[HoldForm[k^2 == -u k'^2]], ",   ",
        TraditionalForm[HoldForm[k'^2 == 1 - k^2]], "   ", smallChip[kRelOK]}],
      13], Style[Row[{"By  ", TraditionalForm[
         HoldForm[Subscript[x, 1] Subsuperscript[p, 1, 2] == 1]], "  and  ",
        TraditionalForm[HoldForm[cn[ς] == t / Subscript[p, 1]]], " :   ",
        TraditionalForm[HoldForm[cn[ς]^2 == Subscript[x, 1] t^2]],
        "   ", smallChip[cnSqOK]}], 13],
     Style[Row[{"By  ", Style["the Pythagorean relations", Italic,
         13, RGBColor[0.75, 0.35, 0.05]], "  and  ",
        TraditionalForm[HoldForm[k'^2 == 1 - k^2]], " :   ",
        TraditionalForm[HoldForm[dn[v]^2 == k'^2 + k^2 cn[v]^2]],
        "   ", smallChip[dnFormOK]}], 13],
     Style[Row[{"Hence, by  ", Style["the Pythagorean relations",
         Italic, 13, RGBColor[0.75, 0.35, 0.05]], " :"}], 13],
     Style[Row[{TraditionalForm[HoldForm[(Subscript[p, 1] sn[ς] dn[ς])^2 ==
          Subsuperscript[p, 1, 2] (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2)]],
        "   ", smallChip[normFirstOK]}], 14], Style[Row[{"By  ",
        TraditionalForm[HoldForm[Subscript[x, 1] Subsuperscript[p, 1, 2] == 1]],
        ",  ", TraditionalForm[HoldForm[cn[ς]^2 == Subscript[x, 1] t^2]],
        ",  and  ", TraditionalForm[HoldForm[k^2 == -u k'^2]],
        " ,  on each sign region:"}], 13],
     Style[TraditionalForm[HoldForm[Subsuperscript[p, 1, 2]
          (1 - cn[ς]^2) (k'^2 + k^2 cn[ς]^2) == - (k'^2 Subscript[x, 1]
           𝒟[Subscript[x, 1], t] / Subsuperscript[x, 1, 2])]], 14],
     Grid[{{Style[lblX[1], 12, GrayLevel[0.3]], smallChip[grid[[1]]]},
       {Style[lblX[-1], 12, GrayLevel[0.3]], smallChip[grid[[2]]]}},
      Alignment → Left, Spacings → {2, 0.7}],
     Style[Row[{"By  ", TraditionalForm[HoldForm[R^2 ==
          Subscript[x, 1] 𝒟[Subscript[x, 1], t]]], " :"}], 13], Style[Row[
```

```
        {TraditionalForm[HoldForm[-(k'^2 Subscript[x, 1] 𝒟[Subscript[x, 1], t] /
                 Subsuperscript[x, 1, 2]) == (I k' R / Abs[Subscript[x, 1]])^2]],
          "   ", smallChip[normLastOK]}], 14], Style[Row[{"By  ",
          Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
          ",  replacing  ", TraditionalForm[HoldForm[ς]], "  by  ",
          TraditionalForm[HoldForm[-ς]], "  preserves  ",
          TraditionalForm[HoldForm[cn[ς]]], "  and reverses the sign of  ",
          TraditionalForm[HoldForm[sn[ς] dn[ς]]], "   ", smallChip[parityOK]}],
        13], Style[Row[{"Thus, we may fix  ",
          Style[TraditionalForm[HoldForm[Subscript[p, 1] sn[ς] dn[ς] ==
              -Subscript[e, 0] I k' R / Abs[Subscript[x, 1]]]],
           RGBColor[0.75, 0.35, 0.05]], ",  so that the two values of  ",
          TraditionalForm[HoldForm[Subscript[e, 0]]],
          "  correspond to the two admissible choices of  ",
          TraditionalForm[HoldForm[ς]], " ."}], 13],
       Style[Row[{"The sign  ", TraditionalForm[HoldForm[-Subscript[e, 0]]],
          "  is a convention, because  ",
          TraditionalForm[HoldForm[Subscript[e, 0]]],
          "  only labels the two admissible choices of  ", TraditionalForm[
           HoldForm[ς]], " ;  it compensates the minus sign in  ",
          Style["the second addition formula", Italic, 12,
           RGBColor[0.10, 0.30, 0.70]], " ,  so that the  ",
          TraditionalForm[HoldForm[Subscript[e, 0]]], "-terms of  ",
          Style["⋆", 12, RGBColor[0.10, 0.35, 0.75]],
          "  carry the same signs as in the case  ", TraditionalForm[
           HoldForm[u > 0]], " ."}], 12, GrayLevel[0.35]]}, Spacings → 1],
     Background → Lighter[RGBColor[0.75, 0.35, 0.05], 0.93],
     FrameMargins →
      12]];];

(* ============================================================*)
(*PART 4:THE W-FORMULAS*)
(* ============================================================*)
Module[{grid, regionOK, ok},
  grid = Table[derCheck[w, sgnTriple], {w, wConfigs}, {sgnTriple, sTriples}];
  regionOK = And @@@ grid;
  ok = snT1OK && snT2OK && snSigOK && denWOK && expWOK && And @@ regionOK;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The derivation:  the formulas for  ", Bold, 15,
          RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[
           HoldForm[Subscript[W, 2]]], Bold, 15, RGBColor[0.10, 0.30, 0.70]],
         Style["  and  ", Bold, 15, RGBColor[0.10, 0.30, 0.70]],
         Style[TraditionalForm[HoldForm[Subscript[W, 1]]], Bold,
          15, RGBColor[0.10, 0.30, 0.70]], Style["  for  ", Bold, 15,
          RGBColor[0.10, 0.30, 0.70]], Style[TraditionalForm[HoldForm[u < 0]],
          Bold, 15, RGBColor[0.10, 0.30, 0.70]], Spacer[20], statusChip[ok]}],
```

```
Style["Given hypotheses and formulas, in addition to those above:",
 Bold, 13],
Style[Row[{"For  ", TraditionalForm[HoldForm[i]], " = 1, 2 :"}],
 Bold, 13], Style[Row[{TraditionalForm[
    HoldForm[Subscript[y, i] Subsuperscript[q, i, 2] == 1]], ",   ",
   TraditionalForm[HoldForm[Subscript[p, i] == Subscript[p, 1]]]}], 13],
Style[TraditionalForm[HoldForm[sn[Subscript[t, i]] == I Subscript[ε, i]
     Sqrt[(Subsuperscript[g, i, 2] - 1) Subsuperscript[p, i, 2]
        Subsuperscript[q, i, 2]] / (k Subscript[p, i] Subscript[q, i])]],
 14], Style[TraditionalForm[HoldForm[cn[Subscript[t, i]] == Subscript[
     ε, i] Sqrt[(Subsuperscript[g, i, 2] - k'^2) Subsuperscript[p, i, 2]
        Subsuperscript[q, i, 2]] / (k Subscript[p, i] Subscript[q, i])]],
 14], Style[Row[{TraditionalForm[HoldForm[{sn, cn, dn}[Derivative[1][e]
        Subscript[t, i]] == {Derivative[1][e] sn[Subscript[t, i]],
       cn[Subscript[t, i]], dn[Subscript[t, i]]}]], "    ", Style[
    "by the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]]}],
 14], "", Style["Verification:", Bold, 13],
Style[Row[{"By  ", Style["the Pythagorean relations",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]], "  and  ",
   TraditionalForm[HoldForm[dn[Subscript[t, i]] == Subscript[g, i]]],
   " :   ", TraditionalForm[HoldForm[sn[Subscript[t, i]]^2 ==
      (1 - Subsuperscript[g, i, 2]) / k^2]], "   ", smallChip[snT1OK]}],
 13], Style[Row[{"By  ", TraditionalForm[HoldForm[
     Subscript[g, i] == 1 / Sqrt[1 + 1 / Subscript[z, i]]]],
   "  and  ", TraditionalForm[HoldForm[k^2 == -u k'^2]], ",  ",
   TraditionalForm[HoldForm[k' == 1 / Sqrt[1 - u]]], " :   ",
   TraditionalForm[HoldForm[(1 - Subsuperscript[g, i, 2]) / k^2 ==
      -((1 - u) / (u (1 + Subscript[z, i])))]], "   ", smallChip[snT2OK]}],
 13], Style[Row[{"By  ", Style["the Pythagorean relations",
    Italic, 13, RGBColor[0.75, 0.35, 0.05]], " :   ", TraditionalForm[
    HoldForm[sn[ς]^2 == 1 - cn[ς]^2 == 1 - Subscript[x, 1] t^2]], "   ",
   smallChip[snSigOK]}], 13], Style[Row[{"Hence  ", TraditionalForm[
    HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[ς]^2 == (Subscript[z,
          i] + Subscript[x, 1] t^2) / (1 + Subscript[z, i])]], "   ",
   smallChip[denWOK], "   ,  which is nonzero by the choice of  ",
   TraditionalForm[HoldForm[t]]}], 13],
Style[Row[{"Because  ", TraditionalForm[HoldForm[± Subscript[t, i]]],
   "  and  ", TraditionalForm[HoldForm[ς]], "  are not poles and  ",
   TraditionalForm[HoldForm[1 - k^2 sn[Subscript[t, i]]^2 sn[ς]^2 ≠ 0]],
   ",  ", Style["the second addition formula", Italic, 13, RGBColor[0.10,
     0.30, 0.70]], ",  at  ", TraditionalForm[HoldForm[v == ς]], ",  ",
   TraditionalForm[HoldForm[w == Derivative[1][e] Subscript[t, i]]],
   ",  holds pointwise and, together with  ",
   Style["the parity formulas", Italic, 13, RGBColor[0.75, 0.35, 0.05]],
   ",  gives"}], 13], Style[Row[{TraditionalForm[
    HoldForm[Subscript[q, i] cn[ς + Derivative[1][e] Subscript[t, i]] ==
      ((1 + Subscript[z, i]) Subscript[q, i] cn[Subscript[t, i]] cn[ς] -
```

```
                Derivative[1][e] (1 + Subscript[z, i]) Subscript[q, i] sn[
                  Subscript[t, i]] dn[Subscript[t, i]] sn[ς] dn[ς]) / (Subscript[
                 z, i] + Subscript[x, 1] t^2)]], "   ", smallChip[expWOK]}], 14],
       Style["Using the given formulas and the normalization above, we get",
        13], Style[Row[{TraditionalForm[
           HoldForm[Subscript[q, i] cn[ς + Derivative[1][e] Subscript[t, i]] ==
             Subscript[ε, i] (t Sqrt[u Subscript[x, 1] Subscript[y, i]
                    (1 + Subscript[z, i]) (1 + u Subscript[z, i])] -
                Derivative[1][e] Subscript[e, 0] Sqrt[u Subscript[y, i]
                   Subscript[z, i] 𝒟[Subscript[x, 1], t]]) /
               (Subscript[y, i] u (Subscript[z, i] + Subscript[x, 1] t^2))]],
          "      ", Style["✱", 15, RGBColor[0.10, 0.30, 0.70]]}], 14],
       Style[Row[{"Setting  ", TraditionalForm[HoldForm[i == 2]], ",  ",
          TraditionalForm[HoldForm[Derivative[1][e] == -Subscript[e, 2]]],
          "  in  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
          "  gives the first formula of  ",
          Style["⋆", 13, s7colTest], " ."}], 13],
       Style[Row[{"Setting  ", TraditionalForm[HoldForm[i == 1]], ",  ",
          TraditionalForm[HoldForm[Derivative[1][e] == Subscript[e, 1]]],
          "  in  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
          "  gives the second formula of  ",
          Style["⋆", 13, s7colTest], " ."}], 13], "",
       Style[Row[{"The identity  ", Style["✱", 13, RGBColor[0.10, 0.30, 0.70]],
          "  is verified separately on each sign region:"}], 13],
       Grid[Table[{Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]],
          Style[Row[{"for all 8 sign combinations of  ",
             TraditionalForm[HoldForm[Derivative[1][e]]], ", ",
             TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
             TraditionalForm[HoldForm[Subscript[ε, i]]], " ∈ {±1}"}],
           12, GrayLevel[0.3]], smallChip[regionOK[[iw]]]}, {iw, 4}],
        Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
     Background → Lighter[RGBColor[0.10, 0.30, 0.70], 0.93],
     FrameMargins →
      12]];];

(* ============================================================*)
(*PART 5:U-FORMULA IN CASES (a) AND (b)*)
(* ============================================================*)
kk = Sqrt[UU] / Sqrt[1 + UU]; kp = 1 / Sqrt[1 + UU];
lblX3[1] = TraditionalForm[HoldForm[Subscript[x, 3] > 0]];
lblX3[-1] = TraditionalForm[HoldForm[Subscript[x, 3] < 0]];
lblY2[1] = TraditionalForm[HoldForm[Subscript[y, 2] > 0]];
lblY2[-1] = TraditionalForm[HoldForm[Subscript[y, 2] < 0]];
lblXi[1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]];
lblXi[-1] = TraditionalForm[HoldForm[Subscript[ξ, 23] == -1]];
lblUZ[1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == 1]];
lblUZ[-1] = TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] == -1]];
```

```
w3Configs = Flatten[
    Table[{Row[{lblX[sx], ",  ", lblX3[s3], ",  ", lblY2[sy], ",  hence  ",
        lblXi[sx s3], ",  ", lblUZ[sx sy]}], sx, s3, sy},
     {sx, {1, -1}}, {s3, {1, -1}}, {sy, {1, -1}}], 2];
thCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, denom,
    lhs, rhs, assum}, p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
   q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
   denom = h3^2 - xi kk^2 x1 h1^2 tt^2;
   lhs = p3 (S23 h2 (tt / p1) + I S23 h1 h3 (-ee0 I kp RR / (XX p1))) / denom;
   rhs = ep2 ep3 suz (XX h2 tt + ee0 kp h1 h3 RR) / (Sqrt[XX X3] denom);
   assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 && RR ≥ 0 &&
     Element[tt | h1 | h2 | h3 | ee0 | ep2 | ep3, Reals] && denom ≠ 0;
   TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{grid, ok}, grid = thCheck /@ w3Configs;
  ok = sSqOK && snWSqOK && snDnWOK &&
    h3RelOK && snSigOK && denUOK && expUOK && And @@ grid;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The derivation:  the formula for  ", Bold, 15,
         RGBColor[0.55, 0.25, 0.70]], Style[TraditionalForm[HoldForm[U]],
         Bold, 15, RGBColor[0.55, 0.25, 0.70]], Style[
         "  in cases (a) and (b) for  ", Bold, 15, RGBColor[0.55, 0.25, 0.70]],
        Style[TraditionalForm[HoldForm[u < 0]], Bold, 15,
         RGBColor[0.55, 0.25, 0.70]], Spacer[20], statusChip[ok]}],
      Style["Given hypotheses and formulas, in addition to those above:",
       Bold, 13], Style[Row[{TraditionalForm[
          HoldForm[Subscript[x, 3] Subsuperscript[p, 3, 2] == 1]], ",   ",
         TraditionalForm[HoldForm[Subscript[y, 2] Subsuperscript[q, 2, 2] == 1]],
         ",   ", TraditionalForm[HoldForm[Subscript[p, 2] == Subscript[p, 1]]],
         ",   ", TraditionalForm[HoldForm[Subscript[q, 3] == Subscript[q, 2]]],
         ",   ", TraditionalForm[HoldForm[Subscript[h, 231]]],
         ", ", TraditionalForm[HoldForm[Subscript[h, 232]]], ", ",
         TraditionalForm[HoldForm[Subscript[h, 233]]], " ∈ ℝ"}], 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[ξ, 23] ==
            Sign[Subscript[x, 1] Subscript[x, 3]]]],
         ",   ", TraditionalForm[HoldForm[Sign[u Subscript[z, 2]] ==
            Sign[Subscript[x, 1] Subscript[y, 2]]]]}], 13],
      Style[TraditionalForm[HoldForm[Subscript[S, 23] ==
          Subscript[ε, 2] Subscript[ε, 3] Sqrt[Subsuperscript[p, 1, 2]
             Subsuperscript[p, 3, 2] Subsuperscript[q, 2, 4] Subscript[ξ, 23]] /
           (Subscript[p, 1] Subscript[p, 3] Subsuperscript[q, 2, 2])]],
       13], Style[TraditionalForm[HoldForm[{sn, cn, dn}[
          Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]] ==
         {I Subscript[S, 23] Subscript[h, 231],
```

```
     Subscript[S, 23] Subscript[h, 232], Subscript[h, 233]}]],
 13], "", Style["Verification:", Bold, 13], Style[
 Row[{"By the given expression for  ",
   TraditionalForm[HoldForm[Subscript[S, 23]]], " :   ", TraditionalForm[
    HoldForm[Subsuperscript[S, 23, 2] == Subscript[ξ, 23]]],
   "   ", smallChip[sSqOK]}], 13],
Style[Row[{"Hence  ", TraditionalForm[HoldForm[
     sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^
       2 == -Subscript[ξ, 23] Subsuperscript[h, 231, 2]]], "   ",
   smallChip[snWSqOK], "   and   ", TraditionalForm[HoldForm[
     sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]
       dn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
           Subscript[t, 3]] == I Subscript[S, 23] Subscript[h, 231]
       Subscript[h, 233]]], "   ", smallChip[snDnWOK]}], 13],
Style[Row[{"By  ", Style["the Pythagorean relations", Italic, 13,
    RGBColor[0.75, 0.35, 0.05]], "  at  ", TraditionalForm[HoldForm[v ==
      Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]],
   " :   ", TraditionalForm[HoldForm[Subsuperscript[h, 233, 2] -
       Subscript[ξ, 23] k^2 Subsuperscript[h, 231, 2] == dn[Subscript[e, 2]
           Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 +
       k^2 sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
            Subscript[t, 3]]^2 == 1]], "   ", smallChip[h3RelOK]}], 13],
Style[Row[{"By  ", Style["the Pythagorean relations", Italic,
    13, RGBColor[0.75, 0.35, 0.05]], " :   ", TraditionalForm[
    HoldForm[sn[ς]^2 == 1 - cn[ς]^2 == 1 - Subscript[x, 1] t^2]],
   "   ", smallChip[snSigOK]}], 13],
Style[Row[{"Hence  ", TraditionalForm[HoldForm[
     1 - k^2 sn[ς]^2 sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
            Subscript[t, 3]]^2 == Subsuperscript[h, 233, 2] - Subscript[ξ,
         23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2]], "   ",
   smallChip[denUOK], "   ,  which is nonzero by the choice of  ",
   TraditionalForm[HoldForm[t]]}], 13],
Style[Row[{"Because  ", TraditionalForm[HoldForm[± (Subscript[e, 2]
         Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3])]], "  and  ",
   TraditionalForm[HoldForm[ς]], "  are not poles and  ",
   TraditionalForm[HoldForm[1 - k^2 sn[ς]^2 sn[Subscript[e, 2]
            Subscript[t, 2] + Subscript[e, 3] Subscript[t, 3]]^2 ≠ 0]],
   ",  ", Style["the second addition formula", Italic, 13,
    RGBColor[0.10, 0.30, 0.70]], ",  at  ",
   TraditionalForm[HoldForm[v == ς]], ",  ", TraditionalForm[HoldForm[w ==
      -Subscript[e, 2] Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]]],
   ",  holds pointwise and, together with  ",
   Style["the parity formulas", Italic, 13,
    RGBColor[0.75, 0.35, 0.05]], ",  gives"}], 13],
Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] cn[ς - Subscript[e, 2]
           Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
      (Subscript[p, 3] (cn[Subscript[e, 2] Subscript[t, 2] +
```

```
                    Subscript[e, 3] Subscript[t, 3]] cn[ς] +
                 sn[Subscript[e, 2] Subscript[t, 2] + Subscript[e, 3]
                     Subscript[t, 3]] dn[Subscript[e, 2] Subscript[t, 2] +
                    Subscript[e, 3] Subscript[t, 3]] sn[ς] dn[ς])) /
            (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2 Subscript[x, 1]
               Subsuperscript[h, 231, 2] t^2)]], "   ", smallChip[expUOK]}],
     14], Style["Using the given hypotheses and formulas and
       the normalization above, we get", 13],
    Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3] cn[ς - Subscript[
                e, 2] Subscript[t, 2] - Subscript[e, 3] Subscript[t, 3]] ==
          Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
           (Abs[Subscript[x, 1]] Subscript[h, 232] t + Subscript[e, 0] k'
                Subscript[h, 231] Subscript[h, 233] Sqrt[Subscript[x, 1]
                  𝒟[Subscript[x, 1], t]]) / (Sqrt[Abs[Subscript[x, 1]
                  Subscript[x, 3]]] (Subsuperscript[h, 233, 2] - Subscript[ξ,
                   23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2))]],
       "      ", Style["◆", 14, RGBColor[0.55, 0.25, 0.70]]}], 14],
    Style[Row[{"which is the third formula of  ",
       Style["⋆", 13, s7colTest], " ."}], 13], "",
    Style[Row[{"The identity  ", Style["◆", 13, RGBColor[0.55, 0.25, 0.70]],
       "  is verified separately on each sign region, with  ",
       TraditionalForm[HoldForm[Subscript[h, 231]]],
       ", ", TraditionalForm[HoldForm[Subscript[h, 232]]],
       ", ", TraditionalForm[HoldForm[Subscript[h, 233]]],
       ", ", TraditionalForm[HoldForm[Subscript[e, 0]]],
       ", ", TraditionalForm[HoldForm[Subscript[ε, 2]]],
       ", ", TraditionalForm[HoldForm[Subscript[ε, 3]]],
       "  symbolic, that is, for all their real values at once,
         which covers all their sign choices:"}], 13],
    Style[Row[{"In case (a), both values of  ", TraditionalForm[
        HoldForm[Subscript[ξ, 23]]], "  occur, because the sign of  ",
       TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3]]],
       "  is not constrained;  in case (b),  ",
       TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
       ",  because  ", TraditionalForm[HoldForm[
         k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
       ",  ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
           -(1 / (1 + Subscript[z, i]))]], ",  and  ", TraditionalForm[
        HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ",  so  ",
       TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
       " .  The eight sign regions cover both cases."}],
     12, GrayLevel[0.35]], Grid[Table[{Style[w3Configs⟦iw, 1⟧,
        12, GrayLevel[0.3]], smallChip[grid⟦iw⟧]}, {iw, 8}],
     Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
 Background → Lighter[RGBColor[0.55, 0.25, 0.70],
   0.93], FrameMargins → 12]];];
```

```
(* ============================================================*)
(*PART 6:U-FORMULA IN CASE (c) FOR u<0*)
(* ============================================================*)
cCheck[{lab_, sx_, s3_, sy_}] :=
  Module[{x1 = sx XX, xi = sx s3, suz = sx sy, p1, p3, q2, S23, lhs, rhs, assum},
   p1 = If[sx == 1, 1 / Sqrt[XX], I / Sqrt[XX]];
   p3 = If[s3 == 1, 1 / Sqrt[X3], I / Sqrt[X3]];
   q2 = If[sy == 1, 1 / Sqrt[YY], I / Sqrt[YY]];
   S23 = ep2 ep3 Sqrt[p1^2 p3^2 q2^4 xi] / (p1 p3 q2^2);
   lhs = S23 e2s I p3 (-e0s I kp RR / (XX p1)) / (kk (1 - x1 tt^2));
   rhs = suz ep2 ep3 e0s e2s kp RR / (kk Sqrt[XX X3] (1 - x1 tt^2));
   assum = XX > 0 && X3 > 0 && YY > 0 && UU > 0 &&
     Element[tt | RR | e0s | e2s | ep2 | ep3, Reals] && 1 - x1 tt^2 ≠ 0;
   TimeConstrained[FullSimplify[lhs - rhs == 0, assum], 180, False] === True];
Module[{cConfigs, grid, ok}, cConfigs = Select[w3Configs, #[[2]] #[[3]] == 1 &];
  grid = cCheck /@ cConfigs;
  ok = snSigOK && shiftC1OK && shiftC2OK && And @@ grid;
  AppendTo[resultsAll, ok];
  Print[
   Panel[Column[{Row[{Style["The derivation:  the formula for  ", Bold, 15,
         RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[U]], Bold,
         15, RGBColor[0.00, 0.50, 0.50]], Style["  in case (c) for  ", Bold, 15,
         RGBColor[0.00, 0.50, 0.50]], Style[TraditionalForm[HoldForm[u < 0]],
         Bold, 15, RGBColor[0.00, 0.50, 0.50]], Spacer[20], statusChip[ok]}],
      Style["Given hypotheses and formulas, in addition to those above:",
       Bold, 13], Style[Row[{TraditionalForm[
          HoldForm[Subscript[t, 2] + Subscript[t, 3] == 2 K m + I K']],
         ",   ", TraditionalForm[HoldForm[m ∈ ℤ]], ",   ",
         TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], ",   ",
         TraditionalForm[HoldForm[Subscript[e, 2] == Subscript[e, 3]]]}], 13],
      Style[Row[{TraditionalForm[HoldForm[cn[v + 2 m K] == (-1)^m cn[v]]], ",   ",
         TraditionalForm[HoldForm[cn[v ∓ I K'] == ±I dn[v] / (k sn[v])]], "    ",
         Style["the shift formulas", Italic, 13, RGBColor[0.00, 0.50, 0.50]]}],
       13], "", Style["Verification:", Bold, 13],
      Style[Row[{"By  ", Style["the Pythagorean relations", Italic,
          13, RGBColor[0.75, 0.35, 0.05]], " :   ", TraditionalForm[
          HoldForm[sn[ς]^2 == 1 - cn[ς]^2 == 1 - Subscript[x, 1] t^2]], "   ",
         smallChip[snSigOK], "   ,  which is nonzero by the choice of  ",
         TraditionalForm[HoldForm[t]]}], 13],
      Style[Row[{"Because  ", TraditionalForm[HoldForm[
           Subscript[e, 2] == Subscript[e, 3]]], "  and  ",
         TraditionalForm[HoldForm[(-1)^(-Subscript[e, 2] m) == (-1)^m]],
         ",  ", Style["the shift formulas", Italic, 13,
          RGBColor[0.00, 0.50, 0.50]], "  give"}], 13],
      Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3]
            cn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
           (-1)^m Subscript[e, 2] I Subscript[p, 3] dn[ς] / (k sn[ς])]],
```

```
          "   ", smallChip[shiftC1OK]}], 14], Style[Row[{"and, by  ",
          TraditionalForm[HoldForm[(-1)^m == Subscript[S, 23]]], " ,"}], 13],
        Style[Row[{TraditionalForm[HoldForm[(-1)^m Subscript[e, 2] I
              Subscript[p, 3] dn[ς] / (k sn[ς]) == Subscript[S, 23] Subscript[e, 2]
              I Subscript[p, 3] sn[ς] dn[ς] / (k sn[ς]^2)]], "   ",
          smallChip[shiftC2OK]}], 14], Style["Using the given hypotheses
          and formulas and the normalization above, we get", 13],
        Style[Row[{TraditionalForm[HoldForm[Subscript[p, 3]
              cn[ς - Subscript[e, 2] (Subscript[t, 2] + Subscript[t, 3])] ==
             Subscript[ε, 2] Subscript[ε, 3] Sign[u Subscript[z, 2]]
              Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1]
                 𝒟[Subscript[x, 1], t]] / (k Sqrt[Abs[Subscript[x, 1]
                    Subscript[x, 3]]] (1 - Subscript[x, 1] t^2))]],
          "      ", Style["▲", 14, RGBColor[0.00, 0.50, 0.50]]}], 14],
        Style[Row[{"which is the fourth formula of  ",
          Style["⋆", 13, s7colTest], " ."}], 13], "",
        Style[Row[{"The identity  ", Style["▲", 13, RGBColor[0.00, 0.50, 0.50]],
          "  is verified separately on each of the four sign
            regions realizable in case (c), with  ",
          TraditionalForm[HoldForm[Subscript[e, 0]]], ", ",
          TraditionalForm[HoldForm[Subscript[e, 2]]], ", ",
          TraditionalForm[HoldForm[Subscript[ε, 2]]], ", ", TraditionalForm[
           HoldForm[Subscript[ε, 3]]], ", ", TraditionalForm[HoldForm[R]],
          "  symbolic, that is, for all their real values at once,
            which covers all their sign choices:"}], 13],
        Style[Row[{"Here  ", TraditionalForm[HoldForm[Subscript[ξ, 23] == 1]],
          ",  because  ", TraditionalForm[HoldForm[
            k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0]],
          ",  ", TraditionalForm[HoldForm[Subsuperscript[g, i, 2] - 1 ==
              -(1 / (1 + Subscript[z, i]))]], ",  and  ", TraditionalForm[
           HoldForm[Subscript[z, i] (1 + Subscript[z, i]) > 0]], ",  so  ",
          TraditionalForm[HoldForm[Subscript[z, 2] Subscript[z, 3] > 0]],
          " ."}], 12, GrayLevel[0.35]],
        Grid[Table[{Style[cConfigs[[iw, 1]], 12, GrayLevel[0.3]],
          smallChip[grid[[iw]]]}, {iw, 4}],
         Alignment → Left, Spacings → {2, 0.7}]}, Spacings → 1],
      Background → Lighter[RGBColor[0.00, 0.50, 0.50], 0.93],
      FrameMargins →
       12]];];

(* ============================================================*)
(*PART 7:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 9: ALL CLAIMS VERIFIED",
       "Section 9: SOME CLAIMS FAILED"], Bold, White, 16], Background →
     If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
    FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses, Case 2 $(u < 0)$ :

$u < 0, \quad k' = \frac{1}{\sqrt{1-u}}, \quad k = k' \sqrt{-u}, \quad k, k' \in (0, 1)$

For $i = 1, 2, 3, 4$ :

$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0, \quad g_i = \mathrm{dn}(t_i) = \frac{1}{\sqrt{1+\frac{1}{z_i}}}$

$p_i, q_i \in \mathbb{R}_{>0} \bigcup i\, \mathbb{R}_{>0}, \quad e_0, e', e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$

$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$

$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) > 0$

Given formulas:

$\mathrm{cn}(v + w) = \frac{\mathrm{cn}(v)\,\mathrm{cn}(w) - \mathrm{sn}(v)\,\mathrm{sn}(w)\,\mathrm{dn}(v)\,\mathrm{dn}(w)}{1 - k^2\,\mathrm{sn}(v)^2\,\mathrm{sn}(w)^2}$ *the second addition formula*

$\{\mathrm{sn}, \mathrm{cn}, \mathrm{dn}\}(-v) = \{-\mathrm{sn}(v), \mathrm{cn}(v), \mathrm{dn}(v)\}$ *the parity formulas*

$\mathrm{sn}(v)^2 + \mathrm{cn}(v)^2 = 1, \quad \mathrm{dn}(v)^2 + k^2\,\mathrm{sn}(v)^2 = 1$ *the Pythagorean relations*

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test these formulas ★ :

$W_2 = \frac{\varepsilon_2 \left(t \sqrt{u x_1 y_2 (1+z_2)(1+u z_2)} + e_0 e_2 \sqrt{u y_2 z_2 \mathcal{D}(x_1,t)}\right)}{y_2 u (z_2 + x_1 t^2)}$

$W_1 = \frac{\varepsilon_1 \left(t \sqrt{u x_1 y_1 (1+z_1)(1+u z_1)} - e_0 e_1 \sqrt{u y_1 z_1 \mathcal{D}(x_1,t)}\right)}{y_1 u (z_1 + x_1 t^2)}$

$U = \frac{\varepsilon_2 \varepsilon_3\, \mathrm{sgn}(u z_2) \left(|x_1| h_{232}\, t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1,t)}\right)}{\sqrt{|x_1 x_3|}\,(h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)}$ *in cases (a) and (b)*

$U = \frac{\varepsilon_2 \varepsilon_3\, \mathrm{sgn}(u z_2)\, e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1,t)}}{k \sqrt{|x_1 x_3|}\,(1 - x_1 t^2)}$ *in case (c)*

## The parameter normalization for $u < 0$ Verified

Given hypotheses and formulas, in addition to those above:

$x_1\ p_1^2 = 1, \quad \mathrm{cn}(\varsigma) = \frac{t}{p_1}$

Verification:

By the given $k'$ and $k$: $k^2 = -u\,(k')^2, \quad (k')^2 = 1 - k^2$ ☑

By $x_1\ p_1^2 = 1$ and $\mathrm{cn}(\varsigma) = \frac{t}{p_1}$: $\mathrm{cn}(\varsigma)^2 = x_1\,t^2$ ☑

By *the Pythagorean relations* and $(k')^2 = 1 - k^2$: $\mathrm{dn}(v)^2 = (k')^2 + k^2\,\mathrm{cn}(v)^2$ ☑

Hence, by *the Pythagorean relations*:

$(p_1\,\mathrm{sn}(\varsigma)\,\mathrm{dn}(\varsigma))^2 = p_1^2\,(1 - \mathrm{cn}(\varsigma)^2)\,((k')^2 + k^2\,\mathrm{cn}(\varsigma)^2)$ ☑

By $x_1\ p_1^2 = 1$, $\mathrm{cn}(\varsigma)^2 = x_1\,t^2$, and $k^2 = -u\,(k')^2$, on each sign region:

$p_1^2\,(1 - \mathrm{cn}(\varsigma)^2)\,((k')^2 + k^2\,\mathrm{cn}(\varsigma)^2) = -\frac{(k')^2\,x_1\,\mathcal{D}(x_1,t)}{x_1^2}$

$x_1 > 0$ ☑

$x_1 < 0$ ☑

By $R^2 = x_1\,\mathcal{D}(x_1, t)$:

$-\frac{(k')^2\,x_1\,\mathcal{D}(x_1,t)}{x_1^2} = \left(\frac{i\,k'\,R}{|x_1|}\right)^2$ ☑

By *the parity formulas*, replacing $\varsigma$ by $-\varsigma$
preserves $\mathrm{cn}(\varsigma)$ and reverses the sign of $\mathrm{sn}(\varsigma)\,\mathrm{dn}(\varsigma)$ ☑

Thus, we may fix $p_1\,\mathrm{sn}(\varsigma)\,\mathrm{dn}(\varsigma) = -\frac{e_0\,i\,k'\,R}{|x_1|}$, so that the two values of
$e_0$ correspond to the two admissible choices of $\varsigma$.

The sign $-e_0$ is a convention, because $e_0$ only labels the two admissible choices of $\varsigma$; it compensates the minus sign in *the second addition formula*, so that the $e_0$–terms of ★ carry the same signs as in the case $u > 0$.

## The derivation: the formulas for $W_2$ and $W_1$ for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

For $i = 1, 2$ :

$y_i\ q_i^2 = 1, \quad p_i = p_1$

$\operatorname{sn}(t_i) = \dfrac{i\ \varepsilon_i\ \sqrt{(g_i^2-1)\ p_i^2\ q_i^2}}{k\, p_i\, q_i}$

$\operatorname{cn}(t_i) = \dfrac{\varepsilon_i\ \sqrt{(g_i^2-(k')^2)\ p_i^2\ q_i^2}}{k\, p_i\, q_i}$

$\{\text{sn, cn, dn }\}(e'\, t_i) = \{e'\operatorname{sn}(t_i), \operatorname{cn}(t_i), \operatorname{dn}(t_i)\}$ *by the parity formulas*

Verification:

By *the Pythagorean relations* and $\operatorname{dn}(t_i) = g_i$ : $\operatorname{sn}(t_i)^2 = \frac{1-g_i^2}{k^2}$ ☑

By $g_i = \dfrac{1}{\sqrt{1+\frac{1}{z_i}}}$ and $k^2 = -u\,(k')^2$, $k' = \dfrac{1}{\sqrt{1-u}}$ : $\dfrac{1-g_i^2}{k^2} = -\dfrac{1-u}{u\,(1+z_i)}$ ☑

By *the Pythagorean relations* : $\operatorname{sn}(\varsigma)^2 = 1 - \operatorname{cn}(\varsigma)^2 = 1 - x_1\, t^2$ ☑

Hence $1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(\varsigma)^2 = \frac{z_i + x_1\, t^2}{1+z_i}$ ☑ , which is nonzero by the choice of $t$

Because $\pm t_i$ and $\varsigma$ are not poles and $1 - k^2 \operatorname{sn}(t_i)^2 \operatorname{sn}(\varsigma)^2 \neq 0$, *the second addition formula* , at $v = \varsigma$, $w = e'\, t_i$, holds pointwise and, together with *the parity formulas* , gives

$q_i \operatorname{cn}(\varsigma + e'\, t_i) = \dfrac{(1+z_i)\, q_i \operatorname{cn}(t_i) \operatorname{cn}(\varsigma) - e'\,(1+z_i)\, q_i \operatorname{sn}(t_i) \operatorname{dn}(t_i) \operatorname{sn}(\varsigma) \operatorname{dn}(\varsigma)}{z_i + x_1\, t^2}$ ☑

Using the given formulas and the normalization above, we get

$q_i \operatorname{cn}(\varsigma + e'\, t_i) = \dfrac{\varepsilon_i \left(t\ \sqrt{u\, x_1\, y_i\, (1+z_i)\, (1+u\, z_i)} - e'\, e_0\ \sqrt{u\, y_i\, z_i\ \mathcal{D}(x_1, t)}\right)}{y_i\, u\, \left(z_i + x_1\, t^2\right)}$ ✸

Setting $i = 2$, $e' = -e_2$ in ✸ gives the first formula of ★ .

Setting $i = 1$, $e' = e_1$ in ✸ gives the second formula of ★ .

The identity ✸ is verified separately on each sign region:

| | | |
|---|---|---|
| $z_i > 0,\ y_i > 0,$ hence $x_1 < 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i > 0,\ y_i < 0,$ hence $x_1 > 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i < -1,\ y_i > 0,$ hence $x_1 > 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |
| $z_i < -1,\ y_i < 0,$ hence $x_1 < 0$ | for all 8 sign combinations of $e'$, $e_0$, $\varepsilon_i \in \{\pm 1\}$ | ☑ |

## The derivation: the formula for $U$ in cases (a) and (b) for $u < 0$

Verified

Given hypotheses and formulas, in addition to those above:

$x_3\ p_3^2 = 1,\quad y_2\ q_2^2 = 1,\quad p_2 = p_1,\quad q_3 = q_2,\quad h_{231}, h_{232}, h_{233} \in \mathbb{R}$

$\xi_{23} = \text{sgn}(x_1\, x_3),\quad \text{sgn}(u\, z_2) = \text{sgn}(x_1\, y_2)$

$S_{23} = \frac{\varepsilon_2\ \varepsilon_3\ \sqrt{p_1^2\ p_3^2\ q_2^4\ \xi_{23}}}{p_1\ p_3\ q_2^2}$

$\{\text{sn, cn, dn}\}(e_2\, t_2 + e_3\, t_3) = \{i\, S_{23}\, h_{231}, S_{23}\, h_{232}, h_{233}\}$

Verification:

By the given expression for $S_{23}$ : $S_{23}^2 = \xi_{23}$ ☑

Hence $\text{sn}(e_2\, t_2 + e_3\, t_3)^2 = -\xi_{23}\ h_{231}^2$ ☑

and $\text{sn}(e_2\, t_2 + e_3\, t_3)\,\text{dn}(e_2\, t_2 + e_3\, t_3) = i\, S_{23}\, h_{231}\, h_{233}$ ☑

By *the Pythagorean relations* at $v = e_2\, t_2 + e_3\, t_3$ :

$h_{233}^2 - \xi_{23}\ k^2\ h_{231}^2 = \text{dn}(e_2\, t_2 + e_3\, t_3)^2 + k^2\,\text{sn}(e_2\, t_2 + e_3\, t_3)^2 = 1$ ☑

By *the Pythagorean relations* : $\text{sn}(\varsigma)^2 = 1 - \text{cn}(\varsigma)^2 = 1 - x_1\, t^2$ ☑

Hence $1 - k^2\,\text{sn}(\varsigma)^2\,\text{sn}(e_2\, t_2 + e_3\, t_3)^2 = h_{233}^2 - \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2$

☑ , which is nonzero by the choice of $t$

Because $\pm(e_2\, t_2 + e_3\, t_3)$ and $\varsigma$ are not poles and

$1 - k^2\,\text{sn}(\varsigma)^2\,\text{sn}(e_2\, t_2 + e_3\, t_3)^2 \neq 0$, *the second addition formula* , at $v = \varsigma$,

$w = -e_2\, t_2 - e_3\, t_3$, holds pointwise and, together with *the parity formulas* , gives

$$p_3\,\text{cn}(\varsigma - e_2\, t_2 - e_3\, t_3) = \frac{p_3\,(\text{cn}(e_2\, t_2 + e_3\, t_3)\,\text{cn}(\varsigma) + \text{sn}(e_2\, t_2 + e_3\, t_3)\,\text{dn}(e_2\, t_2 + e_3\, t_3)\,\text{sn}(\varsigma)\,\text{dn}(\varsigma))}{h_{233}^2 - \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2}$$ ☑

Using the given hypotheses and formulas and the normalization above, we get

$$p_3\,\text{cn}(\varsigma - e_2\, t_2 - e_3\, t_3) = \frac{\varepsilon_2\ \varepsilon_3\ \text{sgn}(u\, z_2)\ \left(|x_1|\, h_{232}\, t + e_0\, k'\, h_{231}\, h_{233}\ \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{\sqrt{|x_1\, x_3|}\ \left(h_{233}^2 - \xi_{23}\ k^2\ x_1\ h_{231}^2\ t^2\right)}$$ ◆

which is the third formula of ★ .

The identity ◆ is verified separately on each sign region, with $h_{231}, h_{232}, h_{233}, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

In case (a), both values of $\xi_{23}$ occur, because the sign of $z_2\, z_3$ is not constrained; in case (b), $\xi_{23} = 1$, because $k^2 = \left(g_2^2 - 1\right)\left(g_3^2 - 1\right) > 0$, $g_i^2 - 1 = -\frac{1}{1 + z_i}$, and $z_i\,(1 + z_i) > 0$, so $z_2\, z_3 > 0$ . The eight sign regions cover both cases.

$x_1 > 0,\ x_3 > 0,\ y_2 > 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 > 0,\ x_3 > 0,\ y_2 < 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 > 0,\ x_3 < 0,\ y_2 > 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 > 0,\ x_3 < 0,\ y_2 < 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 > 0,\ y_2 > 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 > 0,\ y_2 < 0,$ hence $\xi_{23} = -1,\ \text{sgn}(u\, z_2) = 1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 > 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = -1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 < 0,$ hence $\xi_{23} = 1,\ \text{sgn}(u\, z_2) = 1$ ☑

The derivation: the formula for $U$ in case (c) for $u < 0$ Verified

Given hypotheses and formulas, in addition to those above:

$t_2 + t_3 = 2Km + \mathrm{i}K'$, $m \in \mathbb{Z}$, $(-1)^m = S_{23}$, $e_2 = e_3$

$\mathrm{cn}(v + 2mK) = (-1)^m \mathrm{cn}(v)$, $\mathrm{cn}(v \mp \mathrm{i}K') = \frac{\pm \mathrm{i}\,\mathrm{dn}(v)}{k\,\mathrm{sn}(v)}$ *the shift formulas*

Verification:

By *the Pythagorean relations* :

$\mathrm{sn}(\varsigma)^2 = 1 - \mathrm{cn}(\varsigma)^2 = 1 - x_1 t^2$ ☑ , which is nonzero by the choice of $t$

Because $e_2 = e_3$ and $(-1)^{-e_2 m} = (-1)^m$, *the shift formulas* give

$p_3\,\mathrm{cn}(\varsigma - e_2(t_2 + t_3)) = \frac{(-1)^m e_2 \mathrm{i} p_3\,\mathrm{dn}(\varsigma)}{k\,\mathrm{sn}(\varsigma)}$ ☑

and, by $(-1)^m = S_{23}$ ,

$\frac{(-1)^m e_2 \mathrm{i} p_3\,\mathrm{dn}(\varsigma)}{k\,\mathrm{sn}(\varsigma)} = \frac{S_{23} e_2 \mathrm{i} p_3\,\mathrm{sn}(\varsigma)\,\mathrm{dn}(\varsigma)}{k\,\mathrm{sn}(\varsigma)^2}$ ☑

Using the given hypotheses and formulas and the normalization above, we get

$p_3\,\mathrm{cn}(\varsigma - e_2(t_2 + t_3)) = \frac{\varepsilon_2\,\varepsilon_3\,\mathrm{sgn}(u z_2)\,e_0\,e_2\,k'\,\sqrt{x_1\,\mathcal{D}(x_1,t)}}{k\,\sqrt{|x_1 x_3|}\,(1 - x_1 t^2)}$ ▲

which is the fourth formula of ★ .

The identity ▲

is verified separately on each of the four sign regions realizable in case (c), with $e_0, e_2, \varepsilon_2, \varepsilon_3, R$ symbolic, that is, for all their real values at once, which covers all their sign choices:

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$, $g_i^2 - 1 = -\frac{1}{1+z_i}$, and $z_i(1 + z_i) > 0$, so $z_2 z_3 > 0$ .

$x_1 > 0,\ x_3 > 0,\ y_2 > 0$, hence $\xi_{23} = 1$, $\mathrm{sgn}(u z_2) = 1$ ☑

$x_1 > 0,\ x_3 > 0,\ y_2 < 0$, hence $\xi_{23} = 1$, $\mathrm{sgn}(u z_2) = -1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 > 0$, hence $\xi_{23} = 1$, $\mathrm{sgn}(u z_2) = -1$ ☑

$x_1 < 0,\ x_3 < 0,\ y_2 < 0$, hence $\xi_{23} = 1$, $\mathrm{sgn}(u z_2) = 1$ ☑

**Section 9: ALL CLAIMS VERIFIED**

# Section 10

In[•]:=
```
(* ============================================================*)
(*PART 1:INITIALIZATION*)
(* ============================================================*)
ClearAll[UU, XX, X3, Z2, Z3, tt, RR, ee0, ep2, ep3, H1s, H2s, H3s, sX, statusChip,
  smallChip, colGiven, colTest, cBlue, cPurple, cTeal, cOrange, lblZ2, lblZ3,
  lblX, zwin, wsign, mkA, mkAB, mkAP, chkR, bOK, gdict, p6h, p6ok, p7h, p7ok,
  HAat, HBat, uLemPosat, uLemNegat, uLemPosCat, uLemNegCat, uThmABat, uThmCat,
  resultsAll, tf, gr, t4, t2, pan, ttl, rWhich, rSymH, rStar, rThus, rTog,
  rDen, rPre, eH3, dhA, dhB, dHA, dHB, dStar, dStarC, dXiZ, dXiX, dXi1, dZ1,
```

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  dX1, d11, dPreXi, dPre1, dDthm, dDpos1, dDposXi, dDneg1, dDnegXi, zA, zB,
  us, pz, SL, xiZX, xiZX1, k2gPosOK, k2gNegOK, gm1PosOK, gkpPosOK, gm1NegOK,
  gkpNegOK, kNegFormOK, prodNegOK, locIIOK, locPOK, kLocBOK, kLocPOK,
  kpLocPOK, gFormPosOK, gFormNegOK, sgn1zOK, sgnUzPosOK, sgnUzNegOK];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
colGiven = RGBColor[0.85, 0.45, 0.00];
colTest = RGBColor[0.10, 0.35, 0.75];
cBlue = RGBColor[0.10, 0.30, 0.70];
cPurple = RGBColor[0.55, 0.25, 0.70];
cTeal = RGBColor[0.00, 0.50, 0.50];
cOrange = RGBColor[0.75, 0.35, 0.05];
SetAttributes[tf, HoldAll];
tf[x_] := TraditionalForm[HoldForm[x]];
(*Every check below is a rational identity in a fixed set of radical atoms,
decided by Together;
no simplification search is involved.Sign regions:each of z2,
z3 lies in one of the two windows z>0 or z<-1,by z (1+z)>0;
we write x1=sx XX with XX>0,and the sign of x3 is forced by Sign[x1 x3]=
 Sign[z2 z3],the sign of y2 by Sign[y2]=
 Sign[u x1 z2].On each region the atoms sw2=Sqrt[|z2|],
sv2=Sqrt[|1+z2|],etc.are real principal radicals by the window facts,
and sb2=Sqrt[1+u z2],sb3=Sqrt[1+u z3] by the hypothesis 1+u z_i>0;
each de-nesting step used below is certified by squaring,
which is a rational identity,and both sides are nonnegative by
 construction.On each locus the parametrization is certified by
 checking the corresponding guard equality rationally.Every identity
 displayed in the verification panels has its own symbolic check.*)
lblZ2[1] = tf[Subscript[z, 2] > 0];
lblZ2[2] = tf[Subscript[z, 2] < -1];
lblZ3[1] = tf[Subscript[z, 3] > 0];
lblZ3[2] = tf[Subscript[z, 3] < -1];
lblX[1] = tf[Subscript[x, 1] > 0];
lblX[-1] = tf[Subscript[x, 1] < 0];
zwin[1, Z_] := Z; zwin[2, Z_] := -1 - Z; wsign[1] = 1; wsign[2] = -1;
sX = Sqrt[XX X3];
chkR[a_, b_] := Module[{d = Together[a - b]}, d === 0 || Simplify[d] === 0];
mkA[w2_, w3_, uv_] :=
  Module[{z2v = zwin[w2, Z2], z3v = zwin[w3, Z3], s2 = wsign[w2],
    s3 = wsign[w3]}, <|"z2" → z2v, "z3" → z3v, "s2" → s2, "s3" → s3, "xi" → s2 s3,
    "az2" → s2 z2v, "az3" → s3 z3v, "a12" → s2 (1 + z2v), "a13" → s3 (1 + z3v),
```

```
    "sw2" → Sqrt[s2 z2v], "sw3" → Sqrt[s3 z3v], "sv2" → Sqrt[s2 (1 + z2v)],
    "sv3" → Sqrt[s3 (1 + z3v)], "sb2" → Sqrt[1 + uv z2v], "sb3" → Sqrt[1 + uv z3v]|>];
mkAB[w2_, w3_] := Module[{at = mkA[w2, w3, 0]}, at["sb2"] = at["sv3"] / at["sw3"];
   at["sb3"] = at["sv2"] / at["sw2"]; at];
mkAP[w_] := Module[{at = mkA[w, w, 0], sS},
   sS = Sqrt[at["z2"] + at["z3"] + at["z2"] at["z3"]];
   at["sS"] = sS;
   at["sb2"] = at["sw3"] at["sv2"] / sS;
   at["sb3"] = at["sw2"] at["sv3"] / sS; at];
bOK[at_, uv_] :=
  chkR[at["sb2"]^2, 1 + uv at["z2"]] && chkR[at["sb3"]^2, 1 + uv at["z3"]];
gdict[at_, 1] := {at["sv2"] / at["sw2"], at["sv3"] / at["sw3"],
   (1 + at["z2"]) / at["z2"], (1 + at["z3"]) / at["z3"]};
gdict[at_, 2] := {at["sw2"] / at["sv2"], at["sw3"] / at["sv3"],
   at["z2"] / (1 + at["z2"]), at["z3"] / (1 + at["z3"])};
p6h[at_, kk2_, mS_, e2_, e3_, reg_] :=
  Module[{g2, g3, sr1, sr2, sr3, sr4, sr5, d, xi = at["xi"], W},
   {g2, g3} = gdict[at, reg][[1 ;; 2]];
   If[reg == 1, W = at["sw2"] at["sw3"];
    sr1 = at["sb3"] / W;
    sr2 = at["sb2"] / W;
    sr3 = at["sb2"] at["sb3"] / W;
    sr4 = 1 / W;
    sr5 = at["sb2"] at["sb3"] / (at["az2"] at["az3"]), W = at["sv2"] at["sv3"];
    sr1 = at["sb3"] / (mS W);
    sr2 = at["sb2"] / (mS W);
    sr3 = at["sb2"] at["sb3"] / (mS^2 W);
    sr4 = 1 / W;
    sr5 = at["sb2"] at["sb3"] / (mS^2 at["a12"] at["a13"])];
   d = kk2 - (g2^2 - 1) (g3^2 - 1);
   {(e2 g3 sr1 + e3 g2 sr2) / d,
    (sr3 + e2 e3 g2 g3 sr4) / d, (kk2 g2 g3 + e2 e3 xi sr5) / d}];
p6ok[at_, kk2_, kp2_, mS_, reg_] :=
  Module[{g2, g3, d2, d3, sr1, sr2, sr3, sr4, sr5, xi = at["xi"], W},
   {g2, g3, d2, d3} = gdict[at, reg];
   If[reg == 1, W = at["sw2"] at["sw3"];
    sr1 = at["sb3"] / W;
    sr2 = at["sb2"] / W;
    sr3 = at["sb2"] at["sb3"] / W;
    sr4 = 1 / W;
    sr5 = at["sb2"] at["sb3"] / (at["az2"] at["az3"]), W = at["sv2"] at["sv3"];
    sr1 = at["sb3"] / (mS W);
    sr2 = at["sb2"] / (mS W);
    sr3 = at["sb2"] at["sb3"] / (mS^2 W);
    sr4 = 1 / W;
    sr5 = at["sb2"] at["sb3"] / (mS^2 at["a12"] at["a13"])];
   chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[(g2^2 - 1) (g3^2 - kp2) xi, sr1^2] &&
```

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    chkR[(g3^2 - 1) (g2^2 - kp2) xi, sr2^2] && chkR[(g2^2 - kp2) (g3^2 - kp2) xi,
     sr3^2] && chkR[(g2^2 - 1) (g3^2 - 1) xi, sr4^2] &&
    chkR[(g2^2 - 1) (g2^2 - kp2) (g3^2 - 1) (g3^2 - kp2), sr5^2]];
p7h[at_, kkAt_, kp2_, e2_, reg_] :=
  Module[{g2, g3}, {g2, g3} = gdict[at, reg][[1 ;; 2]];
   {e2 (g2^2 - g3^2) / (2 kkAt g2 g3),
    (g2^2 g3^2 - kp2) / (2 kkAt g2 g3), (g2^2 + g3^2) / (2 g2 g3)}];
p7ok[at_, kkAt_, kk2_, reg_] :=
  Module[{g2, g3, d2, d3}, {g2, g3, d2, d3} = gdict[at, reg];
   chkR[g2^2, d2] && chkR[g3^2, d3] && chkR[kkAt^2, kk2]];
HAat[at_, uv_, e2_, e3_] :=
  Module[{d = uv at["z2"] at["z3"] - 1, sA2 = at["sw3"] at["sv3"] at["sb3"],
    sA3 = at["sw2"] at["sv2"] at["sb2"], sB = at["sb2"] at["sb3"],
    sC = at["sw2"] at["sv2"] at["sw3"] at["sv3"]},
   {(at["s2"] at["z2"] sA2 + e2 e3 at["s3"] at["z3"] sA3) / d,
    (at["xi"] at["z2"] at["z3"] sB + e2 e3 sC) / d, (uv at["xi"] sC + e2 e3 sB) / d}];
HBat[at_] := Module[{zProd = at["z2"] at["z3"],
    zSum = at["z2"] + at["z3"], sC = at["sw2"] at["sv2"] at["sw3"] at["sv3"]},
   {zProd (at["z3"] - at["z2"]) / (2 sC), zProd (zSum + 2) / (2 sC),
    (zSum + 2 zProd) at["xi"] / (2 sC)}];
uLemPosat[{h1_, h2_, h3_}, kk2_, xi_, sx_, suz_] :=
  ep2 ep3 suz (XX h2 h3 tt + ee0 h1 RR) / (sX (1 + xi kk2 sx XX h1^2 tt^2));
uLemNegat[{h1_, h2_, h3_}, kk2_, kpAt_, xi_, sx_, suz_] :=
  ep2 ep3 suz (XX h2 tt + ee0 kpAt h1 h3 RR) / (sX (h3^2 - xi kk2 sx XX h1^2 tt^2));
uLemPosCat[kkAt_, sx_, suz_] := ep2 ep3 suz sx / (sX kkAt tt);
uLemNegCat[kkAt_, kpAt_, sx_, suz_, e2_] :=
  ep2 ep3 suz ee0 e2 kpAt RR / (kkAt sX (1 - sx XX tt^2));
uThmABat[{H1_, H2_, H3_}, uv_, at_, sx_, e2_] :=
  ep2 ep3 at["s2"] (at["sw2"] at["sw3"] / sX)
   (XX H2 H3 tt + ee0 e2 H1 RR) / (at["z2"] at["z3"] + uv sx XX H1^2 tt^2);
uThmCat[at_, sy2_] := ep2 ep3 (at["sw2"] at["sw3"] / sX) sy2 / tt;
resultsAll = {};
(*---symbolic checks of the identities
  displayed in the verification panels---*)
SL = zA + zB + zA zB;
k2gPosOK = chkR[us - ((1 + zA) / zA - 1) ((1 + zB) / zB - 1), (us zA zB - 1) / (zA zB)];
k2gNegOK = chkR[-us / (1 - us) - (zA / (1 + zA) - 1) (zB / (1 + zB) - 1),
   - (1 + us SL) / ((1 - us) (1 + zA) (1 + zB))];
gm1PosOK = chkR[(1 + zA) / zA - 1, 1 / zA];
gkpPosOK = chkR[(1 + zA) / zA - (1 - us), (1 + us zA) / zA];
gm1NegOK = chkR[zA / (1 + zA) - 1, -1 / (1 + zA)];
gkpNegOK = chkR[zA / (1 + zA) - 1 / (1 - us), -(1 + us zA) / ((1 - us) (1 + zA))];
kNegFormOK = chkR[-us / (1 - us), us / (us - 1)];
prodNegOK =
  chkR[us^2 zA zB (1 + zA) (1 + zB) - (1 + us zA) (1 + us zB), (us zA zB - 1) (1 + us SL)];
locIIOK = chkR[1 + zA / (zA zB), (1 + zB) / zB] && chkR[1 + zB / (zA zB), (1 + zA) / zA];
locPOK = chkR[1 - zA / SL, zB (1 + zA) / SL] && chkR[1 - zB / SL, zA (1 + zB) / SL];
```

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```
kLocBOK = TrueQ[Simplify[Sqrt[1 / pz] == 1 / Sqrt[pz], pz > 0]];
kLocPOK = chkR[(1 / SL) / (1 + 1 / SL), 1 / ((1 + zA) (1 + zB))];
kpLocPOK = chkR[1 / (1 + 1 / SL), SL / ((1 + zA) (1 + zB))];
gFormPosOK = And @@ Table[With[{zv = zwin[w, Z2], sgnW = wsign[w]},
      chkR[(Sqrt[zv (1 + zv)] / (sgnW zv)) ^ 2, (1 + zv) / zv]], {w, {1, 2}}];
gFormNegOK = And @@ Table[With[{zv = zwin[w, Z2], sgnW = wsign[w]},
      chkR[(Sqrt[zv (1 + zv)] / (sgnW (1 + zv))) ^ 2, zv / (1 + zv)]], {w, {1, 2}}];
sgn1zOK = And @@ Table[With[{zv = zwin[w, Z2]},
      TrueQ[Simplify[Sign[1 + zv] == Sign[zv], Z2 > 0]]], {w, {1, 2}}];
sgnUzPosOK = And @@ Table[With[{zv = zwin[w, Z2]},
      TrueQ[Simplify[Sign[UU zv] == Sign[zv], UU > 0 && Z2 > 0]]], {w, {1, 2}}];
sgnUzNegOK = And @@ Table[With[{zv = zwin[w, Z2]},
      TrueQ[Simplify[Sign[-UU zv] == -Sign[zv], UU > 0 && Z2 > 0]]], {w, {1, 2}}];
xiZX[{lab_, w2_, w3_, sx_}] := Module[{at = mkA[w2, w3, 0], xi}, xi = at["xi"];
   chkR[xi at["z2"] at["z3"], at["az2"] at["az3"]] &&
    chkR[xi (sx XX) (xi sx X3), XX X3]];
xiZX1[{lab_, w2_, w3_, sx_}] := Module[{at = mkA[w2, w3, 0], xi}, xi = at["xi"];
   xiZX[{lab, w2, w3, sx}] &&
    chkR[xi (1 + at["z2"]) (1 + at["z3"]), at["a12"] at["a13"]]];
(*---display helpers---*)
gr[regs_, flags_, txt_] := Grid[Table[{Style[regs[[iw, 1]], 12, GrayLevel[0.3]],
     Style[txt, 12, GrayLevel[0.3]], smallChip[flags[[iw]]]},
    {iw, Length[regs]}], Alignment → Left, Spacings → {2, 0.7}];
t4 = Row[{"for all 4 sign combinations of  ",
    tf[Subscript[e, 2]], ", ", tf[Subscript[e, 3]]}];
t2 = Row[{"for  ", tf[Subscript[e, 2]], " = ±1"}];
pan[col_, rows_] := Panel[Column[rows, Spacings → 1],
   Background → Lighter[col, 0.93], FrameMargins → 12];
ttl[col_, cases_, cond_, ok_] :=
  Row[{Style["The derivation:  the formula for  ", Bold, 15, col],
    Style[tf[U], Bold, 15, col], Style["  in " <> cases <> " for  ", Bold,
     15, col], Style[cond, Bold, 15, col], Spacer[20], statusChip[ok]}];
rWhich = Style["which are verified separately on each sign region:", 13];
rSymH = Style[Row[{"which are verified separately on each sign region, with  ",
     tf[Subscript[H, 1]], ", ", tf[Subscript[H, 2]],
     ", ", tf[Subscript[H, 3]], "  symbolic:"}], 13];
rStar = Style[Row[{"The identity  ", Style["⋆", 13, colTest],
     "  is also verified separately on each sign region, with  ",
     tf[Subscript[H, 1]], ", ", tf[Subscript[H, 2]],
     ", ", tf[Subscript[H, 3]], ", ", tf[Subscript[e, 0]],
     ", ", tf[Subscript[ε, 2]], ", ", tf[Subscript[ε, 3]],
     "  symbolic, that is, for all their real values at once,
       which covers all their sign choices:"}], 13];
rThus[m_] := Style[Row[{"Thus, substituting the identities  ",
     m, "  into the numerator and the denominator of ",
     Style["the given U-formula", Italic, 13, colGiven], ", we get"}], 13];
eH3 =
```

```
  Row[{"  and the cancellation of the factor  ", tf[Subsuperscript[H, 3, 2]],
    "  ( ", tf[Subscript[H, 3] ≠ 0], " ,  as noted above)"}];
rTog[sgn_, chip_, lab_, extra_] :=
  Style[Row[{"Together, the numerator, denominator,
       and prefactor identities, along with  ",
     sgn, "  ", smallChip[chip], extra, " ,  give the formula  ",
     Style["★", 13, colTest], "  in " <> lab <> ":"}], 13];
rDen[d1_, d2_, why_, chip_] :=
  Style[Row[{"Hence, by the denominator identity, the  ", tf[t],
     "-dependent denominators  ", d1, "  and  ", d2, "  vanish for the same  ",
     tf[t], " ,  because  ", why, "   ", smallChip[chip]}], 13];
rPre[d_, by_, n_, chip_] :=
  Column[{Style["and, finally, the prefactor identity:", 13],
    Style[d, 14], Style[Row[{"which holds by  ", by,
       " ,  and is verified separately on each of the ", ToString[n],
       " sign regions   ", smallChip[chip]}], 13]}, Spacings → 1];
dhA = Style[Row[{"the case-(a)  ", tf[Subscript[h, 231]], ", ",
     tf[Subscript[h, 232]], ", ", tf[Subscript[h, 233]]}], 13, colGiven];
dhB = Style[Row[{"the case-(b)  ", tf[Subscript[h, 231]], ", ",
     tf[Subscript[h, 232]], ", ", tf[Subscript[h, 233]]}], 13, colGiven];
dHA = Style[Row[{"the case-(a)  ", tf[Subscript[H, 1]], ", ",
     tf[Subscript[H, 2]], ", ", tf[Subscript[H, 3]]}], 13, colTest];
dHB = Style[Row[{"the case-(b)  ", tf[Subscript[H, 1]], ", ",
     tf[Subscript[H, 2]], ", ", tf[Subscript[H, 3]]}], 13, colTest];
dStar = tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[Subscript[z, 2]]
       Sqrt[Subscript[z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
     ((Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
         Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
          Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]]) / (Subscript[z, 2]
          Subscript[z, 3] + u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2))];
dStarC = tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3]
       Sqrt[Subscript[z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]]
     HoldForm[Sign[Subscript[y, 2]] / t]];
dXiZ = tf[Subscript[ξ, 23] Subscript[z, 2] Subscript[z, 3] ==
    Abs[Subscript[z, 2] Subscript[z, 3]]];
dXiX = tf[Subscript[ξ, 23] Subscript[x, 1] Subscript[x, 3] ==
    Abs[Subscript[x, 1] Subscript[x, 3]]];
dXi1 = tf[Subscript[ξ, 23] (1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
    Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]];
dZ1 = tf[Subscript[z, 2] Subscript[z, 3] == Abs[Subscript[z, 2] Subscript[z, 3]]];
dX1 = tf[Subscript[x, 1] Subscript[x, 3] == Abs[Subscript[x, 1] Subscript[x, 3]]];
d11 = tf[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) ==
    Abs[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]];
dPreXi = tf[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
     HoldForm[Subscript[z, 2] Subscript[z, 3] /
        (Subscript[ξ, 23] Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]])] ==
    Sqrt[Subscript[z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]];
```

```
dPre1 = tf[HoldForm[1 / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]] HoldForm[
       Subscript[z, 2] Subscript[z, 3] / Sqrt[Subscript[z, 2] Subscript[z, 3]]] ==
     Sqrt[Subscript[z, 2] Subscript[z, 3] / (Subscript[x, 1] Subscript[x, 3])]];
dDthm = tf[Subscript[z, 2] Subscript[z, 3] +
     u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2];
dDposXi =
  tf[1 + Subscript[ξ, 23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2];
dDpos1 = tf[1 + k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2];
dDnegXi = tf[Subsuperscript[h, 233, 2] -
     Subscript[ξ, 23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2];
dDneg1 = tf[
    Subsuperscript[h, 233, 2] - k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2];

(* ============================================================*)
(*PART 2:THE GIVEN HYPOTHESES AND THE CLAIMS TO TEST*)
(* ============================================================*)
ClearAll[braceGrp, brc];
braceGrp[items_List, sz_ : 13, col_ : GrayLevel[0]] := DisplayForm[
    RowBox[{GridBox[({ToBoxes[Style[#, sz, col]]} & /@ items), GridBoxAlignment →
        {"Columns" → {{Left}}}, GridBoxSpacings → {"Rows" → {{1.8}}}],
      StyleBox["}", FontColor → GrayLevel[0.55], SpanMaxSize → Infinity]}]];
brc[items_, sz_, col_, lab_] := Grid[{{braceGrp[items, sz, col], lab}},
    Alignment → {Left, Center}, Spacings → {1.5, 0}];
Print[Panel[
    Column[{Style["Given hypotheses:", Bold, 14], Style[Row[{tf[u ≠ 0], ",   ",
        tf[u < 1], ",   ", tf[k], ", ", tf[k'], " ∈ (0, 1)"}], 14],
      Style[Row[{"For  ", tf[i], " = 1, 2, 3, 4 :"}], Bold, 14],
      Style[Row[{tf[Subscript[x, i]], ", ", tf[Subscript[y, i]], ", ",
        tf[Subscript[z, i]], " ∈ ℝ,   ", tf[Subscript[x, i]], ", ",
        tf[Subscript[y, i]], ", ", tf[Subscript[z, i]], " ≠ -1, 0"}], 14],
      Style[Row[{tf[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
          Sign[Subscript[x, 1] Subscript[x, 3]]], ",   ", tf[
         Sign[Subscript[y, 2]] == Sign[u Subscript[x, 1] Subscript[z, 2]]]}], 14],
      Style[Row[{tf[u Subscript[x, i] Subscript[y, i] Subscript[z, i] > 0],
        ",   ", tf[Subscript[z, i] (1 + Subscript[z, i]) > 0],
        ",   ", tf[1 + u Subscript[z, i] > 0]}], 14],
      Style[Row[{tf[Subscript[e, 0]], ", ", tf[Subscript[e, i]], ", ",
        tf[Subscript[ε, i]], ", ", tf[Subscript[ξ, 23]], " ∈ {±1}"}], 14],
      Style[Row[{tf[𝒟[Subscript[x, 1], t] == -(1 - Subscript[x, 1] t^2)
          (1 - u Subscript[x, 1] t^2)], ",   ",
        tf[R == Sqrt[Subscript[x, 1]] 𝒟[Subscript[x, 1], t]]], ",   ",
        tf[Subscript[x, 1] 𝒟[Subscript[x, 1], t] ≥ 0]}], 14],
      "", Style[Row[{"if  ", tf[u > 0], " :   ", tf[k == Sqrt[u]],
        ",   ", tf[k' == Sqrt[1 - u]], ",   ", tf[
         Subscript[g, i] == Sqrt[(1 + Subscript[z, i]) / Subscript[z, i]]]}], 14],
      Style[Row[{"if  ", tf[u < 0], " :   ", tf[k' == 1 / Sqrt[1 - u]],
        ",   ", tf[k == k' Sqrt[-u]], ",   ", tf[
```

```
    Subscript[g, i] == Sqrt[Subscript[z, i] / (1 + Subscript[z, i])]]}], 14],
"", Style[Row[{tf[Subscript[h, 231]], ", ", tf[Subscript[h, 232]],
   ", ", tf[Subscript[h, 233]], ", ", tf[Subscript[H, 1]], ", ",
   tf[Subscript[H, 2]], ", ", tf[Subscript[H, 3]], " ∈ ℝ"}], 14],
Style[Row[{tf[Subscript[h, 231] == 0], "  if and only if  ",
   tf[Subscript[g, 2] == Subscript[g, 3]], "  and  ",
   tf[Subscript[e, 2] == -Subscript[e, 3]], " ;   ",
   tf[Subscript[h, 232] == 0], "  if and only if  ",
   tf[Subscript[g, 2] Subscript[g, 3] == k'], "  and  ",
   tf[Subscript[e, 2] == -Subscript[e, 3]], " ;"}], 14],
Style[Row[{tf[Subscript[h, 233] == 0], "  if and only if  ",
   tf[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
     k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
   "  and  ", tf[Subscript[e, 2] == Subscript[e, 3]],
   " ;   in particular,  ", tf[Subscript[h, 233] ≠ 0], "  if  ",
   tf[k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
   " ."}], 14], "",
Style[Row[{"the equalities  ", tf[u Subscript[z, 2] Subscript[z, 3] == 1],
   "  and  ", tf[1 + u (Subscript[z, 2] + Subscript[z, 3] +
         Subscript[z, 2] Subscript[z, 3]) == 0], "  ",
   Style["cannot hold simultaneously", Bold, 14], " ."}], 14],
"", Style["Given formulas and expressions:", Bold, 14],
Style[tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3]
      Sign[u Subscript[z, 2]] / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
    Piecewise[{{(Abs[Subscript[x, 1]] Subscript[h, 232] Subscript[h, 233]
           t + Subscript[e, 0] Subscript[h, 231] Sqrt[Subscript[x, 1] 𝒟[
              Subscript[x, 1], t]]) / (1 + Subscript[ξ, 23] k^2 Subscript[x,
            1] Subsuperscript[h, 231, 2] t^2), u > 0}, {(Abs[Subscript[x,
             1]] Subscript[h, 232] t + Subscript[e, 0] k' Subscript[h, 231]
           Subscript[h, 233] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]]) /
         (Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2 Subscript[x, 1]
           Subsuperscript[h, 231, 2] t^2), u < 0}}]], 15, Bold, colGiven],
"", brc[{tf[Subscript[h, 231] == (Subscript[e, 2] Subscript[g, 3]
        Sqrt[(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] -
            Superscript[k', 2]) Subscript[ξ, 23]] + Subscript[e, 3]
        Subscript[g, 2] Sqrt[(Subsuperscript[g, 3, 2] - 1) (Subsuperscript[
            g, 2, 2] - Superscript[k', 2]) Subscript[ξ, 23]]) /
     (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))], tf[
   Subscript[h, 232] == (Sqrt[(Subsuperscript[g, 2, 2] - Superscript[k', 2])
         (Subsuperscript[g, 3, 2] - Superscript[k', 2]) Subscript[ξ, 23]] +
       Subscript[e, 2] Subscript[e, 3] Subscript[g, 2]
        Subscript[g, 3] Sqrt[(Subsuperscript[g, 2, 2] - 1)
          (Subsuperscript[g, 3, 2] - 1) Subscript[ξ, 23]]) /
     (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))],
  tf[Subscript[h, 233] == (k^2 Subscript[g, 2] Subscript[g, 3] +
       Subscript[e, 2] Subscript[e, 3] Subscript[ξ, 23]
        Sqrt[(Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 2, 2] -
```

```
            Superscript[k', 2]) (Subsuperscript[g, 3, 2] - 1)
          (Subsuperscript[g, 3, 2] - Superscript[k', 2])]) /
     (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1))]},
 13, colGiven, Row[{Style["(a)    ", Bold, 13], Style[tf[k^2 ≠
      (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)], 13]}]],
"", brc[{Row[{tf[Subscript[h, 231] == Subscript[e, 2]
       (Subsuperscript[g, 2, 2] - Subsuperscript[g, 3, 2]) /
        (2 k Subscript[g, 2] Subscript[g, 3])], ",    ",
    tf[Subscript[h, 232] == (Subsuperscript[g, 2, 2] Subsuperscript[g, 3,
          2] - Superscript[k', 2]) / (2 k Subscript[g, 2] Subscript[g, 3])],
    ",    ", tf[Subscript[h, 233] == (Subsuperscript[g, 2, 2] +
         Subsuperscript[g, 3, 2]) / (2 Subscript[g, 2] Subscript[g, 3])]}]},
 13, colGiven, Row[{Style["(b)    ", Bold, 13], Style[tf[
     k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)], 13],
   Style[",    ", 13], Style[tf[Subscript[e, 2] == -Subscript[e, 3]], 13]}]],
"", brc[{Style[tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[u
           Subscript[z, 2]] / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
      Piecewise[{{Sign[Subscript[x, 1]] / (k t), u > 0}, {Subscript[e, 0]
          Subscript[e, 2] k' Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] /
           (k (1 - Subscript[x, 1] t^2)), u < 0}}]], Bold]},
 14, colGiven, Row[{Style["(c)    ", Bold, 13], Style[tf[
     k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)], 13],
   Style[",    ", 13], Style[tf[Subscript[e, 2] == Subscript[e, 3]], 13]}]],
"", Style[Row[{"Each  ", smallChip[True],
   "  below certifies that the statement displayed beside it has been
     verified by a symbolic computation at that point;  a  ",
   smallChip[False], "  would mean that this verification failed."}],
 12, GrayLevel[0.35]], "", Style[
 Row[{"We want to test the following claims  ",
   Style["⋆", 14, colTest], " :"}], Bold, 14],
Style[Row[{"(1)  in each case, the tested formula for  ", tf[U],
   "  below coincides with the applicable given formula for  ",
   tf[U], " ;"}], 13],
Style[Row[{"(2)  in each case, the  ", tf[t], "-dependent denominators
     of these two formulas vanish for the same values of  ",
   tf[t], " ."}], 13], "", Style[dStar, 15, Bold, colTest],
"", brc[{tf[Subscript[H, 1] == (Abs[Subscript[z, 2]]
        Sqrt[Subscript[z, 3] (1 + Subscript[z, 3]) (1 + u Subscript[z, 3])] +
       Subscript[e, 2] Subscript[e, 3] Abs[Subscript[z, 3]]
        Sqrt[Subscript[z, 2] (1 + Subscript[z, 2]) (1 + u Subscript[z, 2])]) /
     (u Subscript[z, 2] Subscript[z, 3] - 1)],
  tf[Subscript[H, 2] == (Abs[Subscript[z, 2] Subscript[z, 3]]
        Sqrt[(1 + u Subscript[z, 2]) (1 + u Subscript[z, 3])] +
       Subscript[e, 2] Subscript[e, 3] Sqrt[Subscript[z, 2] Subscript[
           z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])]) /
     (u Subscript[z, 2] Subscript[z, 3] - 1)], tf[Subscript[H, 3] ==
    (u Sqrt[Subscript[z, 2] Subscript[z, 3] (1 + Subscript[z, 2])
```

```
                (1 + Subscript[z, 3])] Sign[Subscript[z, 2] Subscript[z, 3]] +
             Subscript[e, 2] Subscript[e, 3] Sqrt[(1 + u Subscript[z, 2]) (1 +
                  u Subscript[z, 3])]) / (u Subscript[z, 2] Subscript[z, 3] - 1)]},
       13, colTest, Row[{Style["(a)    ", Bold, 13], Style[
          tf[u Subscript[z, 2] Subscript[z, 3] ≠ 1], 13]}]],
      "", brc[{tf[Subscript[H, 1] == Subscript[z, 2] Subscript[z, 3]
           (Subscript[z, 3] - Subscript[z, 2]) / (2 Sqrt[Subscript[z, 2]
                Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])])],
        tf[Subscript[H, 2] == Subscript[z, 2] Subscript[z, 3]
           (Subscript[z, 2] + Subscript[z, 3] + 2) / (2 Sqrt[Subscript[z, 2]
                Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])])],
        tf[Subscript[H, 3] == (Subscript[z, 2] + Subscript[z, 3] +
             2 Subscript[z, 2] Subscript[z, 3])
           Sign[Subscript[z, 2] Subscript[z, 3]] / (2 Sqrt[Subscript[z, 2]
                Subscript[z, 3] (1 + Subscript[z, 2]) (1 + Subscript[z, 3])])]},
       13, colTest, Row[{Style["(b)    ", Bold, 13], Style[
          tf[u Subscript[z, 2] Subscript[z, 3] == 1], 13], Style["    ", 13],
         Style[tf[Subscript[e, 2] == -Subscript[e, 3]], 13]}]],
      "", brc[{Style[dStarC, Bold]}, 14, colTest,
       Row[{Style["(c)    ", Bold, 13],
         Style[tf[u Subscript[z, 2] Subscript[z, 3] == 1], 13], Style["    ", 13],
         Style[tf[Subscript[e, 2] == Subscript[e, 3]], 13]}]]},
     Spacings → 1.1], Background → RGBColor[
      0.99,
      0.96,
      0.88], FrameMargins → 14]];

(* ============================================================*)
(*PART 3:CASE (a) FOR u>0*)
(* ============================================================*)
ClearAll[regA, aCheckPos, aPreCk];
regA = Flatten[
   Table[{Row[{lblZ2[w2], ",  ", lblZ3[w3], ",  ", lblX[sx]}], w2, w3, sx},
    {w2, {1, 2}}, {w3, {1, 2}}, {sx, {1, -1}}], 2];
aCheckPos[{lab_, w2_, w3_, sx_}, {e2_, e3_}] :=
  Module[{at = mkA[w2, w3, UU], xi, suz, W, hv, Hv}, xi = at["xi"];
   suz = at["s2"];
   W = at["sw2"] at["sw3"];
   hv = p6h[at, UU, 1, e2, e3, 1]; Hv = HAat[at, UU, e2, e3];
   {bOK[at, UU] && p6ok[at, UU, 1 - UU, 1, 1] && chkR[hv⟦1⟧ W, xi e2 Hv⟦1⟧] &&
     chkR[hv⟦2⟧ W, xi Hv⟦2⟧] && chkR[hv⟦3⟧, Hv⟦3⟧], chkR[XX H2s H3s tt +
       ee0 e2 H1s RR, xi W (XX (xi H2s / W) H3s tt + ee0 (xi e2 H1s / W) RR)] &&
     chkR[at["z2"] at["z3"] + UU sx XX H1s^2 tt^2,
      at["z2"] at["z3"] (1 + xi UU sx XX (xi e2 H1s / W)^2 tt^2)],
    chkR[uLemPosat[{xi e2 H1s / W, xi H2s / W, H3s}, UU, xi, sx, suz],
     uThmABat[{H1s, H2s, H3s}, UU, at, sx, e2]]}];
aPreCk[{lab_, w2_, w3_, sx_}] :=
```

```
  Module[{at = mkA[w2, w3, UU], W}, W = at["sw2"] at["sw3"];
   chkR[at["z2"] at["z3"] / (at["xi"] W sX), W / sX]];
Module[{grid, br, id, fu, pre, xa, ok},
  grid = Table[aCheckPos[w, sgnPair], {w, regA}, {sgnPair, Tuples[{1, -1}, 2]}];
  {br, id, fu} = Table[And @@ grid[[iw, All, j]], {j, 3}, {iw, Length[regA]}];
  pre = And @@ (aPreCk /@ regA); xa = And @@ (xiZX /@ regA);
  ok = (And @@ Join[br, id, fu]) && pre && xa &&
    k2gPosOK && gm1PosOK && gkpPosOK && gFormPosOK && sgnUzPosOK;
  AppendTo[resultsAll, ok];
  Print[pan[cBlue, {ttl[cBlue, "case (a)", tf[u > 0], ok], Style[
      Row[{"By  ", tf[Subscript[ξ, 23] == Sign[Subscript[z, 2] Subscript[z, 3]] ==
          Sign[Subscript[x, 1] Subscript[x, 3]]], " :   ", dXiZ,
        ",   ", dXiX, "   ", smallChip[xa]}], 12, GrayLevel[0.35]],
     "", Style["Verification:", Bold, 13], Style[
      Row[{"By the given  ", tf[k], "  and  ", tf[Subscript[g, i]], " :   ",
        tf[k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) ==
          (u Subscript[z, 2] Subscript[z, 3] - 1) /
           (Subscript[z, 2] Subscript[z, 3])], "   ",
        smallChip[k2gPosOK]}], 13], Style[Row[{"In this case  ",
        tf[k^2 ≠ (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
        " ,  that is,  ", tf[u Subscript[z, 2] Subscript[z, 3] ≠ 1],
        " ,  and the following identities hold:"}], 13],
     Style[Row[{tf[Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]],
        "  ", smallChip[gm1PosOK], ",   ",
        tf[Subsuperscript[g, i, 2] - Superscript[k', 2] ==
          (1 + u Subscript[z, i]) / Subscript[z, i]], "  ", smallChip[gkpPosOK],
        ",   ", tf[Subscript[g, i] == Sqrt[Subscript[z, i] (1 + Subscript[z, i])] /
           Abs[Subscript[z, i]]], "  ", smallChip[gFormPosOK],
        "      ", Style["◆", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
     Style[Row[{"Hence, substituting the identities  ",
        Style["◆", 13, RGBColor[0.55, 0.33, 0.10]], "  and the equality  ",
        dXiZ, "  into ", dhA, " ,  we get, with ", dHA, " ,"}], 13],
     Style[Row[{tf[Subscript[h, 231] == Subscript[ξ, 23] Subscript[e, 2]
           Subscript[H, 1] / Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]],
        ",   ", tf[Subscript[h, 232] == Subscript[ξ, 23]
           Subscript[H, 2] / Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]],
        ",   ", tf[Subscript[h, 233] == Subscript[H, 3]], "      ",
        Style["◆", 14, RGBColor[0.00, 0.45, 0.25]]}], 13], rWhich,
     gr[regA, br, t4], "", rThus[Style["◆", 13, RGBColor[0.00, 0.45, 0.25]]],
     Style["the numerator identity:", 13],
     Style[tf[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
         Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
          Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] == Subscript[ξ, 23]
         Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]] (Abs[Subscript[x, 1]]
            Subscript[h, 232] Subscript[h, 233] t + Subscript[e, 0]
            Subscript[h, 231] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]])],
      14], Style["the denominator identity:", 13], Style[
```

```
       tf[Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
           Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2] Subscript[z, 3] (1 +
            Subscript[ξ, 23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)],
       14], rSymH, gr[regA, id, t4], rDen[dDthm, dDposXi,
       tf[Subscript[z, 2] Subscript[z, 3] ≠ 0], And @@ id],
      "", rPre[dPreXi, Row[{dXiZ, "  and  ", dXiX}], 8, pre], "",
      rTog[tf[Sign[u Subscript[z, 2]] == Sign[Subscript[z, 2]]],
       sgnUzPosOK, "case (a)", ""],
      Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
      rStar, gr[regA, fu, t4]}]]];

(* ============================================================*)
(*PART 4:CASES (b) AND (c) FOR u>0,ON u z2 z3=1*)
(* ============================================================*)
ClearAll[regBpos, bCheckPos, bPreCk, cCheckPos];
regBpos =
  Flatten[Table[{Row[{lblZ2[w], ",  ", lblZ3[w], ",  ", lblX[sx]}], w, sx},
    {w, {1, 2}}, {sx, {1, -1}}], 1];
bCheckPos[{lab_, w_, sx_}, e2_] :=
  Module[{at = mkAB[w, w], uv, kkAt, W, d2, d3, hv, Hv}, uv = 1 / (at["z2"] at["z3"]);
   W = at["sw2"] at["sw3"];
   kkAt = 1 / W;
   {d2, d3} = gdict[at, 1][[3 ;; 4]];
   hv = p7h[at, kkAt, 1 - uv, e2, 1]; Hv = HBat[at];
   {bOK[at, uv] && p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
     chkR[hv[[1]] W, e2 Hv[[1]]] && chkR[hv[[2]] W, Hv[[2]]] && chkR[hv[[3]], Hv[[3]]],
    chkR[XX H2s H3s tt + ee0 e2 H1s RR, W (XX (H2s / W) H3s tt + ee0 (e2 H1s / W) RR)] &&
     chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] (1 + sx XX (e2 H1s / W)^2 kkAt^2 tt^2)],
    chkR[uLemPosat[{e2 H1s / W, H2s / W, H3s}, uv, 1, sx, at["s2"]]],
     uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]]}];
bPreCk[{lab_, w_, sx_}] := Module[{at = mkAB[w, w], W}, W = at["sw2"] at["sw3"];
   chkR[at["z2"] at["z3"] / (W sX), W / sX]];
cCheckPos[{lab_, w_, sx_}] :=
  Module[{at = mkAB[w, w], uv, kkAt, d2, d3}, uv = 1 / (at["z2"] at["z3"]);
   kkAt = 1 / (at["sw2"] at["sw3"]);
   {d2, d3} = gdict[at, 1][[3 ;; 4]];
   p7ok[at, kkAt, uv, 1] && chkR[kkAt^2, (d2 - 1) (d3 - 1)] &&
    chkR[uLemPosCat[kkAt, sx, at["s2"]], uThmCat[at, at["s2"] sx]]];
Module[{gridB, br, id, fu, pre, cR, xa, ok},
  gridB = Table[bCheckPos[w, sgnE2], {w, regBpos}, {sgnE2, {1, -1}}];
  {br, id, fu} = Table[And @@ gridB[[iw, All, j]], {j, 3}, {iw, Length[regBpos]}];
  pre = And @@ (bPreCk /@ regBpos); cR = cCheckPos /@ regBpos;
  xa = And @@ (xiZX[{#[[1]], #[[2]], #[[2]], #[[3]]} & /@ regBpos);
  ok = (And @@ Join[br, id, fu, cR]) && pre && xa &&
    gm1PosOK && k2gPosOK && kLocBOK && gFormPosOK && sgnUzPosOK;
  AppendTo[resultsAll, ok];
```

```
Print[pan[cPurple, {ttl[cPurple, "cases (b) and (c)", tf[u > 0], ok],
   Style[Row[{"Here  ", tf[Subscript[ξ, 23] == 1], ",  because  ",
      tf[k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0],
      "  and  ", tf[Subsuperscript[g, i, 2] - 1 == 1 / Subscript[z, i]], "  ",
      smallChip[gm1PosOK], ",  so  ", tf[Subscript[z, 2] Subscript[z, 3] > 0],
      " ;  hence  ", dZ1, ",   ", dX1, "   ", smallChip[xa]}],
    12, GrayLevel[0.35]], "", Style["Verification:", Bold, 13],
   Style[Row[{"In these cases  ", tf[k^2 == (Subsuperscript[g, 2, 2] - 1)
        (Subsuperscript[g, 3, 2] - 1)], " ,  that is,  ",
      tf[u Subscript[z, 2] Subscript[z, 3] == 1], " ,  by the identity  ",
      tf[k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) ==
        (u Subscript[z, 2] Subscript[z, 3] - 1) /
         (Subscript[z, 2] Subscript[z, 3])], "  ", smallChip[k2gPosOK],
      " ,  and the following identities hold:"}], 13],
   Style[Row[{tf[k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
      "  ", smallChip[kLocBOK], ",   ", tf[Subscript[g, i] ==
        Sqrt[Subscript[z, i] (1 + Subscript[z, i])] / Abs[Subscript[z, i]]],
      "  ", smallChip[gFormPosOK], "      ",
      Style["▲", 14, RGBColor[0.55, 0.33, 0.10]]}], 13], Style[
    Row[{"Hence, in case (b) ( ", tf[Subscript[e, 2] == -Subscript[e, 3]],
      " ),  substituting the identities  ",
      Style["▲", 13, RGBColor[0.55, 0.33, 0.10]], "  and the equality  ",
      dZ1, "  into ", dhB, " ,  we get, with ", dHB, " ,"}], 13],
   Style[Row[{tf[Subscript[h, 231] == Subscript[e, 2] Subscript[H, 1] /
         Sqrt[Subscript[z, 2] Subscript[z, 3]]], ",   ", tf[Subscript[h,
        232] == Subscript[H, 2] / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
      ",   ", tf[Subscript[h, 233] == Subscript[H, 3]], "      ",
      Style["▲", 14, RGBColor[0.00, 0.45, 0.25]]}], 13], rWhich,
   gr[regBpos, br, t2], "", rThus[Style["▲", 13, RGBColor[0.00, 0.45, 0.25]]],
   Style["the numerator identity:", 13],
   Style[tf[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
      Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
       Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] ==
     Sqrt[Subscript[z, 2] Subscript[z, 3]] (Abs[Subscript[x, 1]]
         Subscript[h, 232] Subscript[h, 233] t + Subscript[e, 0]
         Subscript[h, 231] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]])],
    14], Style["the denominator identity:", 13], Style[
    tf[Subscript[z, 2] Subscript[z, 3] + u Subscript[x, 1]
      Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2] Subscript[z, 3]
     (1 + k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
   rSymH, gr[regBpos, id, t2], rDen[dDthm, dDpos1,
    tf[Subscript[z, 2] Subscript[z, 3] ≠ 0], And @@ id],
   "", rPre[dPre1, Row[{dZ1, "  and  ", dX1}], 4, pre], "",
   rTog[tf[Sign[u Subscript[z, 2]] == Sign[Subscript[z, 2]]],
    sgnUzPosOK, "case (b)", ""],
   Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
   rStar, gr[regBpos, fu, t2], "",
```

```
      Style[Row[{"In case (c) ( ", tf[Subscript[e, 2] == Subscript[e, 3]],
         " ),  the ", Style["given", Italic, 13, colGiven], "  and  ",
         Style["tested", Italic, 13, colTest], "  formulas are"}], 13],
      Style[Row[{Style[tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3] Sign[
                 u Subscript[z, 2]] / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
             HoldForm[Sign[Subscript[x, 1]] / (k t)]], 14, colGiven],
         "   and   ", Style[dStarC, 14, colTest], " ,"}], 14],
      Style[Row[{"which coincide because  ",
         tf[Sign[Subscript[y, 2]] == Sign[u Subscript[x, 1] Subscript[z, 2]]],
         ",   ", tf[k == 1 / Sqrt[Subscript[z, 2] Subscript[z, 3]]],
         " ,  and the prefactor identity;  their denominators  ", tf[k t],
         "  and  ", tf[t], "  vanish for the same  ", tf[t], " ,  because  ",
         tf[k ≠ 0], " .  The equality is verified separately on each
           sign region, with  ", tf[Subscript[ε, 2]], ", ",
         tf[Subscript[ε, 3]], "  symbolic:"}], 13], gr[regBpos, cR, ""]}]]];

(* ============================================================*)
(*PART 5:CASE (a) FOR u<0*)
(* ============================================================*)
ClearAll[aCheckNeg, aPreCkNeg, h233ConvCk,
  regZ, regMix, bCheckNegII, bPreCkII, cCheckNegII];
regZ = Flatten[Table[
    {Row[{lblZ2[w2], ",  ", lblZ3[w3]}], w2, w3}, {w2, {1, 2}}, {w3, {1, 2}}], 1];
regMix = Flatten[Table[{Row[{lblZ2[w2], ",  ", lblZ3[3 - w2], ",  ", lblX[sx]}],
     w2, 3 - w2, sx}, {w2, {1, 2}}, {sx, {1, -1}}], 1];
h233ConvCk[{lab_, w2_, w3_}] :=
  Module[{at = mkA[w2, w3, -UU], kk2 = UU / (1 + UU), d2, d3},
   {d2, d3} = gdict[at, 2][[3 ;; 4]];
   chkR[kk2 (d2 + d3) - (kk2 - (d2 - 1) (d3 - 1)),
    (1 - (-UU) at["z2"] at["z3"]) / ((1 + UU) (1 + at["z2"]) (1 + at["z3"]))]];
aCheckNeg[{lab_, w2_, w3_, sx_}, {e2_, e3_}] := Module[{at = mkA[w2, w3, -UU],
    uv = -UU, kk2, mS, kpAt, xi, suz, W, hv, Hv, brh}, kk2 = UU / (1 + UU);
   mS = Sqrt[1 + UU];
   kpAt = 1 / mS;
   xi = at["xi"];
   suz = -at["s2"];
   W = at["sw2"] at["sw3"];
   hv = p6h[at, kk2, mS, e2, e3, 2]; Hv = HAat[at, uv, e2, e3];
   brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
   {bOK[at, uv] && p6ok[at, kk2, 1 / (1 + UU), mS, 2] &&
     chkR[hv[[3]] Hv[[3]], 1] && chkR[hv[[1]] W Hv[[3]], -xi e2 mS Hv[[1]]] &&
     chkR[hv[[2]] W Hv[[3]], -xi Hv[[2]]], chkR[XX H2s H3s tt + ee0 e2 H1s RR,
      -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)] &&
     chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)],
    chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
     uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]]}];
```

```
aPreCkNeg[{lab_, w2_, w3_, sx_}] :=
  Module[{at = mkA[w2, w3, -UU], W}, W = at["sw2"] at["sw3"];
   chkR[at["z2"] at["z3"] / (at["xi"] W sX), W / sX]];
bCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
  Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, hv, Hv, brh},
   uv = 1 / (at["z2"] at["z3"]);
   kk2 = -uv / (1 - uv);
   mS = Sqrt[1 - uv];
   kpAt = 1 / mS;
   xi = at["xi"];
   suz = -at["s2"];
   W = at["sw2"] at["sw3"];
   hv = p6h[at, kk2, mS, e2, -e2, 2];
   Hv = HBat[at];
   brh = {-xi e2 mS H1s / (W H3s), -xi H2s / (W H3s), 1 / H3s};
   {bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] &&
     chkR[hv[[3]] Hv[[3]], 1] && chkR[hv[[1]] W Hv[[3]], -xi e2 mS Hv[[1]]] &&
     chkR[hv[[2]] W Hv[[3]], -xi Hv[[2]]], chkR[XX H2s H3s tt + ee0 e2 H1s RR,
      -xi W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)] &&
     chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - xi kk2 sx XX brh[[1]]^2 tt^2)],
    chkR[uLemNegat[brh, kk2, kpAt, xi, sx, suz],
     uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]]}];
bPreCkII[{lab_, w2_, w3_, sx_}] :=
  Module[{at = mkAB[w2, w3], W}, W = at["sw2"] at["sw3"];
   chkR[at["z2"] at["z3"] / (at["xi"] W sX), W / sX]];
cCheckNegII[{lab_, w2_, w3_, sx_}, e2_] :=
  Module[{at = mkAB[w2, w3], uv, kk2, mS, kpAt, xi, suz, W, hv},
   uv = 1 / (at["z2"] at["z3"]);
   kk2 = -uv / (1 - uv);
   mS = Sqrt[1 - uv];
   kpAt = 1 / mS;
   xi = at["xi"];
   suz = -at["s2"];
   W = at["sw2"] at["sw3"]; hv = p6h[at, kk2, mS, e2, e2, 2];
   bOK[at, uv] && p6ok[at, kk2, 1 / (1 - uv), mS, 2] && chkR[hv[[3]], 0] &&
    chkR[hv[[1]], e2 mS W] && chkR[hv[[2]], W] && chkR[kk2 mS^2 W^2, 1] &&
    chkR[hv[[3]]^2 - xi kk2 sx XX hv[[1]]^2 tt^2, sx XX tt^2] &&
    chkR[uLemNegat[hv, kk2, kpAt, xi, sx, suz], uThmCat[at, suz sx]]];
Module[{gA, br, id, fu, pre, conv, gB,
   brB, idB, fuB, preB, gC, cR, xa, xm, ok, dN, dD, dRel},
  gA = Table[aCheckNeg[w, sgnPair], {w, regA}, {sgnPair, Tuples[{1, -1}, 2]}];
  {br, id, fu} = Table[And @@ gA[[iw, All, j]], {j, 3}, {iw, Length[regA]}];
  pre = And @@ (aPreCkNeg /@ regA); conv = And @@ (h233ConvCk /@ regZ);
  gB = Table[bCheckNegII[w, sgnE2], {w, regMix}, {sgnE2, {1, -1}}];
  {brB, idB, fuB} = Table[And @@ gB[[iw, All, j]], {j, 3}, {iw, Length[regMix]}];
  preB = And @@ (bPreCkII /@ regMix);
```

```
gC = Table[cCheckNegII[w, sgnE2], {w, regMix}, {sgnE2, {1, -1}}];
cR = Table[And @@ gC[[iw]], {iw, Length[regMix]}];
xa = And @@ (xiZX1 /@ regA); xm = And @@ (xiZX1 /@ regMix);
ok = (And @@ Join[br, id, fu, brB, idB, fuB, cR]) && pre && conv &&
  preB && xa && xm && k2gNegOK && kNegFormOK && gm1NegOK && gkpNegOK &&
  gFormNegOK && prodNegOK && sgn1zOK && locIIOK && sgnUzNegOK;
AppendTo[resultsAll, ok];
dN = tf[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
     Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] == -Subscript[ξ, 23]
    Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]] Subsuperscript[H, 3, 2]
    (Abs[Subscript[x, 1]] Subscript[h, 232] t + Subscript[e, 0] k' Subscript[h,
        231] Subscript[h, 233] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]])];
dD = tf[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2]
    Subscript[z, 3] Subsuperscript[H, 3, 2] (Subsuperscript[h, 233, 2] -
      Subscript[ξ, 23] k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)];
dRel =
 Row[{tf[Subscript[h, 233] Subscript[H, 3] == 1], ",   ", tf[Subscript[h, 231] ==
     -Subscript[ξ, 23] Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
       (Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]] Subscript[H, 3])],
   ",   ", tf[Subscript[h, 232] == -Subscript[ξ, 23] Subscript[H, 2] /
       (Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]] Subscript[H, 3])]}];
Print[pan[cTeal, {ttl[cTeal, "case (a)", tf[u < 0], ok],
   Style[Row[{"Here  ", tf[Subscript[h, 233] == 0], "  if and only if  ",
      tf[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) ==
        k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
      "  and  ", tf[Subscript[e, 2] == Subscript[e, 3]],
      " .  When  ", tf[u < 0], " ,"}], 12, GrayLevel[0.35]],
   Style[Row[{tf[k^2 (Subsuperscript[g, 2, 2] + Subsuperscript[g, 3, 2]) -
         (k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)) ==
        (1 - u Subscript[z, 2] Subscript[z, 3]) / ((1 - u) (1 + Subscript[z, 2])
           (1 + Subscript[z, 3]))], "   ", smallChip[conv],
      " ,  verified separately on each of the 4 sign regions of  ",
      tf[Subscript[z, 2]], ", ", tf[Subscript[z, 3]],
      " ."}], 12, GrayLevel[0.35]],
   Style[Row[{"Thus, when  ", tf[u < 0], " ,  ", tf[Subscript[h, 233] == 0],
      "  if and only if  ", tf[u Subscript[z, 2] Subscript[z, 3] == 1],
      "  and  ", tf[Subscript[e, 2] == Subscript[e, 3]],
      " .  Therefore, in cases (i) and (ii) below,  ",
      tf[Subscript[h, 233] ≠ 0], " ,  and the identity  ",
      tf[Subscript[h, 233] Subscript[H, 3] == 1], "  gives  ",
      tf[Subscript[H, 3] ≠ 0], " ,  so the divisions by  ",
      tf[Subscript[H, 3]], "  are legitimate there."}], 12,
    GrayLevel[0.35]], "", Style["Verification:", Bold, 13], Style[
    Row[{"By the given  ", tf[k], "  and  ", tf[Subscript[g, i]], " :   ",
      tf[k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) ==
```

```
      - ((1 + u (Subscript[z, 2] + Subscript[z, 3] +
                Subscript[z, 2] Subscript[z, 3])) /
           ((1 - u) (1 + Subscript[z, 2]) (1 + Subscript[z, 3])))],
   "   ", smallChip[k2gNegOK]}], 13], Style[Row[{"In this case  ",
   tf[k^2 ≠ (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
   " ,  that is,  ", tf[1 + u (Subscript[z, 2] +
        Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) ≠ 0],
   " ;  we consider three possible cases:   (i)  ",
   tf[u Subscript[z, 2] Subscript[z, 3] ≠ 1], " ,   (ii)  ",
   tf[u Subscript[z, 2] Subscript[z, 3] == 1], "  and  ",
   tf[Subscript[e, 2] == - Subscript[e, 3]], " ,   (iii)  ",
   tf[u Subscript[z, 2] Subscript[z, 3] == 1], "  and  ",
   tf[Subscript[e, 2] == Subscript[e, 3]], " ."}], 13], "",
Style[Row[{"(i)  ", tf[u Subscript[z, 2] Subscript[z, 3] ≠ 1], " :"}],
 Bold, 13], Style["The following identities hold:", 13],
Style[Row[{tf[k == Sqrt[u / (u - 1)]], "  ", smallChip[kNegFormOK],
   ",   ", tf[Subsuperscript[g, i, 2] - 1 == - (1 / (1 + Subscript[z, i]))],
   "  ", smallChip[gm1NegOK], ",   ",
   tf[Subsuperscript[g, i, 2] - Superscript[k', 2] ==
     - ((1 + u Subscript[z, i]) / ((1 - u) (1 + Subscript[z, i]))),
   "  ", smallChip[gkpNegOK], ",   ", tf[Subscript[g, i] ==
     Sqrt[Subscript[z, i] (1 + Subscript[z, i])] / Abs[1 + Subscript[z, i]]],
   "  ", smallChip[gFormNegOK], "      ",
   Style["●", 14, RGBColor[0.55, 0.33, 0.10]]}], 13],
Style[Row[{tf[u^2 Subscript[z, 2] Subscript[z, 3] (1 + Subscript[z, 2])
       (1 + Subscript[z, 3]) - (1 + u Subscript[z, 2]) (1 + u Subscript[z, 3]) ==
     (u Subscript[z, 2] Subscript[z, 3] - 1) (1 + u (Subscript[z, 2] +
           Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]))],
   "  ", smallChip[prodNegOK], "      ", Style["●", 14,
    RGBColor[0.55, 0.05, 0.10]]}], 13],
Style[Row[{"and, by  ", tf[Subscript[ξ, 23] == Sign[Subscript[z, 2]
       Subscript[z, 3]] == Sign[Subscript[x, 1] Subscript[x, 3]]],
   "  and  ", tf[Sign[1 + Subscript[z, i]] == Sign[Subscript[z, i]]],
   "  ", smallChip[sgn1zOK], " :   ", dXiZ, ",   ",
   dXiX, ",   ", dXi1, "   ", smallChip[xa]}], 13],
Style[Row[{"Hence, substituting the identities  ",
   Style["●", 13, RGBColor[0.55, 0.33, 0.10]], "  and the equalities  ",
   dXiZ, "  and  ", dXi1, "  into ", dhA, " ,  and using the identity  ",
   Style["●", 13, RGBColor[0.55, 0.05, 0.10]],
   " ,  we get, with ", dHA, " ,"}], 13],
Style[Row[{dRel, "      ", Style["●", 14, RGBColor[0.00, 0.45, 0.25]]}],
 13], rWhich, gr[regA, br, t4], "",
rThus[Style["●", 13, RGBColor[0.00, 0.45, 0.25]]],
Style["the numerator identity:", 13],
Style[dN, 14],
Style["the denominator identity:", 13],
Style[dD, 14], rSymH, gr[regA, id, t4],
```

```
rDen[dDthm, dDnegXi, Row[{tf[Subscript[z, 2] Subscript[z, 3] ≠ 0],
   "  and  ", tf[Subscript[H, 3] ≠ 0]}], And @@ id],
"", rPre[dPreXi, Row[{dXiZ, "  and  ", dXiX}], 8, pre], "",
rTog[tf[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
 sgnUzNegOK, "case (a)", eH3],
Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
rStar, gr[regA, fu, t4], "",
Style[Row[{"(ii)  ", tf[u Subscript[z, 2] Subscript[z, 3] == 1],
   "  and  ", tf[Subscript[e, 2] == -Subscript[e, 3]], " :"}], Bold, 13],
Style[Row[{"Here  ", tf[Subscript[ξ, 23] == -1], ",  because  ", tf[
    u Subscript[z, 2] Subscript[z, 3] == 1], "  and  ", tf[u < 0], ",  so  ",
   tf[Subscript[z, 2] Subscript[z, 3] < 0], " ;  the equalities  ",
   dXiZ, ",   ", dXiX, ",   ", dXi1, "  hold on these regions  ",
   smallChip[xm], " ;  the given expressions remain ",
   dhA, " ,  and the expressions to test are ",
   dHB, " ."}], 12, GrayLevel[0.35]],
Style[Row[{tf[1 + u Subscript[z, 2] == (1 + Subscript[z, 3]) / Subscript[z, 3]],
   ",   ", tf[1 + u Subscript[z, 3] == (1 + Subscript[z, 2]) / Subscript[z, 2]],
   "  ", smallChip[locIIOK], "      ",
   Style["●", 14, RGBColor[0.40, 0.15, 0.60]]}], 13],
Style[Row[{"Hence, substituting the identities  ",
   Style["●", 13, RGBColor[0.55, 0.33, 0.10]],
   "  and  ", Style["●", 13, RGBColor[0.40, 0.15, 0.60]],
   "  and the equalities  ", dXiZ, "  and  ", dXi1,
   "  into ", dhA, " ,  we get, with ", dHB, " ,"}], 13],
Style[Row[{dRel, "      ", Style["●", 14, RGBColor[0.75, 0.10, 0.40]]}],
 13], rWhich, gr[regMix, brB, t2], "",
rThus[Style["●", 13, RGBColor[0.75, 0.10, 0.40]]],
Style["the numerator identity:", 13], Style[dN, 14],
Style["the denominator identity:", 13],
Style[dD, 14], rSymH, gr[regMix, idB, t2],
rDen[dDthm, dDnegXi, Row[{tf[Subscript[z, 2] Subscript[z, 3] ≠ 0],
   "  and  ", tf[Subscript[H, 3] ≠ 0]}], And @@ idB],
"", rPre[dPreXi, Row[{dXiZ, "  and  ", dXiX}], 4, preB], "",
rTog[tf[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
 sgnUzNegOK, "case (b)", eH3],
Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
rStar, gr[regMix, fuB, t2], "",
Style[Row[{"(iii)  ", tf[u Subscript[z, 2] Subscript[z, 3] == 1],
   "  and  ", tf[Subscript[e, 2] == Subscript[e, 3]], " :"}], Bold, 13],
Style[Row[{"As in case (ii),  ", tf[Subscript[ξ, 23] == -1],
   " ;  the given expressions remain ", dhA,
   " ,  and the expression to test is ",
   Style["the case-(c) U-formula", Italic, 12, colTest], " ."}], 12,
 GrayLevel[0.35]], Style[Row[{"Hence, substituting the identities  ",
   Style["●", 13, RGBColor[0.55, 0.33, 0.10]], "  and  ", Style["●", 13,
    RGBColor[0.40, 0.15, 0.60]], "  into ", dhA, " ,  we get"}], 13],
```

```
      Style[Row[{tf[Subscript[h, 233] == 0], ",   ",
         tf[k Subscript[h, 231] == Subscript[e, 2]], ",   ",
         tf[Subscript[h, 232] == Sqrt[Abs[Subscript[z, 2] Subscript[z, 3]]]],
         ",   ", tf[Subsuperscript[h, 233, 2] - Subscript[ξ, 23] k^2 Subscript[
              x, 1] Subsuperscript[h, 231, 2] t^2 == Subscript[x, 1] t^2],
         "      ", Style["●", 14, RGBColor[0.00, 0.55, 0.65]]}], 13],
      Style[Row[{"Thus, by the identities  ", Style["●", 13,
          RGBColor[0.00, 0.55, 0.65]], " ,  the  ", tf[Subscript[e, 0]],
         "-term of ", Style["the given U-formula", Italic, 13, colGiven],
         " vanishes and it reduces to"}], 13], Style[dStarC, 14, colTest],
      Style[Row[{"which is ", Style["the tested case-(c) formula",
          Italic, 13, colTest], " ;  the given denominator reduces to  ",
         tf[Subscript[x, 1] t^2], " ,  which vanishes for the same  ", tf[t],
         "  as the tested denominator  ", tf[t], " ,  namely  ", tf[t == 0],
         " .  The identities  ", Style["●", 13, RGBColor[0.00, 0.55, 0.65]],
         "  and the equality of the ", Style["given", Italic, 13, colGiven],
         "  and  ", Style["tested", Italic, 13, colTest],
         "  formulas are verified separately on each sign region:"}],
       13], gr[regMix, cR, t2]}]]];

(* ============================================================*)
(*PART 6:CASES (b) AND (c) FOR u<0,ON 1+u (z2+z3+z2 z3)=0*)
(* ============================================================*)
ClearAll[pbCheck, pbPreCk, pcCheck];
pbCheck[{lab_, w_, sx_}, e2_] :=
  Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, hv, Hv, brh},
   uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
   kk2 = -uv / (1 - uv);
   mS = at["sv2"] at["sv3"] / at["sS"];
   kpAt = 1 / mS;
   kkAt = 1 / (at["sv2"] at["sv3"]);
   suz = -at["s2"];
   W = at["sw2"] at["sw3"];
   {d2, d3} = gdict[at, 2][[3 ;; 4]];
   hv = p7h[at, kkAt, 1 / (1 - uv), e2, 2]; Hv = HAat[at, uv, e2, -e2];
   brh = {-e2 mS H1s / (W H3s), -H2s / (W H3s), 1 / H3s};
   {bOK[at, uv] && p7ok[at, kkAt, kk2, 2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
     chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] &&
     chkR[hv[[3]] Hv[[3]], 1] && chkR[hv[[1]] W Hv[[3]], -e2 mS Hv[[1]]] &&
     chkR[hv[[2]] W Hv[[3]], -Hv[[2]]], chkR[XX H2s H3s tt + ee0 e2 H1s RR,
      -W H3s^2 (XX brh[[2]] tt + ee0 kpAt brh[[1]] brh[[3]] RR)] &&
     chkR[at["z2"] at["z3"] + uv sx XX H1s^2 tt^2,
      at["z2"] at["z3"] H3s^2 (brh[[3]]^2 - kk2 sx XX brh[[1]]^2 tt^2)],
    chkR[uLemNegat[brh, kk2, kpAt, 1, sx, suz],
     uThmABat[{H1s, H2s, H3s}, uv, at, sx, e2]]}];
pbPreCk[{lab_, w_, sx_}] := Module[{at = mkAP[w], W}, W = at["sw2"] at["sw3"];
   chkR[at["z2"] at["z3"] / (W sX), W / sX]];
```

```
pcCheck[{lab_, w_, sx_}, e2_] :=
  Module[{at = mkAP[w], uv, kk2, mS, kpAt, kkAt, suz, W, d2, d3, Hv, H1v, H2v},
   uv = -1 / (at["z2"] + at["z3"] + at["z2"] at["z3"]);
   kk2 = -uv / (1 - uv);
   mS = at["sv2"] at["sv3"] / at["sS"];
   kpAt = 1 / mS;
   kkAt = 1 / (at["sv2"] at["sv3"]);
   suz = -at["s2"];
   W = at["sw2"] at["sw3"];
   {d2, d3} = gdict[at, 2][[3 ;; 4]];
   Hv = HAat[at, uv, e2, e2];
   H1v = -W at["sS"];
   H2v = -at["sw2"] at["sv2"] at["sw3"] at["sv3"];
   {bOK[at, uv] && chkR[kkAt^2, kk2] && chkR[kk2, (d2 - 1) (d3 - 1)] &&
     chkR[mS^2, 1 - uv] && chkR[kpAt, kkAt at["sS"]] &&
     chkR[Hv[[3]], 0] && chkR[Hv[[1]], H1v] && chkR[Hv[[2]], H2v],
    chkR[ee0 e2 kpAt RR, -(kkAt / W) ee0 e2 H1v RR] && chkR[kkAt (1 - sx XX tt^2),
      (kkAt / (at["z2"] at["z3"])) (at["z2"] at["z3"] + uv sx XX H1v^2 tt^2)],
    chkR[uThmABat[{H1v, H2v, 0}, uv, at, sx, e2],
     uLemNegCat[kkAt, kpAt, sx, suz, e2]]}];
Module[{gridB, br, id, fu, pre, gridC, cB, cI, cR, xa, ok},
  gridB = Table[pbCheck[w, sgnE2], {w, regBpos}, {sgnE2, {1, -1}}];
  {br, id, fu} = Table[And @@ gridB[[iw, All, j]], {j, 3}, {iw, Length[regBpos]}];
  pre = And @@ (pbPreCk /@ regBpos);
  gridC = Table[pcCheck[w, sgnE2], {w, regBpos}, {sgnE2, {1, -1}}];
  {cB, cI, cR} = Table[And @@ gridC[[iw, All, j]], {j, 3}, {iw, Length[regBpos]}];
  xa = And @@ (xiZX1[{#[[1]], #[[2]], #[[2]], #[[3]]}] & /@ regBpos);
  ok = (And @@ Join[br, id, fu, cB, cI, cR]) && pre && xa && gm1NegOK && sgn1zOK &&
    k2gNegOK && gFormNegOK && locPOK && kLocPOK && kpLocPOK && sgnUzNegOK;
  AppendTo[resultsAll, ok];
  Print[pan[cOrange, {ttl[cOrange, "cases (b) and (c)", tf[u < 0], ok],
     Style[Row[{"Here  ", tf[Subscript[ξ, 23] == 1], ",  because  ",
        tf[k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) > 0],
        "  and  ", tf[Subsuperscript[g, i, 2] - 1 == -(1 / (1 + Subscript[z, i]))],
        "  ", smallChip[gm1NegOK], ",  so  ",
        tf[(1 + Subscript[z, 2]) (1 + Subscript[z, 3]) > 0], " ,  and, by  ",
        tf[Sign[1 + Subscript[z, i]] == Sign[Subscript[z, i]]], "  ",
        smallChip[sgn1zOK], " ,  the signs of  ", tf[Subscript[z, 2]],
        "  and  ", tf[Subscript[z, 3]], "  coincide;  hence  ", dZ1, ",   ",
        dX1, ",   ", d11, "   ", smallChip[xa]}], 12, GrayLevel[0.35]],
     Style[Row[{"Moreover,  ", tf[u Subscript[z, 2] Subscript[z, 3] ≠ 1],
        " ,  because the equalities  ",
        tf[u Subscript[z, 2] Subscript[z, 3] == 1], "  and  ", tf[1 + u (Subscript[
              z, 2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]) == 0],
        "  cannot hold simultaneously;  hence the expressions to test are ",
        dHA, " .  Also  ", tf[Subscript[h, 233] ≠ 0],
        "  (as stated in the hypotheses),  and, in case (b), the identity  ",
```

```
   tf[Subscript[h, 233] Subscript[H, 3] == 1], "  gives  ",
   tf[Subscript[H, 3] ≠ 0], " ,  so the divisions by  ",
   tf[Subscript[H, 3]], "  are legitimate there."}], 12, GrayLevel[0.35]],
"", Style["Verification:", Bold, 13], Style[Row[{"In these cases  ",
   tf[k^2 == (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1)],
   " ,  that is,  ", tf[1 + u (Subscript[z, 2] + Subscript[z, 3] +
         Subscript[z, 2] Subscript[z, 3]) == 0], " ,  by the identity  ",
   tf[k^2 - (Subsuperscript[g, 2, 2] - 1) (Subsuperscript[g, 3, 2] - 1) ==
     -((1 + u (Subscript[z, 2] + Subscript[z, 3] +
              Subscript[z, 2] Subscript[z, 3])) /
         ((1 - u) (1 + Subscript[z, 2]) (1 + Subscript[z, 3]))))], "  ",
   smallChip[k2gNegOK], " ,  and the following identities hold:"}], 13],
Style[Row[{tf[Subscript[g, i] == Sqrt[Subscript[z, i] (1 + Subscript[z, i])] /
      Abs[1 + Subscript[z, i]]], "  ", smallChip[gFormNegOK], ",   ", tf[
    1 + u Subscript[z, 2] == Subscript[z, 3] (1 + Subscript[z, 2]) / (Subscript[
          z, 2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])],
   ",   ", tf[1 + u Subscript[z, 3] == Subscript[z, 2]
      (1 + Subscript[z, 3]) / (Subscript[z, 2] + Subscript[z, 3] +
         Subscript[z, 2] Subscript[z, 3])], "  ", smallChip[locPOK],
   ",   ", tf[k == 1 / Sqrt[(1 + Subscript[z, 2]) (1 + Subscript[z, 3])]],
   "  ", smallChip[kLocPOK], ",   ",
   tf[k' == k Sqrt[Subscript[z, 2] + Subscript[z, 3] +
        Subscript[z, 2] Subscript[z, 3]]], "  ", smallChip[kpLocPOK],
   "      ", Style["■", 14, RGBColor[0.55, 0.33, 0.10]]}], 13], Style[
 Row[{"Hence, in case (b) ( ", tf[Subscript[e, 2] == -Subscript[e, 3]],
   " ),  substituting the identities  ",
   Style["■", 13, RGBColor[0.55, 0.33, 0.10]], "  and the equality  ",
   dZ1, "  into ", dhB, " ,  we get, with ", dHA, " ,"}], 13],
Style[Row[{tf[Subscript[h, 233] Subscript[H, 3] == 1], ",   ",
   tf[Subscript[h, 231] == -Subscript[e, 2] Sqrt[1 - u] Subscript[H, 1] /
       (Sqrt[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])],
   ",   ", tf[Subscript[h, 232] == -Subscript[H, 2] /
      (Sqrt[Subscript[z, 2] Subscript[z, 3]] Subscript[H, 3])],
   "      ", Style["■", 14, RGBColor[0.00, 0.45, 0.25]]}],
 13], rWhich, gr[regBpos, br, t2], "",
rThus[Style["■", 13, RGBColor[0.00, 0.45, 0.25]]],
Style["the numerator identity:", 13],
Style[tf[Abs[Subscript[x, 1]] Subscript[H, 2] Subscript[H, 3] t +
    Subscript[e, 0] Subscript[e, 2] Subscript[H, 1]
     Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] ==
   -Sqrt[Subscript[z, 2] Subscript[z, 3]] Subsuperscript[H, 3, 2]
    (Abs[Subscript[x, 1]] Subscript[h, 232] t +
      Subscript[e, 0] k' Subscript[h, 231] Subscript[h, 233]
       Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]])], 14],
Style["the denominator identity:", 13], Style[
 tf[Subscript[z, 2] Subscript[z, 3] +
    u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2 == Subscript[z, 2]
```

```
    Subscript[z, 3] Subsuperscript[H, 3, 2] (Subsuperscript[h, 233, 2] -
      k^2 Subscript[x, 1] Subsuperscript[h, 231, 2] t^2)], 14],
rSymH, gr[regBpos, id, t2], rDen[dDthm, dDneg1,
 Row[{tf[Subscript[z, 2] Subscript[z, 3] ≠ 0],
   "  and  ", tf[Subscript[H, 3] ≠ 0]}], And @@ id],
"", rPre[dPre1, Row[{dZ1, "  and  ", dX1}], 4, pre], "",
rTog[tf[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
 sgnUzNegOK, "case (a)", eH3],
Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
rStar, gr[regBpos, fu, t2], "",
Style[Row[{"In case (c) ( ", tf[Subscript[e, 2] == Subscript[e, 3]],
   " ),  the expressions to test remain ", dHA,
   " ,  and the given expression is ", Style["the case-(c) U-formula",
    Italic, 12, colGiven], " :"}], 12, GrayLevel[0.35]],
Style[tf[U == HoldForm[Subscript[ε, 2] Subscript[ε, 3]
      Sign[u Subscript[z, 2]] / Sqrt[Abs[Subscript[x, 1] Subscript[x, 3]]]]
    HoldForm[Subscript[e, 0] Subscript[e, 2] k' Sqrt[Subscript[x, 1]
        𝒟[Subscript[x, 1], t]] / (k (1 - Subscript[x, 1] t^2))]],
 14, colGiven], Style[Row[{"Substituting the identities  ",
   Style["■", 13, RGBColor[0.55, 0.33, 0.10]],
   "  into ", dHA, " ,  we get"}], 13],
Style[Row[{tf[Subscript[H, 3] == 0], ",   ",
   tf[Subscript[H, 1] == -Sqrt[Subscript[z, 2] Subscript[z, 3] (Subscript[z,
         2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3])]], ",   ",
   tf[Subscript[H, 2] == -Sqrt[Subscript[z, 2] Subscript[z, 3]
        (1 + Subscript[z, 2]) (1 + Subscript[z, 3])]],
   "      ", Style["■", 14, RGBColor[0.40, 0.15, 0.60]]}],
 13], rWhich, gr[regBpos, cB, t2], "",
Style[Row[{"Thus, substituting the identities  ", Style["■", 13,
    RGBColor[0.40, 0.15, 0.60]], "  and the equality  ", tf[k' == k Sqrt[
       Subscript[z, 2] + Subscript[z, 3] + Subscript[z, 2] Subscript[z, 3]]],
   "  into the numerator and the denominator of ", Style[
    "the given case-(c) U-formula", Italic, 13, colGiven], ", we get"}],
 13], Style["the numerator identity:", 13],
Style[tf[Subscript[e, 0] Subscript[e, 2] k'
    Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]] ==
   -(k / Sqrt[Subscript[z, 2] Subscript[z, 3]]) (Abs[Subscript[x, 1]]
       Subscript[H, 2] Subscript[H, 3] t + Subscript[e, 0] Subscript[e, 2]
       Subscript[H, 1] Sqrt[Subscript[x, 1] 𝒟[Subscript[x, 1], t]])],
 14], Style["the denominator identity:", 13], Style[
 tf[k (1 - Subscript[x, 1] t^2) ==
   (k / (Subscript[z, 2] Subscript[z, 3])) (Subscript[z, 2] Subscript[z, 3] +
      u Subscript[x, 1] Subsuperscript[H, 1, 2] t^2)], 14], rSymH,
gr[regBpos, cI, t2], rDen[dDthm, tf[k (1 - Subscript[x, 1] t^2)],
 Row[{tf[k ≠ 0], "  and  ", tf[Subscript[z, 2] Subscript[z, 3] ≠ 0]}],
 And @@ cI], "",
rTog[tf[Sign[u Subscript[z, 2]] == -Sign[Subscript[z, 2]]],
```

```
      sgnUzNegOK, "case (a)", ""],
    Style[Row[{dStar, "      ", Style["⋆", 14]}], 14, colTest],
    Style[Row[{"The identity  ", Style["⋆", 13, colTest],
       "  is also verified separately on each sign region, with  ",
       tf[Subscript[e, 0]], ", ", tf[Subscript[ε, 2]], ", ",
       tf[Subscript[ε, 3]], "  symbolic, that is, for all their
         real values at once, which covers all their
         sign choices:"}], 13], gr[regBpos, cR, t2]}]]];

(* ============================================================*)
(*PART 7:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 10: ALL CLAIMS VERIFIED",
     "Section 10: SOME CLAIMS FAILED"], Bold, White, 16], Background →
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  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses:

$u \neq 0, \quad u < 1, \quad k, k' \in (0, 1)$

For $i = 1, 2, 3, 4$ :

$x_i, y_i, z_i \in \mathbb{R}, \quad x_i, y_i, z_i \neq -1, 0$

$\xi_{23} = \operatorname{sgn}(z_2 z_3) = \operatorname{sgn}(x_1 x_3), \quad \operatorname{sgn}(y_2) = \operatorname{sgn}(u x_1 z_2)$

$u x_i y_i z_i > 0, \quad z_i (1 + z_i) > 0, \quad 1 + u z_i > 0$

$e_0, e_i, \varepsilon_i, \xi_{23} \in \{\pm 1\}$

$\mathcal{D}(x_1, t) = -(1 - x_1 t^2)(1 - u x_1 t^2), \quad R = \sqrt{x_1 \mathcal{D}(x_1, t)}, \quad x_1 \mathcal{D}(x_1, t) \geq 0$

if $u > 0$ : $k = \sqrt{u}, \quad k' = \sqrt{1 - u}, \quad g_i = \sqrt{\frac{1 + z_i}{z_i}}$

if $u < 0$ : $k' = \frac{1}{\sqrt{1 - u}}, \quad k = k' \sqrt{-u}, \quad g_i = \sqrt{\frac{z_i}{1 + z_i}}$

$h_{231}, h_{232}, h_{233}, H_1, H_2, H_3 \in \mathbb{R}$

$h_{231} = 0$ if and only if $g_2 = g_3$ and $e_2 = -e_3$
; $h_{232} = 0$ if and only if $g_2 g_3 = k'$ and $e_2 = -e_3$ ;

$h_{233} = 0$ if and only if $k^2 (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$
and $e_2 = e_3$ ; in particular, $h_{233} \neq 0$ if $k^2 = (g_2^2 - 1)(g_3^2 - 1)$ .

the equalities $u z_2 z_3 = 1$ and
$1 + u (z_2 + z_3 + z_2 z_3) = 0$ cannot hold simultaneously .

Given formulas and expressions:

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_2|}} \left( \begin{cases} \frac{|x_1| h_{232} h_{233} t + e_0 h_{231} \sqrt{x_1 \mathcal{D}(x_1, t)}}{1 + \xi_{23} k^2 x_1 h_{231}^2 t^2} & u > 0 \\ |x_1| h_{232} t + e_0 k' h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)} & \end{cases} \right.$$

$$\sqrt{|x_1 x_3|}\left(\left\{\frac{|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1,t)}}{h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2} \quad u<0\right\}\right)$$

$$\left.\begin{aligned} h_{231} &= \frac{e_2 g_3 \sqrt{(g_2^2-1)(g_3^2-k'^2)\xi_{23}} + e_3 g_2 \sqrt{(g_3^2-1)(g_2^2-k'^2)\xi_{23}}}{k^2-(g_2^2-1)(g_3^2-1)} \\ h_{232} &= \frac{\sqrt{(g_2^2-k'^2)(g_3^2-k'^2)\xi_{23}} + e_2 e_3 g_2 g_3 \sqrt{(g_2^2-1)(g_3^2-1)\xi_{23}}}{k^2-(g_2^2-1)(g_3^2-1)} \\ h_{233} &= \frac{k^2 g_2 g_3 + e_2 e_3 \xi_{23} \sqrt{(g_2^2-1)(g_2^2-k'^2)(g_3^2-1)(g_3^2-k'^2)}}{k^2-(g_2^2-1)(g_3^2-1)} \end{aligned}\right\} \quad \text{(a)}\ k^2 \neq (g_2^2-1)(g_3^2-1)$$

$$\left.h_{231} = \frac{e_2 (g_2^2-g_3^2)}{2 k g_2 g_3},\quad h_{232} = \frac{g_2^2 g_3^2 - k'^2}{2 k g_2 g_3},\quad h_{233} = \frac{g_2^2+g_3^2}{2 g_2 g_3}\right\} \quad \text{(b)}\ k^2 = (g_2^2-1)(g_3^2-1),\ e_2 = -e_3$$

$$U = \frac{\varepsilon_2 \varepsilon_3 \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}}\left(\left\{\begin{array}{ll} \frac{\operatorname{sgn}(x_1)}{k t} & u>0 \\ \frac{e_0 e_2 k' \sqrt{x_1 \mathcal{D}(x_1,t)}}{k(1-x_1 t^2)} & u<0 \end{array}\right.\right) \quad \text{(c)}\ k^2 = (g_2^2-1)(g_3^2-1),\ e_2 = e_3$$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed.

We want to test the following claims ★ :

(1) in each case, the tested formula for $U$ below coincides with the applicable given formula for $U$;

(2) in each case, the $t$ −dependent denominators of these two formulas vanish for the same values of $t$.

$$U = \frac{\left(\varepsilon_2 \varepsilon_3 \operatorname{sgn}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right)\left(|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1,t)}\right)}{z_2 z_3 + u x_1 H_1^2 t^2}$$

$$\left.\begin{aligned} H_1 &= \frac{|z_2| \sqrt{z_3(1+z_3)(1+u z_3)} + e_2 e_3 |z_3| \sqrt{z_2(1+z_2)(1+u z_2)}}{u z_2 z_3 - 1} \\ H_2 &= \frac{|z_2 z_3| \sqrt{(1+u z_2)(1+u z_3)} + e_2 e_3 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}}{u z_2 z_3 - 1} \\ H_3 &= \frac{u \sqrt{z_2 z_3 (1+z_2)(1+z_3)} \operatorname{sgn}(z_2 z_3) + e_2 e_3 \sqrt{(1+u z_2)(1+u z_3)}}{u z_2 z_3 - 1} \end{aligned}\right\} \quad \text{(a)}\ u z_2 z_3 \neq 1$$

$$\left.\begin{aligned} H_1 &= \frac{z_2 z_3 (z_3 - z_2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_2 &= \frac{z_2 z_3 (z_2+z_3+2)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \\ H_3 &= \frac{(z_2+z_3+2 z_2 z_3) \operatorname{sgn}(z_2 z_3)}{2 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}} \end{aligned}\right\} \quad \text{(b)}\ u z_2 z_3 = 1,\ e_2 = -e_3$$

$$\left.U = \left(\varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right) \frac{\operatorname{sgn}(y_2)}{t}\right\} \quad \text{(c)}\ u z_2 z_3 = 1,\ e_2 = e_3$$

## The derivation: the formula for $U$ in case (a) for $u > 0$ Verified

By $\xi_{23} = \text{sgn}(z_2\, z_3) = \text{sgn}(x_1\, x_3)$ : $\xi_{23}\, z_2\, z_3 = |z_2\, z_3|$, $\xi_{23}\, x_1\, x_3 = |x_1\, x_3|$ ☑

Verification:

By the given $k$ and $g_i$ : $k^2 - (g_2^2 - 1)\,(g_3^2 - 1) = \frac{u\, z_2\, z_3 - 1}{z_2\, z_3}$ ☑

In this case $k^2 \neq (g_2^2 - 1)\,(g_3^2 - 1)$, that is, $u\, z_2\, z_3 \neq 1$, and the following identities hold:

$g_i^2 - 1 = \frac{1}{z_i}$ ☑, $g_i^2 - k'^2 = \frac{1 + u\, z_i}{z_i}$ ☑, $g_i = \frac{\sqrt{z_i\,(1+z_i)}}{|z_i|}$ ☑ ◆

Hence, substituting the identities ◆ and the equality $\xi_{23}\, z_2\, z_3 = |z_2\, z_3|$
into the case −(a) $h_{231}, h_{232}, h_{233}$, we get, with the case −(a) $H_1, H_2, H_3$,

$h_{231} = \frac{\xi_{23}\, e_2\, H_1}{\sqrt{|z_2\, z_3|}}$, $h_{232} = \frac{\xi_{23}\, H_2}{\sqrt{|z_2\, z_3|}}$, $h_{233} = H_3$ ◆

which are verified separately on each sign region:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |

Thus, substituting the identities ◆
into the numerator and the denominator of *the given U −formula*, we get
the numerator identity:

$$|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)} = \xi_{23}\, \sqrt{|z_2\, z_3|}\, \left(|x_1|\, h_{232}\, h_{233}\, t + e_0\, h_{231}\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)$$

the denominator identity:

$$z_2\, z_3 + u\, x_1\, H_1^2\, t^2 = z_2\, z_3\, (1 + \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2)$$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |

Hence, by the denominator identity, the $t$ −dependent denominators $z_2\, z_3 + u\, x_1\, H_1^2\, t^2$
and $1 + \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2$ vanish for the same $t$, because $z_2\, z_3 \neq 0$ ☑

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1\, x_3|}}\, \frac{z_2\, z_3}{\xi_{23}\, \sqrt{|z_2\, z_3|}} = \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}$$

$\sqrt{|x_1 x_3|}\ \xi_{23}\ \sqrt{|z_2 z_3|}$ $\sqrt{x_1 x_3}$

which holds by $\xi_{23}\, z_2\, z_3 = |z_2\, z_3|$ and $\xi_{23}\, x_1\, x_3 = |x_1\, x_3|$ , and is verified separately on each of the 8 sign regions ☑

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u\, z_2) = \text{sgn}(z_2)$ ☑ , give the formula ★ in case (a):

$$U = \frac{\left(\varepsilon_2\ \varepsilon_3\ \text{sgn}(z_2)\ \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right)\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\ \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2\, z_3 + u\ x_1\ H_1^2\ t^2} \quad ★$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |

## The derivation: the formula for $U$ in cases (b) and (c) for $u > 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = \frac{1}{z_i}$ ☑, so $z_2\, z_3 > 0$ ; hence $z_2\, z_3 = |z_2\, z_3|$, $x_1\, x_3 = |x_1\, x_3|$ ☑

Verification:

In these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is, $u\, z_2\, z_3 = 1$, by the identity $k^2 - (g_2^2 - 1)(g_3^2 - 1) = \frac{u\, z_2\, z_3 - 1}{z_2\, z_3}$ ☑, and the following identities hold:

$k = \frac{1}{\sqrt{z_2\, z_3}}$ ☑, $g_i = \frac{\sqrt{z_i\,(1 + z_i)}}{|z_i|}$ ☑ ▲

Hence, in case (b) ( $e_2 = -e_3$ ), substituting the identities ▲ and the equality $z_2\, z_3 = |z_2\, z_3|$ into the case −(b) $h_{231}, h_{232}, h_{233}$ , we get, with the case −(b) $H_1, H_2, H_3$ ,

$h_{231} = \frac{e_2\, H_1}{\sqrt{z_2\, z_3}}$, $h_{232} = \frac{H_2}{\sqrt{z_2\, z_3}}$, $h_{233} = H_3$ ▲

which are verified separately on each sign region:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for $e_2 = \pm 1$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for $e_2 = \pm 1$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 > 0$ | for $e_2 = \pm 1$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 < 0$ | for $e_2 = \pm 1$ | ☑ |

Thus, substituting the identities ▲ into the numerator and the denominator of *the given U −formula* , we get the numerator identity:

$$|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)} = \sqrt{z_2\, z_3}\, \left(|x_1|\, h_{232}\, h_{233}\, t + e_0\, h_{231}\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)$$

the denominator identity :

$z_2\, z_3 + u\, x_1\, H_1^2\, t^2 = z_2\, z_3\, (1 + k^2\, x_1\, h_{231}^2\, t^2)$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Hence, by the denominator identity, the $t$ −dependent denominators $z_2\, z_3 + u\, x_1\, H_1^2\, t^2$ and $1 + k^2\, x_1\, h_{231}^2\, t^2$ vanish for the same $t$, because $z_2\, z_3 \neq 0$ ☑

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1\, x_3|}}\ \frac{z_2\, z_3}{\sqrt{z_2\, z_3}} = \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}$$

which holds by $z_2\, z_3 = |z_2\, z_3|$ and $x_1\, x_3 = |x_1\, x_3|$, and is verified separately on each of the 4 sign regions ☑

Together, the numerator, denominator, and prefactor identities, along with $\operatorname{sgn}(u\, z_2) = \operatorname{sgn}(z_2)$ ☑, give the formula ★ in case (b):

$$U = \frac{\left(\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(z_2)\ \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}\right)\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2\, z_3 + u\, x_1\, H_1^2\, t^2} \qquad ★$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

In case (c) ( $e_2 = e_3$ ), the *given* and *tested* formulas are

$$U = \frac{\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(u\, z_2)}{\sqrt{|x_1\, x_3|}}\ \frac{\operatorname{sgn}(x_1)}{k\, t} \quad \text{and} \quad U = \left(\varepsilon_2\ \varepsilon_3\ \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}\right)\frac{\operatorname{sgn}(y_2)}{t},$$

which coincide because $\operatorname{sgn}(y_2) = \operatorname{sgn}(u\, x_1\, z_2)$, $k = \frac{1}{\sqrt{z_2\, z_3}}$, and the prefactor identity; their denominators $k\, t$ and $t$ vanish for the same $t$, because $k \neq 0$. The equality is verified separately on each sign region, with $\varepsilon_2$, $\varepsilon_3$ symbolic:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ ☑

## The derivation: the formula for $U$ in case (a) for $u < 0$ Verified

Here $h_{233} = 0$ if and only if $k^2\, (g_2^2 + g_3^2) = k^2 - (g_2^2 - 1)(g_3^2 - 1)$ and $e_2 = e_3$. When $u < 0$,

$k^2\, (g_2^2 + g_3^2) - (k^2 - (g_2^2 - 1)(g_3^2 - 1)) = \frac{1 - u\, z_2\, z_3}{(1-u)\,(1+z_2)\,(1+z_3)}$

☑, verified separately on each of the 4 sign regions of $z_2, z_3$.

Thus, when $u < 0$, $h_{233} = 0$ if and only if $u z_2 z_3 = 1$ and $e_2 = e_3$. Therefore, in cases (i) and (ii) below, $h_{233} \neq 0$, and the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by $H_3$ are legitimate there.

Verification:

By the given $k$ and $g_i$: $k^2 - (g_2^2 - 1)(g_3^2 - 1) = -\frac{1+u(z_2+z_3+z_2 z_3)}{(1-u)(1+z_2)(1+z_3)}$ ☑

In this case $k^2 \neq (g_2^2 - 1)(g_3^2 - 1)$, that is, $1 + u(z_2 + z_3 + z_2 z_3) \neq 0$; we consider three possible cases: (i) $u z_2 z_3 \neq 1$, (ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$, (iii) $u z_2 z_3 = 1$ and $e_2 = e_3$.

**(i) $u z_2 z_3 \neq 1$:**

The following identities hold:

$k = \sqrt{\frac{u}{u-1}}$ ☑, $g_i^2 - 1 = -\frac{1}{1+z_i}$ ☑, $g_i^2 - k'^2 = -\frac{1+u z_i}{(1-u)(1+z_i)}$ ☑, $g_i = \frac{\sqrt{z_i(1+z_i)}}{|1+z_i|}$ ☑ ●

$u^2 z_2 z_3 (1+z_2)(1+z_3) - (1+u z_2)(1+u z_3) = (u z_2 z_3 - 1)(1 + u(z_2 + z_3 + z_2 z_3))$ ☑ ●

and, by $\xi_{23} = \text{sgn}(z_2 z_3) = \text{sgn}(x_1 x_3)$ and $\text{sgn}(1+z_i) = \text{sgn}(z_i)$ ☑:

$\xi_{23} z_2 z_3 = |z_2 z_3|$, $\xi_{23} x_1 x_3 = |x_1 x_3|$, $\xi_{23}(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ ☑

Hence, substituting the identities ● and the equalities $\xi_{23} z_2 z_3 = |z_2 z_3|$ and $\xi_{23}(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ into the case −(a) $h_{231}, h_{232}, h_{233}$, and using the identity ●, we get, with the case −(a) $H_1, H_2, H_3$,

$h_{233} H_3 = 1$, $h_{231} = -\frac{\xi_{23} e_2 \sqrt{1-u}\, H_1}{\sqrt{|z_2 z_3|}\, H_3}$, $h_{232} = -\frac{\xi_{23} H_2}{\sqrt{|z_2 z_3|}\, H_3}$ ●

which are verified separately on each sign region:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 < -1,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |

Thus, substituting the identities ● into the numerator and the denominator of *the given U −formula*, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\xi_{23} \sqrt{|z_2 z_3|}\, H_3^2 \left(|x_1| h_{232} t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)}\right)$$

the denominator identity:

$$z_2 z_3 + u x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - \xi_{23} k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

| | | |
|---|---|---|
| $z_2 > 0,\ z_3 > 0,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 > 0,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 > 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |
| $z_2 > 0,\ z_3 < -1,\ x_1 < 0$ | for all 4 sign combinations of $e_2, e_3$ | ☑ |

$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑

Hence, by the denominator identity, the $t$ −dependent denominators $z_2 z_3 + u\, x_1\, H_1^2\, t^2$ and $h_{233}^2 - \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2$ vanish for the same $t$, because $z_2 z_3 \neq 0$ and $H_3 \neq 0$ ☑

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}}\ \frac{z_2 z_3}{\xi_{23}\ \sqrt{|z_2 z_3|}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $\xi_{23}\, z_2 z_3 = |z_2 z_3|$ and $\xi_{23}\, x_1 x_3 = |x_1 x_3|$, and is verified separately on each of the 8 sign regions ☑

Together, the numerator, denominator, and prefactor identities, along with $\operatorname{sgn}(u z_2) = -\operatorname{sgn}(z_2)$ ☑ and the cancellation of the factor $H_3^{\frac{2}{3}}$ ($H_3 \neq 0$, as noted above), give the formula ★ in case (a):

$$U = \frac{\left(\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(z_2)\ \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right)\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2 z_3 + u\, x_1\, H_1^2\, t^2} \qquad ★$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for all 4 sign combinations of $e_2, e_3$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for all 4 sign combinations of $e_2, e_3$ ☑

### (ii) $u z_2 z_3 = 1$ and $e_2 = -e_3$:

Here $\xi_{23} = -1$, because $u z_2 z_3 = 1$ and $u < 0$, so $z_2 z_3 < 0$; the equalities $\xi_{23}\, z_2 z_3 = |z_2 z_3|$, $\xi_{23}\, x_1 x_3 = |x_1 x_3|$, $\xi_{23}\,(1 + z_2)(1 + z_3) = |(1 + z_2)(1 + z_3)|$ hold on these regions ☑; the given expressions remain the case −(a) $h_{231}, h_{232}, h_{233}$, and the expressions to test are the case −(b) $H_1, H_2, H_3$.

$1 + u z_2 = \frac{1 + z_3}{z_3}$, $1 + u z_3 = \frac{1 + z_2}{z_2}$ ☑ ●

Hence, substituting the identities ● and ● and the equalities $\xi_{23}\, z_2 z_3 = |z_2 z_3|$ and $\xi_{23}\,(1 + z_2)(1 + z_3) = |(1 + z_2)(1 + z_3)|$ into the case −(a) $h_{231}, h_{232}, h_{233}$, we get, with the case −(b) $H_1, H_2, H_3$,

$$h_{233}\, H_3 = 1,\quad h_{231} = -\frac{\xi_{23}\, e_2\, \sqrt{1 - u}\, H_1}{\sqrt{|z_2 z_3|}\, H_3},\quad h_{232} = -\frac{\xi_{23}\, H_2}{\sqrt{|z_2 z_3|}\, H_3} \qquad ●$$

which are verified separately on each sign region:

$z_2 > 0,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Thus, substituting the identities ● into the numerator and the denominator of *the given U −formula*, we get the numerator identity:

$$|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)} =$$
$$-\xi_{23}\, \sqrt{|z_2\, z_3|}\, H_3^2\, \left(|x_1|\, h_{232}\, t + e_0\, k'\, h_{231}\, h_{233}\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)$$

the denominator identity:

$$z_2\, z_3 + u\, x_1\, H_1^2\, t^2 = z_2\, z_3\, H_3^2\, (h_{233}^2 - \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2)$$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

$z_2 > 0,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Hence, by the denominator identity, the $t$ −dependent denominators $z_2 z_3 + u\, x_1\, H_1^2\, t^2$ and $h_{233}^2 - \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2$ vanish for the same $t$, because $z_2 z_3 \neq 0$ and $H_3 \neq 0$ ☑

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1\, x_3|}}\, \frac{z_2\, z_3}{\xi_{23}\, \sqrt{|z_2\, z_3|}} = \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}$$

which holds by $\xi_{23}\, z_2\, z_3 = |z_2\, z_3|$ and $\xi_{23}\, x_1\, x_3 = |x_1\, x_3|$, and is verified separately on each of the 4 sign regions ☑

Together, the numerator, denominator, and prefactor identities, along with $\text{sgn}(u\, z_2) = -\text{sgn}(z_2)$ ☑ and the cancellation of the factor $H_3^{\frac{2}{3}}$ ($H_3 \neq 0$, as noted above), give the formula ★ in case (b):

$$U = \frac{\left(\varepsilon_2\, \varepsilon_3\, \text{sgn}(z_2)\, \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}\right)\, \left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2\, z_3 + u\, x_1\, H_1^2\, t^2} \quad ★$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

**(iii)** $u\, z_2\, z_3 = 1$ and $e_2 = e_3$ :

As in case (ii), $\xi_{23} = -1$ ; the given expressions remain the case −(a) $h_{231}, h_{232}, h_{233}$, and the expression to test is *the case −(c) U−formula*.

Hence, substituting the identities ● and ● into the case −(a) $h_{231}, h_{232}, h_{233}$, we get

$$h_{233} = 0,\quad k\, h_{231} = e_2,\quad h_{232} = \sqrt{|z_2\, z_3|},\quad h_{233}^2 - \xi_{23}\, k^2\, x_1\, h_{231}^2\, t^2 = x_1\, t^2 \quad ●$$

Thus, by the identities ●, the $e_0$−term of *the given U −formula* vanishes and it reduces to

$$U = \left(\varepsilon_2\, \varepsilon_3\, \sqrt{\frac{z_2\, z_3}{x_1\, x_3}}\right)\, \frac{\text{sgn}(y_2)}{t}$$

which is *the tested case −(c) formula*; the given denominator reduces to $x_1\, t^2$, which vanishes for the same $t$ as the tested denominator $t$, namely $t = 0$. The identities ● and the equality of the *given*

$t$, namely $t = 0$. The identities ■ and the equality of the *given* and *tested* formulas are verified separately on each sign region:

$z_2 > 0,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

## The derivation: the formula for $U$ in cases (b) and (c) for $u < 0$

Verified

Here $\xi_{23} = 1$, because $k^2 = (g_2^2 - 1)(g_3^2 - 1) > 0$ and $g_i^2 - 1 = -\frac{1}{1+z_i}$ ☑, so $(1+z_2)(1+z_3) > 0$, and, by $\operatorname{sgn}(1+z_i) = \operatorname{sgn}(z_i)$ ☑, the signs of $z_2$ and $z_3$ coincide; hence $z_2 z_3 = |z_2 z_3|$, $x_1 x_3 = |x_1 x_3|$, $(1+z_2)(1+z_3) = |(1+z_2)(1+z_3)|$ ☑

Moreover, $u z_2 z_3 \neq 1$, because the equalities $u z_2 z_3 = 1$ and $1 + u(z_2 + z_3 + z_2 z_3) = 0$ cannot hold simultaneously; hence the expressions to test are the case $-$(a) $H_1, H_2, H_3$. Also $h_{233} \neq 0$ (as stated in the hypotheses), and, in case (b), the identity $h_{233} H_3 = 1$ gives $H_3 \neq 0$, so the divisions by $H_3$ are legitimate there.

Verification:

In these cases $k^2 = (g_2^2 - 1)(g_3^2 - 1)$, that is, $1 + u(z_2 + z_3 + z_2 z_3) = 0$, by the identity $k^2 - (g_2^2 - 1)(g_3^2 - 1) = -\frac{1 + u(z_2 + z_3 + z_2 z_3)}{(1-u)(1+z_2)(1+z_3)}$ ☑, and the following identities hold:

$g_i = \frac{\sqrt{z_i (1+z_i)}}{|1+z_i|}$ ☑, $1 + u z_2 = \frac{z_3 (1+z_2)}{z_2 + z_3 + z_2 z_3}$, $1 + u z_3 = \frac{z_2 (1+z_3)}{z_2 + z_3 + z_2 z_3}$ ☑, $k = \frac{1}{\sqrt{(1+z_2)(1+z_3)}}$ ☑, $k' = k\sqrt{z_2 + z_3 + z_2 z_3}$ ☑ ■

Hence, in case (b) ( $e_2 = -e_3$ ), substituting the identities ■ and the equality $z_2 z_3 = |z_2 z_3|$ into the case $-$(b) $h_{231}, h_{232}, h_{233}$, we get, with the case $-$(a) $H_1, H_2, H_3$,

$h_{233} H_3 = 1$, $h_{231} = -\frac{e_2 \sqrt{1-u}\, H_1}{\sqrt{z_2 z_3}\, H_3}$, $h_{232} = -\frac{H_2}{\sqrt{z_2 z_3}\, H_3}$ ■

which are verified separately on each sign region:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Thus, substituting the identities ■ into the numerator and the denominator of *the given $U$ −formula*, we get the numerator identity:

$$|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 \mathcal{D}(x_1, t)} = -\sqrt{z_2 z_3}\ H_3^2 \left(|x_1| h_{232}\, t + e_0 k' h_{231} h_{233} \sqrt{x_1 \mathcal{D}(x_1, t)}\right)$$

the denominator identity:

$$z_2 z_3 + u\, x_1 H_1^2 t^2 = z_2 z_3 H_3^2 (h_{233}^2 - k^2 x_1 h_{231}^2 t^2)$$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑
$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑
$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Hence, by the denominator identity, the $t$ −dependent denominators $z_2 z_3 + u\, x_1\, H_1^2\, t^2$ and $h_{233}^2 - k^2\, x_1\, h_{231}^2\, t^2$ vanish for the same $t$, because $z_2 z_3 \neq 0$ and $H_3 \neq 0$ ☑

and, finally, the prefactor identity:

$$\frac{1}{\sqrt{|x_1 x_3|}}\ \frac{z_2 z_3}{\sqrt{z_2 z_3}} = \sqrt{\frac{z_2 z_3}{x_1 x_3}}$$

which holds by $z_2 z_3 = |z_2 z_3|$ and $x_1 x_3 = |x_1 x_3|$, and is verified separately on each of the 4 sign regions ☑

Together, the numerator, denominator, and prefactor identities, along with $\operatorname{sgn}(u z_2) = -\operatorname{sgn}(z_2)$ ☑ and the cancellation of the factor $H_3^{\frac{2}{3}}$ ($H_3 \neq 0$, as noted above), give the formula ★ in case (a):

$$U = \frac{\left(\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(z_2)\ \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right)\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\ \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2 z_3 + u\, x_1\, H_1^2\, t^2} \qquad ★$$

The identity ★ is also verified separately on each sign region, with $H_1, H_2, H_3, e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

In case (c) ($e_2 = e_3$), the expressions to test remain the case −(a) $H_1, H_2, H_3$, and the given expression is *the case −(c) U−formula*:

$$U = \frac{\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(u z_2)}{\sqrt{|x_1 x_3|}}\ \frac{e_0\, e_2\, k'\ \sqrt{x_1\, \mathcal{D}(x_1, t)}}{k\left(1 - x_1\, t^2\right)}$$

Substituting the identities ■ into the case −(a) $H_1, H_2, H_3$, we get

$$H_3 = 0,\quad H_1 = -\sqrt{z_2 z_3\,(z_2 + z_3 + z_2 z_3)},\quad H_2 = -\sqrt{z_2 z_3\,(1 + z_2)(1 + z_3)} \qquad ■$$

which are verified separately on each sign region:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Thus, substituting the identities ■ and the equality $k' = k\ \sqrt{z_2 + z_3 + z_2 z_3}$ into the numerator and the denominator of *the given case −(c) U−formula*, we get the numerator identity:

$$e_0\, e_2\, k'\ \sqrt{x_1\, \mathcal{D}(x_1, t)} = -\frac{k\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\ \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{\sqrt{z_2 z_3}}$$

the denominator identity:

$$k\,(1 - x_1\, t^2) = \frac{k\,\left(z_2 z_3 + u\, x_1\, H_1^2\, t^2\right)}{z_2 z_3}$$

which are verified separately on each sign region, with $H_1, H_2, H_3$ symbolic:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

Hence, by the denominator identity, the $t$ −dependent denominators $z_2 z_3 + u\, x_1\, H_1^2\, t^2$ and $k\,(1 - x_1 t^2)$ vanish for the same $t$, because $k \neq 0$ and $z_2 z_3 \neq 0$ ☑

Together, the numerator, denominator, and prefactor identities, along with $\operatorname{sgn}(u z_2) = -\operatorname{sgn}(z_2)$ ☑, give the formula ★ in case (a):

$$U = \frac{\left(\varepsilon_2\ \varepsilon_3\ \operatorname{sgn}(z_2)\ \sqrt{\frac{z_2 z_3}{x_1 x_3}}\right)\left(|x_1|\, H_2\, H_3\, t + e_0\, e_2\, H_1\, \sqrt{x_1\, \mathcal{D}(x_1, t)}\right)}{z_2 z_3 + u\, x_1\, H_1^2\, t^2} \qquad ★$$

The identity ★ is also verified separately on each sign region, with $e_0, \varepsilon_2, \varepsilon_3$ symbolic, that is, for all their real values at once, which covers all their sign choices:

$z_2 > 0,\ z_3 > 0,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 > 0,\ z_3 > 0,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 > 0$ for $e_2 = \pm 1$ ☑

$z_2 < -1,\ z_3 < -1,\ x_1 < 0$ for $e_2 = \pm 1$ ☑

**Section 10: ALL CLAIMS VERIFIED**

# Section 11

In[•]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[statusChip, smallChip, givenChip, colTest, cols, bullet, ind, vlift,
  idxB, prmB, tagB, ptB, pt, edg, evec, nv, nvP, uv, wv, tld, gk, thI, thP, cfS,
  cotTag, cotSqTag, dTag, mulR, bx, frac, absR, caseDef, firstBase, vrow, nrow,
  prow, hrow, vgrid, vgap, note, iv, dispRule, trigNames, simpleBoxQ, trigBoxRule,
  disp, sgnBox, polyRow, ratForm, iSet, uPlain, capX, tSym, Apt, Bpt, Cpt, uB,
  uC, wB, wC, wBout, wCout, wCent, wCentLen, eNext, ePrev, nraw, nrawP, nlen,
  nlenP, ndir, nPshow, nPfirst, nPsec, nPname, nvecIdx, nextA, prevA, thNextIdx,
  thPrevIdx, nIdx, pIdx, xRp, yRp, qXr, qYr, e0Tag, sigTag, radTag, flxF, bLen,
  cLen, ll1, ll2, ll3, ll4, mm1, mm2, mm3, mm4, vv1, vv2, vv3, vv4, lens, tw, hh,
  weier, halfRule, zeroT, zeroF, angA, angB, angG, angD, thNext, thPrev, thCent,
  sgnV, betaForm, vclose, outBnum, outBden, outCnum, outCden, outB, outC, eqB,
  eqC, sinBform, sinCform, sgD, bAD, bBD, bGD, bDD, barRule, sig, barA, barB,
  barG, barD, ratA, ratB, ratG, ratD, prodM, prodR, prodS, prodF, quU, quX, quY,
  quZ, biqBar, biqC, qBarD, nameB, nameC, spec, figBlock, cqList, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
   Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
```

```
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
givenChip = Style["•", 12, GrayLevel[0.45]];
colTest = RGBColor[0.10, 0.35, 0.75];
cols = {RGBColor[0.10, 0.35, 0.75],
    RGBColor[0.50, 0.20, 0.65], RGBColor[0.00, 0.55, 0.55],
    RGBColor[0.20, 0.50, 0.20], RGBColor[0.75, 0.35, 0.05],
    RGBColor[0.70, 0.15, 0.40], RGBColor[0.30, 0.25, 0.70],
    RGBColor[0.45, 0.35, 0.15], RGBColor[0.35, 0.35, 0.35]};
bullet = Style["▪ ", 12, colTest]; ind = "    "; vlift = 0.16;
idxB[x_] :=
  If[IntegerQ[x], ToString[x], StyleBox[ToString[x], FontSlant → "Italic"]];
prmB[x_] := SuperscriptBox[idxB[x], "′"];
tagB[x_, y_] := If[IntegerQ[x] && IntegerQ[y], ToString[x] <> ToString[y],
    StyleBox[ToString[x] <> ToString[y], FontSlant → "Italic"]];
ptB[ltr_, i_] := SubscriptBox[StyleBox[ltr, FontSlant → "Italic"], idxB[i]];
pt[ltr_, i_] := DisplayForm[ptB[ltr, i]];
edg[s1_, i1_, s2_, i2_] := DisplayForm[RowBox[{ptB[s1, i1], ptB[s2, i2]}]];
evec[s1_, i1_, s2_, i2_] :=
  DisplayForm[OverscriptBox[RowBox[{ptB[s1, i1], ptB[s2, i2]}],
    AdjustmentBox["⟶", BoxBaselineShift → -vlift]]];
nv[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["n", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], idxB[x]]];
nvP[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["n", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], prmB[x]]];
uv[ltr_] := DisplayForm[OverscriptBox[StyleBox[ltr, FontSlant → "Italic"],
    AdjustmentBox["→", BoxBaselineShift → -vlift]]];
wv[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["w", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], idxB[x]]];
tld[ltr_] :=
  DisplayForm[OverscriptBox[StyleBox[ltr, FontSlant → "Italic"], "~"]];
gk[g_, i_] := DisplayForm[SubscriptBox[g, idxB[i]]];
thI[x_] := DisplayForm[SubscriptBox["ϴ", idxB[x]]];
thP[x_, y_] :=
  DisplayForm[SubscriptBox["ϴ", SuperscriptBox[tagB[x, y], "′"]]];
cfS[k_] :=
  DisplayForm[SubscriptBox[StyleBox["cf", FontSlant → "Italic"], ToString[k]]];
bx[DisplayForm[b_]] := b; bx[x_] := ToBoxes[x, TraditionalForm];
cotSqTag[t_] :=
  DisplayForm[RowBox[{SuperscriptBox["cot", "2"], "(", bx[t], "/", "2", ")"}]];
cotTag[t_] := DisplayForm[RowBox[{"cot", "(", bx[t], "/", "2", ")"}]];
dTag[argX_, argT_] := Row[{DisplayForm[StyleBox["D", FontSlant → "Italic"]],
    "(", argX, ", ", argT, ")"}];
mulR[args___] := Row[Riffle[{args}, ""]];
```

```
absR[e_] := Row[{"|", e, "|"}];
uPlain = DisplayForm[StyleBox["u", FontSlant → "Italic"]];
capX = DisplayForm[StyleBox["X", FontSlant → "Italic"]];
tSym = DisplayForm[StyleBox["t", FontSlant → "Italic"]];
frac[num_, den_] := DisplayForm[FractionBox[bx[num], bx[den]]];
caseDef[lhs_, rows_] :=
  DisplayForm[RowBox[{bx[lhs], " := ", StyleBox["{", SpanMaxSize → Infinity],
     GridBox[Map[bx, rows, {2}], ColumnAlignments → Left,
      ColumnSpacings → 1.6, RowSpacings → 0.9]}]];
note[ntxt_] := Style[ntxt, 12, GrayLevel[0.5]];
firstBase[e_] :=
  If[Head[e] === Column, Append[e, BaselinePosition → {1, 1}], e];
(*a numbered line,an unnumbered line (for a name),
and a heading line of a block*)
nrow[n_, stmt_, mark_, nt_] := {Style["(" <> ToString[n] <> ")", 13],
   Style[firstBase[stmt], 13], mark, note[nt]};
prow[stmt_, mark_, nt_] := {"", Style[firstBase[stmt], 13], mark, note[nt]};
hrow[t_] := {Style[t, 13], SpanFromLeft, SpanFromLeft, SpanFromLeft};
vrow[stmt_, mark_, nt_] :=
  {Spacer[10], Style[firstBase[stmt], 13], mark, note[nt]};
vgap = {"", Spacer[{0, 7}], "", ""};
vgrid[rows_] :=
  Grid[rows, Alignment → {Left, Baseline}, Spacings → {1.2, 0.8}];
(*every symbol that is only displayed is a string,so that no value
 given to a symbol elsewhere in the notebook can enter the display;
the displayed index i is an italic string as well*)
iv = Style["i", Italic];
dispRule = Join[Thread[{α1, α2, α3, α4} → Table[Subscript["α", k], {k, 4}]],
   Thread[{β1, β2, β3, β4} → Table[Subscript["β", k], {k, 4}]],
   Thread[{ɣ1, ɣ2, ɣ3, ɣ4} → Table[Subscript["ɣ", k], {k, 4}]],
   Thread[{δ1, δ2, δ3, δ4} → Table[Subscript["δ", k], {k, 4}]],
   Thread[{θ1, θ2, θ3, θ4} → Table[Subscript["θ", k], {k, 4}]],
   Thread[{ll1, ll2, ll3, ll4} → Table[Subscript[Style["ℓ", Italic], k], {k, 4}]],
   Thread[{mm1, mm2, mm3, mm4} → Table[Subscript[Style["m", Italic], k], {k, 4}]],
   Thread[{vv1, vv2, vv3, vv4} →
     Table[Subscript[Style["v", Italic], k], {k, 4}]]];
trigNames = {"sin", "cos", "tan", "cot", "sec", "csc"};
simpleBoxQ[b_String] := True;
simpleBoxQ[StyleBox[b_, ___]] := simpleBoxQ[b];
simpleBoxQ[(SubscriptBox | SuperscriptBox | OverscriptBox | UnderscriptBox)[
   bs__]] := AllTrue[{bs}, simpleBoxQ];
simpleBoxQ[_] := False;
trigBoxRule =
  {RowBox[{f_String /; MemberQ[trigNames, f], "(", a_, ")"}] /; simpleBoxQ[a] :>
    RowBox[{f, "", a}],
   RowBox[{f_String /; MemberQ[trigNames, f], RowBox[{"(", a_, ")"}]}] /;
     simpleBoxQ[a] :> RowBox[{f, "", a}]};
```

```
disp[e_] :=
  DisplayForm[ToBoxes[e /. dispRule, TraditionalForm] //. trigBoxRule];
sgnBox[e_] := Module[{f = e /. dispRule, sgnE = 1},
   If[Head[f] === Times && NumericQ[First[f]] && TrueQ[First[f] < 0], sgnE = -1;
    f = -f];
   {sgnE, DisplayForm[ToBoxes[f, TraditionalForm] //. trigBoxRule]}];
polyRow[terms_] := Module[{res = {}, sg, mg, tl}, Do[{sg, mg} = sgnBox[terms[[k, 1]]];
    tl = terms[[k, 2]];
    If[k == 1, If[sg < 0, AppendTo[res, "-"]],
     AppendTo[res, If[sg < 0, " - ", " + "]]];
    AppendTo[res, mg];
    If[tl =!= None, AppendTo[res, ""];
     AppendTo[res, tl]], {k, Length[terms]}];
   Row[res]];
ratForm[qd_, ud_, opTag_] :=
  frac[Row[{"(1 + ", mulR[ud, qd], ")(", mulR[qd, cotSqTag[opTag]], " - 1)"}],
   Row[{"(1 + ", qd, ")(1 - ", mulR[ud, qd, cotSqTag[opTag]], ")"}]];
iSet[setX_] := DisplayForm[
   RowBox[{StyleBox["I", FontSlant → "Italic"], "(", bx[setX], ")"}]];
(*the rule lens carries the two lengths that
 close the central face and is used for the figure alone,
never in a verification.A trigonometric identity is decided without
 Simplify:every argument is broken into single symbols by TrigExpand,
each sine and cosine is replaced by its rational form
 in the tangent of the half of its own symbol,
and the numerator is expanded and compared with zero;
zeroF halves the four flat angles first,so that the half sums
 of the quantities of the vertices become integer combinations*)
lens = {ll4 → (ll2 Sin[δ3] - ll1 Sin[δ2 + δ3]) / Sin[δ4],
   ll3 → (ll1 Sin[δ1] - ll2 Sin[δ1 + δ2]) / Sin[δ4]};
weier[e0_] := Module[{e = TrigExpand[e0], vs},
   vs = Union[Cases[e, (Sin | Cos)[x_] ↦ x, Infinity]];
   e = e /. Flatten[Table[{Sin[v] → 2 tw[v] / (1 + tw[v]^2),
        Cos[v] → (1 - tw[v]^2) / (1 + tw[v]^2)}, {v, vs}]];
   Expand[Numerator[Together[e]]]];
halfRule = Flatten[Table[{{α1, α2, α3, α4}[[k]] → 2 hh["a", k],
     {β1, β2, β3, β4}[[k]] → 2 hh["b", k], {γ1, γ2, γ3, γ4}[[k]] → 2 hh["c", k],
     {δ1, δ2, δ3, δ4}[[k]] → 2 hh["d", k]}, {k, 4}]];
zeroT[e_] := AllTrue[Flatten[{weier[e]}], # === 0 &];
zeroF[e_] := zeroT[e /. halfRule];
(*the coordinates;
B_3,B_4,C_3,C_4 are used for the figure alone*)
Apt[2] = {0, 0, 0};
Apt[1] = {ll1, 0, 0};
Apt[3] = ll2 {Cos[δ2], Sin[δ2], 0};
Apt[4] = Apt[1] + ll4 {-Cos[δ1], Sin[δ1], 0};
Bpt[1] = Apt[1] + mm1 {-Cos[α1], Sin[α1] Cos[θ1], Sin[α1] Sin[θ1]};
```

```
Bpt[2] = Apt[2] + mm2 {Cos[α2], Sin[α2] Cos[θ1], Sin[α2] Sin[θ1]};
Bpt[3] = Apt[3] + mm3 {-(Cos[α3] Cos[δ2 + δ3] + Sin[α3] Cos[θ3] Sin[δ2 + δ3]),
      -(Cos[α3] Sin[δ2 + δ3] - Sin[α3] Cos[θ3] Cos[δ2 + δ3]), Sin[α3] Sin[θ3]};
Bpt[4] = Apt[4] + mm4 {Cos[α4] Cos[δ2 + δ3] - Sin[α4] Cos[θ3] Sin[δ2 + δ3],
      Cos[α4] Sin[δ2 + δ3] + Sin[α4] Cos[θ3] Cos[δ2 + δ3], Sin[α4] Sin[θ3]};
Cpt[1] = Apt[1] + vv1 {-(Cos[γ1] Cos[δ1] + Sin[γ1] Cos[θ4] Sin[δ1]),
      Cos[γ1] Sin[δ1] - Sin[γ1] Cos[θ4] Cos[δ1], Sin[γ1] Sin[θ4]};
Cpt[2] = Apt[2] + vv2 {Cos[γ2] Cos[δ2] + Sin[γ2] Cos[θ2] Sin[δ2],
      Cos[γ2] Sin[δ2] - Sin[γ2] Cos[θ2] Cos[δ2], Sin[γ2] Sin[θ2]};
Cpt[3] = Apt[3] + vv3 {-(Cos[γ3] Cos[δ2] - Sin[γ3] Cos[θ2] Sin[δ2]),
      -(Cos[γ3] Sin[δ2] + Sin[γ3] Cos[θ2] Cos[δ2]), Sin[γ3] Sin[θ2]};
Cpt[4] = Apt[4] + vv4 {Cos[γ4] Cos[δ1] - Sin[γ4] Cos[θ4] Sin[δ1],
      -(Cos[γ4] Sin[δ1] + Sin[γ4] Cos[θ4] Cos[δ1]), Sin[γ4] Sin[θ4]};
uB[i_] := (Bpt[i] - Apt[i]) / {mm1, mm2, mm3, mm4}[[i]];
uC[i_] := (Cpt[i] - Apt[i]) / {vv1, vv2, vv3, vv4}[[i]];
eNext[2] = {1, 0, 0};
ePrev[1] = {-Cos[δ1], Sin[δ1], 0};
ePrev[2] = {Cos[δ2], Sin[δ2], 0};
nextA = {2, 1, 4, 3};
prevA = {4, 3, 2, 1};
thNextIdx = {1, 1, 3, 3};
thPrevIdx = {4, 2, 2, 4};
(*the central angle on the alpha-side of A_i and the one on the gamma-side,
and the representative of each shared quantity:x is shared by A_1 and A_2*)
nIdx = {1, 1, 3, 3};
pIdx = {4, 2, 2, 4};
xRp = {1, 1, 3, 3};
yRp = {1, 2, 2, 1};
qXr[i_] := pt["x", xRp[[i]]]; qYr[i_] := pt["y", yRp[[i]]];
(*the branch sign of the flexion parametrized by cot(theta_1/2)*)
e0Tag[k_] :=
  DisplayForm[SubscriptBox[StyleBox["e", FontSlant → "Italic"], "0"]];
sigTag[i_] :=
  DisplayForm[SqrtBox[bx[Row[{mulR[uPlain, qXr[i], qYr[i]], "(1 + ", pt["z", i],
       ")(1 + ", mulR[uPlain, pt["z", i]], ")"}]]]];
radTag[i_, fwd_] := DisplayForm[
   SqrtBox[bx[Row[{mulR[uPlain, If[fwd, qYr[i], qXr[i]], pt["z", i]], " ", dTag[
        If[fwd, qXr[i], qYr[i]], cotTag[thI[If[fwd, nIdx[[i]], pIdx[[i]]]]]}]]]];
flxF[i_, fwd_] :=
  Module[{src = If[fwd, nIdx[[i]], pIdx[[i]]], dst = If[fwd, pIdx[[i]], nIdx[[i]]],
    sg = If[fwd, (-1)^i, (-1)^(i + 1)], qA = If[fwd, qYr[i], qXr[i]],
    qB = If[fwd, qXr[i], qYr[i]]}, Row[{cotTag[thI[dst]], " = ",
     frac[Row[{gk["ε", i], "(", mulR[cotTag[thI[src]], sigTag[i]], If[sg < 0,
         " - ", " + "], mulR[e0Tag[src], pt["e", i], radTag[i, fwd]], ")"}],
      Row[{mulR[qA, uPlain], "(", pt["z", i], " + ",
        mulR[qB, cotSqTag[thI[src]]], ")"}]]}]];
bLen = {mm1, mm2, mm3, mm4}; cLen = {vv1, vv2, vv3, vv4};
```

```
angA = {α1, α2, α3, α4};
angB = {β1, β2, β3, β4};
angG = {γ1, γ2, γ3, γ4};
angD = {δ1, δ2, δ3, δ4};
thNext = {θ1, θ1, θ3, θ3};
thPrev = {θ4, θ2, θ2, θ4};
thCent = {θ1, θ2, θ3, θ4};
sgnV = {-1, 1, -1, 1};
wCent[1] = ll1 eNext[2];
wCent[2] = -ll2 ePrev[2];
wCent[4] = ll4 ePrev[1];
wCentLen = {ll1, ll2, ll3, ll4};
wB[i_] := sgnV[[i]] (Bpt[i] - Apt[i]);
wC[i_] := -sgnV[[i]] (Cpt[i] - Apt[i]);
wBout[i_] := sgnV[[i]] > 0; wCout[i_] := sgnV[[i]] < 0;
nraw[0] = Cross[ll1 eNext[2], ll2 ePrev[2]];
nraw[1] = Cross[ll1 eNext[2], Bpt[2] - Apt[2]];
nraw[2] = Cross[Cpt[2] - Apt[2], ll2 ePrev[2]];
nraw[4] = Cross[ll4 ePrev[1], Cpt[1] - Apt[1]];
nrawP[1] = Cross[Bpt[1] - Apt[1], Cpt[1] - Apt[1]];
nrawP[2] = Cross[Cpt[2] - Apt[2], Bpt[2] - Apt[2]];
nlen[0] = ll1 ll2 Sin[δ2];
nlen[1] = ll1 mm2 Sin[α2];
nlen[2] = ll2 vv2 Sin[γ2];
nlen[4] = ll4 vv1 Sin[γ1];
nlenP[i_] := bLen[[i]] cLen[[i]] Sin[angB[[i]]];
nPfirst[i_] := If[OddQ[i], uB[i], uC[i]];
nPsec[i_] := If[OddQ[i], uC[i], uB[i]];
nPname[i_] := If[OddQ[i], {evec["A", i, "B", i], evec["A", i, "C", i]},
   {evec["A", i, "C", i], evec["A", i, "B", i]}];
nPshow[i_] := bLen[[i]] cLen[[i]] Collect[Cross[nPfirst[i], nPsec[i]],
    {Sin[thNext[[i]]], Cos[thNext[[i]]], Sin[thPrev[[i]]], Cos[thPrev[[i]]]}, Factor];
ndir[0] = {0, 0, 1};
ndir[1] = {0, -Sin[θ1], Cos[θ1]};
ndir[2] = {-Sin[θ2] Sin[δ2], Sin[θ2] Cos[δ2], Cos[θ2]};
ndir[4] = {Sin[δ1] Sin[θ4], Cos[δ1] Sin[θ4], Cos[θ4]};
nvecIdx = {{1, 4}, {1, 2}, {3, 2}, {3, 4}};
betaForm[i_] := Cos[angA[[i]]]
    (Cos[angG[[i]]] Cos[angD[[i]]] + Sin[angG[[i]]] Sin[angD[[i]]] Cos[thPrev[[i]]]) +
   Sin[angA[[i]]] (Cos[thNext[[i]]]
       (Cos[angG[[i]]] Sin[angD[[i]]] - Sin[angG[[i]]] Cos[angD[[i]]] Cos[thPrev[[i]]]) +
      Sin[thNext[[i]]] Sin[angG[[i]]] Sin[thPrev[[i]]]);
vclose = Table[Cos[angB[[k]]] → betaForm[k], {k, 2}];
outBnum[i_] := Cos[angD[[i]]] Cos[angG[[i]]] +
  Sin[angD[[i]]] Sin[angG[[i]]] Cos[thPrev[[i]]] - Cos[angA[[i]]] Cos[angB[[i]]];
outBden[i_] := Sin[angA[[i]]] Sin[angB[[i]]];
outCnum[i_] := Cos[angD[[i]]] Cos[angA[[i]]] +
```

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  Sin[angD[[i]]] Sin[angA[[i]]] Cos[thNext[[i]]] - Cos[angB[[i]]] Cos[angG[[i]]];
outCden[i_] := Sin[angB[[i]]] Sin[angG[[i]]];
outB[i_] := outBnum[i] / outBden[i];
outC[i_] := outCnum[i] / outCden[i];
eqB[i_] := Sin[thNext[[i]]]
    (Cos[angG[[i]]] Sin[angD[[i]]] - Sin[angG[[i]]] Cos[angD[[i]]] Cos[thPrev[[i]]]) -
   Cos[thNext[[i]]] Sin[angG[[i]]] Sin[thPrev[[i]]];
eqC[i_] :=
  Sin[angA[[i]]] (Sin[thNext[[i]]] Cos[thPrev[[i]]] + Cos[thNext[[i]]] Sin[thPrev[[i]]]
       Cos[angD[[i]]]) - Cos[angA[[i]]] Sin[angD[[i]]] Sin[thPrev[[i]]];
sinBform[i_] := eqB[i]; sinCform[i_] := -eqC[i];
sgD[i_] := Subscript["σ", i];
bAD[i_] := Subscript[OverBar["α"], i];
bBD[i_] := Subscript[OverBar["β"], i];
bGD[i_] := Subscript[OverBar["ɣ"], i];
bDD[i_] := Subscript[OverBar["δ"], i];
sig[i_] := (angA[[i]] + angB[[i]] + angG[[i]] + angD[[i]]) / 2;
barA[i_] := sig[i] - angA[[i]];
barB[i_] := sig[i] - angB[[i]];
barG[i_] := sig[i] - angG[[i]];
barD[i_] := sig[i] - angD[[i]];
barRule[i_] := {sgD[i] → sig[i], bAD[i] → barA[i],
   bBD[i] → barB[i], bGD[i] → barG[i], bDD[i] → barD[i]};
ratA[i_] := Sin[angA[[i]]] / Sin[barA[i]];
ratB[i_] := Sin[angB[[i]]] / Sin[barB[i]];
ratG[i_] := Sin[angG[[i]]] / Sin[barG[i]];
ratD[i_] := Sin[angD[[i]]] / Sin[barD[i]];
prodM[i_] := ratA[i] ratB[i] ratG[i] ratD[i];
prodR[i_] := ratA[i] ratD[i];
prodS[i_] := ratG[i] ratD[i];
prodF[i_] := ratA[i] ratG[i];
quU[i_] := 1 - prodM[i];
quX[i_] := 1 / (prodR[i] - 1);
quY[i_] := 1 / (prodS[i] - 1);
quZ[i_] := 1 / (prodF[i] - 1);
biqBar[sg_, bA_, bB_, bD_, bG_, bb_] := {-Sin[bG - bb] Sin[bD - bb],
   Sin[bA] Sin[bB], -Sin[bG] Sin[bD], Sin[sg] Sin[bA - bb]};
biqC[aa_, bb_, dd_, gg_] := {Cos[aa - bb] - Cos[dd - gg], Cos[aa - bb] - Cos[dd + gg],
    Cos[aa + bb] - Cos[dd - gg], Cos[aa + bb] - Cos[dd + gg]} / 2;
(*the four coefficients of biqC as products of
 sines of half sums of the flat angles,in the same order*)
qBarD[i_, isB_] :=
  biqBar[sgD[i], Sequence @@ If[isB, {bAD[i], bBD[i], bDD[i], bGD[i]},
     {bGD[i], bBD[i], bDD[i], bAD[i]}], angB[[i]]];
nameB = {thP[1, 1], thP[1, 2], thP[3, 3], thP[3, 4]};
nameC = {thP[4, 1], thP[2, 2], thP[2, 3], thP[4, 4]};
(*the two corner angles of this section:along A_1C_1 and along A_2C_2*)
```

```
spec = {{1, "C"}, {2, "C"}};
resultsAll = {}; cqList = {};
(*the figure:the orthogonal projection of the
   nine faces of one net onto the plane of the first two axes,
seen from the positive side of the third,
which is drawn as the mark with a dot at A_2;
only the flat and dihedral angles used in this section are marked.The
 data num,rAv,rV,rO,the three nudge tables,the two axis lengths,
pad and shiftL are the only entries to be tuned by hand*)
figBlock = Module[{dg = π / 180., num, vA, vB, vC, vD, grey = GrayLevel[0.45],
    blue = RGBColor[0.15, 0.35, 0.7], rAv = {0.5, 0.58, 0.40, 0.40},
    rV = 0.72, rO = 0.46, aNudge, bNudge, cNudge, xEnd = 9.0, yEnd = 7.8,
    pad = 0.7, shiftL = 1.2, prj, bis, angLab, gapDir, arcTo, arcs, angLabs,
    vtxLabs, thinE, thickE, dots, axes, zmark, xd, yd, allPts, xr, yr},
   num = {ll1 → 4.4, ll2 → 3.8, mm1 → 3.2, mm2 → 3.0, mm3 → 3.1, mm4 → 3.0, vv1 → 3.4,
     vv2 → 3.2, vv3 → 3.0, vv4 → 3.3, δ1 → 105 dg, δ2 → 70 dg, δ3 → 95 dg, δ4 → 90 dg,
     θ1 → 150 dg, θ2 → 145 dg, θ3 → 152 dg, θ4 → 148 dg, α1 → 100 dg, α2 → 125 dg,
     α3 → 92 dg, α4 → 86 dg, ɣ1 → 95 dg, ɣ2 → 90 dg, ɣ3 → 87 dg, ɣ4 → 93 dg};
   aNudge = {{0.2, -0.25}, {-0.30, -0.30}, {0, 0.30}, {0, 0.30}};
   bNudge = {{0, 0.08}, {0, 0.08}, {0, -0.08}, {0, -0.08}};
   cNudge = {{-0.08, 0}, {0.08, 0}, {0.08, 0}, {-0.08, 0}};
   prj[vec_] := N[Take[vec /. lens /. num, 2]];
   vA = Table[prj[Apt[i]], {i, 4}];
   vB = Table[prj[Bpt[i]], {i, 4}];
   vC = Table[prj[Cpt[i]], {i, 4}];
   vD = Table[vA[[i]] + 0.85 (vB[[i]] + vC[[i]] - 2 vA[[i]]), {i, 4}];
   bis[o_, nbr1_, qq_] := Normalize[Normalize[nbr1 - o] + Normalize[qq - o]];
   angLab[o_, nbr1_, qq_, lbl_, rLab_] :=
    Text[Style[lbl, 11, grey], o + rLab bis[o, nbr1, qq]];
   gapDir[o_, nbrs_] :=
    Module[{an, gp, kk}, an = Sort[ArcTan[#[[1]], #[[2]]] & /@ ((# - o) & /@ nbrs)];
     gp = Differences[Append[an, First[an] + 2 π]];
     kk = First[Ordering[gp, -1]];
     {Cos[#], Sin[#]} &[an[[kk]] + gp[[kk]] / 2]];
   arcTo[pa_, pb_, lbl_, tq_, into_, rr_, sw_, le_] :=
    Module[{d, nn, mp, sg, cps, hd}, d = Normalize[pb - pa];
     nn = {-d[[2]], d[[1]]};
     mp = pa + tq (pb - pa);
     sg = Sign[nn.(into - mp)];
     cps = {mp - 0.45 rr sg nn, mp - 0.27 rr sg nn + 0.34 rr d,
       mp + 0.27 rr sg nn + 0.34 rr d, mp + 0.45 rr sg nn};
     If[sw < 0, cps = Reverse[cps]];
     hd = Last[cps];
     {{grey, Thickness[0.0024], Arrowheads[0.016], Arrow[BezierCurve[cps]]},
      Text[Style[lbl, 11, grey], If[le > 0,
        hd + 0.05 Normalize[hd - mp] + 0.35 rr d, mp + 0.50 rr d + 0.30 rr sg nn]]}];
   arcs = Join[{arcTo[vA[[2]], vA[[1]], thI[1], 0.50, Mean[{vB[[1]], vB[[2]]}], 1, 1, 0],
```

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      arcTo[vA[[2]], vA[[3]], thI[2], 0.50, Mean[{vC[[2]], vC[[3]]}], 1, 1, 0],
      arcTo[vA[[4]], vA[[1]], thI[4], 0.50, Mean[{vC[[4]], vC[[1]]}], -1, -1, 1]},
    Table[arcTo[vA[[i]], vC[[i]], nameC[[i]], 0.55, vB[[i]], 1, 1, 0], {i, {1, 2}}]];
  angLabs = Flatten[
    Table[{angLab[vA[[i]], vA[[nextA[[i]]]], vA[[prevA[[i]]]], gk["δ", i], rAv[[i]],
      angLab[vA[[i]], vA[[nextA[[i]]]], vB[[i]], gk["α", i], rAv[[i]],
      angLab[vA[[i]], vB[[i]], vC[[i]], gk["β", i], rAv[[i]],
      angLab[vA[[i]], vA[[prevA[[i]]]], vC[[i]], gk["γ", i], rAv[[i]]}, {i, {1, 2}}]];
  vtxLabs = Join[Table[
     Text[Style[pt["A", i], 12], vA[[i]] + rV bis[vA[[i]], vB[[i]], vC[[i]]] + aNudge[[i]]],
     {i, 4}], Table[Text[Style[pt["B", i], 12],
      vB[[i]] + rO gapDir[vB[[i]], {vA[[i]], vD[[i]], vB[[nextA[[i]]]]}] + bNudge[[i]]],
     {i, 4}], Table[Text[Style[pt["C", i], 12], vC[[i]] +
       rO gapDir[vC[[i]], {vA[[i]], vD[[i]], vC[[prevA[[i]]]]}] + cNudge[[i]], {i, 4}]];
  thinE = {Thickness[0.0024], GrayLevel[0.4], Line[{vB[[2]], vB[[1]]}],
    Line[{vC[[3]], vC[[2]]}], Line[{vB[[4]], vB[[3]]}], Line[{vC[[1]], vC[[4]]}],
    Sequence @@ Table[Line[{vB[[i]], vD[[i]], vC[[i]]}], {i, 4}]};
  thickE = {Thickness[0.006],
    GrayLevel[0.1], Line[{vA[[1]], vA[[2]], vA[[3]], vA[[4]], vA[[1]]}],
    Sequence @@ Table[Line[{vA[[i]], vB[[i]]}], {i, 4}],
    Sequence @@ Table[Line[{vA[[i]], vC[[i]]}], {i, 4}]};
  xd = Normalize[vA[[1]] - vA[[2]]]; yd = {-xd[[2]], xd[[1]]};
  axes = {blue, Thickness[0.0026], Arrowheads[0.018],
    Arrow[{vA[[2]], vA[[2]] + xEnd xd}], Arrow[{vA[[2]], vA[[2]] + yEnd yd}],
    Text[Style[tld["x"], 12, blue], vA[[2]] + (xEnd + 0.26) xd],
    Text[Style[tld["y"], 12, blue], vA[[2]] + (yEnd + 0.26) yd]};
  dots = {White, EdgeForm[{GrayLevel[0.1], Thickness[0.002]}], Sequence @@
     Table[Disk[pp, 0.11], {pp, Join[{vA[[1]], vA[[3]], vA[[4]]}, vB, vC, vD]}]};
  zmark = {White, EdgeForm[{blue, Thickness[0.0026]}],
    Disk[vA[[2]], 0.14], EdgeForm[None], blue, Disk[vA[[2]], 0.05],
    Text[Style[tld["z"], 12, blue], vA[[2]] + 0.10 xd - 0.3 yd]};
  allPts = Join[vA, vB, vC, vD, {vA[[2]] + (xEnd + 0.5) xd, vA[[2]] + (yEnd + 0.5) yd}];
  xr = {Min[First /@ allPts] - pad, Max[First /@ allPts] + pad + shiftL};
  yr = {Min[Last /@ allPts] - pad, Max[Last /@ allPts] + pad};
  Graphics[{axes, thinE, thickE, arcs, angLabs, dots, zmark, vtxLabs},
   ImageSize → 700, PlotRange → {xr, yr}, PlotRangePadding → None,
   Background → None, BaseStyle → {FontFamily → "Times"}]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style["Given hypotheses and definitions:", Bold, 14],
    Style["A 3 x 3 net with, for each i = 1, 2,", Bold, 13],
    Style[Column[{Row[{"the vertices ", pt["A", "i"], ", ",
        pt["B", "i"], ", ", pt["C", "i"], ", and the vertices ",
        pt["A", 3], ", ", pt["A", 4], " of the central face"}],
      Row[{"the flat angles ", gk["α", "i"], ", ", gk["β", "i"], ", ",
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    gk["γ", "i"], ", ", gk["δ", "i"], " ∈ (0, π) at ", pt["A", "i"]}],
  Row[{"the edge lengths ", pt["ℓ", 1], " = |", edg["A", 1, "A", 2],
    "|,   ", pt["ℓ", 2], " = |", edg["A", 2, "A", 3],
    "|,   ", pt["ℓ", 4], " = |", edg["A", 4, "A", 1], "|,   ",
    pt["m", "i"], " = |", edg["A", "i", "B", "i"], "|,   ",
    pt["v", "i"], " = |", edg["A", "i", "C", "i"], "|"}],
  Row[{"the dihedral angles ", thI[1], ", ", thI[2], ", ", thI[4],
    " ∈ [-π, π) \\ {0} along the edges ", edg["A", 1, "A", 2], ", ",
    edg["A", 2, "A", 3], ", ", edg["A", 4, "A", 1]}], Spacer[{0, 6}],
  Grid[{{Row[{disp[sgD[iv]], " = ", disp[(Subscript["α", iv] + Subscript[
            "β", iv] + Subscript["γ", iv] + Subscript["δ", iv]) / 2]}],
     Row[{disp[bAD[iv]], " = ", disp[sgD[iv] - Subscript["α", iv]]}],
     Row[{disp[bBD[iv]], " = ", disp[sgD[iv] - Subscript["β", iv]]}],
     Row[{disp[bGD[iv]], " = ", disp[sgD[iv] - Subscript["γ", iv]]}],
     Row[{disp[bDD[iv]], " = ", disp[sgD[iv] - Subscript["δ", iv]]}]},
    {Row[{pt["a", "i"], " = ", frac[Row[{"sin", gk["α", "i"]}],
        Row[{"sin", disp[bAD[iv]]}]]}], Row[{pt["b", "i"], " = ",
      frac[Row[{"sin", gk["β", "i"]}], Row[{"sin", disp[bBD[iv]]}]]}],
     Row[{pt["c", "i"], " = ", frac[Row[{"sin", gk["γ", "i"]}],
        Row[{"sin", disp[bGD[iv]]}]]}], Row[{pt["d", "i"], " = ",
      frac[Row[{"sin", gk["δ", "i"]}], Row[{"sin", disp[bDD[iv]]}]]}],
     Row[{pt["ε", "i"], " = sign(sin", disp[sgD[iv]], ")"}]},
    {Row[{pt["M", "i"], " = ", mulR[pt["a", "i"],
        pt["b", "i"], pt["c", "i"], pt["d", "i"]]}],
     Row[{pt["r", "i"], " = ", mulR[pt["a", "i"], pt["d", "i"]]}],
     Row[{pt["s", "i"], " = ", mulR[pt["c", "i"], pt["d", "i"]]}],
     Row[{pt["f", "i"], " = ", mulR[pt["a", "i"], pt["c", "i"]]}], ""},
    {Row[{pt["u", "i"], " = 1 - ", pt["M", "i"]}],
     Row[{pt["x", "i"], " = ", frac["1", Row[{pt["r", "i"], " - 1"}]]}],
     Row[{pt["y", "i"], " = ", frac["1", Row[{pt["s", "i"], " - 1"}]]}],
     Row[{pt["z", "i"], " = ", frac["1", Row[{pt["f", "i"], " - 1"}]]}],
     ""}}, Alignment → Left, Spacings → {2.4, 1.1}],
  Spacer[{0, 6}], Row[{gk["α", "i"], " ± ", gk["β", "i"], " ± ",
    gk["γ", "i"], " ± ", gk["δ", "i"], " ≠ 0 (mod 2π)"}],
  Row[{disp[bAD[iv]], ", ", disp[bBD[iv]], ", ", disp[bGD[iv]],
    ", ", disp[bDD[iv]], " ∈ (0, π)"}],
  Row[{pt["ε", "i"], " ∈ {-1, 1}"}], Row[{pt["r", "i"], ", ",
    pt["s", "i"], ", ", pt["f", "i"], ", ", pt["M", "i"],
    " ≠ 0, 1;   ", pt["M", "i"], ", ", pt["f", "i"], " > 0"}], Row[
   {pt["x", "i"], ", ", pt["y", "i"], ", ", pt["z", "i"], " ≠ -1, 0;   ",
    mulR[pt["u", "i"], pt["x", "i"], pt["y", "i"], pt["z", "i"]], " > 0"}],
  Row[{pt["x", "i"], "(1 + ", pt["x", "i"], ") > 0,   ",
    pt["z", "i"], "(1 + ", pt["z", "i"], ") > 0"}],
  Row[{"1 + ", mulR[pt["u", "i"], pt["x", "i"]], " > 0,   1 + ",
    mulR[pt["u", "i"], pt["z", "i"]], " > 0"}]}, Spacings → 0.7], 13],
Spacer[{0, 10}], Style["Normals:", Bold, 14],
Style[
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 Row[{nv[0], " = ", evec["A", 2, "A", 1], " × ", evec["A", 2, "A", 3]}],
 13], Style[Grid[
  {{Row[{nv[1], " = ", evec["A", 2, "A", 1], " × ", evec["A", 2, "B", 2]}],
    Row[{nv[2], " = ", evec["A", 2, "C", 2], " × ", evec["A", 2, "A", 3]}],
    Row[{nv[4], " = ", evec["A", 1, "A", 4], " × ", evec["A", 1, "C", 1]}]}},
  Spacings → {3, 0}], 13], Style[Grid[
  {{Row[{nvP[1], " = ", evec["A", 1, "B", 1], " × ", evec["A", 1, "C", 1]}],
    Row[{nvP[2], " = ", evec["A", 2, "C", 2], " × ",
      evec["A", 2, "B", 2]}]}}, Spacings → {3, 0}], 13],
Spacer[{0, 10}], Style["Dihedral angles:", Bold, 14],
Style[
 caseDef[Row[{"θ(", wv[1], ", ", wv[2], ", ", wv[3], ")"}],
  {{Row[{"∠(", wv[1], ", ", wv[2], ") ∈ (0, π),"}],
    Row[{"if det(", wv[1], ", ", wv[2], ", ", wv[3], ") > 0,"}]},
   {Row[{"-∠(", wv[1], ", ", wv[2], ") ∈ [-π, 0],"}],
    Row[{"if det(", wv[1], ", ", wv[2], ", ", wv[3], ") ≤ 0."}]}}], 13],
Style["central dihedral angles:", Italic, 13],
Style[Grid[{{Row[{thI[1], " = θ(", nv[0], ", ", nv[1], ", ",
      evec["A", 2, "A", 1], ")"}], Row[{thI[2], " = θ(", nv[0],
      ", ", nv[2], ", ", evec["A", 3, "A", 2], ")"}], Row[{thI[4],
      " = θ(", nv[0], ", ", nv[4], ", ", evec["A", 1, "A", 4], ")"}]}},
  Alignment → Left, Spacings → {2, 0.8}], 13], Spacer[{0, 6}],
Style["corner dihedral angles:", Italic, 13],
Style[Grid[{{Row[{thP[4, 1], " = θ(", nv[4], ", ", nvP[1],
      ", ", evec["A", 1, "C", 1], ")"}], Row[{thP[2, 2], " = θ(",
      nv[2], ", ", nvP[2], ", ", evec["C", 2, "A", 2], ")"}]}},
  Alignment → Left, Spacings → {2, 0.8}], 13], Spacer[{0, 10}],
Style["Cartesian coordinates:", Bold, 14],
Style[Row[{"origin at ", pt["A", 2]}], 13],
Style[Row[{"central face in the half-plane ", tld["y"],
   " ≥ 0 of the ", tld["x"], tld["y"], "-plane, the ray ",
   edg["A", 2, "A", 1], " as the ", tld["x"], "-axis"}], 13],
Column[{figBlock, Style["Figure 1", 12, GrayLevel[0.5]]},
 Alignment → Center], Spacer[{0, 8}],
Style[Column[Join[Table[Row[{pt["A", k], " = ", disp[Apt[k]]}], {k, 4}],
   Table[Row[{pt["B", k], " = ", pt["A", k],
      " + ", pt["m", k], " ", disp[uB[k]]}], {k, 2}],
   Table[Row[{pt["C", k], " = ", pt["A", k], " + ", pt["v", k],
      " ", disp[uC[k]]}], {k, 2}]], Spacings → 0.5], 13],
Style[Row[{"The definition gives θ(", wv[1], ", ", wv[2],
   ", ", wv[3], ") = ±∠(", wv[1], ", ", wv[2],
   "), by the sign of the determinant. The cosine is
     even and the sine is odd, so cos θ = ",
   frac[Row[{wv[1], " · ", wv[2]}], mulR[absR[wv[1]], absR[wv[2]]]],
   " and sin θ = ",
   frac[Row[{"det(", wv[1], ", ", wv[2], ", ", wv[3], ")"}],
    mulR[absR[wv[1]], absR[wv[2]], absR[wv[3]]]],
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    " hold whichever sign the determinant has; the formula
      for the sine holds because ", wv[3],
    ", the direction of an edge common to the two faces, is
      orthogonal to ", wv[1], " and ", wv[2],
    ", so that ", wv[1], " × ", wv[2], " is a multiple of ", wv[3],
    ". No formula of the notebook distinguishes the two
      cases, the coordinates above included."}], 12,
  GrayLevel[0.5]], Spacer[{0, 10}], Style["Flexion:", Bold, 14],
Style[Row[{pt["x", 2], " = ", pt["x", 1], ",   ",
    pt["u", 1], " = ", pt["u", 2], " = ", uPlain}], 13],
Style[Row[{pt["e", 1], ", ", pt["e", 2], ", ", e0Tag[1], " ∈ {-1, 1}"}],
 13],
Style[Row[{dTag[capX, tSym], " = (", mulR[capX, DisplayForm[
      SuperscriptBox[StyleBox["t", FontSlant → "Italic"], "2"]]],
    " - 1)(1 - ", mulR[uPlain, capX, DisplayForm[
      SuperscriptBox[StyleBox["t", FontSlant → "Italic"], "2"]]],
    "),      ", iSet[capX], " = {", tSym, " ∈ ℝ : ",
    mulR[capX, dTag[capX, tSym]], " > 0},      for ",
    capX, ", ", tSym, " ∈ ℝ"}], 13], Spacer[{0, 6}],
Style[Column[{flxF[1, True], flxF[2, True]}, Spacings → 0.8], 13],
Spacer[{0, 10}],
Style[Row[{"Each ", smallChip[True],
    " below certifies that the statement displayed beside it has been
      verified by a symbolic computation at that point; a ",
    smallChip[False], " would mean that this verification failed. A ",
    givenChip,
    " marks a given fact, a definition, a result of an earlier block, or
      an inference stated in words beside it."}], 12, GrayLevel[
  0.35]], "", Style[Row[{Style["We want to test the following claims ",
     Bold, 14], Style["⋆", 14, colTest], Style[" :", Bold, 14]}]],
Spacer[{0, 6}], Style["Claim 1.", Bold, 13, colTest],
Sequence @@
 Table[Module[{i = spec⟦pnl, 1⟧, tg, op, qu}, tg = nameC⟦i⟧;
    op = thI[thNextIdx⟦i⟧];
    qu = pt["x", i];
    Row[{bullet, Style[Row[{"(1 - cos ", tg, ")(1 + ", mulR[pt["u", i], qu],
          ")(", mulR[qu, cotSqTag[op]], " - 1) = (1 + cos ", tg, ")(1 + ",
          qu, ")(1 - ", mulR[pt["u", i], qu, cotSqTag[op]], ")"}],
       Bold, 13, colTest]}]], {pnl, 2}], Spacer[{0, 8}],
Style["In particular,", Bold, 13, colTest], Sequence @@
 Table[Module[{i = spec⟦pnl, 1⟧, tg, op, qu}, tg = nameC⟦i⟧;
    op = thI[thNextIdx⟦i⟧];
    qu = pt["x", i];
    Row[{bullet, Style[Row[{tg, " ≠ 0, ±π   if   (", mulR[qu,
          cotSqTag[op]], " - 1)(1 - ", mulR[pt["u", i], qu, cotSqTag[op]],
          ") ≠ 0"}], Bold, 13, colTest]}]], {pnl, 2}], Spacer[{0, 16}],
Style["Claim 2.", Bold, 13, colTest], Style[Row[{"If ", pt["x", 2],
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        " = ", pt["x", 1], ",   ", pt["u", 1], " = ", pt["u", 2], " = ", uPlain,
        ", and ", cotTag[thI[1]], " ∈ ", iSet[pt["x", 1]], ", then"}], Bold, 13,
      colTest], Spacer[{0, 6}], Row[{bullet, Style[Row[{mulR[pt["e", 1],
          cotTag[thP[4, 1]]], " = -", mulR[pt["e", 2], cotTag[thP[2, 2]]],
        " = ", mulR[e0Tag[1], DisplayForm[SqrtBox[bx[ratForm[pt["x", 1],
            uPlain, thI[1]]]]]]}], Bold, 13, colTest]}]}, Spacings → 1.1],
   Background → RGBColor[0.99, 0.96, 0.88], FrameMargins → 14]];

(* ============================================================*)
(*PART 3:THE COORDINATES*)
(* ============================================================*)
Module[{col = cols⟦1⟧, vecs, lenOK, angOK, sideOK, wdir, chk, rows},
  vecs = {{Apt[1] - Apt[2], evec["A", 2, "A", 1]},
    {Apt[3] - Apt[2], evec["A", 2, "A", 3]},
    {Apt[2] - Apt[1], evec["A", 1, "A", 2]}, {Apt[4] - Apt[1],
     evec["A", 1, "A", 4]}, {Bpt[1] - Apt[1], evec["A", 1, "B", 1]},
    {Bpt[2] - Apt[2], evec["A", 2, "B", 2]}, {Cpt[1] - Apt[1],
     evec["A", 1, "C", 1]}, {Cpt[2] - Apt[2], evec["A", 2, "C", 2]}};
  lenOK = MapThread[zeroT[vecs⟦#1, 1⟧.vecs⟦#1, 1⟧ - #2^2] &,
    {{2, 4, 5, 6, 7, 8}, {ll2, ll4, mm1, mm2, vv1, vv2}}];
  angOK = MapThread[zeroT[vecs⟦#1, 1⟧.vecs⟦#2, 1⟧ - #3] &,
    {{4, 1, 3, 1, 4, 2}, {3, 2, 5, 6, 7, 8}, {ll4 ll1 Cos[δ1], ll1 ll2 Cos[δ2],
      ll1 mm1 Cos[α1], ll1 mm2 Cos[α2], ll4 vv1 Cos[γ1], ll2 vv2 Cos[γ2]}}];
  wdir = {0, -Sin[θ1], Cos[θ1]};
  sideOK = {zeroT[Cross[vecs⟦1, 1⟧, vecs⟦6, 1⟧] - ll1 mm2 Sin[α2] wdir],
    zeroT[Cross[vecs⟦1, 1⟧, vecs⟦5, 1⟧] - ll1 mm1 Sin[α1] wdir]};
  chk = And @@ Join[lenOK, angOK, sideOK];
  AppendTo[resultsAll, chk];
  rows =
   Join[Table[nrow[k, Row[{vecs⟦k, 2⟧, " = ", disp[vecs⟦k, 1⟧]}], givenChip,
      "from the coordinates"], {k, 8}], {vgap, nrow[9, Row[{absR[vecs⟦2, 2⟧],
      " = ", pt["ℓ", 2], ",   ", absR[vecs⟦4, 2⟧], " = ", pt["ℓ", 4]}],
     smallChip[lenOK⟦1⟧ && lenOK⟦2⟧], "the lengths of (2) and (4)"],
    nrow[10, Row[{absR[vecs⟦5, 2⟧], " = ", pt["m", 1],
      ",   ", absR[vecs⟦6, 2⟧], " = ", pt["m", 2]}],
     smallChip[lenOK⟦3⟧ && lenOK⟦4⟧], "the lengths of (5) and (6)"],
    nrow[11, Row[{absR[vecs⟦7, 2⟧], " = ", pt["v", 1], ",   ",
      absR[vecs⟦8, 2⟧], " = ", pt["v", 2]}], smallChip[lenOK⟦5⟧ && lenOK⟦6⟧],
     "the lengths of (7) and (8)"], vgap, nrow[12,
     Row[{"cos", gk["δ", 1], " = ", frac[Row[{vecs⟦4, 2⟧, " · ", vecs⟦3, 2⟧}],
        mulR[pt["ℓ", 4], pt["ℓ", 1]]], ",   cos", gk["δ", 2], " = ", frac[
        Row[{vecs⟦1, 2⟧, " · ", vecs⟦2, 2⟧}], mulR[pt["ℓ", 1], pt["ℓ", 2]]]}],
     smallChip[angOK⟦1⟧ && angOK⟦2⟧], "from (1)–(4) and (9); flat
       angles in (0, π) are fixed by their cosines"], nrow[13,
     Row[{"cos", gk["α", 1], " = ", frac[Row[{vecs⟦3, 2⟧, " · ", vecs⟦5, 2⟧}],
        mulR[pt["ℓ", 1], pt["m", 1]]], ",   cos", gk["α", 2], " = ", frac[
        Row[{vecs⟦1, 2⟧, " · ", vecs⟦6, 2⟧}], mulR[pt["ℓ", 1], pt["m", 2]]]}],
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      smallChip[angOK[[3]] && angOK[[4]]], "from (1), (3), (5), (6), and (10)"],
     nrow[14, Row[{"cos", gk["γ", 1], " = ",
        frac[Row[{vecs[[4, 2]], " · ", vecs[[7, 2]]}], mulR[pt["ℓ", 4], pt["v", 1]]],
        ",   cos", gk["γ", 2], " = ", frac[Row[{vecs[[2, 2]], " · ", vecs[[8, 2]]}],
         mulR[pt["ℓ", 2], pt["v", 2]]]}], smallChip[angOK[[5]] && angOK[[6]]],
      "from (2), (4), (7), (8), (9), and (11)"], vgap, nrow[15,
      Row[{"the ", tld["y"], "-coordinates of ", pt["A", 3], ", ", pt["A", 4],
        " are ", disp[ll2 Sin[δ2]], ", ", disp[ll4 Sin[δ1]], " > 0"}],
      givenChip, Row[{"from (2) and (4), because ", gk["δ", 1],
        ", ", gk["δ", 2], " ∈ (0, π)"}]], vgap,
     nrow[16, Row[{vecs[[1, 2]], " × ", vecs[[6, 2]], " = ", disp[ll1 mm2 Sin[α2]],
        " ", disp[wdir]}], smallChip[sideOK[[1]]], "from (1) and (6)"],
     nrow[17, Row[{vecs[[1, 2]], " × ", vecs[[5, 2]], " = ",
        disp[ll1 mm1 Sin[α1]], " ", disp[wdir]}], smallChip[sideOK[[2]]],
      Row[{"from (1) and (5); with (16), ", pt["B", 1], " and ",
        pt["B", 2], " lie on one side of ", edg["A", 1, "A", 2],
        ", because sin", gk["α", 1], ", sin", gk["α", 2], " > 0"}]]}];
  Print[Panel[Column[{Row[
       {Style["the coordinates", Bold, 15, col], Spacer[20], statusChip[chk]}],
      Style[Row[{"Verification: the lengths and the flat angles at ",
         pt["A", 1], ", ", pt["A", 2], " from dot products, and the side of ",
         pt["B", 1], " from two cross products; with ", thI[1], ", ", thI[2],
         ", ", thI[4], " from the next block, these data fix every point."}],
       Bold, 13], vgrid[rows], Style[
       Row[{"Together, these give the lengths and the flat angles ",
         gk["α", "i"], ", ", gk["γ", "i"], ", ", gk["δ", "i"],
         ", i = 1, 2, of the coordinates, with ", pt["B", 1],
         " on the side of ", edg["A", 1, "A", 2], " where ",
         pt["B", 2], " lies."}], 13]}, Spacings → 1],
    Background → Lighter[col, 0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 4:THE NORMALS*)
(* ============================================================*)
Module[{col = cols[[2]], vec, nrm, nOK, oC, oS, chk, rows},
  vec = {{Apt[1] - Apt[2], evec["A", 2, "A", 1]}, {Apt[3] - Apt[2],
     evec["A", 2, "A", 3]}, {Bpt[2] - Apt[2], evec["A", 2, "B", 2]},
    {Cpt[2] - Apt[2], evec["A", 2, "C", 2]}, {Apt[4] - Apt[1],
     evec["A", 1, "A", 4]}, {Cpt[1] - Apt[1], evec["A", 1, "C", 1]}};
  (*each normal:its index and the numbers of its two vectors*)
  nrm = {{0, 1, 2}, {1, 1, 3}, {2, 4, 2}, {4, 5, 6}};
  nOK = Table[zeroT[Cross[vec[[nSpec[[2]], 1]], vec[[nSpec[[3]], 1]]] -
       nlen[nSpec[[1]]] ndir[nSpec[[1]]]] &&
     zeroT[nraw[nSpec[[1]]].nraw[nSpec[[1]]] - nlen[nSpec[[1]]]^2], {nSpec, nrm}];
  oC = And @@ Table[
     zeroT[nraw[0].nraw[k] - nlen[0] nlen[k] Cos[thCent[[k]]]], {k, {1, 2, 4}}];
  oS = And @@ Table[zeroT[Det[{nraw[0], nraw[k], wCent[k]}] -
```

```
      nlen[0] nlen[k] wCentLen⟦k⟧ Sin[thCent⟦k⟧]], {k, {1, 2, 4}}];
  chk = (And @@ nOK) && oC && oS; AppendTo[resultsAll, chk];
  rows = Join[Table[nrow[j, Row[{vec⟦j, 2⟧, " = ", disp[vec⟦j, 1⟧]}],
      givenChip, "from the coordinates"], {j, 6}],
    {vgap}, Table[With[{nSpec = nrm⟦jn⟧}, nrow[6 + jn,
       Row[{nv[nSpec⟦1⟧], " = ", vec⟦nSpec⟦2⟧, 2⟧, " × ", vec⟦nSpec⟦3⟧, 2⟧,
         " = ", disp[nlen[nSpec⟦1⟧]], " ", disp[ndir[nSpec⟦1⟧]], ",   ",
         absR[nv[nSpec⟦1⟧]], " = ", disp[nlen[nSpec⟦1⟧]]}], smallChip[nOK⟦jn⟧],
       Row[{"from (", nSpec⟦2⟧, ") and (", nSpec⟦3⟧, ")"}]]], {jn, 4}],
    {vgap, nrow[11, Row[{nv[0], " · ", nv["i"], " = ",
        mulR[absR[nv[0]], absR[nv["i"]], Row[{"cos", thI["i"]}]]}],
      smallChip[oC], "for i = 1, 2, 4, from (7)-(10)"],
     nrow[12, Row[{"det(", nv[0], ", ", nv["i"], ", ",
        evec["A", "i+1", "A", "i"], ") = ", mulR[absR[nv[0]], absR[nv["i"]],
         pt["ℓ", "i"], Row[{"sin", thI["i"]}]]}], smallChip[oS], Row[
       {"for i = 1, 2, 4, from (7)-(10), with ", pt["A", 5], " := ", pt["A", 1],
        " and ", absR[evec["A", "i+1", "A", "i"]], " = ", pt["ℓ", "i"]}]]}];
  Print[Panel[Column[
     {Row[{Style[Row[{"the normals ", nv[0], ", ", nv[1], ", ", nv[2], ", ",
           nv[4]}], Bold, 15, col], Spacer[20], statusChip[chk]}], Style[Row[
        {"Verification: each normal is the cross product of two edge vectors
           of its own face; (11) and (12) identify the symbols ",
         thI[1], ", ", thI[2], ", ", thI[4], " of the coordinates with
           the dihedral angles of the definition along ",
         edg["A", 1, "A", 2], ", ", edg["A", 2, "A", 3], ", ",
         edg["A", 4, "A", 1], "."}], Bold, 13], vgrid[rows],
      Style[Row[{"Together, these give the four normals and identify ",
         thI[1], ", ", thI[2], ", ", thI[4], "."}], 13]},
     Spacings → 1], Background → Lighter[col, 0.93], FrameMargins →
     12]];];

(* ============================================================*)
(*PARTS 5 AND 6:THE TWO CORNER DIHEDRAL ANGLES*)
(* ============================================================*)
Do[Module[{i = spec⟦pnl, 1⟧, isB = spec⟦pnl, 2⟧ === "B", col = cols⟦pnl + 2⟧,
    tag, nearLet, ang1, fIdx, nms, oPv, oB, oP, oF, oS, chk, rows,
    sForm, wraw, wlen, wName}, tag = If[isB, nameB⟦i⟧, nameC⟦i⟧];
   nearLet = If[isB, "B", "C"];
   ang1 = If[isB, angA⟦i⟧, angG⟦i⟧];
   fIdx = If[isB, nvecIdx⟦i, 1⟧, nvecIdx⟦i, 2⟧];
   nms = nPname[i];
   sForm = If[isB, sinBform[i], sinCform[i]];
   wraw = If[isB, wB[i], wC[i]]; wlen = If[isB, bLen⟦i⟧, cLen⟦i⟧];
   wName = If[isB, If[wBout[i], evec["A", i, "B", i], evec["B", i, "A", i]],
     If[wCout[i], evec["A", i, "C", i], evec["C", i, "A", i]]];
   oPv = zeroT[nrawP[i] - nPshow[i]];
   oB = zeroT[(Bpt[i] - Apt[i]).(Cpt[i] - Apt[i]) - bLen⟦i⟧ cLen⟦i⟧ betaForm[i]];
```

```
oP = zeroT[
   (nrawP[i].nrawP[i] - bLen⟦i⟧^2 cLen⟦i⟧^2 (1 - Cos[angB⟦i⟧]^2)) /. vclose];
oF = zeroT[((nraw[fIdx].nrawP[i]) Sin[ang1] -
      If[isB, outBnum[i], outCnum[i]] nlen[fIdx] bLen⟦i⟧ cLen⟦i⟧) /. vclose];
oS = zeroT[
  Det[{nraw[fIdx], nrawP[i], wraw}] - sForm nlen[fIdx] bLen⟦i⟧ cLen⟦i⟧ wlen];
chk = oPv && oB && oP && oF && oS; AppendTo[resultsAll, chk];
rows = {nrow[1, Row[{nv[fIdx], " = ", disp[nlen[fIdx]], " ",
      disp[ndir[fIdx]], ",   ", absR[nv[fIdx]], " = ", disp[nlen[fIdx]]}],
    givenChip, "from the block on the normals"], vgap, nrow[2,
    Row[{nms⟦1⟧, " = ", disp[If[OddQ[i], bLen⟦i⟧, cLen⟦i⟧] nPfirst[i]]}],
    givenChip, "from the coordinates"],
  nrow[3, Row[{nms⟦2⟧, " = ", disp[If[OddQ[i], cLen⟦i⟧, bLen⟦i⟧] nPsec[i]]}],
    givenChip, "from the coordinates"],
  nrow[4, Row[{nvP[i], " = ", nms⟦1⟧, " × ", nms⟦2⟧, " = ", disp[nPshow[i]]}],
    smallChip[oPv], "from (2) and (3)"],
  nrow[5, Row[{"cos ", disp[angB⟦i⟧], " = ",
      frac[Row[{evec["A", i, "B", i], " · ", evec["A", i, "C", i]}], mulR[
        pt["m", i], pt["v", i]]], " = ", disp[betaForm[i]]}], smallChip[oB],
    Row[{"from (2) and (3); the closed forms below are written with ",
      disp[angB⟦i⟧]}]], nrow[6, Row[{absR[nvP[i]], " = ", mulR[pt["m", i],
      pt["v", i], Row[{"sin", disp[angB⟦i⟧]}]]}]], smallChip[oP],
    Row[{"from (4) and (5), with the positive radical because ",
      disp[angB⟦i⟧], " ∈ (0, π)"}]], vgap,
  nrow[7, Row[{"cos ", tag, " = ", frac[Row[{nv[fIdx], " · ", nvP[i]}], mulR[
        absR[nv[fIdx]], absR[nvP[i]]]]}], givenChip, "by the definition"],
  nrow[8, Row[{"cos ", tag, " = ", If[isB, frac[disp[outBnum[i]],
        disp[outBden[i]]], frac[disp[outCnum[i]], disp[outCden[i]]]]}],
    smallChip[oF], "from (1) and (4)-(7)"], vgap,
  nrow[9, Row[{wName, " = ", disp[wraw]}], givenChip,
    Row[{"the third vector of ", tag}]], nrow[10, Row[{"sin ", tag,
      " = ", frac[Row[{"det(", nv[fIdx], ", ", nvP[i], ", ", wName, ")"}],
       mulR[absR[nv[fIdx]], absR[nvP[i]], absR[wName]]]}],
    givenChip, "by the definition"], nrow[11, Row[
     {"sin ", tag, " = ", frac[disp[sForm], Row[{"sin", disp[angB⟦i⟧]}]]]}],
    smallChip[oS], "from (1), (4), (6), (9), and (10)"]};
Print[
 Panel[Column[{Row[{Style[Row[{tag, ", along ", edg["A", i, nearLet, i]}],
       Bold, 15, col], Spacer[20], statusChip[chk]}],
    Style[Row[{"Verification: the corner normal ", nvP[i],
       " from its two edge vectors, then the cosine and the sine of
         the angle between ", nv[fIdx], " and ", nvP[i],
       ". The closed forms are written with ", disp[angB⟦i⟧],
       ", which the coordinates do not contain: (5) gives its cosine, and
         (6) the length of ", nvP[i], ", which carries sin",
       disp[angB⟦i⟧], "."}], Bold, 13], vgrid[rows], Style[Row[
      {"Together, these give the cosine and the sine of ", tag, "."}], 13]},
```

```
      Spacings → 1], Background → Lighter[col, 0.93],
     FrameMargins → 12]];], {pnl, 2}];

(* ============================================================*)
(*PART 7:CLAIM 1,THE CLEARED IDENTITIES*)
(* ============================================================*)
Module[{col = cols⟦5⟧, oA, oQ3, oR1, oR2, oCQ, oCl1, c1S, c2S, c3S, c4S, cCS, lS,
   xS, uS, chk, rows}, (*the step from the cleared identity and the
   three ratios to the identity of the claim,with the coefficients,
  x,u,the cosine of the corner angle and the square of the half-
   cotangent as symbols*)oCl1 = Expand[Numerator[
      Together[((1 - cCS) (1 + uS xS) (xS lS - 1) - (1 + cCS) (1 + xS) (1 - uS xS lS) +
           (1 + uS xS) / c4S ((1 - cCS) (c3S lS + c4S) + (1 + cCS) (c1S lS + c2S))) /.
        {c3S → - xS c4S, c2S → c4S (1 + xS) / (1 + uS xS),
         c1S → - c4S uS xS (1 + xS) / (1 + uS xS)}]]] === 0;
  {oA, oQ3, oR1, oR2, oCQ} =
   Transpose[Table[Module[{i = spec⟦pnl, 1⟧, isB = spec⟦pnl, 2⟧ === "B", aa, bb,
     dd, gg, cv, nu, de, cosOp, quV, qV}, aa = If[isB, angA⟦i⟧, angG⟦i⟧];
      bb = angB⟦i⟧;
      dd = angD⟦i⟧;
      gg = If[isB, angG⟦i⟧, angA⟦i⟧];
      cv = biqC[aa, bb, dd, gg];
      nu = If[isB, outBnum[i], outCnum[i]];
      de = If[isB, outBden[i], outCden[i]];
      cosOp = Cos[If[isB, thPrev⟦i⟧, thNext⟦i⟧]];
      quV = If[isB, quY[i], quX[i]];
      qV = qBarD[i, isB] /. barRule[i];
      {zeroT[(de - nu) (cv⟦3⟧ (1 + cosOp) + cv⟦4⟧ (1 - cosOp)) +
         (de + nu) (cv⟦1⟧ (1 + cosOp) + cv⟦2⟧ (1 - cosOp))],
       zeroF[quV + cv⟦3⟧ / cv⟦4⟧], zeroF[cv⟦2⟧ / cv⟦4⟧ - (1 + quV) / (1 + quV quU[i])],
       zeroF[cv⟦1⟧ / cv⟦4⟧ + quU[i] quV (1 + quV) / (1 + quV quU[i])],
       And @@ Table[zeroF[cv⟦k⟧ - qV⟦k⟧], {k, 4}]}], {pnl, 2}]];
  cqList = oCQ;
  chk = And @@ Flatten[{oA, oQ3, oR1, oR2, oCQ, oCl1}];
  AppendTo[resultsAll, chk];
  rows = Flatten[
    Table[Module[{i = spec⟦pnl, 1⟧, isB = spec⟦pnl, 2⟧ === "B", tg, op, qu, aa, bb,
      dd, gg, cv, cnd, qD, lb, rf}, (*the lines of Part pnl are numbered pnl.1,
     pnl.2,...*)lb[k_] := ToString[pnl] <> "." <> ToString[k];
      rf[k_] := "(" <> lb[k] <> ")";
      tg = If[isB, nameB⟦i⟧, nameC⟦i⟧];
      op = If[isB, thI[thPrevIdx⟦i⟧], thI[thNextIdx⟦i⟧]];
      qu = If[isB, pt["y", i], pt["x", i]];
      aa = If[isB, angA⟦i⟧, angG⟦i⟧];
      bb = angB⟦i⟧;
      dd = angD⟦i⟧;
      gg = If[isB, angG⟦i⟧, angA⟦i⟧];
```

```
      cv = biqC[aa, bb, dd, gg];
      qD = qBarD[i, isB];
      cnd = Row[{"(", mulR[qu, cotSqTag[op]],
         " - 1)(1 - ", mulR[pt["u", i], qu, cotSqTag[op]], ")"}];
      {hrow[Row[{Style["Part " <> ToString[pnl] <> ".", Bold],
          "  At ", pt["A", i], ", the angle ", tg, ":"}]],
       prow[Row[{cfS[1], " := ", disp[cv[[1]]], ",   ", cfS[2], " := ",
          disp[cv[[2]]], ",   ", cfS[3], " := ", disp[cv[[3]]], ",   ",
          cfS[4], " := ", disp[cv[[4]]]}], givenChip, "definitions"],
       nrow[lb[1], Column[{Row[{cfS[1], " = ", disp[qD[[1]]], ",   ",
            cfS[2], " = ", disp[qD[[2]]], ","}], Row[
           {cfS[3], " = ", disp[qD[[3]]], ",   ", cfS[4], " = ", disp[qD[[4]]]}]},
         Spacings → 0.6], smallChip[oCQ[[pnl]]],
        Row[{"identities in the flat angles at ", pt["A", i]}]], nrow[
        lb[2], Row[{cfS[1], ", ", cfS[2], ", ", cfS[3], ", ", cfS[4], " ≠ 0"}],
        givenChip, Row[{"each factor in ", rf[1],
          " is the sine of half of a sum ±", gk["α", i], " ± ", gk["β", i],
          " ± ", gk["γ", i], " ± ", gk["δ", i], " ≠ 0 (mod 2π)"}]],
       nrow[lb[3], Row[{"(1 - cos ", tg, ")(", mulR[cfS[3], cotSqTag[op]],
          " + ", cfS[4], ") + (1 + cos ", tg, ")(", mulR[cfS[1], cotSqTag[op]],
          " + ", cfS[2], ") = 0"}], smallChip[oA[[pnl]]],
        Row[{"cos ", tg, " from its block and ", cotSqTag[op],
          " = (1 + cos", disp[If[isB, thPrev[[i]], thNext[[i]]]], ")/(1 - cos",
          disp[If[isB, thPrev[[i]], thNext[[i]]]], "), cleared"}]],
       nrow[lb[4], Row[{frac[cfS[3], cfS[4]], " = -", qu, ",   ",
          frac[cfS[2], cfS[4]], " = ", frac[Row[{"1 + ", qu}],
           Row[{"1 + ", mulR[pt["u", i], qu]}]], ",   ", frac[cfS[1], cfS[4]],
          " = -", frac[Row[{mulR[pt["u", i], qu], "(1 + ", qu, ")"}],
           Row[{"1 + ", mulR[pt["u", i], qu]}]]}],
        smallChip[oQ3[[pnl]] && oR1[[pnl]] && oR2[[pnl]]],
        Row[{"identities in the flat angles at ", pt["A", i]}]],
       nrow[lb[5], Row[{"(1 - cos ", tg, ")(1 + ", mulR[pt["u", i], qu],
          ")(", mulR[qu, cotSqTag[op]], " - 1) = (1 + cos ", tg, ")(1 + ",
          qu, ")(1 - ", mulR[pt["u", i], qu, cotSqTag[op]], ")"}],
        smallChip[oCl1 && oA[[pnl]] && oQ3[[pnl]] && oR1[[pnl]] && oR2[[pnl]]],
        Row[{rf[3], " divided by ", cfS[4], ", see ", rf[2], ", with ", rf[4],
          ", multiplied by 1 + ", mulR[pt["u", i], qu]}]], nrow[lb[6],
        Row[{"1 + ", qu, " ≠ 0,   1 + ", mulR[pt["u", i], qu], " > 0"}],
        givenChip, "given"],
       nrow[lb[7], Row[{"if ", cnd, " ≠ 0, then ", tg, " ≠ 0, ±π"}],
        givenChip, Row[{"by ", rf[5], ", cos ", tg, " = 1 would give (1 + ",
          qu, ")(1 - ", mulR[pt["u", i], qu, cotSqTag[op]], ") = 0, and cos ",
          tg, " = -1 would give (1 + ", mulR[pt["u", i], qu], ")(", mulR[qu,
           cotSqTag[op]], " - 1) = 0; both contradict the hypothesis, by ",
          rf[6]}]], vgap}], {pnl, 2}], 1];
Print[
 Panel[Column[{Row[{Style["Claim 1: the cleared identities", Bold, 15, col],
```

```
        Spacer[20], statusChip[chk]}],
      Style["Verification: at each vertex, the coefficients as products
         of sines and the three ratios as identities in the flat
         angles, the cleared identity as a trigonometric identity,
         and the step to the identity of the claim with the
         displayed quantities as symbols; (1.7) and (2.7) read off
         that the corner angles are neither 0 nor ±π.", Bold, 13],
      vgrid[rows], Style["Together, these give Claim 1.", 13]},
     Spacings → 1], Background →
     Lighter[col, 0.93],
    FrameMargins → 12]];];

(* ============================================================*)
(*PART 8:CLAIM 2,THE FORMULA FOR THE TWO CORNER ANGLES*)
(* ============================================================*)
Module[{col = cols⟦6⟧, hy, oCt, oW1, oW2, xI, perV, Fg, xS, uS, L, okPair,
   chk, rows}, xI = DisplayForm[StyleBox["x", FontSlant → "Italic"]];
  oCt = zeroT[Cos[hy] (1 - Cos[2 hy]) - Sin[hy] Sin[2 hy]];
  oW1 = zeroT[Sin[2 hy] (Sin[hy]^2 + Cos[hy]^2) - 2 Cos[hy] Sin[hy]];
  oW2 = zeroT[Cos[2 hy] (Cos[hy]^2 + Sin[hy]^2) - (Cos[hy]^2 - Sin[hy]^2)];
  perV =
   Table[Module[{i = spec⟦pnl, 1⟧, isB = spec⟦pnl, 2⟧ === "B", lb, rf, tag, oppTag,
     oppSym, othTag, othSym, LL, KK, KKK, bb, ang1, oth2, dd, brs, qD, qV, cv, nu,
     de, cosOp, quD, quV, qOth, qOthV, sForm, e0T, sigT, radT, flx, sgnR, eTerms,
     eHand, fmlP, aCo, wCo, vCo, dCo, pq, bsD, yNum, yDen, uyNum, uyDen, kn, kp,
     lq, wRule, EE, F1D, F2D, oE, oK, oCQ, oQ, oU, oR, oP2, oI1, oI2, oI3, oS1, oS2,
     oS3, oM1, oM2, oM3, oCl, stQ, stX, stB, stC1, stC2, stF, stM, stR, stS, cQuot,
     cBric, cR26, cR35, cR36, phiS, f1S, f2S, sbS, sa1S, lS, xT, uT, qd2S, qd4S,
     aaS, wwS, vvS, ddS, ppS, e0S, eiS, qS, sgtS, rdS, bsS, sFS, chkV, rowsV},
    (*the lines of Part pnl are numbered pnl.1,pnl.2,...*)
    lb[k_] := ToString[pnl] <> "." <> ToString[k];
    rf[k_] := "(" <> lb[k] <> ")";
    tag = If[isB, nameB⟦i⟧, nameC⟦i⟧];
    oppTag = If[isB, thI[pIdx⟦i⟧], thI[nIdx⟦i⟧]];
    othTag = If[isB, thI[nIdx⟦i⟧], thI[pIdx⟦i⟧]];
    oppSym = If[isB, thPrev⟦i⟧, thNext⟦i⟧];
    othSym = If[isB, thNext⟦i⟧, thPrev⟦i⟧];
    LL = cotSqTag[oppTag];
    KK = cotSqTag[othTag];
    bb = angB⟦i⟧;
    dd = angD⟦i⟧;
    KKK = DisplayForm[
      RowBox[{SuperscriptBox["cot", "3"], "(", bx[othTag], "/", "2", ")"}]];
    ang1 = If[isB, angA⟦i⟧, angG⟦i⟧];
    oth2 = If[isB, angG⟦i⟧, angA⟦i⟧];
    brs = If[isB,
      {bAD[i], bBD[i], bDD[i], bGD[i]}, {bGD[i], bBD[i], bDD[i], bAD[i]}];
```

```
qD = qBarD[i, isB];
qV = qD /. barRule[i];
cv = biqC[ang1, bb, dd, oth2];
nu = If[isB, outBnum[i], outCnum[i]];
de = If[isB, outBden[i], outCden[i]];
cosOp = Cos[oppSym];
quD = If[isB, qYr[i], qXr[i]];
qOth = If[isB, qXr[i], qYr[i]];
quV = If[isB, quY[i], quX[i]];
qOthV = If[isB, quX[i], quY[i]];
sForm = If[isB, sinBform[i], sinCform[i]];
EE = DisplayForm[StyleBox["Φ", FontSlant → "Italic"]];
F1D = DisplayForm[SubscriptBox["Φ", "1"]];
F2D = DisplayForm[SubscriptBox["Φ", "2"]];
e0T = e0Tag[If[isB, pIdx[[i]], nIdx[[i]]]];
sigT = sigTag[i];
radT = radTag[i, ! isB];
flx = If[isB, (-1) ^ (i + 1), (-1) ^ i];
sgnR = If[-flx < 0, "-", ""];
aCo = -Sin[bDD[i]] Sin[bb - bDD[i]];
dCo = Sin[sgD[i]] Sin[bBD[i]];
wCo = -Sin[brs[[1]]] Sin[bb - brs[[1]]];
vCo = If[isB, -Sin[bGD[i]] Sin[bb - bGD[i]], -Sin[bAD[i]] Sin[bb - bAD[i]]];
pq = Sin[angA[[i]]] Sin[angG[[i]]]; bsD = Sin[bBD[i]] Sin[sgD[i]];
yNum = Sin[brs[[4]]] Sin[brs[[3]]];
yDen = Sin[sgD[i]] Sin[brs[[1]] - bb];
uyNum = Sin[brs[[4]] - bb] Sin[brs[[3]] - bb];
uyDen = Sin[brs[[1]]] Sin[brs[[2]]];
eHand =
 Sin[dd - oth2] kn kp^2 - Sin[oth2] kn^2 kp + Sin[oth2 + dd] kn + Sin[oth2] kp;
eTerms = polyRow[{{Sin[dd - oth2], Row[{cotTag[othTag], " ", LL}]},
    {-Sin[oth2], Row[{KK, " ", cotTag[oppTag]}]},
    {Sin[oth2 + dd], cotTag[othTag]}, {Sin[oth2], cotTag[oppTag]}}];
fmlP = flxF[i, ! isB];
wRule = {Sin[othSym] → 2 kn / (1 + kn^2), Cos[othSym] → (kn^2 - 1) / (kn^2 + 1),
  Sin[oppSym] → 2 kp / (1 + kp^2), Cos[oppSym] → (kp^2 - 1) / (kp^2 + 1)};
(*identities in the angles*)
oE = zeroT[Together[(sForm /. wRule) (1 + kn^2) (1 + kp^2) / 2 - eHand]];
oK = zeroT[Together[((de - nu) /. wRule) (1 + kp^2) - 2 (cv[[1]] kp^2 + cv[[2]])]];
oCQ = cqList[[pnl]];
oQ = zeroF[quV - (yNum / yDen /. barRule[i])];
oU = zeroF[quU[i] quV - (uyNum / uyDen /. barRule[i])];
oR = zeroT[(de + nu) (cv[[1]] (1 + cosOp) + cv[[2]] (1 - cosOp)) +
    (de - nu) (cv[[3]] (1 + cosOp) + cv[[4]] (1 - cosOp))];
oP2 = zeroF[qV[[1]] + quU[i] quV qV[[2]]];
oI1 = zeroF[(aCo - vCo + Sin[ang1] Sin[dd - oth2]) /. barRule[i]];
oI2 = zeroF[(wCo - dCo + Sin[ang1] Sin[oth2 + dd]) /. barRule[i]];
```

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```
oI3 = zeroF[(qD[[4]] / qD[[2]] /. barRule[i]) - (1 + quU[i] quV) / (1 + quV)];
oS1 = zeroF[quU[i] qOthV quZ[i] - (wCo / bsD /. barRule[i])];
oS2 = zeroF[quU[i] quX[i] quY[i] - (aCo / bsD /. barRule[i])];
oS3 = zeroF[quU[i] quX[i] quY[i] (1 + quZ[i]) (1 + quU[i] quZ[i]) -
    (pq^2 / bsD^2 /. barRule[i])];
oM1 = zeroF[(bsD /. barRule[i]) qOthV quU[i] (quZ[i] + quV lq) -
    ((aCo lq + wCo) /. barRule[i])];
oM2 = zeroF[((bsD /. barRule[i])^2 quU[i] quX[i] quY[i]
      (1 + quZ[i]) (1 + quU[i] quZ[i])) - (pq^2 /. barRule[i])];
oM3 = zeroF[((bsD /. barRule[i])^2 quU[i] qOthV quZ[i]) -
    (qD[[2]] qD[[4]] /. barRule[i])];
(*the closure contains both the flat angles and their half sums,
so it is checked after halving the flat angles*)oCl =
 zeroF[Together[((uB[i].uC[i] - Cos[bb]) /. wRule) (1 + kn^2) (1 + kp^2) + 2
     (((aCo kp^2 + wCo) kn^2 - 2 pq kn kp + (vCo kp^2 + dCo)) /. barRule[i])]];
(*the steps between displayed lines,
with the displayed quantities as symbols;
kn and kp stand for the two half-cotangents*)
stQ = Together[(2 phiS / (sbS (1 + kn^2) (1 + lS))) / (2 f2S / (sa1S sbS (1 + lS))) -
    sa1S phiS / ((1 + kn^2) f2S)] === 0;
stX = Expand[Sin[ang1] eHand - (kn (Sin[ang1] Sin[dd - oth2] kp^2 +
          Sin[ang1] Sin[oth2 + dd]) + pq kp (1 - kn^2))] === 0;
stB = Expand[kn ((aaS lS + wwS) kn^2 - 2 ppS kn kp + (vvS lS + ddS)) -
    ((vvS lS + ddS) kn - (2 ppS kn^2 kp - (aaS lS + wwS) kn^3))] === 0;
stC1 = Expand[((2 ppS kn^2 kp - (aaS lS + wwS) kn^3) - kn (aaS lS + wwS) +
      ppS kp (1 - kn^2)) - (1 + kn^2) (ppS kp - (aaS lS + wwS) kn)] === 0;
stC2 =
 Together[(- f2S (sa1S phiS / ((1 + kn^2) f2S)) - ((aaS lS + wwS) kn - ppS kp)) /.
    phiS → (1 + kn^2) (ppS kp - (aaS lS + wwS) kn) / sa1S] === 0;
stF = And @@ Table[Together[
     (flx (ep kn qS - kp sgtS) /. kn → ep (kp sgtS + flx e0S eiS rdS) / qS) -
      e0S eiS rdS] === 0, {ep, {-1, 1}}];
stM =
 And @@ Table[Together[(flx e0S eiS sFS - (kn (aaS lS + wwS) - kp ppS)) - flx bsS
       ep ((e0S eiS rdS - flx (ep kn qS - kp sgtS)) /. {qS → (aaS lS + wwS) / bsS,
           sgtS → ep ppS / bsS, rdS → ep sFS / bsS})] === 0, {ep, {-1, 1}}];
stR = Together[((qd4S (xT lS - 1)) / (qd2S (1 - uT xT lS)) - (1 + uT xT) (xT lS - 1) /
        ((1 + xT) (1 - uT xT lS))) /. qd4S → qd2S (1 + uT xT) / (1 + xT)] === 0;
stS = Together[(Sqrt[f1S f2S] / f2S)^2 - f1S / f2S] === 0;
cQuot = stQ && oE && oK && oCQ && oCt; cBric = stB && oCl;
cR26 = stC1 && stC2 && stX && oI1 && oI2 && cBric && cQuot;
cR35 = stM && stF && oM1 && oM2 && oM3 && oS1 && oS2 && oS3 && cR26;
cR36 = stS && stR && oQ && oP2 && oI3 && cR35;
chkV = oE && oK && oCQ && oQ && oU && oR && oP2 &&
  oI1 && oI2 && oI3 && oS1 && oS2 && oS3 && oM1 && oM2 && oM3 && oCl &&
  stQ && stX && stB && stC1 && stC2 && stF && stM && stR && stS;
rowsV = {hrow[
```

```
 Row[{Style["Part " <> ToString[pnl] <> ".", Bold], "  At ", pt["A", i],
    ", the angle ", tag, " through ", oppTag, " and ", othTag, ":"}]],
nrow[lb[1], Row[{tag, " ≠ 0, ±π"}], givenChip,
 Row[{"Claim 1, because (1) and (2) give ",
    dTag[quD, cotTag[oppTag]], " ≠ 0"}]], nrow[lb[2],
 Row[{"sin ", tag, " = ", frac[disp[sForm], Row[{"sin", disp[bb]}]],
    ",      cos ", tag, " = ", frac[disp[nu], disp[de]]}],
 givenChip, Row[{"from the block on ", tag}]],
nrow[lb[3], Row[{"sin ", tag, " sin", disp[bb], " ",
    frac[Row[{"(1 + ", KK, ")(1 + ", LL, ")"}], "2"], " = ", eTerms}],
 smallChip[oE], Row[{rf[2], ", with (3)"}]],
nrow[lb[4], Row[{"(1 - cos ", tag, ") sin", disp[ang1],
    " sin", disp[bb], " ", frac[Row[{"1 + ", LL}], "2"], " = ",
    polyRow[{{qD[[1]], LL}, {qD[[2]], None}}]}], smallChip[oK && oCQ],
 Row[{rf[2], ", with (3) and ", rf[1], " of the block on Claim 1"}]],
prow[Row[{EE, " := ", eTerms}], givenChip,
 Row[{"a name for the right side of ", rf[3]}]],
prow[Row[{F2D, " := ", polyRow[{{qD[[1]], LL}, {qD[[2]], None}}]}],
 givenChip, Row[{"a name for the right side of ", rf[4]}]],
nrow[lb[5], Row[{cotTag[tag], " = ", frac[Row[{"sin ", tag}], Row[
      {"1 - cos ", tag}]], " = ", frac[Row[{"sin", disp[ang1], " ", EE}],
     Row[{"(1 + ", KK, ") ", F2D}]]}], smallChip[cQuot],
 Row[{"the quotient of ", rf[3], " and ", rf[4], ", by (3); here ",
    F2D, " ≠ 0 by ", rf[4], ", because ", tag, " ≠ 0 by ", rf[1]}]],
vgap, nrow[lb[6], Row[{quD, " = ", frac[disp[yNum], disp[yDen]],
    ",      ", mulR[uPlain, quD], " = ",
    frac[disp[uyNum], disp[uyDen]]}], smallChip[oQ && oU],
 Row[{"identities in the flat angles at ", pt["A", i]}]], nrow[lb[7],
 Row[{F2D, " = ", disp[qD[[2]]], " (1 - ", mulR[uPlain, quD, LL], ")"}],
 smallChip[oP2], Row[{"by ", rf[6]}]],
prow[Row[{F1D, " := ", polyRow[{{-qD[[3]], LL}, {-qD[[4]], None}}]}],
 givenChip, "a name"],
nrow[lb[8], Row[{F1D, " = ", disp[qD[[4]]], " (", mulR[quD, LL], " - 1)"}],
 smallChip[oQ], Row[{"by ", rf[6]}]],
nrow[lb[9], Row[{frac[disp[qD[[4]]], disp[qD[[2]]]], " = ", frac[
     Row[{"1 + ", mulR[uPlain, quD]}], Row[{"1 + ", quD}]]}], smallChip[
  oI3], Row[{"an identity in the flat angles at ", pt["A", i]}]],
nrow[lb[10], Row[{cotSqTag[tag], " = ", frac[F1D, F2D]}],
 smallChip[oR && oCQ], Row[{"from ", rf[2],
    ", cleared, with ", rf[1], " of the block on Claim 1"}]],
nrow[lb[11], Row[{mulR[F1D, F2D], " = ", mulR[disp[qD[[4]]], disp[qD[[2]]],
    " ", dTag[quD, cotTag[oppTag]]}], smallChip[oQ && oP2],
 Row[{rf[7], " times ", rf[8]}]], vgap, nrow[lb[12],
 Row[{"sin", disp[brs[[1]]], " > 0,   sin", disp[brs[[2]]], " > 0"}],
 givenChip, Row[{disp[brs[[1]]], ", ", disp[brs[[2]]], " ∈ (0, π)"}]],
nrow[lb[13], Row[{"if ", quD, " < 0:   ", mulR[quD, LL],
    " - 1 < 0,   so ", mulR[quD, dTag[quD, cotTag[oppTag]]],
```

```
    " > 0 gives 1 - ", mulR[uPlain, quD, LL], " > 0"}],
  givenChip, Row[{LL, " ≥ 0"}]], nrow[lb[14],
  Row[{"if ", quD, " > 0 and 1 - ", mulR[uPlain, quD, LL], " ≤ 0:   ",
    mulR[quD, dTag[quD, cotTag[oppTag]]], " > 0 gives ", mulR[quD, LL],
    " < 1 ≤ ", mulR[uPlain, quD, LL], ",   so ", uPlain, " ≥ 1"}],
  givenChip, Row[{"against ", uPlain, " = 1 - ", pt["M", i], ",   ",
    pt["M", i], " > 0"}]], nrow[lb[15], Row[{F2D, " > 0"}], givenChip,
  Row[{rf[7], ", ", rf[12], "-", rf[14], ", and (2)"}]], vgap, nrow[
  lb[16], Column[{Row[{"(", polyRow[{{aCo, LL}, {wCo, None}}], ") ", KK}],
    Row[{"   - 2 ", mulR[disp[pq], cotTag[othTag], cotTag[oppTag]]}],
    Row[{"   + (", polyRow[{{vCo, LL}, {dCo, None}}], ") = 0"}]},
   Spacings → 0.6], smallChip[oCl],
  Row[{"the closure ", evec["A", i, "B", i], " · ", evec["A", i, "C", i],
    " = ", mulR[pt["m", i], pt["v", i], Row[{"cos", disp[bb]}]],
    " (Bricard's equation), with (3)"}]],
 nrow[lb[17], Row[{disp[aCo], " - ", disp[vCo], " = ",
    disp[-Sin[ang1] Sin[dd - oth2]]}], smallChip[oI1],
  Row[{"an identity in the flat angles at ", pt["A", i]}]],
 nrow[lb[18], Row[{disp[wCo], " - ", disp[dCo],
    " = ", disp[-Sin[ang1] Sin[oth2 + dd]]}],
  smallChip[oI2], "the same"], nrow[lb[19],
  Row[{"sin", disp[ang1], " ", EE, " = ", cotTag[othTag], " (", polyRow[
     {{Sin[ang1] Sin[dd - oth2], LL}, {Sin[ang1] Sin[oth2 + dd], None}}],
    ") + ", mulR[disp[pq], cotTag[oppTag]], " (1 - ", KK, ")"}],
  smallChip[stX], Row[{EE, " expanded"}]],
 nrow[lb[20], Row[{"sin", disp[ang1], " ", EE, " = ",
    cotTag[othTag], " ((", polyRow[{{vCo, LL}, {dCo, None}}],
    ") - (", polyRow[{{aCo, LL}, {wCo, None}}], ")) + ",
    mulR[disp[pq], cotTag[oppTag]], " (1 - ", KK, ")"}], smallChip[
   stX && oI1 && oI2], Row[{rf[19], ", with ", rf[17], " and ", rf[18]}]],
 nrow[lb[21], Row[{"(", polyRow[{{vCo, LL}, {dCo, None}}], ") ",
    cotTag[othTag], " = 2 ", mulR[disp[pq], KK, cotTag[oppTag]], " - (",
    polyRow[{{aCo, LL}, {wCo, None}}], ") ", KKK}], smallChip[cBric],
  Row[{rf[16], " times ", cotTag[othTag]}]], nrow[lb[22], Column[
   {Row[{frac[Row[{"sin", disp[ang1], " ", EE}], Row[{"1 + ", KK}]],
      " = ", mulR[cotTag[oppTag], disp[pq]], " - ", cotTag[othTag], " (",
      polyRow[{{aCo, LL}, {wCo, None}}], "),"}], Row[{"      that is   ",
      cotTag[othTag], " (", polyRow[{{aCo, LL}, {wCo, None}}], ") - ",
      mulR[cotTag[oppTag], disp[pq]], " = -", mulR[F2D, cotTag[tag]]}]},
   Spacings → 0.6], smallChip[cR26], Row[{rf[20], " and ",
    rf[21], ", divided by 1 + ", KK, "; then ", rf[5]}]], vgap,
 nrow[lb[23], fmlP, givenChip, Row[{"the parametrization by ", oppTag}]],
 nrow[lb[24], Column[{Row[{mulR[e0T, pt["e", i], radT], " ="}],
    Row[{"      ", If[flx < 0, "-", ""], mulR[gk["ε", i], cotTag[othTag],
       qOth, uPlain], "(", pt["z", i], " + ", mulR[quD, LL], ")", If[flx <
        0, " + ", " - "], mulR[cotTag[oppTag], sigT]}]}, Spacings → 0.6],
  smallChip[stF], Row[{rf[23], ", solved for its radical"}]],
```

```
      nrow[lb[25], Row[{mulR[uPlain, qOth, pt["z", i]], " = ",
         frac[disp[wCo], disp[bsD]]}], smallChip[oS1],
       Row[{"an identity in the flat angles at ", pt["A", i]}]],
      nrow[lb[26], Row[{mulR[uPlain, qXr[i], qYr[i]], " = ",
         frac[disp[aCo], disp[bsD]]}], smallChip[oS2], "the same"],
      nrow[lb[27], Row[{mulR[uPlain, qXr[i], qYr[i]], "(1 + ",
         pt["z", i], ")(1 + ", mulR[uPlain, pt["z", i]], ") = ",
         frac[disp[pq^2], disp[bsD^2]]}], smallChip[oS3], "the same"],
      nrow[lb[28], Row[{disp[bsD], " ", mulR[qOth, uPlain], "(", pt["z", i],
         " + ", mulR[quD, LL], ") = ", polyRow[{{aCo, LL}, {wCo, None}}]}],
       smallChip[oM1], Row[{rf[25], " and ", rf[26]}]], nrow[lb[29],
       Row[{disp[bsD], " ", sigT, " = ", mulR[gk["ε", i], disp[pq]]}],
       smallChip[oM2], Row[{rf[27], ", as squares; ", gk["ε", i],
         " = sign sin", disp[sgD[i]], " and ", disp[bBD[i]], " ∈ (0, π)"}]],
      nrow[lb[30], Row[{disp[bsD], " ", radT, " = ", mulR[gk["ε", i],
          DisplayForm[SqrtBox[bx[mulR[F1D, F2D]]]]]}], smallChip[oM3], Row[
        {rf[25], " and ", rf[11], ", as squares; the signs as in ", rf[29]}]],
      nrow[lb[31], Column[{Row[{If[flx < 0, "-", ""], mulR[e0T, pt["e", i]],
           " ", DisplayForm[SqrtBox[bx[mulR[F1D, F2D]]]], " ="}], Row[
          {"      ", cotTag[othTag], " (", polyRow[{{aCo, LL}, {wCo, None}}],
           ") - ", mulR[cotTag[oppTag], disp[pq]], " = -",
           mulR[F2D, cotTag[tag]]}]}, Spacings → 0.6], smallChip[cR35],
       Row[{rf[24], " multiplied by ", disp[bsD], " and divided by ",
         gk["ε", i], ", with ", rf[28], "-", rf[30], "; then ", rf[22]}]],
      nrow[lb[32], Row[{cotTag[tag], " = ", sgnR, mulR[e0T, pt["e", i]],
         " ", DisplayForm[SqrtBox[FractionBox[SubscriptBox["Φ", "1"],
            SubscriptBox["Φ", "2"]]]], " = ", sgnR, mulR[e0T, pt["e", i]], " ",
         DisplayForm[SqrtBox[bx[ratForm[quD, uPlain, oppTag]]]]}],
       smallChip[cR36], Row[{rf[31], " divided by ", F2D,
         " > 0, see ", rf[15], "; then ", rf[7], "-", rf[9]}]]};
   {chkV, rowsV}], {pnl, 2}];
(*the radicand of each corner angle at its own vertex;
under the flexion the two coincide*)
Fg[qq_, uq_] := (1 + uq qq) (qq L - 1) / ((1 + qq) (1 - uq qq L));
okPair = Expand[Numerator[Together[(Fg[xS[1], uS[1]] - Fg[xS[2], uS[2]]) /.
      {xS[2] → xS[1], uS[1] → uS[0], uS[2] → uS[0]}]]] === 0;
chk = oCt && oW1 && oW2 && perV[[1, 1]] && perV[[2, 1]] && okPair;
AppendTo[resultsAll, chk];
rows =
 Join[{nrow[1, Row[{pt["x", 2], " = ", pt["x", 1], ",   ", pt["u", 1], " = ",
      pt["u", 2], " = ", uPlain}], givenChip,
    "the flexion; the lines below are written with these values"],
   nrow[2, Row[{cotTag[thI[1]], " ∈ ", iSet[pt["x", 1]]}],
    givenChip, "the admissible set"], vgap, nrow[3,
    Row[{cotTag[xI], " = ", frac[Row[{"sin", xI}], Row[{"1 - cos", xI}]],
      ",      sin", xI, " = ", frac[Row[{"2 ", cotTag[xI]}],
       Row[{"1 + ", cotSqTag[xI]}]], ",      cos", xI, " = ",
```

```
      frac[Row[{cotSqTag[xI], " - 1"}], Row[{cotSqTag[xI], " + 1"}]]}],
    smallChip[oCt && oW1 && oW2], Row[{"for ", xI, " ≠ 0 (mod 2π)"}]], vgap},
  perV[[1, 2]], {vgap}, perV[[2, 2]], {vgap, nrow[4,
    Row[{ratForm[pt["x", 1], pt["u", 1], thI[1]], " = ", ratForm[pt["x", 2],
       pt["u", 2], thI[1]], " = ", ratForm[pt["x", 1], uPlain, thI[1]]}],
    smallChip[okPair], "the radicands of (1.32) and (2.32), under (1)"],
   nrow[5, Row[{mulR[pt["e", 1], cotTag[thP[4, 1]]], " = -",
      mulR[pt["e", 2], cotTag[thP[2, 2]]], " = ", mulR[e0Tag[1],
       DisplayForm[SqrtBox[bx[ratForm[pt["x", 1], uPlain, thI[1]]]]]]}],
    smallChip[chk], "(1.32) and (2.32), with (4)"]}];
Print[
 Panel[Column[{Row[{Style["Claim 2: the formula for the two corner angles",
       Bold, 15, col], Spacer[20], statusChip[chk]}],
    Style[Row[{"Verification: at each vertex, the sine and 1 - cos
          of the corner angle through the two half-angle
          cotangents give its half-cotangent as a quotient;
          Bricard's equation reduces the numerator to
          the radical of the parametrization by ", thI[1],
        ", which fixes the sign. Identities in the angles are checked
          as trigonometric identities, and each step between
          lines with the displayed quantities as symbols."}],
     Bold, 13], vgrid[rows], Style["Together, these give Claim 2.",
     13]}, Spacings → 1],
  Background → Lighter[col, 0.93], FrameMargins →
   12]];];

(* ============================================================*)
(*PART 9:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 11: ALL CLAIMS VERIFIED",
     "Section 11: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses and definitions:

A 3 x 3 net with, for each i = 1, 2,

the vertices $A_i$, $B_i$, $C_i$, and the vertices $A_3$, $A_4$ of the central face

the flat angles $\alpha_i, \beta_i, \gamma_i, \delta_i \in (0, \pi)$ at $A_i$

the edge lengths $\ell_1 = |A_1 A_2|$, $\ell_2 = |A_2 A_3|$, $\ell_4 = |A_4 A_1|$, $m_i = |A_i B_i|$, $v_i = |A_i C_i|$

the dihedral angles $\theta_1, \theta_2, \theta_4 \in [-\pi, \pi) \setminus \{0\}$ along the edges $A_1 A_2$, $A_2 A_3$, $A_4 A_1$

$\sigma_i = \frac{1}{2}(\alpha_i + \beta_i + \gamma_i + \delta_i)$ $\quad \overline{\alpha}_i = \sigma_i - \alpha_i$ $\quad \overline{\beta}_i = \sigma_i - \beta_i$ $\quad \overline{\gamma}_i = \sigma_i - \gamma_i$ $\quad \overline{\delta}_i = \sigma_i - \delta_i$

$a_i = \frac{\sin \alpha_i}{\sin \overline{\alpha}_i}$ $\quad b_i = \frac{\sin \beta_i}{\sin \overline{\beta}_i}$ $\quad c_i = \frac{\sin \gamma_i}{\sin \overline{\gamma}_i}$ $\quad d_i = \frac{\sin \delta_i}{\sin \overline{\delta}_i}$ $\quad \varepsilon_i = \operatorname{sign}(\sin \sigma_i)$

$M_i = a_i b_i c_i d_i$ $\quad r_i = a_i d_i$ $\quad s_i = c_i d_i$ $\quad f_i = a_i c_i$

$u_i = 1 - M_i$ $\quad x_i = \frac{1}{\ldots}$ $\quad y_i = \frac{1}{\ldots}$ $\quad z_i = \frac{1}{\ldots}$

$u_i = 1 - M_i$ $\quad x_i = \frac{}{r_i - 1}$ $\quad y_i = \frac{}{s_i - 1}$ $\quad z_i = \frac{}{f_i - 1}$

$\alpha_i \pm \beta_i \pm \gamma_i \pm \delta_i \neq 0 \pmod{2\pi}$

$\overline{\alpha}_i, \overline{\beta}_i, \overline{\gamma}_i, \overline{\delta}_i \in (0, \pi)$

$\varepsilon_i \in \{-1, 1\}$

$r_i, s_i, f_i, M_i \neq 0, 1; \quad M_i, f_i > 0$

$x_i, y_i, z_i \neq -1, 0; \quad u_i x_i y_i z_i > 0$

$x_i(1 + x_i) > 0, \quad z_i(1 + z_i) > 0$

$1 + u_i x_i > 0, \quad 1 + u_i z_i > 0$

Normals:

$\vec{n}_0 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 A_3}$

$\vec{n}_1 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 B_2}$ $\qquad \vec{n}_2 = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 A_3}$ $\qquad \vec{n}_4 = \overrightarrow{A_1 A_4} \times \overrightarrow{A_1 C_1}$

$\vec{n}_{1'} = \overrightarrow{A_1 B_1} \times \overrightarrow{A_1 C_1}$ $\qquad \vec{n}_{2'} = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 B_2}$

Dihedral angles:

$$\theta(\vec{w}_1, \vec{w}_2, \vec{w}_3) := \begin{cases} \angle(\vec{w}_1, \vec{w}_2) \in (0, \pi), & \text{if } \det(\vec{w}_1, \vec{w}_2, \vec{w}_3) > 0, \\ -\angle(\vec{w}_1, \vec{w}_2) \in [-\pi, 0], & \text{if } \det(\vec{w}_1, \vec{w}_2, \vec{w}_3) \leq 0. \end{cases}$$

*central dihedral angles:*

$\theta_1 = \theta(\vec{n}_0, \vec{n}_1, \overrightarrow{A_2 A_1})$ $\qquad \theta_2 = \theta(\vec{n}_0, \vec{n}_2, \overrightarrow{A_3 A_2})$ $\qquad \theta_4 = \theta(\vec{n}_0, \vec{n}_4, \overrightarrow{A_1 A_4})$

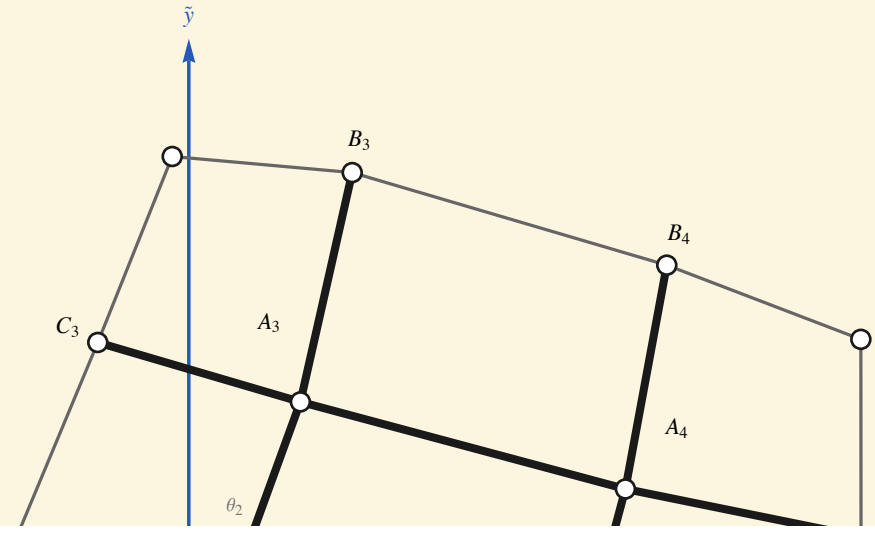
*corner dihedral angles:*

$\theta_{41'} = \theta(\vec{n}_4, \vec{n}_{1'}, \overrightarrow{A_1 C_1})$ $\qquad \theta_{22'} = \theta(\vec{n}_2, \vec{n}_{2'}, \overrightarrow{C_2 A_2})$

Cartesian coordinates:

origin at $A_2$

central face in the half $-$plane $\tilde{y} \geq 0$ of the $\tilde{x}\tilde{y}-$plane, the ray $A_2 A_1$ as the $\tilde{x}-$axis



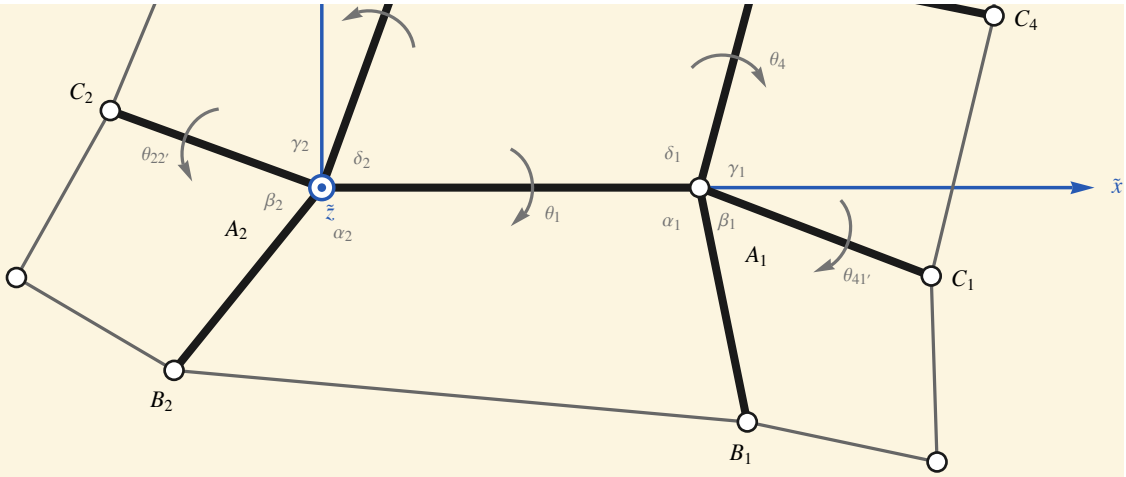



Figure 1

$A_1 = \{\ell_1, 0, 0\}$
$A_2 = \{0, 0, 0\}$
$A_3 = \{\ell_2 \cos\delta_2, \ell_2 \sin\delta_2, 0\}$
$A_4 = \{\ell_1 - \ell_4 \cos\delta_1, \ell_4 \sin\delta_1, 0\}$
$B_1 = A_1 + m_1\ \{-\cos\alpha_1, \sin\alpha_1 \cos\theta_1, \sin\alpha_1 \sin\theta_1\}$
$B_2 = A_2 + m_2\ \{\cos\alpha_2, \sin\alpha_2 \cos\theta_1, \sin\alpha_2 \sin\theta_1\}$
$C_1 = A_1 + v_1\ \{-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1, \cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4, \sin\gamma_1 \sin\theta_4\}$
$C_2 = A_2 + v_2\ \{\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2, \cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2, \sin\gamma_2 \sin\theta_2\}$

The definition gives $\theta(\vec{w}_1, \vec{w}_2, \vec{w}_3) = \pm\angle(\vec{w}_1, \vec{w}_2$ ), by the sign of the determinant. The cosine is even and the sine is odd, so cos $\theta = \frac{\vec{w}_1 \cdot \vec{w}_2}{|\vec{w}_1|\ |\vec{w}_2|}$ and sin $\theta$ $\frac{\det(\vec{w}_1, \vec{w}_2, \vec{w}_3)}{|\vec{w}_1|\ |\vec{w}_2|\ |\vec{w}_3|}$ hold whichever sign the determinant has; the formula for the sine holds because $w_3$, the direction of an edge common to the two faces, is orthogonal to $w_1$ and $w_2$, so that $w_1 \times w_2$ is a multiple of $w_3$. No formula of the notebook distinguishes the two cases, the coordinates above i

Flexion:

$x_2 = x_1, \quad u_1 = u_2 = u$

$e_1, e_2, e_0 \in \{-1, 1\}$

$D(X, t) = (Xt^2 - 1)(1 - uXt^2), \qquad I(X) = \{t \in \mathbb{R} : X\ D(X, t) > 0\}, \qquad \text{for } X,\ t \in \mathbb{R}$

$$\cot(\theta_4/2) = \frac{\varepsilon_1 (\cot(\theta_1/2)\ \sqrt{u x_1 y_1 (1 + z_1)(1 + u z_1)} - e_0\ e_1\ \sqrt{u y_1 z_1\ D(x_1, \cot(\theta_1/2))}\ )}{y_1 u (z_1 + x_1 \cot^2(\theta_1/2))}$$

$$\cot(\theta_2/2) = \frac{\varepsilon_2 (\cot(\theta_1/2)\ \sqrt{u x_1 y_2 (1 + z_2)(1 + u z_2)} + e_0\ e_2\ \sqrt{u y_2 z_2\ D(x_1, \cot(\theta_1/2))}\ )}{y_2 u (z_2 + x_1 \cot^2(\theta_1/2))}$$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at ✕ would mean that this verification failed. A • marks a given fact, a definition, a result of an earlier block, or an inference stated in words beside it.

We want to test the following claims ★ :

Claim 1.

- $(1-\cos\ \theta_{41'})(1+u_1 x_1)(x_1 \cot^2(\theta_1/2)-1) = (1+\cos\ \theta_{41'})(1+x_1)(1-u_1 x_1 \cot^2(\theta_1/2))$
- $(1-\cos\ \theta_{22'})(1+u_2 x_2)(x_2 \cot^2(\theta_1/2)-1) = (1+\cos\ \theta_{22'})(1+x_2)(1-u_2 x_2 \cot^2(\theta_1/2))$

In particular,

- $\theta_{41'} \neq 0, \pm\pi$ if $(x_1 \cot^2(\theta_1/2)-1)(1-u_1 x_1 \cot^2(\theta_1/2)) \neq 0$
- $\theta_{22'} \neq 0, \pm\pi$ if $(x_2 \cot^2(\theta_1/2)-1)(1-u_2 x_2 \cot^2(\theta_1/2)) \neq 0$

Claim 2.

If $x_2 = x_1$, $u_1 = u_2 = u$, and $\cot(\theta_1/2) \in I(x_1)$, then

- $e_1 \cot(\theta_{41'}/2) = -e_2 \cot(\theta_{22'}/2) = e_0 \sqrt{\frac{(1+u x_1)(x_1 \cot^2(\theta_1/2)-1)}{(1+x_1)(1-u x_1 \cot^2(\theta_1/2))}}$

## the coordinates

Verified

Verification: the lengths and the flat angles at $A_1$, $A_2$ from dot products, and the side of $B_1$ from two cross products; with $\theta_1$, $\theta_2$, $\theta_4$ from the next block, these data fix every point.

(1) $\overrightarrow{A_2 A_1} = \{\ell_1, 0, 0\}$ • from the coordinates

(2) $\overrightarrow{A_2 A_3} = \{\ell_2 \cos\delta_2, \ell_2 \sin\delta_2, 0\}$ • from the coordinates

(3) $\overrightarrow{A_1 A_2} = \{-\ell_1, 0, 0\}$ • from the coordinates

(4) $\overrightarrow{A_1 A_4} = \{\ell_4 (-\cos\delta_1), \ell_4 \sin\delta_1, 0\}$ • from the coordinates

(5) $\overrightarrow{A_1 B_1} = \{m_1 (-\cos\alpha_1), m_1 \sin\alpha_1 \cos\theta_1, m_1 \sin\alpha_1 \sin\theta_1\}$ • from the coordinates

(6) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2, m_2 \sin\alpha_2 \cos\theta_1, m_2 \sin\alpha_2 \sin\theta_1\}$ • from the coordinates

(7) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1), v_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4), v_1 \sin\gamma_1 \sin\theta_4\}$ • from the coordinates

(8) $\overrightarrow{A_2 C_2} = \{v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2), v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2), v_2 \sin\gamma_2 \sin\theta_2\}$ • from the coordinates

(9) $|\overrightarrow{A_2 A_3}| = \ell_2$, $|\overrightarrow{A_1 A_4}| = \ell_4$ ☑ the lengths of (2) and (4)

(10) $|\overrightarrow{A_1 B_1}| = m_1$, $|\overrightarrow{A_2 B_2}| = m_2$ ☑ the lengths of (5) and (6)

(11) $|\overrightarrow{A_1 C_1}| = v_1$, $|\overrightarrow{A_2 C_2}| = v_2$ ☑ the lengths of (7) and (8)

(11) $|A_1 C_1| = v_1$, $|A_2 C_2| = v_2$ ☑ the lengths of (7) and (8)

(12) $\cos \delta_1 = \frac{\overrightarrow{A_1 A_4} \cdot \overrightarrow{A_1 A_2}}{\ell_4 \ell_1}$, $\cos \delta_2 = \frac{\overrightarrow{A_2 A_1} \cdot \overrightarrow{A_2 A_3}}{\ell_1 \ell_2}$ ☑ from (1)–(4) and (9); flat angles in $(0, \pi)$ are fixed by their cosines

(13) $\cos \alpha_1 = \frac{\overrightarrow{A_1 A_2} \cdot \overrightarrow{A_1 B_1}}{\ell_1 m_1}$, $\cos \alpha_2 = \frac{\overrightarrow{A_2 A_1} \cdot \overrightarrow{A_2 B_2}}{\ell_1 m_2}$ ☑ from (1), (3), (5), (6), and (10)

(14) $\cos \gamma_1 = \frac{\overrightarrow{A_1 A_4} \cdot \overrightarrow{A_1 C_1}}{\ell_4 v_1}$, $\cos \gamma_2 = \frac{\overrightarrow{A_2 A_3} \cdot \overrightarrow{A_2 C_2}}{\ell_2 v_2}$ ☑ from (2), (4), (7), (8), (9), and (11)

(15) the $\tilde{y}$−coordinates of $A_3$, $A_4$ are $\ell_2 \sin \delta_2$, $\ell_4 \sin \delta_1 > 0$ • from (2) and (4), because $\delta_1, \delta_2 \in (0, \pi)$

(16) $\overrightarrow{A_2 A_1} \times \overrightarrow{A_2 B_2} = m_2 \ell_1 \sin \alpha_2 \{0, -\sin \theta_1, \cos \theta_1\}$ ☑ from (1) and (6)

(17) $\overrightarrow{A_2 A_1} \times \overrightarrow{A_1 B_1} = m_1 \ell_1 \sin \alpha_1 \{0, -\sin \theta_1, \cos \theta_1\}$ ☑ from (1) and (5); with (16), $B_1$ and $B_2$ lie on one side of $A_1 A_2$, because $\sin \alpha_1, \sin \alpha_2 > 0$

Together, these give the lengths and the flat angles $\alpha_i, \gamma_i, \delta_i$, i = 1, 2, of the coordinates, with $B_1$ on the side of $A_1 A_2$ where $B_2$ lies.

the normals $n \vec{}_0, \vec{n}_1, \vec{n}_2, \vec{n}_4$ **Verified**

Verification: each normal is the cross product of two edge vectors of its own face; (11) and (12) identify the symbols $\theta_1$, $\theta_2$, $\theta_4$ of the coordinates with the dihedral angles of the definition along $A_1 A_2$, $A_2 A_3$, $A_4 A_1$.

(1) $\overrightarrow{A_2 A_1} = \{\ell_1, 0, 0\}$ • from the coordinates

(2) $\overrightarrow{A_2 A_3} = \{\ell_2 \cos\delta_2, \ell_2 \sin\delta_2, 0\}$ • from the coordinates

(3) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2, m_2 \sin\alpha_2 \cos\theta_1, m_2 \sin\alpha_2 \sin\theta_1\}$ • from the coordinates

(4) $\overrightarrow{A_2 C_2} = \{v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2), v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2), v_2 \sin\gamma_2 \sin\theta_2\}$ • from the coordinates

(5) $\overrightarrow{A_1 A_4} = \{\ell_4 (-\cos\delta_1), \ell_4 \sin\delta_1, 0\}$ • from the coordinates

(6) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1), v_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4), v_1 \sin\gamma_1 \sin\theta_4\}$ • from the coordinates

(7) $\vec{n}_0 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 A_3} = \ell_1 \ell_2 \sin\delta_2$ ☑ from (1) and (2)
$\{0, 0, 1\}$, $|\vec{n}_0| = \ell_1 \ell_2 \sin\delta_2$

(8) $\vec{n}_1 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 B_2} =$ ☑ from (1) and (3)
$m_2 \ell_1 \sin\alpha_2 \{0, -\sin\theta_1, \cos\theta_1\}$, $|\vec{n}_1| = m_2 \ell_1 \sin\alpha_2$

(9) $\vec{n}_2 = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 A_3} = v_2 \ell_2 \sin\gamma_2$ ☑ from (4) and (2)
$\{-\sin\delta_2 \sin\theta_2, \cos\delta_2 \sin\theta_2, \cos\theta_2\}$, $|\vec{n}_2| = v_2 \ell_2 \sin\gamma_2$

(10) $\vec{n}_4 = \overrightarrow{A_1 A_4} \times \overrightarrow{A_1 C_1} = v_1 \ell_4 \sin\gamma_1$ ☑ from (5) and (6)
$\{\sin\delta_1 \sin\theta_4, \cos\delta_1 \sin\theta_4, \cos\theta_4\}$, $|\vec{n}_4| = v_1 \ell_4 \sin\gamma_1$

(11) $\vec{n}_0 \cdot \vec{n}_i = |\vec{n}_0| \, |\vec{n}_i| \cos\theta_i$ ☑ for i = 1, 2, 4, from (7)–(10)

(12) $\det(\vec{n}_0, \vec{n}_i, \overrightarrow{A_{i+1} A_i}) = |\vec{n}_0| \, |\vec{n}_i| \, \ell_i \sin\theta_i$ ☑ for i = 1, 2, 4, from (7)–(10), with $A_5 := A_1$ and $|\overrightarrow{A_{i+1} A_i}| = \ell_i$

Together, these give the four normals and identify $\theta_1$, $\theta_2$, $\theta_4$.

$\theta_{41}$′, along $A_1 C_1$ **Verified**

Verification: the corner normal $n \vec{}_{1'}$ from its two edge vectors, then the cosine and the sine of the angle between

from its two edge vectors , then the cosine and the sine of the angle between $\vec{n}_4$ and $\vec{n}_{1'}$. The closed forms are written with $\beta_1$ , which the coordinates do not contain: (5) gives its cosine, and (6) the length of $\vec{n}_{1'}$, which carries sin $\beta_1$.

(1) $\vec{n}_4 = v_1 \ell_4 \sin\gamma_1 \{\sin\delta_1 \sin\theta_4, \cos\delta_1 \sin\theta_4, \cos\theta_4\}$ , $|\vec{n}_4| = v_1 \ell_4 \sin\gamma_1$ — from the block on the normals

(2) $\overrightarrow{A_1 B_1} = \{m_1 (-\cos\alpha_1), m_1 \sin\alpha_1 \cos\theta_1, m_1 \sin\alpha_1 \sin\theta_1\}$ — from the coordinates

(3) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1), v_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4), v_1 \sin\gamma_1 \sin\theta_4\}$ — from the coordinates

(4) $\vec{n}_{1'} = \overrightarrow{A_1 B_1} \times \overrightarrow{A_1 C_1} =$ ☑ from (2) and (3)

$\{m_1 v_1 (\sin\theta_1 (\sin\alpha_1 \sin\gamma_1 \cos\delta_1 \cos\theta_4 - \sin\alpha_1 \cos\gamma_1 \sin\delta_1) + \sin\alpha_1 \sin\gamma_1 \sin\theta_4 \cos\theta_1), m_1 v_1 (\sin\theta_1 (\sin\alpha_1 (-\cos\gamma_1) \cos\delta_1 - \sin\alpha_1 \sin\gamma_1 \sin\delta_1 \cos\theta_4) + \cos\alpha_1 \sin\gamma_1 \sin\theta_4), m_1 v_1 (\cos\alpha_1 \sin\gamma_1 \cos\delta_1 \cos\theta_4 + \cos\theta_1 (\sin\alpha_1 \sin\gamma_1 \sin\delta_1 \cos\theta_4 + \sin\alpha_1 \cos\gamma_1 \cos\delta_1) - \cos\alpha_1 \cos\gamma_1 \sin\delta_1)\}$

(5) $\cos\beta_1 = \frac{\overrightarrow{A_1 B_1} \cdot \overrightarrow{A_1 C_1}}{m_1 v_1} = \cos\alpha_1 (\sin\gamma_1 \sin\delta_1 \cos\theta_4 + \cos\gamma_1 \cos\delta_1) + \sin\alpha_1 (\cos\theta_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4) + \sin\gamma_1 \sin\theta_1 \sin\theta_4)$ ☑ from (2) and (3); the closed forms below are written with $\beta_1$

(6) $|\vec{n}_{1'}| = m_1 \; v_1 \; \sin\beta_1$ ☑ from (4) and (5), with the positive radical because $\beta_1 \in (0, \pi)$

(7) $\cos\theta_{41'} = \frac{\vec{n}_4 \cdot \vec{n}_{1'}}{|\vec{n}_4| \; |\vec{n}_{1'}|}$ — by the definition

(8) $\cos\theta_{41'} = \frac{\sin\alpha_1 \sin\delta_1 \cos\theta_1 + \cos\alpha_1 \cos\delta_1 - \cos\beta_1 \cos\gamma_1}{\sin\beta_1 \sin\gamma_1}$ ☑ from (1) and (4)–(7)

(9) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1), v_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4), v_1 \sin\gamma_1 \sin\theta_4\}$ — the third vector of $\theta_{41'}$

(10) $\sin\theta_{41'} = \frac{\det(\vec{n}_4, \vec{n}_{1'}, \overrightarrow{A_1 C_1})}{|\vec{n}_4| \; |\vec{n}_{1'}| \; |\overrightarrow{A_1 C_1}|}$ — by the definition

(11) $\sin\theta_{41'} = \frac{1}{}$ ☑ from (1), (4), (6), (9), and (10)

(11) $\sin\theta_{41'} = \frac{}{\sin\beta_1}$ ☑ from (1), (4), (8), (9), and (10)

$(\cos\alpha_1 \sin\delta_1 \sin\theta_4 - \sin\alpha_1 (\cos\delta_1 \sin\theta_4 \cos\theta_1 + \sin\theta_1 \cos\theta_4))$

Together, these give the cosine and the sine of $\theta_{41'}$.

## $\theta_{22'}$, along $A_2 C_2$ Verified

Verification: the corner normal $\vec{n}_{2'}$ from its two edge vectors, then the cosine and the sine of the angle between $\vec{n}_2$ and $\vec{n}_{2'}$. The closed forms are written with $\beta_2$, which the coordinates do not contain: (5) gives its cosine, and (6) the length of $\vec{n}_{2'}$, which carries sin $\beta_2$.

(1) $\vec{n}_2 = v_2 \ell_2 \sin\gamma_2 \{-\sin\delta_2 \sin\theta_2, \cos\delta_2 \sin\theta_2, \cos\theta_2\}$, $|\vec{n}_2| = v_2 \ell_2 \sin\gamma_2$ • from the block on the normals

(2) $\overrightarrow{A_2 C_2} = \{v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2), v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2), v_2 \sin\gamma_2 \sin\theta_2\}$ • from the coordinates

(3) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2, m_2 \sin\alpha_2 \cos\theta_1, m_2 \sin\alpha_2 \sin\theta_1\}$ • from the coordinates

(4) $\vec{n}_{2'} = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 B_2} = \{m_2 v_2 (\sin\theta_1 (\sin\alpha_2 \cos\gamma_2 \sin\delta_2 - \sin\alpha_2 \sin\gamma_2 \cos\delta_2 \cos\theta_2) - \sin\alpha_2 \sin\gamma_2 \sin\theta_2 \cos\theta_1), m_2 v_2 (\sin\theta_1 (\sin\alpha_2 (-\cos\gamma_2) \cos\delta_2 - \sin\alpha_2 \sin\gamma_2 \sin\delta_2 \cos\theta_2) + \cos\alpha_2 \sin\gamma_2 \sin\theta_2), m_2 v_2 (\cos\alpha_2 \sin\gamma_2 \cos\delta_2 \cos\theta_2 + \cos\theta_1 (\sin\alpha_2 \sin\gamma_2 \sin\delta_2 \cos\theta_2 + \sin\alpha_2 \cos\gamma_2 \cos\delta_2) - \cos\alpha_2 \cos\gamma_2 \sin\delta_2)\}$ ☑ from (2) and (3)

(5) $\cos\beta_2 = \frac{\overrightarrow{A_2 B_2} \cdot \overrightarrow{A_2 C_2}}{m_2 v_2} = \cos\alpha_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2) + \sin\alpha_2 (\cos\theta_1 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2) + \sin\gamma_2 \sin\theta_1 \sin\theta_2)$ ☑ from (2) and (3); the closed forms below are written with $\beta_2$

(6) $|\vec{n}_{2'}| = m_2 v_2 \sin\beta_2$ ☑ from (4) and (5), with the positive radical because $\beta_2 \in (0, \pi)$

(7) $\cos\theta_{22'} = \frac{\vec{n}_2 \cdot \vec{n}_{2'}}{|\vec{n}_2| |\vec{n}_{2'}|}$ • by the definition

(8) $\cos\theta_{22'} = \frac{\sin\alpha_2 \sin\delta_2 \cos\theta_1 + \cos\alpha_2 \cos\delta_2 - \cos\beta_2 \cos\gamma_2}{\sin\beta_2 \sin\gamma_2}$ ☑ from (1) and (4)–(7)

(9) $\overrightarrow{C_2 A_2} =$ • the third vector of $\theta_{22'}$

$\{-(v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2)),$
$-(v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2)),$
$v_2 (-\sin\gamma_2) \sin\theta_2\}$

(10) $\sin\theta_{22'} = \frac{\det(\vec{n}_2, \vec{n}_{2'}, \overrightarrow{C_2 A_2})}{|\vec{n}_2|\,|\vec{n}_{2'}|\,|\overrightarrow{C_2 A_2}|}$ • by the definition

(11) $\sin\theta_{22'} = \frac{1}{\sin\beta_2}$ ☑ from (1), (4), (6), (9), and (10)

$(\cos\alpha_2 \sin\delta_2 \sin\theta_2 - \sin\alpha_2 (\cos\delta_2 \sin\theta_2 \cos\theta_1 + \sin\theta_1 \cos\theta_2))$

Together, these give the cosine and the sine of $\theta_{22'}$.

## Claim 1: the cleared identities

Verified

Verification: at each vertex, the coefficients as products of sines and the three ratios as identities in the flat angles, the cleared identity as a trigonometric identity, and the step to the identity of the claim with the displayed quantities as symbols; (1.7) and (2.7) read off that the corner angles are neither 0 nor $\pm\pi$.

Part 1. At $A_1$, the angle $\theta_{41'}$:

$cf_1 := \frac{1}{2}(\cos(\beta_1 - \gamma_1) - \cos(\alpha_1 - \delta_1)),$ • definitions
$cf_2 := \frac{1}{2}(\cos(\beta_1 - \gamma_1) - \cos(\alpha_1 + \delta_1)),$
$cf_3 := \frac{1}{2}(\cos(\beta_1 + \gamma_1) - \cos(\alpha_1 - \delta_1)),$
$cf_4 := \frac{1}{2}(\cos(\beta_1 + \gamma_1) - \cos(\alpha_1 + \delta_1))$

(1.1) $cf_1 = -\sin(\beta_1 - \overline{\alpha}_1)\sin(\beta_1 - \overline{\delta}_1)$, $cf_2 = \sin\overline{\beta}_1 \sin\overline{\gamma}_1,$ ☑ identities in the flat angles at $A_1$
$cf_3 = -\sin\overline{\alpha}_1 \sin\overline{\delta}_1,$
$cf_4 = -\sin\sigma_1 \sin(\beta_1 - \overline{\gamma}_1)$

(1.2) $cf_1, cf_2, cf_3, cf_4 \neq 0$ • each factor in (1.1) is the sine of half of a sum $\pm\alpha_1 \pm \beta_1 \pm \gamma_1 \pm \delta_1 \neq 0 \pmod{2\pi}$

(1.3) $(1 - \cos\theta_{41'})(cf_3 \cot^2(\theta_1/2) + cf_4) + (1 + \cos\theta_{41'})(cf_1 \cot^2(\theta_1/2) + cf_2) = 0$ ☑ $\cos\theta_{41'}$ from its block and $\cot^2(\theta_1/2) = (1 + \cos\theta_1)/(1 - \cos\theta_1)$, cleared

(1.4) $\frac{cf_3}{cf_4} = -x_1, \quad \frac{cf_2}{cf_4} = \frac{1 + x_1}{1 + u_1 x_1}, \quad \frac{cf_1}{cf_4} = -\frac{u_1 x_1 (1 + x_1)}{1 + u_1 x_1}$ ☑ identities in the flat angles at $A_1$

(1.5) $(1 - \cos\theta_{41'})(1 + u_1 x_1)(x_1 \cot^2(\theta_1/2) - 1) = (1 + \cos\theta_{41'})(1 + x_1)(1 - u_1 x_1 \cot^2(\theta_1/2))$ ☑ (1.3) divided by $cf_4$, see (1.2), with (1.4), multiplied by $1 + u_1 x_1$

(1.6) $1 + x_1 \neq 0, \quad 1 + u_1 x_1 > 0$ • given

(1.7) if $(x_1 \cot^2(\theta_1/2) - 1)(1 - u_1 x_1 \cot^2(\theta_1/2)) \neq 0$, then $\theta_{41'} \neq 0, \pm\pi$
- by (1.5), $\cos \theta_{41'} = 1$ would give $(1 + x_1)(1 - u_1 x_1 \cot^2(\theta_1/2)) = 0$, and $\cos \theta_{41'} = -1$ would give $(1 + u_1 x_1)(x_1 \cot^2(\theta_1/2) - 1) = 0$; both contradict the hypothesis, by (1.6)

Part 2. At $A_2$, the angle $\theta_{22'}$:

$cf_1 := \frac{1}{2}(\cos(\beta_2 - \gamma_2) - \cos(\alpha_2 - \delta_2))$, $cf_2 := \frac{1}{2}(\cos(\beta_2 - \gamma_2) - \cos(\alpha_2 + \delta_2))$, $cf_3 := \frac{1}{2}(\cos(\beta_2 + \gamma_2) - \cos(\alpha_2 - \delta_2))$, $cf_4 := \frac{1}{2}(\cos(\beta_2 + \gamma_2) - \cos(\alpha_2 + \delta_2))$
- definitions

(2.1) $cf_1 = -\sin(\beta_2 - \overline{\alpha}_2)\sin(\beta_2 - \overline{\delta}_2)$, $cf_2 = \sin\overline{\beta}_2 \sin\overline{\gamma}_2$, $cf_3 = -\sin\overline{\alpha}_2 \sin\overline{\delta}_2$, $cf_4 = -\sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)$
- ☑ identities in the flat angles at $A_2$

(2.2) $cf_1, cf_2, cf_3, cf_4 \neq 0$
- each factor in (2.1) is the sine of half of a sum $\pm\alpha_2 \pm \beta_2 \pm \gamma_2 \pm \delta_2 \neq 0 \pmod{2\pi}$

(2.3) $(1 - \cos\theta_{22'})(cf_3 \cot^2(\theta_1/2) + cf_4) + (1 + \cos\theta_{22'})(cf_1 \cot^2(\theta_1/2) + cf_2) = 0$
- ☑ $\cos\theta_{22'}$ from its block and $\cot^2(\theta_1/2) = (1 + \cos\theta_1)/(1 - \cos\theta_1)$, cleared

(2.4) $\frac{cf_3}{cf_4} = -x_2$, $\frac{cf_2}{cf_4} = \frac{1 + x_2}{1 + u_2 x_2}$, $\frac{cf_1}{cf_4} = -\frac{u_2 x_2(1 + x_2)}{1 + u_2 x_2}$
- ☑ identities in the flat angles at $A_2$

(2.5) $(1 - \cos\theta_{22'})(1 + u_2 x_2)(x_2 \cot^2(\theta_1/2) - 1) = (1 + \cos\theta_{22'})(1 + x_2)(1 - u_2 x_2 \cot^2(\theta_1/2))$
- ☑ (2.3) divided by $cf_4$, see (2.2), with (2.4), multiplied by $1 + u_2 x_2$

(2.6) $1 + x_2 \neq 0$, $1 + u_2 x_2 > 0$
- given

(2.7) if $(x_2 \cot^2(\theta_1/2) - 1)(1 - u_2 x_2 \cot^2(\theta_1/2)) \neq 0$, then $\theta_{22'} \neq 0, \pm\pi$
- by (2.5), $\cos\theta_{22'} = 1$ would give $(1 + x_2)(1 - u_2 x_2 \cot^2(\theta_1/2)) = 0$, and $\cos\theta_{22'} = -1$ would give $(1 + u_2 x_2)(x_2 \cot^2(\theta_1/2) - 1) = 0$; both contradict the hypothesis, by (2.6)

Together, these give Claim 1.

## Claim 2: the formula for the two corner angles

Verified

Verification: at each vertex, the sine and 1 − cos of the corner angle through the two half −angle cotangents give its half −cotangent as a quotient; Bricard's equation reduces the numerator to the radical of the parametrization by $\theta_1$, which fixes the sign. Identities in the angles are checked as trigonometric identities, and each step between lines with the displayed quantities as symbols.

(1) $x_2 = x_1$, $u_1 = u_2 = u$
- the flexion; the lines below are

(2) $\cot(\theta_1/2) \in I(x_1)$
- the admissible set

(3) $\cot(x/2) = \frac{\sin x}{1-\cos x}$, $\sin x = \frac{2\cot(x/2)}{1+\cot^2(x/2)}$, $\cos x = \frac{\cot^2(x/2)-1}{\cot^2(x/2)+1}$ ☑ for $x \neq 0 \pmod{2\pi}$

Part 1. At $A_1$, the angle $\theta_{41'}$ through $\theta_1$ and $\theta_4$:

(1.1) $\theta_{41'} \neq 0, \pm\pi$ • Claim 1, because (1) and (2) gi

(1.2) $\sin\theta_{41'} = \frac{1}{\sin\beta_1}(\cos\alpha_1 \sin\delta_1 \sin\theta_4 - \sin\alpha_1(\cos\delta_1 \sin\theta_4 \cos\theta_1 + \sin\theta_1 \cos\theta_4))$, $\cos\theta_{41'} = \frac{\sin\alpha_1 \sin\delta_1 \cos\theta_1 + \cos\alpha_1 \cos\delta_1 - \cos\beta_1 \cos\gamma_1}{\sin\beta_1 \sin\gamma_1}$ • from the block on $\theta_{41'}$

(1.3) $\sin\theta_{41'} \sin\beta_1 \frac{(1+\cot^2(\theta_4/2))(1+\cot^2(\theta_1/2))}{2} = -\sin(\alpha_1-\delta_1)\cot(\theta_4/2)\cot^2(\theta_1/2) - \sin\alpha_1 \cot^2(\theta_4/2)\cot(\theta_1/2) + \sin(\alpha_1+\delta_1)\cot(\theta_4/2) + \sin\alpha_1 \cot(\theta_1/2)$ ☑ (1.2), with (3)

(1.4) $(1-\cos\theta_{41'})\sin\gamma_1 \sin\beta_1 \frac{1+\cot^2(\theta_1/2)}{2} = -\sin(\beta_1-\overline{\alpha}_1)\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2) + \sin\overline{\beta}_1 \sin\overline{\gamma}_1$ ☑ (1.2), with (3) and (1.1) of the b

$\Phi := -\sin(\alpha_1-\delta_1)\cot(\theta_4/2)\cot^2(\theta_1/2) - \sin\alpha_1 \cot^2(\theta_4/2)\cot(\theta_1/2) + \sin(\alpha_1+\delta_1)\cot(\theta_4/2) + \sin\alpha_1 \cot(\theta_1/2)$ • a name for the right side of (1.

$\Phi_2 := -\sin(\beta_1-\overline{\alpha}_1)\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2) + \sin\overline{\beta}_1 \sin\overline{\gamma}_1$ • a name for the right side of (1.

(1.5) $\cot(\theta_{41'}/2) = \frac{\sin\theta_{41'}}{1-\cos\theta_{41'}} = \frac{\sin\gamma_1 \Phi}{(1+\cot^2(\theta_4/2))\Phi_2}$ ☑ the quotient of (1.3) and (1.4), $\Phi_2 \neq 0$ by (1.4), because $\theta_{41'}$

(1.6) $x_1 = \frac{\sin\overline{\alpha}_1 \sin\overline{\delta}_1}{-\sin\sigma_1 \sin(\beta_1-\overline{\gamma}_1)}$, $u x_1 = \frac{\sin(\beta_1-\overline{\alpha}_1)\sin\left(\beta_1-\overline{\delta}_1\right)}{\sin\overline{\beta}_1 \sin\overline{\gamma}_1}$ ☑ identities in the flat angles at

(1.7) $\Phi_2 = \sin\overline{\beta}_1 \sin\overline{\gamma}_1 (1 - u x_1 \cot^2(\theta_1/2))$ ☑ by (1.6)

$\Phi_1 := \sin\overline{\alpha}_1 \sin\overline{\delta}_1 \cot^2(\theta_1/2) + \sin\sigma_1 \sin(\beta_1-\overline{\gamma}_1)$ • a name

(1.8) $\Phi_1 = -\sin\sigma_1 \sin(\beta_1-\overline{\gamma}_1)(x_1 \cot^2(\theta_1/2) - 1)$ ☑ by (1.6)

(1.9) $\frac{-\sin\sigma_1 \sin(\beta_1-\overline{\gamma}_1)}{\sin\overline{\beta}_1 \sin\overline{\gamma}_1} = \frac{1+u x_1}{1+x_1}$ ☑ an identity in the flat angles at

(1.10) $\cot^2(\theta_{41'}/2) = \frac{\Phi_1}{\Phi_2}$ ☑ from (1.2), cleared, with (1.1) o

(1.11) $\Phi_1 \Phi_2 = -\sin\sigma_1 \sin(\beta_1-\overline{\gamma}_1)\sin\overline{\beta}_1 \sin\overline{\gamma}_1$ ☑ (1.7) times (1.8)

$D(x_1, \cot(\theta_1/2))$

(1.12) $\sin\overline{\gamma}_1 > 0$, $\sin\overline{\beta}_1 > 0$ • $\overline{\gamma}_1, \overline{\beta}_1 \in (0, \pi)$

(1.13) if $x_1 < 0$: $x_1 \cot^2(\theta_1/2) - 1 < 0$, so $x_1 D(x_1, \cot(\theta_1/2)) > 0$ gives $1 - u x_1 \cot^2(\theta_1/2) > 0$ • $\cot^2(\theta_1/2) \geq 0$

(1.14) if $x_1 > 0$ and $1 - u x_1 \cot^2(\theta_1/2) \leq 0$: $x_1 D(x_1, \cot(\theta_1/2)) > 0$ gives $x_1 \cot^2(\theta_1/2) < 1 \leq u x_1 \cot^2(\theta_1/2)$, so $u \geq 1$ • against $u = 1 - M_1$, $M_1 > 0$

(1.15) $\Phi_2 > 0$ • (1.7), (1.12)–(1.14), and (2)

(1.16) $(-\sin\overline{\delta}_1 \sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2) - \sin\overline{\gamma}_1 \sin(\beta_1-\overline{\gamma}_1)$ ☑ the closure $\overrightarrow{A_1 B_1} \cdot \overrightarrow{A_1 C_1} =$

) $\cot^2(\theta_1/2)$ $m_1 v_1 \cos\beta_1$ (Bricard's equa

$$\cot(\theta_4/2)$$

$$-2\ \sin\alpha_1 \sin\gamma_1 \cot(\theta_4/2)\cot(\theta_1/2)$$

$$+\left(-\sin\overline{\alpha}_1 \sin(\beta_1-\overline{\alpha}_1)\cot^2(\theta_1/2)+\sin\sigma_1 \sin\overline{\beta}_1\right)=0$$

(1.17) $\sin\overline{\delta}_1\left(-\sin\left(\beta_1-\overline{\delta}_1\right)\right)-$ $\sin\overline{\alpha}_1\left(-\sin(\beta_1-\overline{\alpha}_1)\right)=\sin\gamma_1\sin(\alpha_1-\delta_1)$ ☑ an identity in the flat angles at

(1.18) $\sin\overline{\gamma}_1\left(-\sin(\beta_1-\overline{\gamma}_1)\right)-\sin\sigma_1\sin\overline{\beta}_1=-\sin\gamma_1\sin(\alpha_1+\delta_1)$ ☑ the same

(1.19) $\sin\gamma_1\ \Phi=\cot(\theta_4/2)\ ($ $-\sin\gamma_1\sin(\alpha_1-\delta_1)\cot^2(\theta_1/2)+\sin\gamma_1\sin(\alpha_1+\delta_1)$ $)+\sin\alpha_1\sin\gamma_1\cot(\theta_1/2)\ (1-\cot^2(\theta_4/2))$ ☑ $\Phi$ expanded

(1.20) $\sin\gamma_1\ \Phi=\cot(\theta_4/2)\ (($ $-\sin\overline{\alpha}_1\sin(\beta_1-\overline{\alpha}_1)\cot^2(\theta_1/2)+\sin\sigma_1\sin\overline{\beta}_1)-($ $-\sin\overline{\delta}_1\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$ $))+\sin\alpha_1\sin\gamma_1\cot(\theta_1/2)\ (1-\cot^2(\theta_4/2))$ ☑ (1.19), with (1.17) and (1.18)

(1.21) $\left(-\sin\overline{\alpha}_1\sin(\beta_1-\overline{\alpha}_1)\cot^2(\theta_1/2)+\sin\sigma_1\sin\overline{\beta}_1\right)$ $\cot(\theta_4/2)=2\ \sin\alpha_1\sin\gamma_1\cot^2(\theta_4/2)\cot(\theta_1/2)-($ $-\sin\overline{\delta}_1\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$ $)\cot^3(\theta_4/2)$ ☑ (1.16) times $\cot(\theta_4/2)$

(1.22) $\frac{\sin\gamma_1\ \Phi}{1+\cot^2(\theta_4/2)}=\cot(\theta_1/2)\sin\alpha_1\sin\gamma_1-\cot(\theta_4/2)\ ($ $-\sin\overline{\delta}_1\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)),$ that is $\cot(\theta_4/2)\ ($ $-\sin\overline{\delta}_1\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$ $)-\cot(\theta_1/2)\sin\alpha_1\sin\gamma_1=-\Phi_2\cot(\theta_{41'}/2)$ ☑ (1.20) and (1.21), divided by 1

(1.23) $\cot(\theta_4/2)=$ $\left(\begin{array}{c}\varepsilon_1\left(\cot(\theta_1/2)\ \sqrt{u x_1 y_1(1+z_1)(1+u z_1)}\right.\\ \left.-\ e_0\ e_1\ \sqrt{u y_1 z_1\ D(x_1,\cot(\theta_1/2))}\ \right)\end{array}\right/$ $(y_1 u\,(z_1+x_1\cot^2(\theta_1/2)))$ • the parametrization by $\theta_1$

(1.24) $e_0\ e_1\ \sqrt{u y_1 z_1\ D(x_1,\cot(\theta_1/2))}=$ $-\varepsilon_1\cot(\theta_4/2)y_1 u\,(z_1+x_1\cot^2(\theta_1/2)$ $)+\cot(\theta_1/2)\ \sqrt{u x_1 y_1(1+z_1)(1+u z_1)}$ ☑ (1.23), solved for its radical

(1.25) $u y_1 z_1=\frac{\sin\overline{\gamma}_1(-\sin(\beta_1-\overline{\gamma}_1))}{\sin\sigma_1\sin\overline{\beta}_1}$ ☑ an identity in the flat angles at

(1.26) $u x_1 y_1=\frac{\sin\overline{\delta}_1\left(-\sin\left(\beta_1-\overline{\delta}_1\right)\right)}{\sin\sigma_1\sin\overline{\beta}_1}$ ☑ the same

(1.27) $u x_1 y_1(1+z_1)(1+u z_1)=\frac{\sin^2(\alpha_1)\sin^2(\gamma_1)}{\sin^2(\sigma_1)\sin^2(\overline{\beta}_1)}$ ☑ the same

(1.28) $\sin\sigma_1\sin\overline{\beta}_1\ y_1 u\,(z_1+x_1\cot^2(\theta_1/2))=$ $-\sin\overline{\delta}_1\sin\left(\beta_1-\overline{\delta}_1\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$ ☑ (1.25) and (1.26)

(1.29) $\sin\sigma_1\sin\overline{\beta}_1$ $\sqrt{u x_1 y_1(1+z_1)(1+u z_1)}=\varepsilon_1\sin\alpha_1\sin\gamma_1$ ☑ (1.27), as squares; $\varepsilon_1$ = sign si

(1.30) $\sin\sigma_1\sin\overline{\beta}_1\ \sqrt{u y_1 z_1\ D(x_1,\cot(\theta_1/2))}=\varepsilon_1\sqrt{\Phi_1\Phi_2}$ ☑ (1.25) and (1.11), as squares; t

(1.31) $-e_0 e_1\ \sqrt{\Phi_1\Phi_2}=$ $\cot(\theta_4/2)\ ($ ☑ (1.24) multiplied by sin $\sigma_1$ sin $\overline{\beta}$ and divided by $\varepsilon_1$, with (1.28

$-\sin\overline{\delta}_1 \sin\left(\beta_1 - \overline{\delta}_1\right)\cot^2(\theta_1/2) - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1)$
$) - \cot(\theta_1/2)\sin\alpha_1 \sin\gamma_1 = -\Phi_2 \cot(\theta_{41'}/2)$

(1.32) $\cot(\theta_{41'}/2) = e_0\, e_1 \sqrt{\frac{\Phi_1}{\Phi_2}}$ ☑ (1.31) divided by $\Phi_2 > 0$, see (1

$= e_0\, e_1 \sqrt{\frac{(1 + u x_1)(x_1 \cot^2(\theta_1/2) - 1)}{(1 + x_1)(1 - u x_1 \cot^2(\theta_1/2))}}$

Part 2. At $A_2$, the angle $\theta_{22'}$ through $\theta_1$ and $\theta_2$:

(2.1) $\theta_{22'} \neq 0, \pm\pi$ • Claim 1, because (1) and (2) gi

(2.2) $\sin\theta_{22'} = \frac{\cos\alpha_2 \sin\delta_2 \sin\theta_2 - \sin\alpha_2 (\cos\delta_2 \sin\theta_2 \cos\theta_1 + \sin\theta_1 \cos\theta_2)}{\sin\beta_2}$ • from the block on $\theta_{22'}$

, $\cos\theta_{22'} = \frac{\sin\alpha_2 \sin\delta_2 \cos\theta_1 + \cos\alpha_2 \cos\delta_2 - \cos\beta_2 \cos\gamma_2}{\sin\beta_2 \sin\gamma_2}$

(2.3) $\sin\theta_{22'} \sin\beta_2 \frac{(1 + \cot^2(\theta_2/2))(1 + \cot^2(\theta_1/2))}{2} =$ ☑ (2.2), with (3)

$-\sin(\alpha_2 - \delta_2)\cot(\theta_2/2)\cot^2(\theta_1/2)$
$- \sin\alpha_2 \cot^2(\theta_2/2)\cot(\theta_1/2) +$
$\sin(\alpha_2 + \delta_2)\cot(\theta_2/2) + \sin\alpha_2 \cot(\theta_1/2)$

(2.4) $(1 - \cos\theta_{22'})\sin\gamma_2 \sin\beta_2 \frac{1 + \cot^2(\theta_1/2)}{2} =$ ☑ (2.2), with (3) and (2.1) of the

$-\sin(\beta_2 - \overline{\alpha}_2)\sin\left(\beta_2 - \overline{\delta}_2\right)\cot^2(\theta_1/2) + \sin\overline{\beta}_2 \sin\overline{\gamma}_2$

$\Phi := -\sin(\alpha_2 - \delta_2)\cot(\theta_2/2)\cot^2(\theta_1/2)$ • a name for the right side of (2
$- \sin\alpha_2 \cot^2(\theta_2/2)\cot(\theta_1/2) +$
$\sin(\alpha_2 + \delta_2)\cot(\theta_2/2) + \sin\alpha_2 \cot(\theta_1/2)$

$\Phi_2 :=$ • a name for the right side of (2
$-\sin(\beta_2 - \overline{\alpha}_2)\sin\left(\beta_2 - \overline{\delta}_2\right)\cot^2(\theta_1/2) + \sin\overline{\beta}_2 \sin\overline{\gamma}_2$

(2.5) $\cot(\theta_{22'}/2) = \frac{\sin\theta_{22'}}{1 - \cos\theta_{22'}} = \frac{\sin\gamma_2\, \Phi}{(1 + \cot^2(\theta_2/2))\,\Phi_2}$ ☑ the quotient of (2.3) and (2.4), $\Phi_2 \neq 0$ by (2.4), because $\theta_{22}$

(2.6) $x_1 = \frac{\sin\overline{\alpha}_2 \sin\overline{\delta}_2}{-\sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)}$, $u x_1 = \frac{\sin(\beta_2 - \overline{\alpha}_2)\sin\left(\beta_2 - \overline{\delta}_2\right)}{\sin\overline{\beta}_2 \sin\overline{\gamma}_2}$ ☑ identities in the flat angles at

(2.7) $\Phi_2 = \sin\overline{\beta}_2 \sin\overline{\gamma}_2 (1 - u x_1 \cot^2(\theta_1/2))$ ☑ by (2.6)

$\Phi_1 := \sin\overline{\alpha}_2 \sin\overline{\delta}_2 \cot^2(\theta_1/2) + \sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)$ • a name

(2.8) $\Phi_1 = -\sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)(x_1 \cot^2(\theta_1/2) - 1)$ ☑ by (2.6)

(2.9) $\frac{-\sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)}{\sin\overline{\beta}_2 \sin\overline{\gamma}_2} = \frac{1 + u x_1}{1 + x_1}$ ☑ an identity in the flat angles at

(2.10) $\cot^2(\theta_{22'}/2) = \frac{\Phi_1}{\Phi_2}$ ☑ from (2.2), cleared, with (2.1)

(2.11) $\Phi_1 \Phi_2 = -\sin\sigma_2 \sin(\beta_2 - \overline{\gamma}_2)\sin\overline{\beta}_2 \sin\overline{\gamma}_2$ ☑ (2.7) times (2.8)
$D(x_1, \cot(\theta_1/2))$

(2.12) $\sin\overline{\gamma}_2 > 0$, $\sin\overline{\beta}_2 > 0$ • $\overline{\gamma}_2, \overline{\beta}_2 \in (0, \pi)$

(2.13) if $x_1 < 0$: $x_1 \cot^2(\theta_1/2) - 1 < 0$, so • $\cot^2(\theta_1/2) \geq 0$
$x_1\ D(x_1, \cot(\theta_1/2)) > 0$ gives $1 - u x_1 \cot^2(\theta_1/2) > 0$

(2.14) if $x_1 > 0$ and $1 - u x_1 \cot^2(\theta_1/2)$ • against $u = 1 - M_2$, $M_2 > 0$
$\leq 0$: $x_1\ D(x_1, \cot(\theta_1/2)) > 0$ gives
$x_1 \cot^2(\theta_1/2) < 1 \leq u x_1 \cot^2(\theta_1/2)$, so $u \geq 1$

(2.15) $\Phi_2 > 0$ • (2.7), (2.12)–(2.14), and (2)

(2.16) $(-\sin\overline{\delta}_2 \sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)$ $)\cot^2(\theta_2/2)$ $-2\sin\alpha_2\sin\gamma_2\cot(\theta_2/2)\cot(\theta_1/2)$ $+(-\sin\overline{\alpha}_2\sin(\beta_2-\overline{\alpha}_2)\cot^2(\theta_1/2)+\sin\sigma_2\sin\overline{\beta}_2)=0$ ☑ the closure $\overrightarrow{A_2B_2}\cdot\overrightarrow{A_2C_2}=$ $m_2\ v_2\cos\beta_2$ (Bricard's equ

(2.17) $\sin\overline{\delta}_2\left(-\sin\left(\beta_2-\overline{\delta}_2\right)\right)-$ $\sin\overline{\alpha}_2(-\sin(\beta_2-\overline{\alpha}_2))=\sin\gamma_2\sin(\alpha_2-\delta_2)$ ☑ an identity in the flat angles at

(2.18) $\sin\overline{\gamma}_2(-\sin(\beta_2-\overline{\gamma}_2))-$ $\sin\sigma_2\sin\overline{\beta}_2=-\sin\gamma_2\sin(\alpha_2+\delta_2)$ ☑ the same

(2.19) $\sin\gamma_2\ \Phi=\cot(\theta_2/2)($ $-\sin\gamma_2\sin(\alpha_2-\delta_2)\cot^2(\theta_1/2)+\sin\gamma_2\sin(\alpha_2+\delta_2)$ $)+\sin\alpha_2\sin\gamma_2\cot(\theta_1/2)\ (1-\cot^2(\theta_2/2))$ ☑ $\Phi$ expanded

(2.20) $\sin\gamma_2\ \Phi=\cot(\theta_2/2)\ ((-\sin\overline{\alpha}_2\sin(\beta_2-\overline{\alpha}_2)\cot^2(\theta_1/2)+\sin\sigma_2\sin\overline{\beta}_2)-($ $-\sin\overline{\delta}_2\sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)$ $))+\sin\alpha_2\sin\gamma_2\cot(\theta_1/2)\ (1-\cot^2(\theta_2/2))$ ☑ (2.19), with (2.17) and (2.18)

(2.21) $(-\sin\overline{\alpha}_2\sin(\beta_2-\overline{\alpha}_2)\cot^2(\theta_1/2)+\sin\sigma_2\sin\overline{\beta}_2)$ $\cot(\theta_2/2)=2\sin\alpha_2\sin\gamma_2\cot^2(\theta_2/2)\cot(\theta_1/2)-($ $-\sin\overline{\delta}_2\sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)$ $)\cot^3(\theta_2/2)$ ☑ (2.16) times $\cot(\theta_2/2)$

(2.22) $\frac{\sin\gamma_2\ \Phi}{1+\cot^2(\theta_2/2)}=\cot(\theta_1/2)\sin\alpha_2\sin\gamma_2-\cot(\theta_2/2)\ ($ $-\sin\overline{\delta}_2\sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)),$ that is $\cot(\theta_2/2)\ ($ $-\sin\overline{\delta}_2\sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)$ $)-\cot(\theta_1/2)\sin\alpha_2\sin\gamma_2=-\Phi_2\cot(\theta_{22'}/2)$ ☑ (2.20) and (2.21), divided by 1

(2.23) $\cot(\theta_2/2)=$ $\left(\begin{array}{c}\varepsilon_2\,(\cot(\theta_1/2)\ \sqrt{u x_1 y_2(1+z_2)(1+u z_2)}\\ +\ e_0\ e_2\ \sqrt{u y_2 z_2\ D(x_1,\cot(\theta_1/2))}\ )\end{array}\right)\Big/$ $(y_2 u(z_2+x_1\cot^2(\theta_1/2)))$ • the parametrization by $\theta_1$

(2.24) $e_0\ e_2\ \sqrt{u y_2 z_2\ D(x_1,\cot(\theta_1/2))}=$ $\varepsilon_2\cot(\theta_2/2)y_2 u(z_2+x_1\cot^2(\theta_1/2)$ $)-\cot(\theta_1/2)\ \sqrt{u x_1 y_2(1+z_2)(1+u z_2)}$ ☑ (2.23), solved for its radical

(2.25) $u y_2 z_2=\frac{\sin\overline{\gamma}_2(-\sin(\beta_2-\overline{\gamma}_2))}{\sin\sigma_2\sin\overline{\beta}_2}$ ☑ an identity in the flat angles at

(2.26) $u x_1 y_2=\frac{\sin\overline{\delta}_2\left(-\sin\left(\beta_2-\overline{\delta}_2\right)\right)}{\sin\sigma_2\sin\overline{\beta}_2}$ ☑ the same

(2.27) $u x_1 y_2(1+z_2)(1+u z_2)=\frac{\sin^2(\alpha_2)\sin^2(\gamma_2)}{\sin^2(\sigma_2)\sin^2(\overline{\beta}_2)}$ ☑ the same

(2.28) $\sin\sigma_2\sin\overline{\beta}_2\ \ y_2 u(z_2+x_1\cot^2(\theta_1/2))=$ $-\sin\overline{\delta}_2\sin\left(\beta_2-\overline{\delta}_2\right)\cot^2(\theta_1/2)-\sin\overline{\gamma}_2\sin(\beta_2-\overline{\gamma}_2)$ ☑ (2.25) and (2.26)

(2.29) $\sin\sigma_2\sin\overline{\beta}_2$ $\sqrt{u x_1 y_2(1+z_2)(1+u z_2)}=\varepsilon_2\sin\alpha_2\sin\gamma_2$ ☑ (2.27), as squares; $\varepsilon_2$ = sign si

(2.30) $\sin\sigma_2 \sin\overline{\beta}_2 \sqrt{u y_2 z_2 D(x_1, \cot(\theta_1/2))} = \varepsilon_2 \sqrt{\Phi_1 \Phi_2}$ ☑ (2.25) and (2.11), as squares; t

(2.31) $e_0 e_2 \sqrt{\Phi_1 \Phi_2} =$ ☑ (2.24) multiplied by sin $\sigma_2$ sin $\overline{\beta}$ and divided by $\varepsilon_2$, with (2.2

$\cot(\theta_2/2)$ (
$-\sin\overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right)\cot^2(\theta_1/2) - \sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)$
) $-\cot(\theta_1/2)\sin\alpha_2 \sin\gamma_2 = -\Phi_2 \cot(\theta_{22'}/2)$

(2.32) $\cot(\theta_{22'}/2) = -e_0 e_2 \sqrt{\frac{\Phi_1}{\Phi_2}}$ ☑ (2.31) divided by $\Phi_2 > 0$, see (

$= -e_0 e_2 \sqrt{\frac{(1 + u x_1)(x_1 \cot^2(\theta_1/2) - 1)}{(1 + x_1)(1 - u x_1 \cot^2(\theta_1/2))}}$

(4) $\frac{(1 + u_1 x_1)(x_1 \cot^2(\theta_1/2) - 1)}{(1 + x_1)(1 - u_1 x_1 \cot^2(\theta_1/2))} =$ ☑ the radicands of (1.32) and (2.

$\frac{(1 + u_2 x_2)(x_2 \cot^2(\theta_1/2) - 1)}{(1 + x_2)(1 - u_2 x_2 \cot^2(\theta_1/2))} = \frac{(1 + u x_1)(x_1 \cot^2(\theta_1/2) - 1)}{(1 + x_1)(1 - u x_1 \cot^2(\theta_1/2))}$

(5) $e_1 \cot(\theta_{41'}/2) = -e_2 \cot(\theta_{22'}/2)$ ☑ (1.32) and (2.32), with (4)

$= e_0 \sqrt{\frac{(1 + u x_1)(x_1 \cot^2(\theta_1/2) - 1)}{(1 + x_1)(1 - u x_1 \cot^2(\theta_1/2))}}$

Together, these give Claim 2.

**Section 11: ALL CLAIMS VERIFIED**

# Section 12

In[•]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[statusChip, smallChip, givenChip, colTest, cols, bullet, ind, vlift,
  idxB, prmB, tagB, ptB, pt, ptTB, ptT, edg, evec, nv, nvP, wv, tld, gk, thI, thP,
  bx, cotSqTag, cotTag, dTag, mulR, absR, uPlain, capX, tSym, frac, caseDef,
  note, firstBase, vrow, nrow, prow, hrow, vgap, vgrid, iv, dispRule, trigNames,
  simpleBoxQ, trigBoxRule, disp, sgnBox, polyRow, iSet, lens, angSyms, hh, zz,
  hRule, zeroE, Apt, Bpt, Cpt, uB, uC, eNext, ePrev, nextA, prevA, thNextIdx,
  thPrevIdx, nIdx, pIdx, xRp, yRp, qXr, qYr, e0Tag, sigTag, radTag, flxF, bLen,
  cLen, ll1, ll2, ll3, ll4, mm1, mm2, mm3, mm4, vv1, vv2, vv3, vv4, angA, angB,
  angG, angD, thNext, thPrev, thCent, sgnV, nraw, nrawP, nlen, nSrc, ndir, nPsrc,
  nPshow, wCent, wCentLen, wName, betaForm, outBnum, outBden, eqB, sgD, bAD, bBD,
  bGD, bDD, barRule, sig, barA, barB, barG, barD, ratA, ratB, ratG, ratD, prodM,
  prodR, prodS, prodF, quU, quX, quY, quZ, quYt, quZt, vRat, kn, kp, coA, coW, coV,
  coD, coP, coQ1, coQ2, coA2, coV2, coP2, coQ3, coQ4, coC0, phiE, bric, c2Num,
  c2Den, c2Psi, sn, sinSq, sqR, tSq, omD, phD, sigTagT, radTagT, radTagTt, rhsTt,
  flxT, formalRed, linkList, monoD, phiRows, nameB, figBlock, resultsAll];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
   Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
   RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
```

```
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
    Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["×", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
givenChip = Style["•", 12, GrayLevel[0.45]];
colTest = RGBColor[0.10, 0.35, 0.75];
cols = {RGBColor[0.10, 0.35, 0.75],
    RGBColor[0.50, 0.20, 0.65], RGBColor[0.00, 0.55, 0.55],
    RGBColor[0.20, 0.50, 0.20], RGBColor[0.75, 0.35, 0.05],
    RGBColor[0.70, 0.15, 0.40], RGBColor[0.30, 0.25, 0.70],
    RGBColor[0.45, 0.35, 0.15], RGBColor[0.35, 0.35, 0.35]};
bullet = Style["▪ ", 12, colTest]; ind = "    "; vlift = 0.16;
idxB[x_] :=
  If[IntegerQ[x], ToString[x], StyleBox[ToString[x], FontSlant → "Italic"]];
prmB[x_] := SuperscriptBox[idxB[x], "′"];
tagB[x_, y_] := If[IntegerQ[x] && IntegerQ[y], ToString[x] <> ToString[y],
    StyleBox[ToString[x] <> ToString[y], FontSlant → "Italic"]];
ptB[ltr_, i_] := SubscriptBox[StyleBox[ltr, FontSlant → "Italic"], idxB[i]];
pt[ltr_, i_] := DisplayForm[ptB[ltr, i]];
ptTB[ltr_, i_] :=
 SubscriptBox[OverscriptBox[StyleBox[ltr, FontSlant → "Italic"], "~"], idxB[i]];
ptT[ltr_, i_] := DisplayForm[ptTB[ltr, i]];
edg[s1_, i1_, s2_, i2_] := DisplayForm[RowBox[{ptB[s1, i1], ptB[s2, i2]}]];
evec[s1_, i1_, s2_, i2_] :=
  DisplayForm[OverscriptBox[RowBox[{ptB[s1, i1], ptB[s2, i2]}],
    AdjustmentBox["⟶", BoxBaselineShift → -vlift]]];
nv[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["n", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], idxB[x]]];
nvP[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["n", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], prmB[x]]];
wv[x_] :=
  DisplayForm[SubscriptBox[OverscriptBox[StyleBox["w", FontSlant → "Italic"],
     AdjustmentBox["→", BoxBaselineShift → -vlift]], idxB[x]]];
tld[ltr_] :=
  DisplayForm[OverscriptBox[StyleBox[ltr, FontSlant → "Italic"], "~"]];
gk[g_, i_] := DisplayForm[SubscriptBox[g, idxB[i]]];
thI[x_] := DisplayForm[SubscriptBox["ϴ", idxB[x]]];
thP[x_, y_] :=
  DisplayForm[SubscriptBox["ϴ", SuperscriptBox[tagB[x, y], "′"]]];
bx[DisplayForm[b_]] := b; bx[x_] := ToBoxes[x, TraditionalForm];
cotSqTag[t_] :=
  DisplayForm[RowBox[{SuperscriptBox["cot", "2"], "(", bx[t], "/", "2", ")"}]];
cotTag[t_] := DisplayForm[RowBox[{"cot", "(", bx[t], "/", "2", ")"}]];
dTag[argX_, argT_] := Row[{DisplayForm[StyleBox["D", FontSlant → "Italic"]],
```

```
    "(", argX, ", ", argT, ")"}];
mulR[args___] := Row[Riffle[{args}, ""]];
absR[e_] := Row[{"|", e, "|"}];
uPlain = DisplayForm[StyleBox["u", FontSlant → "Italic"]];
capX = DisplayForm[StyleBox["X", FontSlant → "Italic"]];
tSym = DisplayForm[StyleBox["t", FontSlant → "Italic"]];
frac[num_, den_] := DisplayForm[FractionBox[bx[num], bx[den]]];
caseDef[lhs_, rows_] :=
  DisplayForm[RowBox[{bx[lhs], " := ", StyleBox["{", SpanMaxSize → Infinity],
     GridBox[Map[bx, rows, {2}], ColumnAlignments → Left,
      ColumnSpacings → 1.6, RowSpacings → 0.9]}]];
note[ntxt_] := Style[ntxt, 12, GrayLevel[0.5]];
firstBase[e_] :=
  If[Head[e] === Column, Append[e, BaselinePosition → {1, 1}], e];
vrow[stmt_, mark_, nt_] :=
  {Spacer[10], Style[firstBase[stmt], 13], mark, note[nt]};
(*a numbered line,an unnumbered line (for a name),
and a heading line of a block*)
nrow[n_, stmt_, mark_, nt_] := {Style["(" <> ToString[n] <> ")", 13],
   Style[firstBase[stmt], 13], mark, note[nt]};
prow[stmt_, mark_, nt_] := {"", Style[firstBase[stmt], 13], mark, note[nt]};
hrow[t_] := {Style[t, 13], SpanFromLeft, SpanFromLeft, SpanFromLeft};
vgap = {"", Spacer[{0, 7}], "", ""};
vgrid[rows_] :=
  Grid[rows, Alignment → {Left, Baseline}, Spacings → {1.2, 0.8}];
(*every symbol that is only displayed is a string,so that no value
 given to a symbol elsewhere in the notebook can enter the display;
the displayed index i is an italic string as well*)
iv = Style["i", Italic];
dispRule = Join[Thread[{α1, α2, α3, α4} → Table[Subscript["α", k], {k, 4}]],
   Thread[{β1, β2, β3, β4} → Table[Subscript["β", k], {k, 4}]],
   Thread[{ɣ1, ɣ2, ɣ3, ɣ4} → Table[Subscript["ɣ", k], {k, 4}]],
   Thread[{δ1, δ2, δ3, δ4} → Table[Subscript["δ", k], {k, 4}]],
   Thread[{θ1, θ2, θ3, θ4} → Table[Subscript["θ", k], {k, 4}]],
   Thread[{ll1, ll2, ll3, ll4} → Table[Subscript[Style["ℓ", Italic], k], {k, 4}]],
   Thread[{mm1, mm2, mm3, mm4} → Table[Subscript[Style["m", Italic], k], {k, 4}]],
   Thread[{vv1, vv2, vv3, vv4} →
     Table[Subscript[Style["v", Italic], k], {k, 4}]]];
trigNames = {"sin", "cos", "tan", "cot", "sec", "csc"};
simpleBoxQ[b_String] := True;
simpleBoxQ[StyleBox[b_, ___]] := simpleBoxQ[b];
simpleBoxQ[(SubscriptBox | SuperscriptBox | OverscriptBox | UnderscriptBox)[
   bs__]] := AllTrue[{bs}, simpleBoxQ];
simpleBoxQ[_] := False;
trigBoxRule =
  {RowBox[{f_String /; MemberQ[trigNames, f], "(", a_, ")"}] /; simpleBoxQ[a] :>
    RowBox[{f, "", a}],
```

```
    RowBox[{f_String /; MemberQ[trigNames, f], RowBox[{"(", a_, ")"}]}] /;
      simpleBoxQ[a] :> RowBox[{f, "", a}]};
disp[e_] :=
  DisplayForm[ToBoxes[e /. dispRule, TraditionalForm] //. trigBoxRule];
sgnBox[e_] := Module[{f = e /. dispRule, sgnE = 1},
    If[Head[f] === Times && NumericQ[First[f]] && TrueQ[First[f] < 0], sgnE = -1;
     f = -f];
    {sgnE, DisplayForm[ToBoxes[f, TraditionalForm] //. trigBoxRule]}];
polyRow[terms_] := Module[{res = {}, sg, mg, tl}, Do[{sg, mg} = sgnBox[terms[[k, 1]]];
     tl = terms[[k, 2]];
     If[k == 1, If[sg < 0, AppendTo[res, "-"]],
      AppendTo[res, If[sg < 0, " - ", " + "]]];
     AppendTo[res, mg];
     If[tl =!= None, AppendTo[res, ""];
      AppendTo[res, tl]], {k, Length[terms]}];
    Row[res]];
iSet[setX_] := DisplayForm[
    RowBox[{StyleBox["I", FontSlant → "Italic"], "(", bx[setX], ")"}]];
(*the rule lens carries the two lengths that
 close the central face and is used for the figure alone,
never in a verification.A trigonometric identity is decided
 without Simplify:every angle is written as twice a half angle,
each trigonometric function of an integer combination of the half
 angles is written through the exponentials of the half angles,
and the resulting rational function,
or each component of a vector,must be 0*)
lens = {ll4 → (ll2 Sin[δ3] - ll1 Sin[δ2 + δ3]) / Sin[δ4],
    ll3 → (ll1 Sin[δ1] - ll2 Sin[δ1 + δ2]) / Sin[δ4]};
angSyms = {α1, α2, β1, β2, γ1, γ2, δ1, δ2, θ1, θ2, θ4};
hRule = Thread[angSyms → Table[2 hh[k], {k, Length[angSyms]}]];
zeroE[e_] :=
  Module[{f = e /. hRule, hv, mon}, hv = Union[Cases[f, hh[_], {0, Infinity}]];
    mon[x_] := Product[zz[h]^Coefficient[x, h], {h, hv}];
    f =
     f /. {Sin[x_] :> (mon[x] - 1 / mon[x]) / (2 I), Cos[x_] :> (mon[x] + 1 / mon[x]) / 2,
       Tan[x_] :> (mon[x] - 1 / mon[x]) / (I (mon[x] + 1 / mon[x])),
       Cot[x_] :> I (mon[x] + 1 / mon[x]) / (mon[x] - 1 / mon[x]),
       Sec[x_] :> 2 / (mon[x] + 1 / mon[x]), Csc[x_] :> 2 I / (mon[x] - 1 / mon[x])};
    AllTrue[Flatten[{Together[f]}], # === 0 &]];
(*the coordinates;
B_3,B_4,C_3,C_4 are used for the figure alone*)
Apt[2] = {0, 0, 0};
Apt[1] = {ll1, 0, 0};
Apt[3] = ll2 {Cos[δ2], Sin[δ2], 0};
Apt[4] = Apt[1] + ll4 {-Cos[δ1], Sin[δ1], 0};
Bpt[1] = Apt[1] + mm1 {-Cos[α1], Sin[α1] Cos[θ1], Sin[α1] Sin[θ1]};
Bpt[2] = Apt[2] + mm2 {Cos[α2], Sin[α2] Cos[θ1], Sin[α2] Sin[θ1]};
```

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```
Bpt[3] = Apt[3] + mm3 {-(Cos[α3] Cos[δ2 + δ3] + Sin[α3] Cos[θ3] Sin[δ2 + δ3]),
      -(Cos[α3] Sin[δ2 + δ3] - Sin[α3] Cos[θ3] Cos[δ2 + δ3]), Sin[α3] Sin[θ3]};
Bpt[4] = Apt[4] + mm4 {Cos[α4] Cos[δ2 + δ3] - Sin[α4] Cos[θ3] Sin[δ2 + δ3],
      Cos[α4] Sin[δ2 + δ3] + Sin[α4] Cos[θ3] Cos[δ2 + δ3], Sin[α4] Sin[θ3]};
Cpt[1] = Apt[1] + vv1 {-(Cos[γ1] Cos[δ1] + Sin[γ1] Cos[θ4] Sin[δ1]),
      Cos[γ1] Sin[δ1] - Sin[γ1] Cos[θ4] Cos[δ1], Sin[γ1] Sin[θ4]};
Cpt[2] = Apt[2] + vv2 {Cos[γ2] Cos[δ2] + Sin[γ2] Cos[θ2] Sin[δ2],
      Cos[γ2] Sin[δ2] - Sin[γ2] Cos[θ2] Cos[δ2], Sin[γ2] Sin[θ2]};
Cpt[3] = Apt[3] + vv3 {-(Cos[γ3] Cos[δ2] - Sin[γ3] Cos[θ2] Sin[δ2]),
      -(Cos[γ3] Sin[δ2] + Sin[γ3] Cos[θ2] Cos[δ2]), Sin[γ3] Sin[θ2]};
Cpt[4] = Apt[4] + vv4 {Cos[γ4] Cos[δ1] - Sin[γ4] Cos[θ4] Sin[δ1],
      -(Cos[γ4] Sin[δ1] + Sin[γ4] Cos[θ4] Cos[δ1]), Sin[γ4] Sin[θ4]};
uB[i_] := (Bpt[i] - Apt[i]) / {mm1, mm2, mm3, mm4}[[i]];
uC[i_] := (Cpt[i] - Apt[i]) / {vv1, vv2, vv3, vv4}[[i]];
eNext[2] = {1, 0, 0};
ePrev[1] = {-Cos[δ1], Sin[δ1], 0};
ePrev[2] = {Cos[δ2], Sin[δ2], 0};
nextA = {2, 1, 4, 3};
prevA = {4, 3, 2, 1};
thNextIdx = {1, 1, 3, 3};
thPrevIdx = {4, 2, 2, 4};
(*the central angle on the alpha-side of A_i and the one on the gamma-side,
and the representative of each shared quantity:x is shared by A_1 and A_2*)
nIdx = {1, 1, 3, 3};
pIdx = {4, 2, 2, 4};
xRp = {1, 1, 3, 3};
yRp = {1, 2, 2, 1};
qXr[i_] := pt["x", xRp[[i]]]; qYr[i_] := pt["y", yRp[[i]]];
(*the branch sign of the flexion parametrized by cot(theta_1/2)*)
e0Tag[k_] :=
  DisplayForm[SubscriptBox[StyleBox["e", FontSlant → "Italic"], "0"]];
sigTag[i_] :=
  DisplayForm[SqrtBox[bx[Row[{mulR[uPlain, qXr[i], qYr[i]], "(1 + ", pt["z", i],
       ")(1 + ", mulR[uPlain, pt["z", i]], ")"}]]]];
radTag[i_, fwd_] := DisplayForm[
   SqrtBox[bx[Row[{mulR[uPlain, If[fwd, qYr[i], qXr[i]], pt["z", i]], " ", dTag[
        If[fwd, qXr[i], qYr[i]], cotTag[thI[If[fwd, nIdx[[i]], pIdx[[i]]]]]}]]]];
flxF[i_, fwd_] :=
  Module[{src = If[fwd, nIdx[[i]], pIdx[[i]]], dst = If[fwd, pIdx[[i]], nIdx[[i]]],
    sg = If[fwd, (-1)^i, (-1)^(i + 1)], qA = If[fwd, qYr[i], qXr[i]],
    qB = If[fwd, qXr[i], qYr[i]]}, Row[{cotTag[thI[dst]], " = ",
     frac[Row[{gk["ε", i], "(", mulR[cotTag[thI[src]], sigTag[i]], If[sg < 0,
         " - ", " + "], mulR[e0Tag[src], pt["e", i], radTag[i, fwd]], ")"}],
      Row[{mulR[qA, uPlain], "(", pt["z", i], " + ",
        mulR[qB, cotSqTag[thI[src]]], ")"}]}]}]];
bLen = {mm1, mm2, mm3, mm4}; cLen = {vv1, vv2, vv3, vv4};
angA = {α1, α2, α3, α4};
```

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```
angB = {β1, β2, β3, β4};
angG = {ɣ1, ɣ2, ɣ3, ɣ4};
angD = {δ1, δ2, δ3, δ4};
thNext = {θ1, θ1, θ3, θ3};
thPrev = {θ4, θ2, θ2, θ4};
thCent = {θ1, θ2, θ3, θ4};
sgnV = {-1, 1, -1, 1};
nraw[0] = Cross[ll1 eNext[2], ll2 ePrev[2]];
nraw[1] = Cross[ll1 eNext[2], Bpt[2] - Apt[2]];
nraw[2] = Cross[Cpt[2] - Apt[2], ll2 ePrev[2]];
nraw[4] = Cross[ll4 ePrev[1], Cpt[1] - Apt[1]];
nrawP[1] = Cross[Bpt[1] - Apt[1], Cpt[1] - Apt[1]];
nrawP[2] = Cross[Cpt[2] - Apt[2], Bpt[2] - Apt[2]];
nlen[0] = ll1 ll2 Sin[δ2];
nlen[1] = ll1 mm2 Sin[α2];
nlen[2] = ll2 vv2 Sin[ɣ2];
nlen[4] = ll4 vv1 Sin[ɣ1];
(*for each normal:its unit vector;
for each corner normal:its two edge vectors in the order of the definition,
and its components;
the third vectors of the central dihedral angles*)
ndir[0] = {0, 0, 1};
ndir[1] = {0, -Sin[θ1], Cos[θ1]};
ndir[2] = {-Sin[θ2] Sin[δ2], Sin[θ2] Cos[δ2], Cos[θ2]};
ndir[4] = {Sin[δ1] Sin[θ4], Cos[δ1] Sin[θ4], Cos[θ4]};
nPsrc[1] =
 {{Bpt[1] - Apt[1], evec["A", 1, "B", 1]}, {Cpt[1] - Apt[1], evec["A", 1, "C", 1]}};
nPsrc[2] = {{Cpt[2] - Apt[2], evec["A", 2, "C", 2]},
  {Bpt[2] - Apt[2], evec["A", 2, "B", 2]}};
nPshow[i_] := bLen[[i]] cLen[[i]]
   Collect[Cross[nPsrc[i][[1, 1]], nPsrc[i][[2, 1]]] / (bLen[[i]] cLen[[i]]),
    {Sin[thNext[[i]]], Cos[thNext[[i]]], Sin[thPrev[[i]]], Cos[thPrev[[i]]]}, Factor];
wCent[1] = ll1 eNext[2];
wCent[2] = -ll2 ePrev[2];
wCent[4] = ll4 ePrev[1];
wCentLen[1] = ll1;
wCentLen[2] = ll2;
wCentLen[4] = ll4;
wName[1] = evec["A", 2, "A", 1];
wName[2] = evec["A", 3, "A", 2];
wName[4] = evec["A", 1, "A", 4];
betaForm[i_] := Cos[angA[[i]]]
    (Cos[angG[[i]]] Cos[angD[[i]]] + Sin[angG[[i]]] Sin[angD[[i]]] Cos[thPrev[[i]]]) +
   Sin[angA[[i]]] (Cos[thNext[[i]]]
       (Cos[angG[[i]]] Sin[angD[[i]]] - Sin[angG[[i]]] Cos[angD[[i]]] Cos[thPrev[[i]]]) +
      Sin[thNext[[i]]] Sin[angG[[i]]] Sin[thPrev[[i]]]);
outBnum[i_] := Cos[angD[[i]]] Cos[angG[[i]]] +
```

```
  Sin[angD[[i]]] Sin[angG[[i]]] Cos[thPrev[[i]]] - Cos[angA[[i]]] Cos[angB[[i]]];
outBden[i_] := Sin[angA[[i]]] Sin[angB[[i]]];
eqB[i_] := Sin[thNext[[i]]]
     (Cos[angG[[i]]] Sin[angD[[i]]] - Sin[angG[[i]]] Cos[angD[[i]]] Cos[thPrev[[i]]]) -
   Cos[thNext[[i]]] Sin[angG[[i]]] Sin[thPrev[[i]]];
sgD[i_] := Subscript["σ", i];
bAD[i_] := Subscript[OverBar["α"], i];
bBD[i_] := Subscript[OverBar["β"], i];
bGD[i_] := Subscript[OverBar["ɣ"], i];
bDD[i_] := Subscript[OverBar["δ"], i];
sig[i_] := (angA[[i]] + angB[[i]] + angG[[i]] + angD[[i]]) / 2;
barA[i_] := sig[i] - angA[[i]];
barB[i_] := sig[i] - angB[[i]];
barG[i_] := sig[i] - angG[[i]];
barD[i_] := sig[i] - angD[[i]];
barRule[i_] := {sgD[i] → sig[i], bAD[i] → barA[i],
   bBD[i] → barB[i], bGD[i] → barG[i], bDD[i] → barD[i]};
ratA[i_] := Sin[angA[[i]]] / Sin[barA[i]];
ratB[i_] := Sin[angB[[i]]] / Sin[barB[i]];
ratG[i_] := Sin[angG[[i]]] / Sin[barG[i]];
ratD[i_] := Sin[angD[[i]]] / Sin[barD[i]];
prodM[i_] := ratA[i] ratB[i] ratG[i] ratD[i];
prodR[i_] := ratA[i] ratD[i];
prodS[i_] := ratG[i] ratD[i];
prodF[i_] := ratA[i] ratG[i];
quU[i_] := 1 - prodM[i];
quX[i_] := 1 / (prodR[i] - 1);
quY[i_] := 1 / (prodS[i] - 1);
quZ[i_] := 1 / (prodF[i] - 1);
(*the two corner angles of this section,
along A_1 B_1 and along A_2 B_2,and the numbers of their
 formulas:the tilde numbers and the four sine ratios of the angles a,b,g,
d in the roles of alpha,beta,gamma,delta.With kn=cot(theta_1/2) and kp=
  cot of the half of the other central angle at the vertex,
phiE is the polynomial of the sine of the corner angle,
bric is the left side of Bricard's equation at the vertex,
with the coefficients coA,coW,coV,coD,coP;
coQ1,coQ2 are the coefficients of the polynomial of 1-
 cos of the corner angle and-coQ3,-coQ4 those of 1+cos;
coC0 enters the long identity of the block of Claim 3;
coA2,coV2,coP2 are coA,coV,
coP of the substituted angles.The coefficients are written with the
 display symbols of the half sums and are compared through barRule*)
nameB = {thP[1, 1], thP[1, 2]};
quYt[i_] := -(1 + quY[i]) / (1 + quU[i] quY[i]);
quZt[i_] := -(1 + quZ[i]) / (1 + quU[i] quZ[i]);
vRat[a_, b_, g_, d_] := With[{sgm = (a + b + g + d) / 2}, {Sin[a] / Sin[sgm - a],
```

```
    Sin[b] / Sin[sgm - b], Sin[g] / Sin[sgm - g], Sin[d] / Sin[sgm - d]}];
coA[i_] := -Sin[bDD[i]] Sin[angB[[i]] - bDD[i]];
coW[i_] := -Sin[bAD[i]] Sin[angB[[i]] - bAD[i]];
coV[i_] := -Sin[bGD[i]] Sin[angB[[i]] - bGD[i]];
coD[i_] := Sin[sgD[i]] Sin[bBD[i]];
coP[i_] := Sin[angA[[i]]] Sin[angG[[i]]];
coQ1[i_] := -Sin[bGD[i] - angB[[i]]] Sin[bDD[i] - angB[[i]]];
coQ2[i_] := Sin[bAD[i]] Sin[bBD[i]];
coA2[i_] := -Sin[bAD[i]] Sin[angG[[i]] - bAD[i]];
coV2[i_] := -Sin[bBD[i]] Sin[angG[[i]] - bBD[i]];
coP2[i_] := Sin[angB[[i]]] Sin[angD[[i]]];
coQ3[i_] := -Sin[bGD[i]] Sin[bDD[i]];
coQ4[i_] := Sin[sgD[i]] Sin[bAD[i] - angB[[i]]];
coC0[i_] := Sin[angG[[i]] - bBD[i]] Sin[bAD[i] - angB[[i]]];
phiE[i_] := Sin[angD[[i]] - angG[[i]]] kn kp^2 -
   Sin[angG[[i]]] kn^2 kp + Sin[angG[[i]] + angD[[i]]] kn + Sin[angG[[i]]] kp;
bric[i_] := (coA[i] kp^2 + coW[i]) kn^2 - 2 coP[i] kn kp + (coV[i] kp^2 + coD[i]);
c2Num[i_] := polyRow[{{coP[i], cotTag[thI[thPrevIdx[[i]]]]},
    {-coA[i], Row[{cotTag[thI[1]], " ", cotSqTag[thI[thPrevIdx[[i]]]]}]},
    {-coW[i], cotTag[thI[1]]}}];
c2Den[i_] := polyRow[{{coQ1[i], cotSqTag[thI[thPrevIdx[[i]]]]}, {coQ2[i], None}}];
c2Psi[i_] :=
  polyRow[{{-coQ3[i], cotSqTag[thI[thPrevIdx[[i]]]]}, {-coQ4[i], None}}];
sn[v_] := Row[{"sin", v}];
sinSq[v_] := Row[{DisplayForm[SuperscriptBox["sin", "2"]], "", v}];
sqR[e_] := DisplayForm[SuperscriptBox[RowBox[{"(", bx[e], ")"}], "2"]];
tSq = DisplayForm[SuperscriptBox[StyleBox["t", FontSlant → "Italic"], "2"]];
omD = DisplayForm[StyleBox["ω", FontSlant → "Italic"]];
phD[""] = DisplayForm["Φ"];
phD[j_Integer] := DisplayForm[SubscriptBox["Φ", ToString[j]]];
sigTagT[i_] :=
  DisplayForm[SqrtBox[bx[Row[{mulR[uPlain, pt["x", 1], ptT["y", i]], "(1 + ",
       ptT["z", i], ")(1 + ", mulR[uPlain, ptT["z", i]], ")"}]]]];
radTagT[i_] :=
  DisplayForm[SqrtBox[bx[Row[{mulR[uPlain, ptT["y", i], ptT["z", i]],
       " ", dTag[pt["x", 1], cotTag[thI[1]]]}]]]];
radTagTt[i_] := DisplayForm[SqrtBox[bx[Row[
      {mulR[uPlain, ptT["y", i], ptT["z", i]], " ", dTag[pt["x", 1], tSym]}]]]];
rhsTt[i_] := frac[Row[{gk["ε", i], "(", mulR[tSym, sigTagT[i]],
     If[i == 1, " - ", " + "], mulR[e0Tag[1], pt["e", i], radTagTt[i]], ")"}],
   Row[{mulR[ptT["y", i], uPlain], "(", ptT["z", i], " + ",
     mulR[pt["x", 1], tSq], ")"}]];
flxT[i_] := Row[{cotTag[nameB[[i]]], " = ",
    frac[Row[{gk["ε", i], "(", mulR[cotTag[thI[1]], sigTagT[i]],
       If[i == 1, " - ", " + "], mulR[e0Tag[1], pt["e", i], radTagT[i]], ")"}],
     Row[{mulR[ptT["y", i], uPlain], "(", ptT["z", i], " + ",
       mulR[pt["x", 1], cotSqTag[thI[1]]], ")"}]]}];
```

```
(*the reductions of the two long identities,
checked with the sines as independent symbols:sA,sB,sG,sD,
sS for the sines of the barred angles and of the half sum,
xA,xG,xD for sin(beta-abar),sin(beta-gbar),sin(beta-dbar),
pP,pQ for sin(alpha) sin(gamma),sin(beta) sin(delta);
d9,d10,d11 are the left sides minus the right sides of the three product-
 to-sum identities;
linkList rewrites the coefficients in these symbols.monoD displays
 t^a cot^b of the half of the central angle k,phiRows restates
  the polynomials of the block of Claim 2 with t=cot(theta_1/2)*)
formalRed = Module[{sA, sB, sG, sD, sS, xA, xG, xD, pP, pQ, fa, fw, fv, fd, fq1, fq2,
    fq3, fq4, fa2, fv2, fc0, fb, fR, fph1, fph2, fpsi, d9, d10, d11}, fa = -sD xD;
   fw = -sA xA;
   fv = -sG xG;
   fd = sS sB;
   fq1 = -xG xD;
   fq2 = sA sB;
   fq3 = -sG sD;
   fq4 = -sS xA;
   fa2 = sA xD;
   fv2 = -sB xG;
   fc0 = -xG xA;
   fb = (fa kp^2 + fw) kn^2 - 2 pP kn kp + (fv kp^2 + fd);
   fph1 = pP kp - kn (fa kp^2 + fw);
   fph2 = fq1 kp^2 + fq2;
   fpsi = -fq3 kp^2 - fq4;
   fR = (pQ - pP) kn + (fa kn^2 + fv) kp;
   d9 = sA sG + sB sD - pP - pQ;
   d10 = xA xG + sB sD - pP;
   d11 = sA sG - pP - sS xD;
   {(fa2 kn^2 + fv2) fph1 - fR fph2 + (fq1 kp + fa2 kn) fb -
     (sA xD d10 kn^2 kp - xG xD d9 kn kp^2 + sA sB (d9 - d10 - d11) kn + sB xG d11 kp),
    (fa2 kn^2 + fv2) fpsi - fR fph1 - (fa kn kp + fc0) fb -
     (sD xD (d9 - d10) kn^2 kp^2 + sA xA (d9 - d10 - d11) kn^2 -
       sG xG d10 kp^2 + (pP d9 + (sS xD - pP) d10 + (xA xG - pP) d11) kn kp)};
linkList[i_] := {coA2[i] - Sin[bAD[i]] Sin[angB[[i]] - bDD[i]],
    coV2[i] + Sin[bBD[i]] Sin[angB[[i]] - bGD[i]],
    coQ1[i] + Sin[angB[[i]] - bGD[i]] Sin[angB[[i]] - bDD[i]],
    coQ4[i] + Sin[sgD[i]] Sin[angB[[i]] - bAD[i]],
    coC0[i] + Sin[angB[[i]] - bGD[i]] Sin[angB[[i]] - bAD[i]]} /. barRule[i];
monoD[a_, b_, k_] :=
  Row[Riffle[Join[Which[a == 0, {}, a == 1, {tSym}, True, {DisplayForm[
        SuperscriptBox[StyleBox["t", FontSlant → "Italic"], ToString[a]]]}],
     Which[b == 0, {}, b == 1, {cotTag[thI[k]]}, True,
      {DisplayForm[RowBox[{SuperscriptBox["cot", ToString[b]],
          "(", bx[thI[k]], "/", "2", ")"}]]}]], " "]];
phiRows[i_] := Module[{k = thPrevIdx[[i]], lT, LT}, lT = cotTag[thI[k]];
```

```
    LT = cotSqTag[thI[k]];
    {prow[Row[
       {phD[""], " := ", polyRow[{{Sin[angD⟦i⟧ - angG⟦i⟧], Row[{tSym, " ", LT}]},
          {-Sin[angG⟦i⟧], Row[{tSq, " ", lT}]}, {Sin[angG⟦i⟧ + angD⟦i⟧], tSym},
          {Sin[angG⟦i⟧], lT}}]}], givenChip,
      Row[{"as in the block of Claim 2, with ", tSym, " = ", cotTag[thI[1]]}]],
     prow[Column[{Row[{phD[0], " := (", polyRow[{{coA[i], LT}, {coW[i], None}}],
          ") ", tSq}], Row[{"          - 2 ", mulR[disp[coP[i]], tSym, lT]}],
       Row[{"          + (", polyRow[{{coV[i], LT}, {coD[i], None}}], ")"}]},
      Spacings → 0.6], givenChip,
      Row[{"as there; ", phD[0], " = 0 by its (", i, ".4)"}]],
     prow[Row[{phD[1], " := ", polyRow[{{coP[i], lT}, {-coA[i],
           Row[{tSym, " ", LT}]}, {-coW[i], tSym}}]}], givenChip, "as there"],
     prow[Row[{phD[2], " := ", c2Den[i]}], givenChip, "as there"]}];
resultsAll = {};
(*the figure:the orthogonal projection of the
   nine faces of one net onto the plane of the first two axes,
seen from the positive side of the third,
which is drawn as the mark with a dot at A_2;
only the flat and dihedral angles used in this section are marked.The
 data num,rAv,rV,rO,the three nudge tables,the two axis lengths,
pad and shiftL are the only entries to be tuned by hand*)
figBlock = Module[{dg = π / 180., num, vA, vB, vC, vD, grey = GrayLevel[0.45],
    blue = RGBColor[0.15, 0.35, 0.7], rAv = {0.5, 0.58, 0.40, 0.40},
    rV = 0.72, rO = 0.46, aNudge, bNudge, cNudge, xEnd = 9.0, yEnd = 7.8,
    pad = 0.7, shiftL = 1.2, prj, bis, angLab, gapDir, arcTo, arcs, angLabs,
    vtxLabs, thinE, thickE, dots, axes, zmark, xd, yd, allPts, xr, yr},
   num = {ll1 → 4.4, ll2 → 3.8, mm1 → 3.2, mm2 → 3.0, mm3 → 3.1, mm4 → 3.0, vv1 → 3.4,
     vv2 → 3.2, vv3 → 3.0, vv4 → 3.3, δ1 → 105 dg, δ2 → 70 dg, δ3 → 95 dg, δ4 → 90 dg,
     θ1 → 150 dg, θ2 → 145 dg, θ3 → 152 dg, θ4 → 148 dg, α1 → 100 dg, α2 → 125 dg,
     α3 → 92 dg, α4 → 86 dg, γ1 → 95 dg, γ2 → 90 dg, γ3 → 87 dg, γ4 → 93 dg};
   aNudge = {{0.2, -0.25}, {-0.30, -0.30}, {0, 0.30}, {0, 0.30}};
   bNudge = {{0, 0.08}, {0, 0.08}, {0, -0.08}, {0, -0.08}};
   cNudge = {{-0.08, 0}, {0.08, 0}, {0.08, 0}, {-0.08, 0}};
   prj[vec_] := N[Take[vec /. lens /. num, 2]];
   vA = Table[prj[Apt[i]], {i, 4}];
   vB = Table[prj[Bpt[i]], {i, 4}];
   vC = Table[prj[Cpt[i]], {i, 4}];
   vD = Table[vA⟦i⟧ + 0.85 (vB⟦i⟧ + vC⟦i⟧ - 2 vA⟦i⟧), {i, 4}];
   bis[o_, nbr1_, qq_] := Normalize[Normalize[nbr1 - o] + Normalize[qq - o]];
   angLab[o_, nbr1_, qq_, lbl_, rLab_] :=
    Text[Style[lbl, 11, grey], o + rLab bis[o, nbr1, qq]];
   gapDir[o_, nbrs_] :=
    Module[{an, gp, kk}, an = Sort[ArcTan[#⟦1⟧, #⟦2⟧] & /@ ((# - o) & /@ nbrs)];
     gp = Differences[Append[an, First[an] + 2 π]];
     kk = First[Ordering[gp, -1]];
     {Cos[#], Sin[#]} &[an⟦kk⟧ + gp⟦kk⟧ / 2]];
```

```
arcTo[pa_, pb_, lbl_, tq_, into_, rr_, sw_, le_] :=
 Module[{d, nn, mp, sg, cps, hd}, d = Normalize[pb - pa];
  nn = {-d[[2]], d[[1]]};
  mp = pa + tq (pb - pa);
  sg = Sign[nn.(into - mp)];
  cps = {mp - 0.45 rr sg nn, mp - 0.27 rr sg nn + 0.34 rr d,
    mp + 0.27 rr sg nn + 0.34 rr d, mp + 0.45 rr sg nn};
  If[sw < 0, cps = Reverse[cps]];
  hd = Last[cps];
  {{grey, Thickness[0.0024], Arrowheads[0.016], Arrow[BezierCurve[cps]]},
   Text[Style[lbl, 11, grey], If[le > 0,
     hd + 0.05 Normalize[hd - mp] + 0.35 rr d, mp + 0.50 rr d + 0.30 rr sg nn]]}];
arcs = Join[{arcTo[vA[[2]], vA[[1]], thI[1], 0.50, Mean[{vB[[1]], vB[[2]]}], 1, 1, 0],
   arcTo[vA[[2]], vA[[3]], thI[2], 0.50, Mean[{vC[[2]], vC[[3]]}], 1, 1, 0],
   arcTo[vA[[4]], vA[[1]], thI[4], 0.50, Mean[{vC[[4]], vC[[1]]}], -1, -1, 1]},
  Table[arcTo[vA[[i]], vB[[i]], nameB[[i]], 0.55, vC[[i]], 1, 1, 0], {i, {1, 2}}]];
angLabs = Flatten[
  Table[{angLab[vA[[i]], vA[[nextA[[i]]]], vA[[prevA[[i]]]], gk["δ", i], rAv[[i]],
    angLab[vA[[i]], vA[[nextA[[i]]]], vB[[i]], gk["α", i], rAv[[i]],
    angLab[vA[[i]], vB[[i]], vC[[i]], gk["β", i], rAv[[i]],
    angLab[vA[[i]], vA[[prevA[[i]]]], vC[[i]], gk["γ", i], rAv[[i]]}, {i, {1, 2}}]];
vtxLabs = Join[Table[
   Text[Style[pt["A", i], 12], vA[[i]] + rV bis[vA[[i]], vB[[i]], vC[[i]]] + aNudge[[i]],
   {i, 4}], Table[Text[Style[pt["B", i], 12],
    vB[[i]] + rO gapDir[vB[[i]], {vA[[i]], vD[[i]], vB[[nextA[[i]]]]}] + bNudge[[i]],
   {i, 4}], Table[Text[Style[pt["C", i], 12], vC[[i]] +
     rO gapDir[vC[[i]], {vA[[i]], vD[[i]], vC[[prevA[[i]]]]}] + cNudge[[i]], {i, 4}]];
thinE = {Thickness[0.0024], GrayLevel[0.4], Line[{vB[[2]], vB[[1]]}],
  Line[{vC[[3]], vC[[2]]}], Line[{vB[[4]], vB[[3]]}], Line[{vC[[1]], vC[[4]]}],
  Sequence @@ Table[Line[{vB[[i]], vD[[i]], vC[[i]]}], {i, 4}]};
thickE = {Thickness[0.006],
  GrayLevel[0.1], Line[{vA[[1]], vA[[2]], vA[[3]], vA[[4]], vA[[1]]}],
  Sequence @@ Table[Line[{vA[[i]], vB[[i]]}], {i, 4}],
  Sequence @@ Table[Line[{vA[[i]], vC[[i]]}], {i, 4}]};
xd = Normalize[vA[[1]] - vA[[2]]]; yd = {-xd[[2]], xd[[1]]};
axes = {blue, Thickness[0.0026], Arrowheads[0.018],
  Arrow[{vA[[2]], vA[[2]] + xEnd xd}], Arrow[{vA[[2]], vA[[2]] + yEnd yd}],
  Text[Style[tld["x"], 12, blue], vA[[2]] + (xEnd + 0.26) xd],
  Text[Style[tld["y"], 12, blue], vA[[2]] + (yEnd + 0.26) yd]};
dots = {White, EdgeForm[{GrayLevel[0.1], Thickness[0.002]}], Sequence @@
   Table[Disk[pp, 0.11], {pp, Join[{vA[[1]], vA[[3]], vA[[4]]}, vB, vC, vD]}]};
zmark = {White, EdgeForm[{blue, Thickness[0.0026]}],
  Disk[vA[[2]], 0.14], EdgeForm[None], blue, Disk[vA[[2]], 0.05],
  Text[Style[tld["z"], 12, blue], vA[[2]] + 0.10 xd - 0.3 yd]};
allPts = Join[vA, vB, vC, vD, {vA[[2]] + (xEnd + 0.5) xd, vA[[2]] + (yEnd + 0.5) yd}];
xr = {Min[First /@ allPts] - pad, Max[First /@ allPts] + pad + shiftL};
yr = {Min[Last /@ allPts] - pad, Max[Last /@ allPts] + pad};
```

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```
    Graphics[{axes, thinE, thickE, arcs, angLabs, dots, zmark, vtxLabs},
     ImageSize → 700, PlotRange → {xr, yr}, PlotRangePadding → None,
     Background → None, BaseStyle → {FontFamily → "Times"}]];

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[Panel[Column[{Style["Given hypotheses and definitions:", Bold, 14],
     Style["A 3 x 3 net with, for each i = 1, 2,", Bold, 13],
     Style[Column[{Row[{"the vertices ", pt["A", "i"], ", ",
          pt["B", "i"], ", ", pt["C", "i"], ", and the vertices ",
          pt["A", 3], ", ", pt["A", 4], " of the central face"}],
        Row[{"the flat angles ", gk["α", "i"], ", ", gk["β", "i"], ", ",
          gk["γ", "i"], ", ", gk["δ", "i"], " ∈ (0, π) at ", pt["A", "i"]}],
        Row[{"the edge lengths ", pt["ℓ", 1], " = |", edg["A", 1, "A", 2],
          "|,   ", pt["ℓ", 2], " = |", edg["A", 2, "A", 3],
          "|,   ", pt["ℓ", 4], " = |", edg["A", 4, "A", 1], "|,   ",
          pt["m", "i"], " = |", edg["A", "i", "B", "i"], "|,   ",
          pt["v", "i"], " = |", edg["A", "i", "C", "i"], "|"}],
        Row[{"the dihedral angles ", thI[1], ", ", thI[2], ", ", thI[4],
          " ∈ [-π, π) \\ {0} along the edges ", edg["A", 1, "A", 2], ", ",
          edg["A", 2, "A", 3], ", ", edg["A", 4, "A", 1]}], Spacer[{0, 6}],
        Grid[{{Row[{disp[sgD[iv]], " = ", disp[(Subscript["α", iv] + Subscript[
                  "β", iv] + Subscript["γ", iv] + Subscript["δ", iv]) / 2]}],
          Row[{disp[bAD[iv]], " = ", disp[sgD[iv] - Subscript["α", iv]]}],
          Row[{disp[bBD[iv]], " = ", disp[sgD[iv] - Subscript["β", iv]]}],
          Row[{disp[bGD[iv]], " = ", disp[sgD[iv] - Subscript["γ", iv]]}],
          Row[{disp[bDD[iv]], " = ", disp[sgD[iv] - Subscript["δ", iv]]}]},
         {Row[{pt["a", "i"], " = ", frac[Row[{"sin", gk["α", "i"]}],
             Row[{"sin", disp[bAD[iv]]}]]}], Row[{pt["b", "i"], " = ",
           frac[Row[{"sin", gk["β", "i"]}], Row[{"sin", disp[bBD[iv]]}]]}],
          Row[{pt["c", "i"], " = ", frac[Row[{"sin", gk["γ", "i"]}],
             Row[{"sin", disp[bGD[iv]]}]]}], Row[{pt["d", "i"], " = ",
           frac[Row[{"sin", gk["δ", "i"]}], Row[{"sin", disp[bDD[iv]]}]]}],
          Row[{pt["ε", "i"], " = sign(sin", disp[sgD[iv]], ")"}]},
         {Row[{pt["M", "i"], " = ", mulR[pt["a", "i"],
             pt["b", "i"], pt["c", "i"], pt["d", "i"]]}],
          Row[{pt["r", "i"], " = ", mulR[pt["a", "i"], pt["d", "i"]]}],
          Row[{pt["s", "i"], " = ", mulR[pt["c", "i"], pt["d", "i"]]}],
          Row[{pt["f", "i"], " = ", mulR[pt["a", "i"], pt["c", "i"]]}], ""},
         {Row[{pt["u", "i"], " = 1 - ", pt["M", "i"]}],
          Row[{pt["x", "i"], " = ", frac["1", Row[{pt["r", "i"], " - 1"}]]}],
          Row[{pt["y", "i"], " = ", frac["1", Row[{pt["s", "i"], " - 1"}]]}],
          Row[{pt["z", "i"], " = ", frac["1", Row[{pt["f", "i"], " - 1"}]]}],
          ""}, {Row[{ptT["y", "i"], " = ", frac[Row[{"-(1 + ", pt["y", "i"],
               ")"}], Row[{"1 + ", mulR[pt["u", "i"], pt["y", "i"]]}]]}],
          Row[{ptT["z", "i"], " = ", frac[Row[{"-(1 + ", pt["z", "i"], ")"}],
```

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          Row[{"1 + ", mulR[pt["u", "i"], pt["z", "i"]]}]]}], "", "", ""}},
     Alignment → Left, Spacings → {2.4, 1.1}], Spacer[{0, 6}],
    Row[{gk["α", "i"], " ± ", gk["β", "i"], " ± ", gk["γ", "i"],
      " ± ", gk["δ", "i"], " ≠ 0 (mod 2π)"}],
    Row[{disp[bAD[iv]], ", ", disp[bBD[iv]], ", ", disp[bGD[iv]], ", ",
      disp[bDD[iv]], " ∈ (0, π)"}], Row[{pt["ε", "i"], " ∈ {-1, 1}"}],
    Row[{pt["x", "i"], ", ", pt["y", "i"], ", ", pt["z", "i"],
      ", ", ptT["y", "i"], ", ", ptT["z", "i"], " ≠ -1, 0;   ",
      mulR[pt["u", "i"], pt["x", "i"], pt["y", "i"], pt["z", "i"]],
      " > 0"}]}, Spacings → 0.7], 13],
Spacer[{0, 10}], Style["Normals:", Bold, 14],
Style[
 Row[{nv[0], " = ", evec["A", 2, "A", 1], " × ", evec["A", 2, "A", 3]}],
 13], Style[Grid[
  {{Row[{nv[1], " = ", evec["A", 2, "A", 1], " × ", evec["A", 2, "B", 2]}],
    Row[{nv[2], " = ", evec["A", 2, "C", 2], " × ", evec["A", 2, "A", 3]}],
    Row[{nv[4], " = ", evec["A", 1, "A", 4], " × ", evec["A", 1, "C", 1]}]}},
  Spacings → {3, 0}], 13], Style[Grid[
  {{Row[{nvP[1], " = ", evec["A", 1, "B", 1], " × ", evec["A", 1, "C", 1]}],
    Row[{nvP[2], " = ", evec["A", 2, "C", 2], " × ",
      evec["A", 2, "B", 2]}]}}, Spacings → {3, 0}], 13],
Spacer[{0, 10}], Style["Dihedral angles:", Bold, 14],
Style[
 caseDef[Row[{"θ(", wv[1], ", ", wv[2], ", ", wv[3], ")"}],
  {{Row[{"∠(", wv[1], ", ", wv[2], ") ∈ (0, π),"}],
    Row[{"if det(", wv[1], ", ", wv[2], ", ", wv[3], ") > 0,"}]},
   {Row[{"-∠(", wv[1], ", ", wv[2], ") ∈ [-π, 0],"}],
    Row[{"if det(", wv[1], ", ", wv[2], ", ", wv[3], ") ≤ 0."}]}}], 13],
Style["central dihedral angles:", Italic, 13],
Style[
 Grid[{{Row[{thI[1], " = θ(", nv[0], ", ", nv[1], ", ",
      evec["A", 2, "A", 1], ")"}], Row[{thI[2], " = θ(", nv[0],
      ", ", nv[2], ", ", evec["A", 3, "A", 2], ")"}], Row[{thI[4],
      " = θ(", nv[0], ", ", nv[4], ", ", evec["A", 1, "A", 4], ")"}]}},
  Alignment → Left, Spacings → {2, 0.8}], 13], Spacer[{0, 6}],
Style["corner dihedral angles:", Italic, 13],
Style[
 Grid[{{Row[{thP[1, 1], " = θ(", nv[1], ", ", nvP[1], ", ",
      evec["B", 1, "A", 1], ")"}], Row[{thP[1, 2], " = θ(",
      nv[1], ", ", nvP[2], ", ", evec["A", 2, "B", 2], ")"}]}},
  Alignment → Left, Spacings → {2, 0.8}], 13], Spacer[{0, 10}],
Style["Cartesian coordinates:", Bold, 14],
Style[Row[{"origin at ", pt["A", 2]}], 13],
Style[Row[{"central face in the half-plane ", tld["y"],
   " ≥ 0 of the ", tld["x"], tld["y"], "-plane, the ray ",
   edg["A", 2, "A", 1], " as the ", tld["x"], "-axis"}], 13],
Column[{figBlock, Style["Figure 1", 12, GrayLevel[0.5]]},
```

```
 Alignment → Center], Spacer[{0, 8}],
Style[Column[Join[Table[Row[{pt["A", k], " = ", disp[Apt[k]]}], {k, 4}],
   Table[Row[{pt["B", k], " = ", pt["A", k],
      " + ", pt["m", k], " ", disp[uB[k]]}], {k, 2}],
   Table[Row[{pt["C", k], " = ", pt["A", k], " + ", pt["v", k],
      " ", disp[uC[k]]}], {k, 2}]], Spacings → 0.5], 13],
Style[Row[{"The definition gives θ(", wv[1], ", ", wv[2],
   ", ", wv[3], ") = ±∠(", wv[1], ", ", wv[2],
   "), by the sign of the determinant. The cosine is
     even and the sine is odd, so cos θ = ",
   frac[Row[{wv[1], " · ", wv[2]}], mulR[absR[wv[1]], absR[wv[2]]]],
   " and sin θ = ",
   frac[Row[{"det(", wv[1], ", ", wv[2], ", ", wv[3], ")"}],
    mulR[absR[wv[1]], absR[wv[2]], absR[wv[3]]]],
   " hold whichever sign the determinant has; the formula
     for the sine holds because ", wv[3],
   ", the direction of an edge common to the two faces, is
     orthogonal to ", wv[1], " and ", wv[2],
   ", so that ", wv[1], " × ", wv[2], " is a multiple of ", wv[3],
   ". No formula of the notebook distinguishes the two
     cases, the coordinates above included."}], 12,
 GrayLevel[0.5]], Spacer[{0, 10}], Style["Flexion:", Bold, 14],
Style[Row[{pt["x", 2], " = ", pt["x", 1], ",   ",
   pt["u", 1], " = ", pt["u", 2], " = ", uPlain}], 13],
Style[Row[{pt["e", 1], ", ", pt["e", 2], ", ", e0Tag[1], " ∈ {-1, 1}"}],
 13],
Style[Row[{dTag[capX, tSym], " = (", mulR[capX, tSq], " - 1)(1 - ",
   mulR[uPlain, capX, tSq], "),      ", iSet[capX], " = {", tSym,
   " ∈ ℝ : ", mulR[capX, dTag[capX, tSym]], " > 0},      for ",
   capX, ", ", tSym, " ∈ ℝ"}], 13], Spacer[{0, 6}],
Style[Column[{flxF[1, True], flxF[2, True]}, Spacings → 0.8], 13],
Style[Row[{"for ", cotTag[thI[1]], " ∈ ", iSet[pt["x", 1]],
   " with ", cotSqTag[thI[1]], " ≠ -", pt["z", 1], "/",
   pt["x", 1], ", -", pt["z", 2], "/", pt["x", 1]}], 13],
Spacer[{0, 10}], Style[Row[{"Each ", smallChip[True],
   " below certifies that the statement displayed beside it has been
     verified by a symbolic computation at that point; a ",
   smallChip[False], " would mean that this verification failed. A ",
   givenChip,
   " marks a given fact, a definition, a result of an earlier block, or
     an inference stated in words beside it."}], 12, GrayLevel[
  0.35]], "", Style[Row[{Style["We want to test the following claims ",
    Bold, 14], Style["⋆", 14, colTest], Style[" :", Bold, 14]}]],
Spacer[{0, 6}], Style["Claim 1.", Bold, 13, colTest],
Style["For each i = 1, 2, the substitution",
 Bold, 13, colTest], Row[{bullet,
  Style[Row[{"(", gk["α", "i"], ", ", gk["β", "i"], ", ", gk["γ", "i"], ", ",
```

```
        gk["δ", "i"], ")  ↦  (", gk["δ", "i"], ", ", gk["ɣ", "i"], ", ",
        gk["β", "i"], ", ", gk["α", "i"], ")"}], Bold, 13, colTest]}],
    Style["takes the numbers of the vertex", Bold, 13, colTest],
    Row[{bullet, Style[Row[{"(", pt["u", "i"], ", ", pt["x", "i"], ", ",
        pt["y", "i"], ", ", pt["z", "i"], ", ", pt["ε", "i"], ")  ↦  (",
        pt["u", "i"], ", ", pt["x", "i"], ", ", ptT["y", "i"], ", ",
        ptT["z", "i"], ", ", pt["ε", "i"], ")"}], Bold, 13, colTest]}],
    Spacer[{0, 16}], Style["Claim 2.", Bold, 13, colTest],
    Style[
     Row[{"If ", nameB⟦1⟧, " ≠ 0, respectively ", nameB⟦2⟧, " ≠ 0, then"}],
     Bold, 13, colTest], Spacer[{0, 6}],
    Sequence @@ Table[Row[{bullet, Style[Row[{cotTag[nameB⟦pnl⟧], " = ",
          frac[c2Num[pnl], c2Den[pnl]]}], Bold, 13, colTest]}], {pnl, 2}],
    Spacer[{0, 16}], Style["Claim 3.", Bold, 13, colTest],
    Sequence @@ Table[Row[{bullet,
       Style[Row[{nameB⟦pnl⟧, " ≠ 0   if   ", cotSqTag[thI[1]], " ≠ -",
          ptT["z", pnl], "/", pt["x", pnl]}], Bold, 13, colTest]}], {pnl, 2}],
    Spacer[{0, 16}], Style["Claim 4.", Bold, 13, colTest],
    Style[Row[{"If ", pt["x", 2], " = ", pt["x", 1], ",   ",
       pt["u", 1], " = ", pt["u", 2], " = ", uPlain, ",   ",
       cotTag[thI[1]], " ∈ ", iSet[pt["x", 1]], ", and ",
       cotSqTag[thI[1]], " ≠ -", pt["z", "i"], "/", pt["x", 1], ", -",
       ptT["z", "i"], "/", pt["x", 1], " for i = 1, 2, then"}],
     Bold, 13, colTest], Spacer[{0, 6}], Sequence @@
     Table[Row[{bullet, Style[flxT[pnl], Bold, 13, colTest]}], {pnl, 2}]},
   Spacings → 1.1], Background →
   RGBColor[
    0.99, 0.96,
    0.88], FrameMargins → 14]];

(* ============================================================*)
(*PART 3:CLAIM 1*)
(* ============================================================*)
Module[{col = cols⟦1⟧, oks = {}, chk, rows},
  chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
    verdict];
  rows = Flatten[Table[Module[{a = pt["a", i], b = pt["b", i], c = pt["c", i],
      d = pt["d", i], nw = vRat[angD⟦i⟧, angG⟦i⟧, angB⟦i⟧, angA⟦i⟧], lb},
     lb[k_] := ToString[i] <> "." <> ToString[k];
     {hrow[Row[{Style["Part " <> ToString[i] <> ".", Bold], "  At ", pt["A", i],
         ":"}]], nrow[lb[1], Row[{"(", gk["δ", i], " + ", gk["ɣ", i],
         " + ", gk["β", i], " + ", gk["α", i], ")/2 = ", gk["σ", i]}],
       smallChip[chk[(angD⟦i⟧ + angG⟦i⟧ + angB⟦i⟧ + angA⟦i⟧) / 2 - sig[i]]],
       Row[{"the half sum does not change, so each barred angle stays with
         its angle, and ", pt["ε", i], " does not change"}]],
      nrow[lb[2], Row[{"(", a, ", ", b, ", ", c, ", ", d, ")  ↦  (",
        d, ", ", c, ", ", b, ", ", a, ")"}],
```

```
        smallChip[And @@ (chk /@ (nw - {ratD[i], ratG[i], ratB[i], ratA[i]}))],
        "each is the sine of an angle over the sine of its barred angle"],
       nrow[lb[3], Row[{pt["u", i], "  ↦  1 - ", mulR[d, c, b, a],
          " = ", pt["u", i]}], smallChip[chk[(1 - Times @@ nw) - quU[i]]],
        "the product does not change"], nrow[lb[4],
        Row[{pt["x", i], "  ↦  ", frac["1", Row[{mulR[d, a], " - 1"}]],
          " = ", pt["x", i]}], smallChip[chk[1 / (nw[[1]] nw[[4]] - 1) - quX[i]]],
        Row[{"the product ", mulR[a, d], " does not change"}]], nrow[
        lb[5], Row[{pt["y", i], "  ↦  ", frac["1", Row[{mulR[b, a], " - 1"}]],
          " = ", frac[Row[{"-(1 + ", pt["y", i], ")"}],
           Row[{"1 + ", mulR[pt["u", i], pt["y", i]]}]], " = ", ptT["y", i]}],
        smallChip[chk[1 / (nw[[3]] nw[[4]] - 1) - quYt[i]]],
        Row[{"because ", mulR[a, b], " = ", pt["M", i], "/(", mulR[c, d],
          ") and ", mulR[c, d], " = 1 + 1/", pt["y", i]}]], nrow[lb[6],
        Row[{pt["z", i], "  ↦  ", frac["1", Row[{mulR[d, b], " - 1"}]],
          " = ", frac[Row[{"-(1 + ", pt["z", i], ")"}],
           Row[{"1 + ", mulR[pt["u", i], pt["z", i]]}]], " = ", ptT["z", i]}],
        smallChip[chk[1 / (nw[[1]] nw[[3]] - 1) - quZt[i]]],
        Row[{"because ", mulR[b, d], " = ", pt["M", i], "/(", mulR[a, c],
          ") and ", mulR[a, c], " = 1 + 1/", pt["z", i]}]], vgap}], {i, 2}], 1];
  AppendTo[resultsAll, And @@ oks];
  Print[Panel[Column[{Row[{Style["Claim 1: the substitution", Bold, 15, col],
       Spacer[20], statusChip[And @@ oks]}],
     Style["Verification: the numbers of the given panel for the
        substituted angles, compared with those of the vertex; each
        line is an identity in the four flat angles.", Bold, 13],
     vgrid[rows], Style["Together, these give Claim 1.", 13]},
    Spacings → 1], Background → Lighter[col, 0.93],
   FrameMargins → 12]];];

(* ============================================================*)
(*PART 4:THE NORMALS AND THE CENTRAL DIHEDRAL ANGLES*)
(* ============================================================*)
Module[{col = cols[[9]], oks = {}, chk, vec, nrm, nOK, oC, oS, rows},
  chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
    verdict];
  vec = {{Apt[1] - Apt[2], evec["A", 2, "A", 1]}, {Apt[3] - Apt[2],
     evec["A", 2, "A", 3]}, {Bpt[2] - Apt[2], evec["A", 2, "B", 2]},
    {Cpt[2] - Apt[2], evec["A", 2, "C", 2]}, {Apt[4] - Apt[1],
     evec["A", 1, "A", 4]}, {Cpt[1] - Apt[1], evec["A", 1, "C", 1]}};
  (*each normal:its index and the numbers of its two vectors*)
  nrm = {{0, 1, 2}, {1, 1, 3}, {2, 4, 2}, {4, 5, 6}};
  nOK = Table[chk[Cross[vec[[nSpec[[2]], 1]], vec[[nSpec[[3]], 1]]] -
       nlen[nSpec[[1]]] ndir[nSpec[[1]]]] &&
     chk[nraw[nSpec[[1]]].nraw[nSpec[[1]]] - nlen[nSpec[[1]]]^2], {nSpec, nrm}];
  oC = And @@
    Table[chk[nraw[0].nraw[k] - nlen[0] nlen[k] Cos[thCent[[k]]]], {k, {1, 2, 4}}];
```

```
  oS = And @@ Table[chk[Det[{nraw[0], nraw[k], wCent[k]}] -
       nlen[0] nlen[k] wCentLen[k] Sin[thCent[[k]]]], {k, {1, 2, 4}}];
  rows = Join[Table[nrow[j, Row[{vec[[j, 2]], " = ", disp[vec[[j, 1]]]}],
      givenChip, "from the coordinates"], {j, 6}],
    {vgap}, Table[With[{nSpec = nrm[[jn]]}, nrow[6 + jn,
       Row[{nv[nSpec[[1]]], " = ", vec[[nSpec[[2]], 2]], " × ", vec[[nSpec[[3]], 2]],
         " = ", disp[nlen[nSpec[[1]]]], " ", disp[ndir[nSpec[[1]]]],
         ",   ", absR[nv[nSpec[[1]]]], " = ", disp[nlen[nSpec[[1]]]]}],
       smallChip[nOK[[jn]]], Row[{"from (", nSpec[[2]], ") and (", nSpec[[3]],
         "); the length because ", disp[nlen[nSpec[[1]]]], " > 0"}]]], {jn, 4}],
    {vgap, nrow[11, Row[{nv[0], " · ", nv["i"], " = ",
        mulR[absR[nv[0]]], absR[nv["i"]], Row[{"cos", thI["i"]}]]}],
      smallChip[oC], "for i = 1, 2, 4, from (7)-(10)"],
     nrow[12, Row[{"det(", nv[0], ", ", nv["i"], ", ",
        evec["A", "i+1", "A", "i"], ") = ", mulR[absR[nv[0]]], absR[nv["i"]],
         pt["ℓ", "i"], Row[{"sin", thI["i"]}]]}], smallChip[oS], Row[
       {"for i = 1, 2, 4, from (7)-(10), with ", pt["A", 5], " := ", pt["A", 1],
        " and ", absR[evec["A", "i+1", "A", "i"]], " = ", pt["ℓ", "i"]}]]}];
  AppendTo[resultsAll, And @@ oks];
  Print[Panel[
    Column[{Row[{Style[Row[{"the normals ", nv[0], ", ", nv[1], ", ", nv[2],
           ", ", nv[4], " and the central dihedral angles"}],
         Bold, 15, col], Spacer[20], statusChip[And @@ oks]}],
      Style[Row[{"Verification: each normal from two edge vectors of
           its face; (11) and (12) identify ", thI[1], ", ", thI[2],
         ", ", thI[4], " with the dihedral angles of the definition."}],
       Bold, 13], vgrid[rows], Style["Together, these give the four
         normals and the three central dihedral angles.", 13]},
     Spacings → 1], Background → Lighter[col, 0.93], FrameMargins →
     12]];];

(* ============================================================*)
(*PART 5:THE CORNER DIHEDRAL ANGLES FROM THE COORDINATES*)
(* ============================================================*)
Module[{col = cols[[9]], oks = {}, chk, rows},
  chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
    verdict];
  rows =
   Flatten[Table[Module[{be = angB[[i]], mi = bLen[[i]], vi = cLen[[i]], tg = nameB[[i]],
       w3 = If[i == 1, Apt[1] - Bpt[1], Bpt[2] - Apt[2]], w3d = If[i == 1, evec["B", 1,
          "A", 1], evec["A", 2, "B", 2]], cl, lb, rf}, cl = Cos[be] → betaForm[i];
      lb[k_] := ToString[i] <> "." <> ToString[k];
      rf[k_] := "(" <> lb[k] <> ")";
      {hrow[Row[{Style["Part " <> ToString[i] <> ".", Bold], "  At ", pt["A", i],
          ", the angle ", tg, " along ", edg["A", i, "B", i], ":"}]],
       nrow[lb[1], Row[{nPsrc[i][[1, 2]], " = ", disp[nPsrc[i][[1, 1]]]}],
        givenChip, "from the coordinates"],
```

```
        nrow[lb[2], Row[{nPsrc[i]⟦2, 2⟧, " = ", disp[nPsrc[i]⟦2, 1⟧]}],
         givenChip, "from the coordinates"],
        nrow[lb[3], Row[{nvP[i], " = ", nPsrc[i]⟦1, 2⟧, " × ", nPsrc[i]⟦2, 2⟧,
           " = ", disp[nPshow[i]]}], smallChip[chk[nrawP[i] - nPshow[i]]], Row[
          {"from ", rf[1], " and ", rf[2]}]], nrow[lb[4], Row[{"cos ", disp[be],
           " = ", frac[Row[{evec["A", i, "B", i], " · ", evec["A", i, "C", i]}],
            mulR[pt["m", i], pt["v", i]]], " = ", disp[betaForm[i]]}],
         smallChip[chk[(Bpt[i] - Apt[i]).(Cpt[i] - Apt[i]) - mi vi betaForm[i]]],
         Row[{disp[be], " is the angle of the corner face at ",
           pt["A", i], "; the closed forms below are written with it"}]],
        nrow[lb[5], Row[{absR[nvP[i]], " = ", mulR[pt["m", i],
            pt["v", i], Row[{"sin ", disp[be]}]]}]],
         smallChip[chk[(nrawP[i].nrawP[i] - (mi vi)^2 (1 - Cos[be]^2)) /. cl]],
         Row[{"from ", rf[3], " and ", rf[4],
           ", with the positive radical because ", disp[be], " ∈ (0, π)"}]],
        vgap, nrow[lb[6], Row[{"cos ", tg, " = ",
           frac[Row[{nv[1], " · ", nvP[i]}], mulR[absR[nv[1]], absR[nvP[i]]]]}],
         givenChip, "by the definition"], nrow[lb[7], Row[{"cos ", tg,
           " = ", frac[disp[outBnum[i]], disp[outBden[i]]]}], smallChip[chk[
           ((nraw[1].nrawP[i]) Sin[angA⟦i⟧] - outBnum[i] nlen[1] mi vi) /. cl]],
         Row[{"from ", nv[1], " of the block above and ", rf[3], "-", rf[6]}]],
        vgap, nrow[lb[8], Row[{w3d, " = ", disp[w3]}], givenChip,
         Row[{"the third vector of ", tg, ", of length ", pt["m", i]}]],
        nrow[lb[9], Row[{"sin ", tg, " = ", frac[Row[{"det(", nv[1], ", ", nvP[i],
              ", ", w3d, ")"}], mulR[absR[nv[1]], absR[nvP[i]], absR[w3d]]]}],
         givenChip, "by the definition"], nrow[lb[10],
         Row[{"sin ", tg, " = ", frac[disp[eqB[i]], Row[{"sin ", disp[be]}]]}],
         smallChip[chk[Det[{nraw[1], nrawP[i], w3}] - eqB[i] nlen[1] mi vi mi]],
         Row[{"from ", nv[1], " of the block above, ", rf[3], ", ",
           rf[5], ", ", rf[8], ", and ", rf[9]}]], vgap}], {i, 2}], 1];
  AppendTo[resultsAll, And @@ oks];
  Print[Panel[
    Column[{Row[{Style[Row[{"the corner dihedral angles ", nameB⟦1⟧, " and ",
           nameB⟦2⟧}], Bold, 15, col], Spacer[20], statusChip[And @@ oks]}],
      Style[Row[{"Verification: at each vertex, the corner normal
           from its two edge vectors, then the cosine
           and the sine of its angle with ", nv[1], "; ",
         gk["β", "i"], " enters through its cosine and the length of ",
         nvP["i"], "."}], Bold, 13], vgrid[rows],
      Style[Row[{"Together, these give the cosine and the sine of ",
         nameB⟦1⟧, " and ", nameB⟦2⟧, "."}], 13]}, Spacings → 1],
    Background → Lighter[col, 0.93], FrameMargins → 12]];]

(* ============================================================*)
(*PART 6:CLAIM 2*)
(* ============================================================*)
Module[{col = cols⟦2⟧, oks = {}, chk, stp, gT = DisplayForm["θ"], w1, perV, rows},
```

```
chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
  verdict];
stp[e_] := Module[{verdict = TrueQ[Together[e] === 0]}, AppendTo[oks, verdict];
  verdict];
w1 = And @@ (chk /@ {Cot[θ1 / 2] (1 - Cos[θ1]) - Sin[θ1], Sin[θ1] (1 + Cot[θ1 / 2]^2) -
      2 Cot[θ1 / 2], Cos[θ1] (1 + Cot[θ1 / 2]^2) - (Cot[θ1 / 2]^2 - 1)});
perV = Table[Module[{k = thPrevIdx[[i]], tg = nameB[[i]], br = barRule[i],
    kT = cotTag[thI[1]], KT = cotSqTag[thI[1]], lT, LT, wR, eT, lb, rf,
    c1, c2, c4, c5, c6, c7, s3, s8, phS, f0S, f1S, f2S, saS, sbS, lS},
   lb[lineNo_] := ToString[i] <> "." <> ToString[lineNo];
   rf[lineNo_] := "(" <> lb[lineNo] <> ")";
   lT = cotTag[thI[k]]; LT = cotSqTag[thI[k]];
   wR = {Sin[θ1] → 2 kn / (1 + kn^2), Cos[θ1] → (kn^2 - 1) / (kn^2 + 1), Sin[
     thPrev[[i]]] → 2 kp / (1 + kp^2), Cos[thPrev[[i]]] → (kp^2 - 1) / (kp^2 + 1)};
   eT = polyRow[{{Sin[angD[[i]] - angG[[i]]], Row[{kT, " ", LT}]},
     {-Sin[angG[[i]]], Row[{KT, " ", lT}]},
     {Sin[angG[[i]] + angD[[i]]], kT}, {Sin[angG[[i]]], lT}}];
   c1 = chk[(eqB[i] /. wR) (1 + kn^2) (1 + kp^2) / 2 - phiE[i]];
   c2 = chk[((outBden[i] - outBnum[i]) /. wR) (1 + kp^2) / 2 -
     (coQ1[i] kp^2 + coQ2[i] /. br)];
   c4 = chk[((uB[i].uC[i] - Cos[angB[[i]]]) /. wR)
     (1 + kn^2) (1 + kp^2) + 2 (bric[i] /. br)];
   c5 = chk[(coV[i] - coA[i] - Sin[angA[[i]]] Sin[angD[[i]] - angG[[i]]]) /. br];
   c6 = chk[(coD[i] - coW[i] - Sin[angA[[i]]] Sin[angG[[i]] + angD[[i]]]) /. br];
   c7 = chk[(Sin[angA[[i]]] phiE[i] -
     (1 + kn^2) (coP[i] kp - kn (coA[i] kp^2 + coW[i])) - kn bric[i]) /. br];
   (*the two steps between displayed lines,
   with the displayed quantities as symbols*)
   s3 = stp[(2 phS / (sbS (1 + kn^2) (1 + lS))) / (2 f2S / (saS sbS (1 + lS))) -
     saS phS / ((1 + kn^2) f2S)];
   s8 = stp[(((1 + kn^2) f1S + kn f0S) / ((1 + kn^2) f2S) /. f0S → 0) - f1S / f2S];
   {c1 && c2 && c4 && c5 && c6 && c7 && s3 && s8,
    {hrow[Row[{Style["Part " <> ToString[i] <> ".", Bold], "  At ", pt["A", i],
       ", the angle ", tg, " through ", thI[1], " and ", thI[k], ":"}]],
     nrow[lb[1], Row[{"sin", tg, " sin", disp[angB[[i]]], " ", frac[Row[
       {"(1 + ", KT, ")(1 + ", LT, ")"}], "2"], " = ", eT}], smallChip[c1],
      Row[{"sin", tg, " of the block of the corner angles, with (1)"}]],
     nrow[lb[2], Row[{"(1 - cos", tg, ") sin", disp[angA[[i]]],
       " sin", disp[angB[[i]]], " ", frac[Row[{"1 + ", LT}], "2"],
       " = ", c2Den[i]}], smallChip[c2],
      Row[{"cos", tg, " of the block of the corner angles, with (1)"}]],
     prow[Row[{phD[""], " := ", eT}], givenChip,
      Row[{"a name for the right side of ", rf[1]}]],
     prow[Row[{phD[2], " := ", c2Den[i]}], givenChip,
      Row[{"a name for the right side of ", rf[2]}]], nrow[lb[3],
      Row[{cotTag[tg], " = ", frac[Row[{"sin", tg}], Row[{"1 - cos", tg}]],
       " = ", frac[Row[{"sin", disp[angA[[i]]], " ", phD[""]}],
```

```
          Row[{"(1 + ", KT, ") ", phD[2]}]]}], smallChip[s3 && c1 && c2 && w1],
       Row[{"the quotient of ", rf[1], " and ", rf[2], ", by (1); here ",
         phD[2], " ≠ 0 by ", rf[2], ", because ", tg, " ≠ 0"}]],
      vgap, nrow[lb[4], Column[{Row[{phD[0], " := (",
           polyRow[{{coA[i], LT}, {coW[i], None}}], ") ", KT}],
         Row[{"          - 2 ", mulR[disp[coP[i]], kT, lT]}],
         Row[{"          + (", polyRow[{{coV[i], LT}, {coD[i], None}}],
           ") = 0"}]}, Spacings → 0.6], smallChip[c4],
       Row[{"the closure ", evec["A", i, "B", i], " · ", evec["A", i, "C", i],
         " = ", mulR[pt["m", i], pt["v", i]], " cos",
         disp[angB[[i]]], " (Bricard's equation), with (1)"}]],
      nrow[lb[5], Row[{disp[Sin[angA[[i]]] Sin[angD[[i]] - angG[[i]]]], " = ",
         polyRow[{{-coA[i], None}, {coV[i], None}}]}], smallChip[c5],
       Row[{"by sin x sin y = (cos(x - y) - cos(x + y))/2, each side is ",
         disp[(Cos[angA[[i]] + angG[[i]] - angD[[i]] -
             Cos[angA[[i]] - angG[[i]] + angD[[i]]]) / 2]}]],
      nrow[lb[6], Row[{disp[Sin[angA[[i]]] Sin[angG[[i]] + angD[[i]]]],
         " = ", polyRow[{{coD[i], None}, {-coW[i], None}}]}], smallChip[c6],
       Row[{"in the same way, each side is ", disp[(Cos[angG[[i]] +
             angD[[i]] - angA[[i]]] - Cos[angA[[i]] + angG[[i]] + angD[[i]]]) / 2]}]],
      prow[Row[{phD[1], " := ", c2Num[i]}], givenChip,
       "a name for the numerator of the formula in Claim 2"],
      nrow[lb[7], Row[{"sin", disp[angA[[i]]], " ", phD[""], " - (1 + ",
         KT, ") ", phD[1], " = ", mulR[kT, phD[0]]}], smallChip[c7],
       Row[{"expand the three products: the coefficients of ",
         kT, " ", LT, " and of ", kT, " cancel by ", rf[5],
         " and ", rf[6], ", the other four term by term"}]],
      nrow[lb[8], Row[{cotTag[tg], " = ", frac[phD[1], phD[2]]}],
       smallChip[s8 && s3 && c7 && c4], Row[{rf[3], ", where sin", disp[angA[[i]]],
         " ", phD[""], " = (1 + ", KT, ") ", phD[1], " by ", rf[7],
         ", because ", phD[0], " = 0 by ", rf[4]}]], vgap}}], {i, 2}];
rows = Join[{nrow[1,
    Row[{cotTag[gT], " = ", frac[Row[{"sin", gT}], Row[{"1 - cos", gT}]],
      ",      sin", gT, " = ", frac[Row[{"2", cotTag[gT]}],
       Row[{"1 + ", cotSqTag[gT]}]], ",      cos", gT, " = ",
      frac[Row[{cotSqTag[gT], " - 1"}], Row[{cotSqTag[gT], " + 1"}]]}],
    smallChip[w1], Row[{"for ", gT, " ≠ 0 (mod 2π)"}]],
   vgap}, perV[[1, 2]], perV[[2, 2]]];
AppendTo[resultsAll, And @@ oks];
Print[Panel[Column[{Row[
     {Style["Claim 2: the corner dihedral angles through the central ones",
       Bold, 15, col], Spacer[20], statusChip[And @@ oks]}],
    Style[Row[{"Verification: at each vertex, the sine and 1 - cos of
         the corner angle through the two half-angle cotangents;
         Bricard's equation reduces their quotient to ",
       phD[1], "/", phD[2], ". Each step between lines is checked
         with the displayed quantities as symbols."}], Bold, 13],
```

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      vgrid[rows], Style["Together, these give Claim 2.", 13]},
     Spacings → 1], Background →
     Lighter[col, 0.93],
    FrameMargins → 12]];];

(* ============================================================*)
(*PART 7:CLAIM 3*)
(* ============================================================*)
Module[{col = cols⟦3⟧, oks = {}, chk, fchk, rows},
  chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
    verdict];
  fchk[e_] := Module[{verdict = TrueQ[Expand[e] === 0]}, AppendTo[oks, verdict];
    verdict];
  rows = Flatten[
    Table[Module[{k = thPrevIdx⟦i⟧, tg = nameB⟦i⟧, br = barRule[i], uu = quU[i],
       xx = quX[i], yt = quYt[i], zt = quZt[i], sgm = sig[i], gB = barG[i],
       S = gk["σ", i], gBd = disp[bGD[i]], xi = pt["x", i], ui = pt["u", i],
       ytd = ptT["y", i], ztd = ptT["z", i], ep = pt["ε", i], lT, aKv, aKv2, pqD,
       pq2D, lb, rf}, lb[lineNo_] := ToString[i] <> "." <> ToString[lineNo];
      rf[lineNo_] := "(" <> lb[lineNo] <> ")";
      lT = cotTag[thI[k]];
      aKv = polyRow[{{coA[i], tSq}, {coV[i], None}}];
      aKv2 = polyRow[{{coA2[i], tSq}, {coV2[i], None}}];
      pqD = disp[coP[i]];
      pq2D = disp[coP2[i]];
      Join[{hrow[Row[{Style["Part " <> ToString[i] <> ".", Bold], "  At ",
           pt["A", i], ":"}]], nrow[lb[1], Row[{mulR[sn[S], sn[gBd], ytd, ui],
           "(", ztd, " + ", mulR[xi, tSq], ") = ", aKv2}], smallChip[chk[
           Sin[sgm] Sin[gB] yt uu (zt + xx kn^2) - (coA2[i] kn^2 + coV2[i] /. br)]],
        "an identity in the flat angles at the vertex"], nrow[lb[2],
        Row[{aKv2, " ≠ 0   if   ", tSq, " ≠ -", ztd, "/", xi}], givenChip,
        Row[{"by ", rf[1], ", because ", ytd, " ≠ 0, ", ui, " ≠ 0 by ",
          mulR[ui, xi, pt["y", i], pt["z", i]], " > 0, sin", S, " ≠ 0 by ", ep,
          " ∈ {-1, 1}, and sin", gBd, " > 0"}]], vgap}, phiRows[i], {nrow[
        lb[3], Row[{disp[Sin[bAD[i]] Sin[bGD[i]] + Sin[bBD[i]] Sin[bDD[i]]],
          " = ", disp[Sin[angA⟦i⟧] Sin[angG⟦i⟧] + Sin[angB⟦i⟧] Sin[angD⟦i⟧]]}],
        smallChip[chk[(Sin[bAD[i]] Sin[bGD[i]] + Sin[bBD[i]] Sin[bDD[i]] -
            Sin[angA⟦i⟧] Sin[angG⟦i⟧] - Sin[angB⟦i⟧] Sin[angD⟦i⟧]) /. br]],
        Row[{"by sin x sin y = (cos(x - y) - cos(x + y))/2, each side is ",
          disp[(Cos[angA⟦i⟧ - angG⟦i⟧] + Cos[angB⟦i⟧ - angD⟦i⟧] - Cos[
              angA⟦i⟧ + angG⟦i⟧] - Cos[angB⟦i⟧ + angD⟦i⟧]) / 2]}]], nrow[lb[4],
        Row[{disp[Sin[angB⟦i⟧ - bAD[i]] Sin[angB⟦i⟧ - bGD[i]] + Sin[bBD[i]]
            Sin[bDD[i]]], " = ", disp[Sin[angA⟦i⟧] Sin[angG⟦i⟧]]}],
        smallChip[chk[(Sin[angB⟦i⟧ - bAD[i]] Sin[angB⟦i⟧ - bGD[i]] +
            Sin[bBD[i]] Sin[bDD[i]] - Sin[angA⟦i⟧] Sin[angG⟦i⟧]) /. br]],
        Row[{"in the same way, each side is ", disp[
           (Cos[angA⟦i⟧ - angG⟦i⟧] - Cos[angA⟦i⟧ + angG⟦i⟧]) / 2]}]], nrow[lb[5],
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      Row[{disp[Sin[bAD[i]] Sin[bGD[i]] - Sin[angA⟦i⟧] Sin[angG⟦i⟧]],
        " = ", disp[Sin[sgD[i]] Sin[angB⟦i⟧ - bDD[i]]]}],
      smallChip[chk[(Sin[bAD[i]] Sin[bGD[i]] - Sin[angA⟦i⟧] Sin[angG⟦i⟧] -
           Sin[sgD[i]] Sin[angB⟦i⟧ - bDD[i]]) /. br]],
      Row[{"in the same way, each side is ", disp[
         (Cos[angA⟦i⟧ + angG⟦i⟧] - Cos[angB⟦i⟧ + angD⟦i⟧]) / 2]}]],
     nrow[lb[6], Row[{disp[Sin[angG⟦i⟧ - bAD[i]]], " = -",
        disp[Sin[angB⟦i⟧ - bDD[i]]], ",   ", disp[Sin[angG⟦i⟧ - bBD[i]]],
        " = ", disp[Sin[angB⟦i⟧ - bGD[i]]]}],
      smallChip[And @@ (chk /@ ({Sin[angG⟦i⟧ - bAD[i]] + Sin[angB⟦i⟧ - bDD[i]],
            Sin[angG⟦i⟧ - bBD[i]] - Sin[angB⟦i⟧ - bGD[i]]} /. br))],
      Row[{"because ", disp[angG⟦i⟧ - bAD[i]], " = ", disp[bDD[i] - angB⟦i⟧],
        " and ", disp[angG⟦i⟧ - bBD[i]], " = ", disp[angB⟦i⟧ - bGD[i]]}]],
     nrow[lb[7], Row[{"(1 + cos", tg, ") sin", disp[angA⟦i⟧], " sin",
        disp[angB⟦i⟧], " ", frac[Row[{"1 + ", cotSqTag[thI[k]]}], "2"],
        " = ", c2Psi[i], " =: Ψ"}], smallChip[
       chk[((outBden[i] + outBnum[i]) /. {Cos[thPrev⟦i⟧] → (kp^2 - 1) /
              (kp^2 + 1)}) (1 + kp^2) / 2 - (-coQ3[i] kp^2 - coQ4[i] /. br)]],
      Row[{"cos", tg, " of the block of the corner angles, with
          (1) of the block of Claim 2"}]],
     nrow[lb[8], Column[{Row[{"(", aKv2, ") Ψ - (", mulR[pq2D, tSym],
          " + (", aKv, ") ", lT, " - ", mulR[pqD, tSym], ") ", phD[1]}],
        Row[{"          = (", polyRow[{{coA[i], Row[{tSym, " ", lT}]},
            {coC0[i], None}}], ") ", phD[0]}]}, Spacings → 0.6],
      smallChip[chk[((coA2[i] kn^2 + coV2[i]) (-coQ3[i] kp^2 - coQ4[i]) -
            ((coP2[i] - coP[i]) kn + (coA[i] kn^2 + coV[i]) kp)
             (coP[i] kp - kn (coA[i] kp^2 + coW[i])) - (coA[i] kn kp + coC0[i])
             bric[i]) /. br] && chk[linkList[i]] && fchk[formalRed⟦2⟧]],
      Row[{"expand both sides, with ", rf[6], ": the coefficients of ",
        monoD[3, 3, k], ", ", monoD[3, 1, k], ", ", monoD[1, 3, k],
        ", 1 agree term by term, those of ", monoD[2, 2, k], ", ", monoD[2,
         0, k], ", ", monoD[0, 2, k], ", after taking out the factors ",
        disp[Sin[bDD[i]] Sin[angB⟦i⟧ - bDD[i]]], ", ",
        disp[Sin[bAD[i]] Sin[angB⟦i⟧ - bAD[i]]], ", ",
        disp[Sin[bGD[i]] Sin[angB⟦i⟧ - bGD[i]]], ", by ", rf[3],
        " and ", rf[4], ", ", rf[3], " to ", rf[5], ", ", rf[4],
        ", and that of ", monoD[1, 1, k], " by ", rf[3], " to ", rf[5]}]],
     nrow[lb[9], Row[{tg, " ≠ 0   if   ", tSq, " ≠ -", ztd, "/", xi}],
      givenChip, Row[{"if ", tg, " = 0, then, in the block of Claim 2, ",
        phD[""], " = ", phD[2], " = 0 by ", rf[1], " and ", rf[2], ", and ",
        phD[1], " = 0 by ", rf[7], ", because ", phD[0], " = 0 by ", rf[4],
        "; here Ψ = sin", disp[angA⟦i⟧], " sin", disp[angB⟦i⟧], "(1 + ",
        cotSqTag[thI[k]], ") > 0 by ", rf[7], ", so ", rf[8], " gives (",
        aKv2, ") Ψ = 0, against ", rf[2]}]], vgap}]], {i, 2}], 1];
AppendTo[resultsAll, And @@ oks];
Print[
 Panel[Column[{Row[{Style["Claim 3: the corner angles are not 0", Bold,
```

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        15, col], Spacer[20], statusChip[And @@ oks]}],
      Style[Row[{"Verification: at each vertex, with ", tSym, " = ",
         cotTag[thI[1]], ": if the corner angle were 0, the identity (i.8)
           would make the left side of (i.2) vanish."}], Bold, 13],
      vgrid[rows], Style["Together, these give Claim 3.", 13]},
     Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 8:CLAIM 4*)
(* ============================================================*)
Module[{col = cols⟦4⟧, oks = {}, chk, fchk, stp, rows},
  chk[e_] := Module[{verdict = zeroE[e]}, AppendTo[oks, verdict];
    verdict];
  fchk[e_] := Module[{verdict = TrueQ[Expand[e] === 0]}, AppendTo[oks, verdict];
    verdict];
  stp[e_] := Module[{verdict = TrueQ[Together[e] === 0]}, AppendTo[oks, verdict];
    verdict];
  rows = Flatten[Table[Module[{k = thPrevIdx⟦i⟧, tg = nameB⟦i⟧, br = barRule[i],
       sg = (-1)^i, uu = quU[i], xx = quX[i], yy = quY[i], zv = quZ[i],
       yt = quYt[i], zt = quZt[i], sgm = sig[i], bB = barB[i], gB = barG[i],
       S = gk["σ", i], bBd = disp[bBD[i]], gBd = disp[bGD[i]], x1 = pt["x", 1],
       y = pt["y", i], z = pt["z", i], ytd = ptT["y", i], ztd = ptT["z", i],
       ep = pt["ε", i], lT, dt, sgS, aKv, aKv2, pqD, pq2D, lb, rf, c1, c2,
       c4, c5, c8, s3, s7, s9, s10, aA, aA2, pP, pP2, lTs, cS, rT, Sg, SgT,
       Rd, RdT, Q, QT, bs, bsT, om, e0S, eiS, f0S, f1S, f2S, q1S, a2S, tS},
      lb[lineNo_] := ToString[i] <> "." <> ToString[lineNo];
      rf[lineNo_] := "(" <> lb[lineNo] <> ")";
      lT = cotTag[thI[k]];
      dt = dTag[x1, tSym];
      sgS = If[sg < 0, "-", ""];
      aKv = polyRow[{{coA[i], tSq}, {coV[i], None}}];
      aKv2 = polyRow[{{coA2[i], tSq}, {coV2[i], None}}];
      pqD = disp[coP[i]];
      pq2D = disp[coP2[i]];
      (*identities in the angles*)
      c1 = chk[uu xx yy (1 + zv) (1 + uu zv) - (coP[i] / (Sin[sgm] Sin[bB]))^2];
      c2 =
       chk[Sin[sgm] Sin[bB] yy uu (zv + xx kn^2) - (coA[i] kn^2 + coV[i] /. br)];
      c4 = chk[uu xx yt (1 + zt) (1 + uu zt) - (coP2[i] / (Sin[sgm] Sin[gB]))^2];
      c5 = chk[uu yt zt Sin[gB]^2 - uu yy zv Sin[bB]^2];
      c8 = chk[((coA2[i] kn^2 + coV2[i]) (coP[i] kp - kn (coA[i] kp^2 + coW[i])) -
            ((coP2[i] - coP[i]) kn + (coA[i] kn^2 + coV[i]) kp)
             (coQ1[i] kp^2 + coQ2[i]) + (coQ1[i] kp + coA2[i] kn) bric[i]) /. br] &&
        chk[linkList[i]] && fchk[formalRed⟦1⟧];
      (*the steps between displayed lines,with the displayed
       quantities as symbols:aA,aA2 the two polynomials in t,
      pP,pP2 the two products of sines,lTs and rT the half-
```

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 cotangent of the central angle and the right side of the formula,
Sg,SgT,Rd,RdT the four radicals,Q,QT the two denominators,bs,
bsT the two products of sines before them,om the quantity omega;
each for both values of the sign epsilon*)s3 = And @@
  Table[stp[((aA lTs - pP tS - sg om) /. {lTs → e (tS Sg + sg e0S eiS Rd) / Q,
        aA → bs Q, om → e0S eiS e bs Rd}) /. Sg → e pP / bs], {e, {-1, 1}}];
s7 = And @@
  Table[stp[((aA2 rT - pP2 tS - sg om) /. {rT → e (tS SgT + sg e0S eiS RdT) / QT,
        aA2 → bsT QT, om → e0S eiS e bs Rd}) /.
      {SgT → e pP2 / bsT, RdT → bs Rd / bsT}], {e, {-1, 1}}];
s9 =
 stp[((((aA2 f1S - (pP2 tS + aA lTs - pP tS) f2S + (q1S lTs + a2S tS) f0S) /. f0S →
           0) / f2S) - (aA2 cS - pP2 tS - (aA lTs - pP tS))) /. cS → f1S / f2S];
s10 = stp[((pP2 tS + aA lTs - pP tS) / aA2 - rT) /.
   {lTs → (pP tS + sg om) / aA, rT → (pP2 tS + sg om) / aA2}];
Join[{hrow[If[i == 1, Row[{Style["Part 1.", Bold], "  At ",
       pt["A", 1], ", where ", pt["u", 1], " = ", uPlain, ":"}], Row[
      {Style["Part 2.", Bold], "  At ", pt["A", 2], ", where ", pt["x", 2],
       " = ", pt["x", 1], " and ", pt["u", 2], " = ", uPlain, ":"}]]],
  nrow[lb[1], Row[{mulR[uPlain, x1, y], "(1 + ", z, ")(1 + ",
     mulR[uPlain, z], ") = ", sqR[frac[pqD, mulR[sn[S], sn[bBd]]]]}],
   smallChip[c1], Row[{"hence ", ep, "", sigTag[i], " = ",
     frac[pqD, mulR[sn[S], sn[bBd]]], ", because ", pqD,
     " > 0, sin", bBd, " > 0 and ", ep, " = sign(sin", S, ")"}]],
  nrow[lb[2], Row[{mulR[sn[S], sn[bBd], y, uPlain], "(", z,
     " + ", mulR[x1, tSq], ") = ", aKv}], smallChip[c2],
   Row[{"the denominator of the formula for ", cotTag[thI[k]],
     ", times ", mulR[sn[S], sn[bBd]]}]], nrow[lb[3],
   Row[{"(", aKv, ") ", lT, " - ", mulR[pqD, tSym], " = ", sgS, omD}],
   smallChip[s3 && c1 && c2], Row[{"the formula for ", cotTag[thI[k]],
     " of the given panel, whose denominator is not 0 because ",
     tSq, " ≠ -", z, "/", x1, " by assumption and ", y,
     ", ", uPlain, " ≠ 0, multiplied by the left side of ",
     rf[2], ", with ", rf[1]}]], vgap,
  nrow[lb[4], Row[{mulR[uPlain, x1, ytd], "(1 + ", ztd, ")(1 + ",
     mulR[uPlain, ztd], ") = ", sqR[frac[pq2D, mulR[sn[S], sn[gBd]]]]}],
   smallChip[c4], Row[{"hence ", ep, "", sigTagT[i], " = ",
     frac[pq2D, mulR[sn[S], sn[gBd]]], ", in the same way"}]],
  nrow[lb[5], Row[{mulR[uPlain, ytd, ztd], " ", sinSq[gBd],
     " = ", mulR[uPlain, y, z], " ", sinSq[bBd]}], smallChip[c5],
   Row[{"hence sin", gBd, " ", radTagT[i], " = sin", bBd, " ",
     radTag[i, True], ", because ", mulR[uPlain, y, z], " ", dt,
     " = ", mulR[uPlain, x1, y, z], " · ", mulR[x1, dt], "/",
     DisplayForm[SuperscriptBox[bx[x1], "2"]], " > 0"}]],
  nrow[lb[6], Row[{mulR[sn[S], sn[gBd], ytd, uPlain], "(", ztd,
     " + ", mulR[x1, tSq], ") = ", aKv2, " ≠ 0"}], givenChip,
   Row[{rf[1], " and ", rf[2], " of the block of Claim 3, ",
```

```
        If[i == 1, Row[{"with ", pt["u", 1], " = ", uPlain}], Row[{"with ", pt[
            "x", 2], " = ", pt["x", 1], " and ", pt["u", 2], " = ", uPlain}]],
        ", because ", tSq, " ≠ -", ztd, "/", x1, " by assumption"}]],
      nrow[lb[7], Row[{"(", aKv2, ") ", rhsTt[i], " - ", mulR[pq2D, tSym],
        " = ", sgS, omD}], smallChip[s7 && c4 && c5],
       Row[{"by ", rf[6], ", the factor before the fraction is ",
        mulR[sn[S], sn[gBd]], " times its denominator; then ", rf[4],
        ", ", rf[5], ", and the definition of ", omD}]], vgap},
     phiRows[i], {nrow[lb[8], Column[{Row[{"(", aKv2, ") ", phD[1],
          " - (", mulR[pq2D, tSym], " + (", aKv, ") ", lT, " - ",
          mulR[pqD, tSym], ") ", phD[2]}], Row[{"          = -(",
          polyRow[{{coQ1[i], lT}, {coA2[i], tSym}}], ") ", phD[0]}]},
       Spacings → 0.6], smallChip[c8], Row[{"expand both sides, with ",
        rf[6], " of the block of Claim 3: the coefficients of ",
        monoD[3, 2, k], ", ", monoD[3, 0, k], ", ", monoD[2, 3, k],
        ", ", monoD[0, 3, k], " agree term by term, and those of ",
        monoD[2, 1, k], ", ", monoD[1, 2, k], ", ", monoD[1, 0, k],
        ", ", monoD[0, 1, k], ", after taking out the factors ",
        disp[Sin[bAD[i]] Sin[angB[[i]] - bDD[i]]], ", ",
        disp[Sin[angB[[i]] - bGD[i]] Sin[angB[[i]] - bDD[i]]],
        ", ", disp[Sin[bAD[i]] Sin[bBD[i]]], ", ",
        disp[Sin[bBD[i]] Sin[angB[[i]] - bGD[i]]], ", by its ", rf[4],
        ", ", rf[3], ", ", rf[3], " to ", rf[5], ", ", rf[5]}]],
      nrow[lb[9], Row[{"(", aKv2, ") ", cotTag[tg], " - ",
        mulR[pq2D, tSym], " = (", aKv, ") ", lT, " - ", mulR[pqD, tSym]}],
       smallChip[s9 && c8], Row[{rf[8], " divided by ", phD[2],
        ", which is not 0 by Claim 3 and ", rf[2],
        " of the block of Claim 2, because ", phD[1], "/", phD[2],
        " = ", cotTag[tg], " by Claim 2 and ", phD[0], " = 0"}]],
      nrow[lb[10], Row[{cotTag[tg], " = ", rhsTt[i]}],
       smallChip[s10 && s3 && s7 && s9],
       Row[{"the right sides of ", rf[7], " and ", rf[9], " are equal by ",
        rf[3], ", hence so are their left sides; divide by ", aKv2,
        ", which is not 0 by ", rf[6]}]], vgap}]], {i, 2}], 1];
AppendTo[resultsAll, And @@ oks];
Print[
 Panel[Column[{Row[{Style["Claim 4: the formulas", Bold, 15, col], Spacer[20],
     statusChip[And @@ oks]}],
   Style[Row[{"Verification: at each vertex, ", tSym, " = ", cotTag[thI[1]],
     " and ", omD, " = ", mulR[e0Tag[1], pt["e", "i"], pt["ε", "i"]],
     " sin", gk["σ", "i"], " sin", disp[bBD[iv]], "",
     DisplayForm[SqrtBox[bx[Row[{mulR[uPlain, pt["y", "i"], pt["z", "i"]],
          " ", dTag[pt["x", 1], tSym]}]]]],
     ". Lines (i.3) and (i.7) reduce Claim 4 to line (i.9),
      which has no radical, and line (i.8) proves
      (i.9). Each step between lines is checked with
      the displayed quantities as symbols."}], Bold, 13],
```

```
      vgrid[rows], Style["Together, these give Claim 4.", 13]},
     Spacings → 1], Background →
     Lighter[col, 0.93],
    FrameMargins → 12]];];

(* ============================================================*)
(*PART 9:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 12: ALL CLAIMS VERIFIED",
     "Section 12: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses and definitions:

A 3 x 3 net with, for each i = 1, 2,

the vertices $A_i$, $B_i$, $C_i$, and the vertices $A_3$, $A_4$ of the central face

the flat angles $\alpha_i, \beta_i, \gamma_i, \delta_i \in (0, \pi)$ at $A_i$

the edge lengths $\ell_1 = |A_1 A_2|$, $\ell_2 = |A_2 A_3|$, $\ell_4 = |A_4 A_1|$, $m_i = |A_i B_i|$, $v_i = |A_i C_i|$

the dihedral angles $\theta_1, \theta_2, \theta_4 \in [-\pi, \pi) \setminus \{0\}$ along the edges $A_1 A_2$, $A_2 A_3$, $A_4 A_1$

$\sigma_i = \frac{1}{2}(\alpha_i + \beta_i + \gamma_i + \delta_i)$ $\quad \overline{\alpha}_i = \sigma_i - \alpha_i$ $\quad \overline{\beta}_i = \sigma_i - \beta_i$ $\quad \overline{\gamma}_i = \sigma_i - \gamma_i$ $\quad \overline{\delta}_i = \sigma_i - \delta_i$

$a_i = \frac{\sin \alpha_i}{\sin \overline{\alpha}_i}$ $\quad b_i = \frac{\sin \beta_i}{\sin \overline{\beta}_i}$ $\quad c_i = \frac{\sin \gamma_i}{\sin \overline{\gamma}_i}$ $\quad d_i = \frac{\sin \delta_i}{\sin \overline{\delta}_i}$ $\quad \varepsilon_i = \text{sign}(\sin \sigma_i)$

$M_i = a_i b_i c_i d_i$ $\quad r_i = a_i d_i$ $\quad s_i = c_i d_i$ $\quad f_i = a_i c_i$

$u_i = 1 - M_i$ $\quad x_i = \frac{1}{r_i - 1}$ $\quad y_i = \frac{1}{s_i - 1}$ $\quad z_i = \frac{1}{f_i - 1}$

$\tilde{y}_i = \frac{-(1 + y_i)}{1 + u_i y_i}$ $\quad \tilde{z}_i = \frac{-(1 + z_i)}{1 + u_i z_i}$

$\alpha_i \pm \beta_i \pm \gamma_i \pm \delta_i \neq 0 \pmod{2\pi}$

$\overline{\alpha}_i, \overline{\beta}_i, \overline{\gamma}_i, \overline{\delta}_i \in (0, \pi)$

$\varepsilon_i \in \{-1, 1\}$

$x_i, y_i, z_i, \tilde{y}_i, \tilde{z}_i \neq -1, 0$; $\quad u_i x_i y_i z_i > 0$

Normals:

$\vec{n}_0 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 A_3}$

$\vec{n}_1 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 B_2}$ $\quad \vec{n}_2 = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 A_3}$ $\quad \vec{n}_4 = \overrightarrow{A_1 A_4} \times \overrightarrow{A_1 C_1}$

$\vec{n}_{1'} = \overrightarrow{A_1 B_1} \times \overrightarrow{A_1 C_1}$ $\quad \vec{n}_{2'} = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 B_2}$

Dihedral angles:

$$\theta(\vec{w}_1, \vec{w}_2, \vec{w}_3) := \begin{cases} \angle(\vec{w}_1, \vec{w}_2) \in (0, \pi), & \text{if } \det(\vec{w}_1, \vec{w}_2, \vec{w}_3) > 0, \\ -\angle(\vec{w}_1, \vec{w}_2) \in [-\pi, 0], & \text{if } \det(\vec{w}_1, \vec{w}_2, \vec{w}_3) \leq 0. \end{cases}$$

*central dihedral angles:*

$\theta_1 = \theta(\vec{n}_0, \vec{n}_1, \overrightarrow{A_2 A_1})$ $\theta_2 = \theta(\vec{n}_0, \vec{n}_2, \overrightarrow{A_3 A_2})$ $\theta_4 = \theta(\vec{n}_0, \vec{n}_4, \overrightarrow{A_1 A_4})$

*corner dihedral angles:*

$\theta_{11'} = \theta(\vec{n}_1, \vec{n}_{1'}, \overrightarrow{B_1 A_1})$ $\theta_{12'} = \theta(\vec{n}_1, \vec{n}_{2'}, \overrightarrow{A_2 B_2})$

Cartesian coordinates:

origin at $A_2$

central face in the half $-$plane $\tilde{y} \geq 0$ of the $\tilde{x}\tilde{y}-$plane, the ray $A_2 A_1$ as the $\tilde{x}-$axis

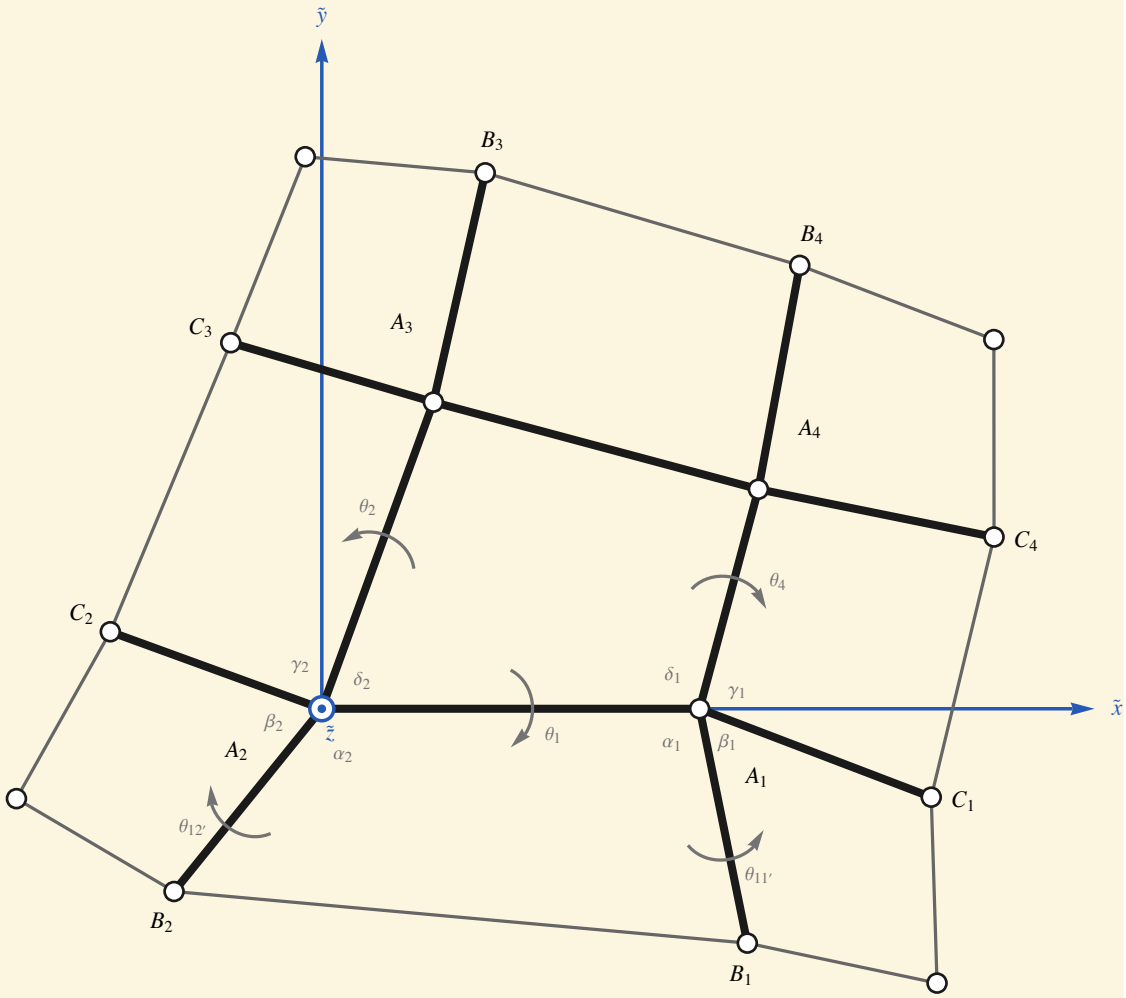



Figure 1

$A_1 = \{\ell_1, 0, 0\}$
$A_2 = \{0, 0, 0\}$
$A_3 = \{\ell_2 \cos \delta_2, \ell_2 \sin \delta_2, 0\}$
$A_4 = \{\ell_1 - \ell_4 \cos \delta_1, \ell_4 \sin \delta_1, 0\}$
$B_1 = A_1 + m_1 \{-\cos \alpha_1, \sin \alpha_1 \cos \theta_1, \sin \alpha_1 \sin \theta_1\}$
$B_2 = A_2 + m_2 \{\cos \alpha_2, \sin \alpha_2 \cos \theta_1, \sin \alpha_2 \sin \theta_1\}$
$C_1 = A_1 + v_1 \{-\sin \gamma_1 \sin \delta_1 \cos \theta_4 - \cos \gamma_1 \cos \delta_1, \cos \gamma_1 \sin \delta_1 - \sin \gamma_1 \cos \delta_1 \cos \theta_4, \sin \gamma_1 \sin \theta_4\}$

$C_2 = A_2 + v_2 \{\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2, \cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2, \sin\gamma_2 \sin\theta_2\}$

The definition gives $\theta(\vec{w}_1, \vec{w}_2, \vec{w}_3) = \pm\angle(\vec{w}_1, \vec{w}_2$ ), by the sign of the determinant. The cosine is even and the sine is odd, so cos $\theta = \frac{\vec{w}_1 \cdot \vec{w}_2}{|\vec{w}_1|\,|\vec{w}_2|}$ and sin $\theta$ $\frac{\det(\vec{w}_1, \vec{w}_2, \vec{w}_3)}{|\vec{w}_1|\,|\vec{w}_2|\,|\vec{w}_3|}$ hold whichever sign the determinant has; the formula for the sine holds because $w_3$, the direction of an edge common to the two faces, is orthogonal to $w_1$ and $w_2$, so that $w_1 \times w_2$ is a multiple of $w_3$. No formula of the notebook distinguishes the two cases, the coordinates above i

Flexion:

$x_2 = x_1, \quad u_1 = u_2 = u$

$e_1, e_2, e_0 \in \{-1, 1\}$

$D(X, t) = (Xt^2 - 1)(1 - uXt^2), \qquad I(X) = \{t \in \mathbb{R} : X\, D(X, t) > 0\}, \qquad \text{for } X, t \in \mathbb{R}$

$$\cot(\theta_4/2) = \frac{\varepsilon_1 (\cot(\theta_1/2)\ \sqrt{u x_1 y_1 (1 + z_1)(1 + u z_1)} - e_0\ e_1\ \sqrt{u y_1 z_1\ D(x_1, \cot(\theta_1/2))}\ )}{y_1 u (z_1 + x_1 \cot^2(\theta_1/2))}$$

$$\cot(\theta_2/2) = \frac{\varepsilon_2 (\cot(\theta_1/2)\ \sqrt{u x_1 y_2 (1 + z_2)(1 + u z_2)} + e_0\ e_2\ \sqrt{u y_2 z_2\ D(x_1, \cot(\theta_1/2))}\ )}{y_2 u (z_2 + x_1 \cot^2(\theta_1/2))}$$

for $\cot(\theta_1/2) \in I(x_1)$ with $\cot^2(\theta_1/2) \neq -z_1/x_1, -z_2/x_1$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at ✗ would mean that this verification failed. A • marks a given fact, a definition, a result of an earlier block, or an inference stated in words beside it.

We want to test the following claims ★ :

Claim 1.

For each i = 1, 2, the substitution

- $(\alpha_i, \beta_i, \gamma_i, \delta_i) \mapsto (\delta_i, \gamma_i, \beta_i, \alpha_i)$

takes the numbers of the vertex

- $(u_i, x_i, y_i, z_i, \varepsilon_i) \mapsto (u_i, x_i, \tilde{y}_i, \tilde{z}_i, \varepsilon_i)$

Claim 2.

If $\theta_{11'} \neq 0$, respectively $\theta_{12'} \neq 0$, then

- $$\cot(\theta_{11'}/2) = \frac{\sin\alpha_1 \sin\gamma_1\ \cot(\theta_4/2) + \sin\bar{\delta}_1 \sin(\beta_1 - \bar{\delta}_1)\ \cot(\theta_1/2) \cot^2(\theta_4/2) + \sin\bar{\alpha}_1 \sin(\beta_1 - \bar{\alpha}_1)\ \cot(\theta_1/2)}{-\sin(\beta_1 - \bar{\gamma}_1) \sin(\beta_1 - \bar{\delta}_1) \cot^2(\theta_4/2) + \sin\bar{\alpha}_1 \sin\bar{\beta}_1}$$
- $$\cot(\theta_{12'}/2) = \frac{\sin\alpha_2 \sin\gamma_2\ \cot(\theta_2/2) + \sin\bar{\delta}_2 \sin(\beta_2 - \bar{\delta}_2)\ \cot(\theta_1/2) \cot^2(\theta_2/2) + \sin\bar{\alpha}_2 \sin(\beta_2 - \bar{\alpha}_2)\ \cot(\theta_1/2)}{-\sin(\beta_2 - \bar{\gamma}_2) \sin(\beta_2 - \bar{\delta}_2) \cot^2(\theta_2/2) + \sin\bar{\alpha}_2 \sin\bar{\beta}_2}$$

Claim 3.

- $\theta_{11'} \neq 0$ if $\cot^2(\theta_1/2) \neq -\tilde{z}_1/x_1$
- $\theta_{12'} \neq 0$ if $\cot^2(\theta_1/2) \neq -\tilde{z}_2/x_2$

Claim 4.

If $x_2 = x_1$, $u_1 = u_2 = u$, $\cot(\theta_1/2) \in I(x_1)$, and $\cot^2(\theta_1/2) \neq -z_i/x_1, -\tilde{z}_i/x_1$ for i = 1, 2, then

- $\cot(\theta_{11'}/2) = \dfrac{\varepsilon_1 (\cot(\theta_1/2) \sqrt{u x_1 \tilde{y}_1 (1+\tilde{z}_1)(1+u\tilde{z}_1)} - e_0 e_1 \sqrt{u \tilde{y}_1 \tilde{z}_1 D(x_1, \cot(\theta_1/2))})}{\tilde{y}_1 u (\tilde{z}_1 + x_1 \cot^2(\theta_1/2))}$
- $\cot(\theta_{12'}/2) = \dfrac{\varepsilon_2 (\cot(\theta_1/2) \sqrt{u x_1 \tilde{y}_2 (1+\tilde{z}_2)(1+u\tilde{z}_2)} + e_0 e_2 \sqrt{u \tilde{y}_2 \tilde{z}_2 D(x_1, \cot(\theta_1/2))})}{\tilde{y}_2 u (\tilde{z}_2 + x_1 \cot^2(\theta_1/2))}$

## Claim 1: the substitution

Verified

Verification: the numbers of the given panel for the substituted angles, compared with those of the vertex; each line is an identity in the four flat angles.

Part 1. At $A_1$:

(1.1) $(\delta_1 + \gamma_1 + \beta_1 + \alpha_1)/2 = \sigma_1$ ☑ the half sum does not change, so each barred angle stays with its angle, and $\varepsilon_1$ does not change

(1.2) $(a_1, b_1, c_1, d_1) \mapsto (d_1, c_1, b_1, a_1)$ ☑ each is the sine of an angle over the sine of its barred angle

(1.3) $u_1 \mapsto 1 - d_1 c_1 b_1 a_1 = u_1$ ☑ the product does not change

(1.4) $x_1 \mapsto \dfrac{1}{d_1 a_1 - 1} = x_1$ ☑ the product $a_1 d_1$ does not change

(1.5) $y_1 \mapsto \dfrac{1}{b_1 a_1 - 1} = \dfrac{-(1+y_1)}{1+u_1 y_1} = \tilde{y}_1$ ☑ because $a_1 b_1 = M_1/(c_1 d_1)$ and $c_1 d_1 = 1 + 1/y_1$

(1.6) $z_1 \mapsto \dfrac{1}{d_1 b_1 - 1} = \dfrac{-(1+z_1)}{1+u_1 z_1} = \tilde{z}_1$ ☑ because $b_1 d_1 = M_1/(a_1 c_1)$ and $a_1 c_1 = 1 + 1/z_1$

Part 2. At $A_2$:

(2.1) $(\delta_2 + \gamma_2 + \beta_2 + \alpha_2)/2 = \sigma_2$ ☑ the half sum does not change, so each barred angle stays with its angle, and $\varepsilon_2$ does not change

(2.2) $(a_2, b_2, c_2, d_2) \mapsto (d_2, c_2, b_2, a_2)$ ☑ each is the sine of an angle over the sine of its barred angle

(2.3) $u_2 \mapsto 1 - d_2 c_2 b_2 a_2 = u_2$ ☑ the product does not change

(2.4) $x_2 \mapsto \dfrac{1}{d_2 a_2 - 1} = x_2$ ☑ the product $a_2 d_2$ does not change

(2.5) $y_2 \mapsto \dfrac{1}{b_2 a_2 - 1} = \dfrac{-(1+y_2)}{1+u_2 y_2} = \tilde{y}_2$ ☑ because $a_2 b_2 = M_2/(c_2 d_2)$ and $c_2 d_2 = 1 + 1/y_2$

(2.6) $z_2 \mapsto \dfrac{1}{d_2 b_2 - 1} = \dfrac{-(1+z_2)}{1+u_2 z_2} = \tilde{z}_2$ ☑ because $b_2 d_2 = M_2/(a_2 c_2)$ and $a_2 c_2 = 1 + 1/z_2$

Together, these give Claim 1.

the normals $n\vec{}_0, \vec{n}_1, \vec{n}_2, \vec{n}_4$ and the central dihedral angles

Verified

Verification: each normal from two edge vectors of its face; (11) and (12) identify $\theta_1$, $\theta_2$, $\theta_4$ with the dihedral angles of the definition.

(1) $\overrightarrow{A_2 A_1} = \{\ell_1, 0, 0\}$ • from the coordinates

(2) $\overrightarrow{A_2 A_3} = \{\ell_2 \cos\delta_2, \ell_2 \sin\delta_2, 0\}$ • from the coordinates

(3) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2, m_2 \sin\alpha_2 \cos\theta_1, m_2 \sin\alpha_2 \sin\theta_1\}$ • from the coordinates

(4) $\overrightarrow{A_2 C_2} = \{v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2), v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2), v_2 \sin\gamma_2 \sin\theta_2\}$ • from the coordinates

(5) $\overrightarrow{A_1 A_4} = \{\ell_4 (-\cos\delta_1), \ell_4 \sin\delta_1, 0\}$ • from the coordinates

(6) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin\gamma_1 \sin\delta_1 \cos\theta_4 - \cos\gamma_1 \cos\delta_1), v_1 (\cos\gamma_1 \sin\delta_1 - \sin\gamma_1 \cos\delta_1 \cos\theta_4), v_1 \sin\gamma_1 \sin\theta_4\}$ • from the coordinates

(7) $\vec{n}_0 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 A_3} = \ell_1 \ell_2 \sin\delta_2 \{0, 0, 1\}$, $|\vec{n}_0| = \ell_1 \ell_2 \sin\delta_2$ ☑ from (1) and (2); the length because $\ell_1 \ell_2 \sin\delta_2 > 0$

(8) $\vec{n}_1 = \overrightarrow{A_2 A_1} \times \overrightarrow{A_2 B_2} = m_2 \ell_1 \sin\alpha_2 \{0, -\sin\theta_1, \cos\theta_1\}$, $|\vec{n}_1| = m_2 \ell_1 \sin\alpha_2$ ☑ from (1) and (3); the length because $m_2 \ell_1 \sin\alpha_2 > 0$

(9) $\vec{n}_2 = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 A_3} = v_2 \ell_2 \sin\gamma_2 \{-\sin\delta_2 \sin\theta_2, \cos\delta_2 \sin\theta_2, \cos\theta_2\}$, $|\vec{n}_2| = v_2 \ell_2 \sin\gamma_2$ ☑ from (4) and (2); the length because $v_2 \ell_2 \sin\gamma_2 > 0$

(10) $\vec{n}_4 = \overrightarrow{A_1 A_4} \times \overrightarrow{A_1 C_1} = v_1 \ell_4 \sin\gamma_1 \{\sin\delta_1 \sin\theta_4, \cos\delta_1 \sin\theta_4, \cos\theta_4\}$, $|\vec{n}_4| = v_1 \ell_4 \sin\gamma_1$ ☑ from (5) and (6); the length because $v_1 \ell_4 \sin\gamma_1 > 0$

(11) $\vec{n}_0 \cdot \vec{n}_i = |\vec{n}_0| \, |\vec{n}_i| \cos\theta_i$ ☑ for i = 1, 2, 4, from (7)–(10)

(12) $\det(\vec{n}_0, \vec{n}_i, \overrightarrow{A_{i+1} A_i}) = |\vec{n}_0| \, |\vec{n}_i| \, \ell_i \sin\theta_i$ ☑ for i = 1, 2, 4, from (7)–(10), with $A_5 := A_1$ and $|\overrightarrow{A_{i+1} A_i}| = \ell_i$

Together, these give the four normals and the three central dihedral angles.

the corner dihedral angles $\theta_{11}'$ and $\theta_{12}'$

Verified

Verification: at each vertex, the corner normal from its two edge vectors, then the cosine and the sine of its angle with $\vec{n}_1$; $\beta_i$ enters through its cosine and the length of $\vec{n}_i'$.

Part 1. At $A_1$, the angle $\theta_{11'}$ along $A_1 B_1$:

(1.1) $\overrightarrow{A_1 B_1} = \{m_1 (-\cos \alpha_1),$
$m_1 \sin \alpha_1 \cos \theta_1, m_1 \sin \alpha_1 \sin \theta_1\}$
- from the coordinates

(1.2) $\overrightarrow{A_1 C_1} = \{v_1 (-\sin \gamma_1 \sin \delta_1 \cos \theta_4 -$
$\cos \gamma_1 \cos \delta_1), v_1$
$(\cos \gamma_1 \sin \delta_1 - \sin \gamma_1 \cos \delta_1 \cos \theta_4),$
$v_1 \sin \gamma_1 \sin \theta_4\}$
- from the coordinates

(1.3) $\vec{n}_{1'} = \overrightarrow{A_1 B_1} \times \overrightarrow{A_1 C_1} = \{m_1 v_1$
$(\sin \theta_1 (\sin \alpha_1 \sin \gamma_1 \cos \delta_1 \cos \theta_4 -$
$\sin \alpha_1 \cos \gamma_1 \sin \delta_1) +$
$\sin \alpha_1 \sin \gamma_1 \sin \theta_4 \cos \theta_1),$
$m_1 v_1 (\sin \theta_1 (\sin \alpha_1 (-\cos \gamma_1) \cos \delta_1 -$
$\sin \alpha_1 \sin \gamma_1$
$\sin \delta_1 \cos \theta_4) +$
$\cos \alpha_1 \sin \gamma_1 \sin \theta_4), m_1 v_1$
$(\cos \alpha_1 \sin \gamma_1 \cos \delta_1 \cos \theta_4 + \cos \theta_1$
$(\sin \alpha_1 \sin \gamma_1 \sin \delta_1 \cos \theta_4 +$
$\sin \alpha_1 \cos \gamma_1 \cos \delta_1) -$
$\cos \alpha_1 \cos \gamma_1 \sin \delta_1)\}$
☑ from (1.1) and (1.2)

(1.4) $\cos \beta_1 = \frac{\overrightarrow{A_1 B_1} \cdot \overrightarrow{A_1 C_1}}{m_1 v_1} =$
$\cos \alpha_1 (\sin \gamma_1 \sin \delta_1 \cos \theta_4 +$
$\cos \gamma_1 \cos \delta_1) +$
$\sin \alpha_1 (\cos \theta_1 (\cos \gamma_1 \sin \delta_1 -$
$\sin \gamma_1 \cos \delta_1 \cos \theta_4) +$
$\sin \gamma_1 \sin \theta_1 \sin \theta_4)$
☑ $\beta_1$ is the angle of the corner face at $A_1$; the closed forms below are written with it

(1.5) $|\vec{n}_{1'}| = m_1 v_1 \sin \beta_1$
☑ from (1.3) and (1.4), with the positive radical because $\beta_1 \in (0, \pi)$

(1.6) $\cos \theta_{11'} = \frac{\vec{n}_1 \cdot \vec{n}_{1'}}{|\vec{n}_1| |\vec{n}_{1'}|}$
- by the definition

(1.7) $\cos \theta_{11'} =$
$\frac{-\cos \alpha_1 \cos \beta_1 + \sin \gamma_1 \sin \delta_1 \cos \theta_4 + \cos \gamma_1 \cos \delta_1}{\sin \alpha_1 \sin \beta_1}$
☑ from $\vec{n}_1$ of the block above and (1.3)–(1.6)

(1.8) $\overrightarrow{B_1 A_1} = \{m_1 \cos \alpha_1, m_1 \sin \alpha_1 (-\cos \theta_1),$
$m_1 (-\sin \alpha_1) \sin \theta_1\}$
- the third vector of $\theta_{11'}$, of length $m_1$

(1.9) $\sin \theta_{11'} = \frac{\det(\vec{n}_1, \vec{n}_{1'}, \overrightarrow{B_1 A_1})}{|\vec{n}_1| |\vec{n}_{1'}| |\overrightarrow{B_1 A_1}|}$
- by the definition

(1.10) $\sin \theta_{11'} = \frac{1}{\sin \beta_1}$
$(\sin \theta_1 (\cos \gamma_1 \sin \delta_1 - \sin \gamma_1 \cos \delta_1$
$\cos \theta_4) -$
$\sin \gamma_1 \sin \theta_4 \cos \theta_1)$
☑ from $\vec{n}_1$ of the block above, (1.3), (1.5), (1.8), and (1.9)

Part 2. At $A_2$, the angle $\theta_{12'}$ along $A_2 B_2$:

(2.1) $\overrightarrow{A_2 C_2} =$
- from the coordinates

$$\{v_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2),\ v_2 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2),\ v_2 \sin\gamma_2 \sin\theta_2\}$$

(2.2) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2,\ m_2 \sin\alpha_2 \cos\theta_1,\ m_2 \sin\alpha_2 \sin\theta_1\}$ — • from the coordinates

(2.3) $\vec{n}_{2'} = \overrightarrow{A_2 C_2} \times \overrightarrow{A_2 B_2} = \{m_2 v_2 (\sin\theta_1 (\sin\alpha_2 \cos\gamma_2 \sin\delta_2 - \sin\alpha_2 \sin\gamma_2 \cos\delta_2 \cos\theta_2) - \sin\alpha_2 \sin\gamma_2 \sin\theta_2 \cos\theta_1),\ m_2 v_2 (\sin\theta_1 (\sin\alpha_2 (-\cos\gamma_2) \cos\delta_2 - \sin\alpha_2 \sin\gamma_2 \sin\delta_2 \cos\theta_2) + \cos\alpha_2 \sin\gamma_2 \sin\theta_2),\ m_2 v_2 (\cos\alpha_2 \sin\gamma_2 \cos\delta_2 \cos\theta_2 + \cos\theta_1 (\sin\alpha_2 \sin\gamma_2 \sin\delta_2 \cos\theta_2 + \sin\alpha_2 \cos\gamma_2 \cos\delta_2) - \cos\alpha_2 \cos\gamma_2 \sin\delta_2)\}$ — ☑ from (2.1) and (2.2)

(2.4) $\cos\beta_2 = \frac{\overrightarrow{A_2 B_2} \cdot \overrightarrow{A_2 C_2}}{m_2 v_2} = \cos\alpha_2 (\sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2) + \sin\alpha_2 (\cos\theta_1 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2) + \sin\gamma_2 \sin\theta_1 \sin\theta_2)$ — ☑ $\beta_2$ is the angle of the corner face at $A_2$; the closed forms below are written with it

(2.5) $|\vec{n}_{2'}| = m_2\ v_2\ \sin\beta_2$ — ☑ from (2.3) and (2.4), with the positive radical because $\beta_2 \in (0, \pi)$

(2.6) $\cos\theta_{12'} = \frac{\vec{n}_1 \cdot \vec{n}_{2'}}{|\vec{n}_1|\ |\vec{n}_{2'}|}$ — • by the definition

(2.7) $\cos\theta_{12'} = \frac{-\cos\alpha_2 \cos\beta_2 + \sin\gamma_2 \sin\delta_2 \cos\theta_2 + \cos\gamma_2 \cos\delta_2}{\sin\alpha_2 \sin\beta_2}$ — ☑ from $\vec{n}_1$ of the block above and (2.3)–(2.6)

(2.8) $\overrightarrow{A_2 B_2} = \{m_2 \cos\alpha_2,\ m_2 \sin\alpha_2 \cos\theta_1,\ m_2 \sin\alpha_2 \sin\theta_1\}$ — • the third vector of $\theta_{12'}$, of length $m_2$

(2.9) $\sin\theta_{12'} = \frac{\det(\vec{n}_1, \vec{n}_{2'}, \overrightarrow{A_2 B_2})}{|\vec{n}_1|\ |\vec{n}_{2'}|\ |\overrightarrow{A_2 B_2}|}$ — • by the definition

(2.10) $\sin\theta_{12'} = \frac{1}{\sin\beta_2} (\sin\theta_1 (\cos\gamma_2 \sin\delta_2 - \sin\gamma_2 \cos\delta_2 \cos\theta_2) - \sin\gamma_2 \sin\theta_2 \cos\theta_1)$ — ☑ from $\vec{n}_1$ of the block above, (2.3), (2.5), (2.8), and (2.9)

Together, these give the cosine and the sine of $\theta_{11'}$ and $\theta_{12'}$.

## Claim 2: the corner dihedral angles through the central ones

Verified

Verification : at each vertex , the sine and 1 − cos of the corner angle through the two half −angle cotangents; Bricard's equation reduces their quotient to $\Phi_1/\Phi_2$ . Each step between lines is checked with the displayed quantities as symbols.

(1) $\cot(\theta/2) = \frac{\sin\theta}{1-\cos\theta}$, $\sin\theta = \frac{2\cot(\theta/2)}{1+\cot^2(\theta/2)}$, $\cos\theta = \frac{\cot^2(\theta/2)-1}{\cot^2(\theta/2)+1}$ ☑ for $\theta \neq 0 \pmod{2\pi}$

Part 1. At $A_1$, the angle $\theta_{11'}$ through $\theta_1$ and $\theta_4$:

(1.1) $\sin\theta_{11'}\sin\beta_1 \frac{(1+\cot^2(\theta_1/2))(1+\cot^2(\theta_4/2))}{2} =$
$-\sin(\gamma_1-\delta_1)\cot(\theta_1/2)\cot^2(\theta_4/2)$
$-\sin\gamma_1\cot^2(\theta_1/2)\cot(\theta_4/2) +$
$\sin(\gamma_1+\delta_1)\cot(\theta_1/2) + \sin\gamma_1\cot(\theta_4/2)$ ☑ $\sin\theta_{11'}$ of the block of the corner angles, with (1)

(1.2) $(1-\cos\theta_{11'})\sin\alpha_1\sin\beta_1 \frac{1+\cot^2(\theta_4/2)}{2}$
$= -\sin(\beta_1-\overline{\gamma}_1)\sin(\beta_1-\overline{\delta}_1)$
$\cot^2(\theta_4/2) + \sin\overline{\alpha}_1\sin\overline{\beta}_1$ ☑ $\cos\theta_{11'}$ of the block of the corner angles, with (1)

$\Phi :=$
$-\sin(\gamma_1-\delta_1)\cot(\theta_1/2)\cot^2(\theta_4/2)$
$-\sin\gamma_1\cot^2(\theta_1/2)\cot(\theta_4/2) +$
$\sin(\gamma_1+\delta_1)\cot(\theta_1/2) + \sin\gamma_1\cot(\theta_4/2)$ • a name for the right side of (1.1)

$\Phi_2 := -\sin(\beta_1-\overline{\gamma}_1)\sin(\beta_1-\overline{\delta}_1)$
$\cot^2(\theta_4/2) + \sin\overline{\alpha}_1\sin\overline{\beta}_1$ • a name for the right side of (1.2)

(1.3) $\cot(\theta_{11'}/2) = \frac{\sin\theta_{11'}}{1-\cos\theta_{11'}} = \frac{\sin\alpha_1\,\Phi}{(1+\cot^2(\theta_1/2))\,\Phi_2}$ ☑ the quotient of (1.1) and (1.2), by (1); here $\Phi_2 \neq 0$ by (1.2), because $\theta_{11'} \neq 0$

(1.4) $\Phi_0 := (-\sin\overline{\delta}_1\sin(\beta_1-\overline{\delta}_1)\cot^2(\theta_4/2)$
$-\sin\overline{\alpha}_1\sin(\beta_1-\overline{\alpha}_1))\cot^2(\theta_1/2)$
$-2\sin\alpha_1\sin\gamma_1\cot(\theta_1/2)\cot(\theta_4/2)$
$+(-\sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$
$\cot^2(\theta_4/2) + \sin\sigma_1\sin\overline{\beta}_1) = 0$ ☑ the closure $\overrightarrow{A_1B_1}\cdot\overrightarrow{A_1C_1} = m_1 v_1 \cos\beta_1$ (Bricard's equation ), with (1)

(1.5) $-\sin\alpha_1\sin(\gamma_1-\delta_1) =$
$\sin\overline{\delta}_1\sin(\beta_1-\overline{\delta}_1) - \sin\overline{\gamma}_1\sin(\beta_1-\overline{\gamma}_1)$ ☑ by $\sin x\sin y = (\cos(x-y) - \cos(x+y))/2$, each side is $\frac{1}{2}(\cos(\alpha_1+\gamma_1-\delta_1) - \cos(\alpha_1-\gamma_1+\delta_1))$

(1.6) $\sin\alpha_1\sin(\gamma_1+\delta_1) =$
$\sin\sigma_1\sin\overline{\beta}_1 + \sin\overline{\alpha}_1\sin(\beta_1-\overline{\alpha}_1)$ ☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_1-\gamma_1-\delta_1) - \cos(\alpha_1+\gamma_1+\delta_1))$

$\Phi_1 :=$
$\sin\alpha_1\sin\gamma_1\cot(\theta_4/2) +$
$\sin\overline{\delta}_1\sin(\beta_1-\overline{\delta}_1)\cot(\theta_1/2)\cot^2(\theta_4/2)$
$+\sin\overline{\alpha}_1\sin(\beta_1-\overline{\alpha}_1)\cot(\theta_1/2)$ • a name for the numerator of the formula in Claim 2

(1.7) $\sin\alpha_1\,\Phi - (1+\cot^2(\theta_1/2))\,\Phi_1 = \cot(\theta_1/2)\Phi_0$ ☑ expand the three products: the coefficients of $\cot(\theta_1/2)\cot^2(\theta_4/2)$ and of $\cot(\theta_1/2)$ cancel by (1.5) and (1.6), the other four term by term

(1.8) $\cot(\theta_{11'}/2) = \frac{\Phi_1}{\Phi_2}$ ☑ (1.3), where $\sin\alpha_1\,\Phi = (1+\cot^2(\theta_1/2))$ $\Phi_2$ by (1.7), because $\Phi_2 \neq 0$ by (1.4)

$\Phi_1$ by (1.7), because $\Phi_0 = 0$ by (1.4)

Part 2. At $A_2$, the angle $\theta_{12'}$ through $\theta_1$ and $\theta_2$:

(2.1) $\sin\theta_{12'} \sin\beta_2 \frac{(1+\cot^2(\theta_1/2))(1+\cot^2(\theta_2/2))}{2} =$
$-\sin(\gamma_2-\delta_2)$
$\cot(\theta_1/2)\cot^2(\theta_2/2) - \sin\gamma_2$
$\cot^2(\theta_1/2)\cot(\theta_2/2) + \sin(\gamma_2+\delta_2)$
$\cot(\theta_1/2) + \sin\gamma_2 \cot(\theta_2/2)$
☑ $\sin\theta_{12'}$ of the block of the corner angles, with (1)

(2.2) $(1-\cos\theta_{12'})\sin\alpha_2 \sin\beta_2 \frac{1+\cot^2(\theta_2/2)}{2}$
$= -\sin(\beta_2-\overline{\gamma}_2)\sin(\beta_2-\overline{\delta}_2)$
$\cot^2(\theta_2/2) + \sin\overline{\alpha}_2 \sin\overline{\beta}_2$
☑ $\cos\theta_{12'}$ of the block of the corner angles, with (1)

$\Phi :=$
$-\sin(\gamma_2-\delta_2)$
$\cot(\theta_1/2)\cot^2(\theta_2/2) - \sin\gamma_2$
$\cot^2(\theta_1/2)\cot(\theta_2/2) + \sin(\gamma_2+\delta_2)$
$\cot(\theta_1/2) + \sin\gamma_2 \cot(\theta_2/2)$
• a name for the right side of (2.1)

$\Phi_2 := -\sin(\beta_2-\overline{\gamma}_2)\sin(\beta_2-\overline{\delta}_2)$
$\cot^2(\theta_2/2) + \sin\overline{\alpha}_2 \sin\overline{\beta}_2$
• a name for the right side of (2.2)

(2.3) $\cot(\theta_{12'}/2) = \frac{\sin\theta_{12'}}{1-\cos\theta_{12'}} = \frac{\sin\alpha_2\,\Phi}{(1+\cot^2(\theta_1/2))\,\Phi_2}$
☑ the quotient of (2.1) and (2.2), by (1); here $\Phi_2 \neq 0$ by (2.2), because $\theta_{12'} \neq 0$

(2.4) $\Phi_0 := (-\sin\overline{\delta}_2 \sin(\beta_2-\overline{\delta}_2)\cot^2(\theta_2/2)$
$- \sin\overline{\alpha}_2 \sin(\beta_2-\overline{\alpha}_2))\cot^2(\theta_1/2)$
$- 2\sin\alpha_2 \sin\gamma_2 \cot(\theta_1/2)\cot(\theta_2/2)$
$+ (-\sin\overline{\gamma}_2 \sin(\beta_2-\overline{\gamma}_2)$
$\cot^2(\theta_2/2) + \sin\sigma_2 \sin\overline{\beta}_2) = 0$
☑ the closure $\overrightarrow{A_2 B_2} \cdot \overrightarrow{A_2 C_2} = m_2 v_2 \cos\beta_2$ (Bricard's equation), with (1)

(2.5) $-\sin\alpha_2 \sin(\gamma_2-\delta_2) =$
$\sin\overline{\delta}_2 \sin(\beta_2-\overline{\delta}_2) - \sin\overline{\gamma}_2 \sin(\beta_2-\overline{\gamma}_2)$
☑ by $\sin x \sin y = (\cos(x-y) - \cos(x+y))/2$, each side is $\frac{1}{2}(\cos(\alpha_2+\gamma_2-\delta_2) - \cos(\alpha_2-\gamma_2+\delta_2))$

(2.6) $\sin\alpha_2 \sin(\gamma_2+\delta_2) =$
$\sin\sigma_2 \sin\overline{\beta}_2 + \sin\overline{\alpha}_2 \sin(\beta_2-\overline{\alpha}_2)$
☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_2-\gamma_2-\delta_2) - \cos(\alpha_2+\gamma_2+\delta_2))$

$\Phi_1 :=$
$\sin\alpha_2 \sin\gamma_2 \cot(\theta_2/2) +$
$\sin\overline{\delta}_2 \sin(\beta_2-\overline{\delta}_2) \cot(\theta_1/2)\cot^2(\theta_2/2)$
$+ \sin\overline{\alpha}_2 \sin(\beta_2-\overline{\alpha}_2) \cot(\theta_1/2)$
• a name for the numerator of the formula in Claim 2

(2.7) $\sin\alpha_2\ \Phi - (1+\cot^2(\theta_1/2))\ \Phi_1 = \cot(\theta_1/2)\Phi_0$
☑ expand the three products: the coefficients of $\cot(\theta_1/2)\cot^2(\theta_2/2)$ and of $\cot(\theta_1/2)$ cancel by (2.5) and (2.6), the other four term by term

(2.8) $\cot(\theta_{12'}/2) = \frac{\Phi_1}{\Phi_2}$
☑ (2.3), where $\sin\alpha_2\,\Phi = (1+\cot^2(\theta_1/2))\,\Phi_1$ by (2.7), because $\Phi_0 = 0$ by (2.4)

Together, these give Claim 2.

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## Claim 3: the corner angles are not 0

Verified

Verification: at each vertex, with $t = \cot(\theta_1/2)$
: if the corner angle were 0, the identity (i.8) would make the left side of (i.2) vanish.

Part 1. At $A_1$:

(1.1) $\sin\sigma_1 \sin\overline{\gamma}_1 \tilde{y}_1 u_1 (\tilde{z}_1 + x_1 t^2) = -\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin(\gamma_1 - \overline{\beta}_1)$

☑ an identity in the flat angles at the vertex

(1.2) $-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin(\gamma_1 - \overline{\beta}_1) \neq 0$ if $t^2 \neq -\tilde{z}_1/x_1$

- by (1.1), because $\tilde{y}_1 \neq 0$, $u_1 \neq 0$ by $u_1 x_1 y_1 z_1 > 0$, $\sin\sigma_1 \neq 0$ by $\varepsilon_1 \in \{-1, 1\}$, and $\sin\overline{\gamma}_1 > 0$

$\Phi := -\sin(\gamma_1 - \delta_1)\, t \cot^2(\theta_4/2) - \sin\gamma_1\, t^2 \cot(\theta_4/2) + \sin(\gamma_1 + \delta_1)\, t + \sin\gamma_1 \cot(\theta_4/2)$

- as in the block of Claim 2, with $t = \cot(\theta_1/2)$

$\Phi_0 := (-\sin\overline{\delta}_1 \sin(\beta_1 - \overline{\delta}_1) \cot^2(\theta_4/2) - \sin\overline{\alpha}_1 \sin(\beta_1 - \overline{\alpha}_1))\, t^2 - 2 \sin\alpha_1 \sin\gamma_1 t \cot(\theta_4/2) + (-\sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1) \cot^2(\theta_4/2) + \sin\sigma_1 \sin\overline{\beta}_1)$

- as there; $\Phi_0 = 0$ by its (1.4)

$\Phi_1 := \sin\alpha_1 \sin\gamma_1 \cot(\theta_4/2) + \sin\overline{\delta}_1 \sin(\beta_1 - \overline{\delta}_1)\, t \cot^2(\theta_4/2) + \sin\overline{\alpha}_1 \sin(\beta_1 - \overline{\alpha}_1)\, t$

- as there

$\Phi_2 := -\sin(\beta_1 - \overline{\gamma}_1) \sin(\beta_1 - \overline{\delta}_1) \cot^2(\theta_4/2) + \sin\overline{\alpha}_1 \sin\overline{\beta}_1$

- as there

(1.3) $\sin\overline{\alpha}_1 \sin\overline{\gamma}_1 + \sin\overline{\beta}_1 \sin\overline{\delta}_1 = \sin\alpha_1 \sin\gamma_1 + \sin\beta_1 \sin\delta_1$

☑ by $\sin x \sin y = (\cos(x - y) - \cos(x + y))/2$, each side is $\frac{1}{2}(\cos(\alpha_1 - \gamma_1) - \cos(\alpha_1 + \gamma_1) + \cos(\beta_1 - \delta_1) - \cos(\beta_1 + \delta_1))$

(1.4) $\sin(\beta_1 - \overline{\alpha}_1) \sin(\beta_1 - \overline{\gamma}_1) + \sin\overline{\beta}_1 \sin\overline{\delta}_1 = \sin\alpha_1 \sin\gamma_1$

☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_1 - \gamma_1) - \cos(\alpha_1 + \gamma_1))$

(1.5) $\sin\overline{\alpha}_1 \sin\overline{\gamma}_1 - \sin\alpha_1 \sin\gamma_1 = \sin\sigma_1 \sin(\beta_1 - \overline{\delta}_1)$

☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_1 + \gamma_1) - \cos(\beta_1 + \delta_1))$

(1.6) $\sin(\gamma_1 - \overline{\alpha}_1) = -\sin(\beta_1 - \overline{\delta}_1)$, $\sin(\gamma_1 - \overline{\beta}_1) = \sin(\beta_1 - \overline{\gamma}_1)$

☑ because $\gamma_1 - \overline{\alpha}_1 = \overline{\delta}_1 - \beta_1$ and $\gamma_1 - \overline{\beta}_1 = \beta_1 - \overline{\gamma}_1$

(1.7) $(1 + \cos\theta_{11'}) \sin\alpha_1 \sin\beta_1 \frac{1 + \cot^2(\theta_4/2)}{2} = \sin\overline{\gamma}_1 \sin\overline{\delta}_1 \cot^2(\theta_4/2) + \sin\sigma_1 \sin(\beta_1 - \overline{\alpha}_1) =: \Psi$

☑ $\cos\theta_{11'}$ of the block of the corner angles, with (1) of the block of Claim 2

(1.8) $(-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin(\gamma_1 - \overline{\beta}_1))\, \Psi - (\sin\beta_1 \sin\delta_1 t + (-\sin\overline{\delta}_1 \sin(\beta_1 - \overline{\delta}_1) t^2 - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1))$

☑ expand both sides, with (1.6): the coefficients of $t^3 \cot^3(\theta_4/2)$, $t^3 \cot(\theta_4/2)$, $t \cot^3(\theta_4/2)$, 1 agree term by term, those of $t^2 \cot^2(\theta_4/2)$, $t^2$, $\cot^2(\theta_4/2)$

$\cot(\theta_4/2) - \sin\alpha_1 \sin\gamma_1 t)\ \Phi_1$
$= (-\sin\bar{\delta}_1 \sin(\beta_1 - \bar{\delta}_1)\ t\cot(\theta_4/2)$
$- \sin(\beta_1 - \bar{\alpha}_1)\sin(\gamma_1 - \bar{\beta}_1))\ \Phi_0$

, after taking out the factors $\sin\bar{\delta}_1 \sin(\beta_1 - \bar{\delta}_1)$, $\sin\bar{\alpha}_1 \sin(\beta_1 - \bar{\alpha}_1)$, $\sin\bar{\gamma}_1 \sin(\beta_1 - \bar{\gamma}_1)$, by (1.3) and (1.4), (1.3) to (1.5), (1.4), and that of $t\cot(\theta_4/2)$ by (1.3) to (1.5)

(1.9) $\theta_{11'} \neq 0$ if $t^2 \neq -\tilde{z}_1/x_1$

- if $\theta_{11'} = 0$, then, in the block of Claim 2, $\Phi = \Phi_2 = 0$ by (1.1) and (1.2), and $\Phi_1 = 0$ by (1.7), because $\Phi_0 = 0$ by (1.4); here $\Psi = \sin\alpha_1 \sin\beta_1 (1 + \cot^2(\theta_4/2)) > 0$ by (1.7), so (1.8) gives $(-\sin\bar{\alpha}_1 \sin(\gamma_1 - \bar{\alpha}_1) t^2 - \sin\bar{\beta}_1 \sin(\gamma_1 - \bar{\beta}_1))\ \Psi = 0$, against (1.2)

Part 2. At $A_2$:

(2.1) $\sin\sigma_2 \sin\bar{\gamma}_2 \tilde{y}_2 u_2 (\tilde{z}_2 + x_2 t^2) = -\sin\bar{\alpha}_2 \sin(\gamma_2 - \bar{\alpha}_2)\ t^2 - \sin\bar{\beta}_2 \sin(\gamma_2 - \bar{\beta}_2)$

☑ an identity in the flat angles at the vertex

(2.2) $-\sin\bar{\alpha}_2 \sin(\gamma_2 - \bar{\alpha}_2)\ t^2 - \sin\bar{\beta}_2 \sin(\gamma_2 - \bar{\beta}_2) \neq 0$ if $t^2 \neq -\tilde{z}_2/x_2$

- by (2.1), because $\tilde{y}_2 \neq 0$, $u_2 \neq 0$ by $u_2 x_2 y_2 z_2 > 0$, $\sin\sigma_2 \neq 0$ by $\varepsilon_2 \in \{-1, 1\}$, and $\sin\bar{\gamma}_2 > 0$

$\Phi := -\sin(\gamma_2 - \delta_2)\ t\cot^2(\theta_2/2) - \sin\gamma_2\ t^2 \cot(\theta_2/2) + \sin(\gamma_2 + \delta_2)\ t + \sin\gamma_2\ \cot(\theta_2/2)$

- as in the block of Claim 2, with $t = \cot(\theta_1/2)$

$\Phi_0 := (-\sin\bar{\delta}_2 \sin(\beta_2 - \bar{\delta}_2)\ \cot^2(\theta_2/2) - \sin\bar{\alpha}_2 \sin(\beta_2 - \bar{\alpha}_2))\ t^2 - 2\ \sin\alpha_2 \sin\gamma_2\, t\cot(\theta_2/2) + (-\sin\bar{\gamma}_2 \sin(\beta_2 - \bar{\gamma}_2)\ \cot^2(\theta_2/2) + \sin\sigma_2 \sin\bar{\beta}_2)$

- as there; $\Phi_0 = 0$ by its (2.4)

$\Phi_1 := \sin\alpha_2 \sin\gamma_2\ \cot(\theta_2/2) + \sin\bar{\delta}_2 \sin(\beta_2 - \bar{\delta}_2)\ t\cot^2(\theta_2/2) + \sin\bar{\alpha}_2 \sin(\beta_2 - \bar{\alpha}_2)\ t$

- as there

$\Phi_2 := -\sin(\beta_2 - \bar{\gamma}_2) \sin(\beta_2 - \bar{\delta}_2)\ \cot^2(\theta_2/2) + \sin\bar{\alpha}_2 \sin\bar{\beta}_2$

- as there

(2.3) $\sin\bar{\alpha}_2 \sin\bar{\gamma}_2 + \sin\bar{\beta}_2 \sin\bar{\delta}_2 = \sin\alpha_2 \sin\gamma_2 + \sin\beta_2 \sin\delta_2$

☑ by $\sin x \sin y = (\cos(x - y) - \cos(x + y))/2$, each side is $\frac{1}{2}(\cos(\alpha_2 - \gamma_2) - \cos(\alpha_2 + \gamma_2) + \cos(\beta_2 - \delta_2) - \cos(\beta_2 + \delta_2))$

(2.4) $\sin(\beta_2 - \bar{\alpha}_2) \sin(\beta_2 - \bar{\gamma}_2) + \sin\bar{\beta}_2 \sin\bar{\delta}_2 = \sin\alpha_2 \sin\gamma_2$

☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_2 - \gamma_2) - \cos(\alpha_2 + \gamma_2))$

(2.5) $\sin\bar{\alpha}_2 \sin\bar{\gamma}_2 - \sin\alpha_2 \sin\gamma_2 = \sin\sigma_2 \sin(\beta_2 - \bar{\delta}_2)$

☑ in the same way, each side is $\frac{1}{2}(\cos(\alpha_2 + \gamma_2) - \cos(\beta_2 + \delta_2))$

(2.6) $\sin(\gamma_2 - \bar{\alpha}_2) = -\sin(\beta_2 - \bar{\delta}_2)$, $\sin(\gamma_2 - \bar{\beta}_2) = \sin(\beta_2 - \bar{\gamma}_2)$

☑ because $\gamma_2 - \bar{\alpha}_2 = \bar{\delta}_2 - \beta_2$ and $\gamma_2 - \bar{\beta}_2 = \beta_2 - \bar{\gamma}_2$

(2.7) $(1 + \cos\theta_{12'}) \sin\alpha_2 \sin\beta_2$

☑ $\cos\theta_{12'}$ of the block of the corner angles with

$\frac{1+\cot^2(\theta_2/2)}{2} = \sin\overline{\gamma}_2 \sin\overline{\delta}_2 \cot^2(\theta_2/2) + \sin\sigma_2 \sin(\beta_2 - \overline{\alpha}_2) =: \Psi$

of the block of the corner angles, with (1) of the block of Claim 2

(2.8) $(-\sin\overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin\overline{\beta}_2 \sin(\gamma_2 - \overline{\beta}_2)) \Psi - (\sin\beta_2 \sin\delta_2 t + (-\sin\overline{\delta}_2 \sin(\beta_2 - \overline{\delta}_2) t^2 - \sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)) \cot(\theta_2/2) - \sin\alpha_2 \sin\gamma_2 t) \Phi_1 = (-\sin\overline{\delta}_2 \sin(\beta_2 - \overline{\delta}_2)\, t \cot(\theta_2/2) - \sin(\beta_2 - \overline{\alpha}_2) \sin(\gamma_2 - \overline{\beta}_2)) \Phi_0$

☑ expand both sides, with (2.6): the coefficients of $t^3 \cot^3(\theta_2/2)$, $t^3 \cot(\theta_2/2)$, $t \cot^3(\theta_2/2)$, 1 agree term by term, those of $t^2 \cot^2(\theta_2/2)$, $t^2$, $\cot^2(\theta_2/2)$, after taking out the factors $\sin\overline{\delta}_2 \sin(\beta_2 - \overline{\delta}_2)$, $\sin\overline{\alpha}_2 \sin(\beta_2 - \overline{\alpha}_2)$, $\sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)$, by (2.3) and (2.4), (2.3) to (2.5), (2.4), and that of $t \cot(\theta_2/2)$ by (2.3) to (2.5)

(2.9) $\theta_{12'} \neq 0$ if $t^2 \neq -\tilde{z}_2/x_2$

- if $\theta_{12'} = 0$, then, in the block of Claim 2, $\Phi = \Phi_2 = 0$ by (2.1) and (2.2), and $\Phi_1 = 0$ by (2.7), because $\Phi_0 = 0$ by (2.4); here $\Psi = \sin\alpha_2 \sin\beta_2 (1 + \cot^2(\theta_2/2)) > 0$ by (2.7), so (2.8) gives $(-\sin\overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin\overline{\beta}_2 \sin(\gamma_2 - \overline{\beta}_2)) \Psi = 0$, against (2.2)

Together, these give Claim 3.

## Claim 4: the formulas

Verified

Verification: at each vertex, $t = \cot(\theta_1/2)$ and $\omega = e_0 e_i \varepsilon_i \sin\sigma_i \sin\overline{\beta}_i \sqrt{u y_i z_i\, D(x_1, t)}$. Lines (i.3) and (i.7) reduce Claim 4 to line (i.9), which has no radical, and line (i.8) proves (i.9). Each step between lines is checked with the displayed quantities as symbols.

Part 1. At $A_1$, where $u_1 = u$:

(1.1) $u x_1 y_1 (1 + z_1)(1 + u z_1) = \left(\frac{\sin\alpha_1 \sin\gamma_1}{\sin\sigma_1 \sin\overline{\beta}_1}\right)^2$

☑ hence $\varepsilon_1 \sqrt{u x_1 y_1 (1 + z_1)(1 + u z_1)} = \frac{\sin\alpha_1 \sin\gamma_1}{\sin\sigma_1 \sin\overline{\beta}_1}$, because $\sin\alpha_1 \sin\gamma_1 > 0$, $\sin\overline{\beta}_1 > 0$ and $\varepsilon_1 = \text{sign}(\sin\sigma_1)$

(1.2) $\sin\sigma_1 \sin\overline{\beta}_1\, y_1\, u(z_1 + x_1 t^2) = -\sin\overline{\delta}_1 \sin(\beta_1 - \overline{\delta}_1)\, t^2 - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1)$

☑ the denominator of the formula for $\cot(\theta_4/2)$, times $\sin\sigma_1 \sin\overline{\beta}_1$

(1.3) $(-\sin\overline{\delta}_1 \sin(\beta_1 - \overline{\delta}_1)\, t^2 - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1)) \cot(\theta_4/2) - \sin\alpha_1 \sin\gamma_1 t = -\omega$

☑ the formula for $\cot(\theta_4/2)$ of the given panel, whose denominator is not 0 because $t^2 \neq -z_1/x_1$ by assumption and $y_1$, $u \neq 0$, multiplied by the left side of (1.2), with (1.1)

(1.4) $u x_1 \tilde{y}_1 (1 + \tilde{z}_1)(1 + u \tilde{z}_1) = \left(\frac{\sin\beta_1 \sin\delta_1}{\sin\sigma_1 \sin\overline{\gamma}_1}\right)^2$

☑ hence $\varepsilon_1 \sqrt{u x_1 \tilde{y}_1 (1 + \tilde{z}_1)(1 + u \tilde{z}_1)} = \frac{\sin\beta_1 \sin\delta_1}{\sin\sigma_1 \sin\overline{\gamma}_1}$, in the same way

(1.5) $u \tilde{y}_1 \tilde{z}_1 \sin^2\overline{\gamma}_1 = u y_1 z_1 \sin^2\overline{\beta}_1$

☑ hence $\sin\overline{\gamma}_1 \sqrt{u \tilde{y}_1 \tilde{z}_1\, D(x_1, \cot(\theta_1/2))}$

$= \sin\beta_1\ \sqrt{u\,y_1 z_1}\ D(x_1, \cot(\theta_1/2))$
, because $u\,y_{\ 1} z_1\ D(x_1, t) = u x_1 y_1 z_1 \cdot x_1\ D(x_1, t)/x_1^{\ 2} > 0$

(1.6) $\sin\sigma_1\ \sin\overline{\gamma}_1\ \tilde{y}_1\ u\,(\tilde{z}_1 + x_1 t^2) = -\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1)\ t^2 - \sin\overline{\beta}_1 \sin\left(\gamma_1 - \overline{\beta}_1\right) \neq 0$

- (1.1) and (1.2) of the block of Claim 3, with $u_{\ 1} = u$, because $t^{\ 2} \neq -\tilde{z}_1/x_1$ by assumption

(1.7) $(-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1)\ t^2 - \sin\overline{\beta}_1 \sin\left(\gamma_1 - \overline{\beta}_1\right))$
$\left(\varepsilon_1\,(t\ \sqrt{u x_1 \tilde{y}_1 (1+\tilde{z}_1)(1+u\tilde{z}_1)} - e_0\ e_1\ \sqrt{u\tilde{y}_1\tilde{z}_1\ D(x_1, t)}\ )\right)/$
$(\tilde{y}_1\,u\,(\tilde{z}_1 + x_1 t^2))$
$-\sin\beta_1 \sin\delta_1 t = -\omega$

☑ by (1.6), the factor before the fraction is $\sin\sigma_1\ \sin\overline{\gamma}_1$ times its denominator; then (1.4), (1.5), and the definition of $\omega$

$\Phi := -\sin(\gamma_1 - \delta_1)\ t\cot^2(\theta_4/2) - \sin\gamma_1\ t^2\cot(\theta_4/2) + \sin(\gamma_1 + \delta_1)\ t + \sin\gamma_1\ \cot(\theta_4/2)$

- as in the block of Claim 2, with $t = \cot(\theta_1/2)$

$\Phi_0 := (-\sin\overline{\delta}_1 \sin\left(\beta_1 - \overline{\delta}_1\right) \cot^2(\theta_4/2) - \sin\overline{\alpha}_1 \sin(\beta_1 - \overline{\alpha}_1))\ t^2 - 2\ \sin\alpha_1 \sin\gamma_1\, t\cot(\theta_4/2) + (-\sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1) \cot^2(\theta_4/2) + \sin\sigma_1 \sin\overline{\beta}_1)$

- as there; $\Phi_0 = 0$ by its (1.4)

$\Phi_1 := \sin\alpha_1 \sin\gamma_1\ \cot(\theta_4/2) + \sin\overline{\delta}_1 \sin\left(\beta_1 - \overline{\delta}_1\right)\ t\cot^2(\theta_4/2) + \sin\overline{\alpha}_1 \sin(\beta_1 - \overline{\alpha}_1)\ t$

- as there

$\Phi_2 := -\sin(\beta_1 - \overline{\gamma}_1) \sin\left(\beta_1 - \overline{\delta}_1\right) \cot^2(\theta_4/2) + \sin\overline{\alpha}_1 \sin\overline{\beta}_1$

- as there

(1.8) $(-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin\left(\gamma_1 - \overline{\beta}_1\right))\ \Phi_1 - (\sin\beta_1 \sin\delta_1 t + (-\sin\overline{\delta}_1 \sin\left(\beta_1 - \overline{\delta}_1\right) t^2 - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1))\ \cot(\theta_4/2) - \sin\alpha_1 \sin\gamma_1 t)\ \Phi_2 = -(-\sin(\beta_1 - \overline{\gamma}_1) \sin\left(\beta_1 - \overline{\delta}_1\right) \cot(\theta_4/2) - \sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t)\ \Phi_0$

☑ expand both sides, with (1.6) of the block of Claim 3: the coefficients of $t^{\ 3}\cot^2(\theta_4/2)$, $t^3$, $t^2\cot^3(\theta_4/2)$, $\cot^3(\theta_4/2)$ agree term by term, and those of $t^2\cot(\theta_4/2)$, $t\cot^2(\theta_4/2)$, $t$, $\cot(\theta_4/2)$, after taking out the factors $\sin\overline{\alpha}_1 \sin\left(\beta_1 - \overline{\delta}_1\right)$, $\sin(\beta_1 - \overline{\gamma}_1) \sin\left(\beta_1 - \overline{\delta}_1\right)$, $\sin\overline{\alpha}_1 \sin\overline{\beta}_1$, $\sin\overline{\beta}_1 \sin(\beta_1 - \overline{\gamma}_1)$, by its (1.4), (1.3), (1.3) to (1.5), (1.5)

(1.9) $(-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin\left(\gamma_1 - \overline{\beta}_1\right))\ \cot(\theta_{11'}/2) - \sin\beta_1 \sin\delta_1 t = (-\sin\overline{\delta}_1 \sin\left(\beta_1 - \overline{\delta}_1\right) t^2 - \sin\overline{\gamma}_1 \sin(\beta_1 - \overline{\gamma}_1))\ \cot(\theta_4/2) - \sin\alpha_1 \sin\gamma_1 t$

☑ (1.8) divided by $\Phi_2$, which is not 0 by Claim 3 and (1.2) of the block of Claim 2, because $\Phi_1/\Phi_2 = \cot(\theta_{11'}/2)$ by Claim 2 and $\Phi_0 = 0$

(1.10) $\cot(\theta_{11'}/2) =$
$\left(\varepsilon_1\,(t\ \sqrt{u x_1 \tilde{y}_1 (1+\tilde{z}_1)(1+u\tilde{z}_1)}\right)/$

☑ the right sides of (1.7) and (1.9) are equal by (1.3), hence so are their left sides; divide by $-\sin\overline{\alpha}_1 \sin(\gamma_1 - \overline{\alpha}_1) t^2 - \sin\overline{\beta}_1 \sin\left(\gamma_1 - \overline{\beta}_1\right)$

$\left( -e_0\ e_1\ \sqrt{u y_1 z_1\ D(x_1, t)}\ \right)$ / $(\tilde{y}_1\, u\, (\tilde{z}_1 + x_1 t^2))$ , which is not 0 by (1.6)

Part 2. At $A_2$, where $x_2 = x_1$ and $u_2 = u$:

(2.1) $u x_1 y_2 (1 + z_2)(1 + u z_2) = \left(\frac{\sin\alpha_2 \sin\gamma_2}{\sin\sigma_2 \sin\overline{\beta}_2}\right)^2$ ☑ hence $\varepsilon_2 \sqrt{u x_1 y_2 (1 + z_2)(1 + u z_2)} = \frac{\sin\alpha_2 \sin\gamma_2}{\sin\sigma_2 \sin\overline{\beta}_2}$, because $\sin\alpha_2 \sin\gamma_2 > 0$, $\sin\overline{\beta}_2 > 0$ and $\varepsilon_2 = \text{sign}(\sin\sigma_2)$

(2.2) $\sin\sigma_2 \sin\overline{\beta}_2\ y_2\ u\,(z_2 + x_1 t^2) = -\sin\overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right) t^2 - \sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)$ ☑ the denominator of the formula for $\cot(\theta_2/2)$, times $\sin\sigma_2 \sin\overline{\beta}_2$

(2.3) $\left(-\sin\overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right) t^2 - \sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)\right)$ $\cot(\theta_2/2) - \sin\alpha_2 \sin\gamma_2\, t = \omega$ ☑ the formula for $\cot(\theta_2/2)$ of the given panel, whose denominator is not 0 because $t^2 \neq -z_2/x_1$ by assumption and $y_2$, $u \neq 0$, multiplied by the left side of (2.2), with (2.1)

(2.4) $u x_1 \tilde{y}_2 (1 + \tilde{z}_2)(1 + u\tilde{z}_2) = \left(\frac{\sin\beta_2 \sin\delta_2}{\sin\sigma_2 \sin\overline{\gamma}_2}\right)^2$ ☑ hence $\varepsilon_2 \sqrt{u x_1 \tilde{y}_2 (1 + \tilde{z}_2)(1 + u\tilde{z}_2)} = \frac{\sin\beta_2 \sin\delta_2}{\sin\sigma_2 \sin\overline{\gamma}_2}$, in the same way

(2.5) $u\tilde{y}_2 \tilde{z}_2\ \sin^2\overline{\gamma}_2 = u y_2 z_2\ \sin^2\overline{\beta}_2$ ☑ hence $\sin\overline{\gamma}_2 \sqrt{u\tilde{y}_2\tilde{z}_2\ D(x_1, \cot(\theta_1/2))} = \sin\overline{\beta}_2 \sqrt{u y_2 z_2\ D(x_1, \cot(\theta_1/2))}$, because $u y_2 z_2\ D(x_1, t) = u x_1 y_2 z_2 \cdot x_1\ D(x_1, t)/x_1^2 > 0$

(2.6) $\sin\sigma_2 \sin\overline{\gamma}_2\ \tilde{y}_2\ u\,(\tilde{z}_2 + x_1 t^2) = -\sin\overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin\overline{\beta}_2 \sin\left(\gamma_2 - \overline{\beta}_2\right) \neq 0$ • (2.1) and (2.2) of the block of Claim 3, with $x_2 = x_1$ and $u_2 = u$, because $t^2 \neq -\tilde{z}_2/x_1$ by assumption

(2.7) $\left(-\sin\overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin\overline{\beta}_2 \sin\left(\gamma_2 - \overline{\beta}_2\right)\right)$ $\left(\varepsilon_2\,(t\ \sqrt{u x_1 \tilde{y}_2 (1 + \tilde{z}_2)(1 + u\tilde{z}_2)} + e_0\ e_2\ \sqrt{u\tilde{y}_2\tilde{z}_2\ D(x_1, t)}\ )\right) \Big/ (\tilde{y}_2\, u\,(\tilde{z}_2 + x_1 t^2))$ $- \sin\beta_2 \sin\delta_2\, t = \omega$ ☑ by (2.6), the factor before the fraction is $\sin\sigma_2 \sin\overline{\gamma}_2$ times its denominator; then (2.4), (2.5), and the definition of $\omega$

$\Phi := -\sin(\gamma_2 - \delta_2)\ t \cot^2(\theta_2/2) - \sin\gamma_2\ t^2 \cot(\theta_2/2) + \sin(\gamma_2 + \delta_2)\ t + \sin\gamma_2\ \cot(\theta_2/2)$ • as in the block of Claim 2, with $t = \cot(\theta_1/2)$

$\Phi_0 := \left(-\sin\overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right)\cot^2(\theta_2/2) - \sin\overline{\alpha}_2 \sin(\beta_2 - \overline{\alpha}_2)\right) t^2 - 2\ \sin\alpha_2 \sin\gamma_2\, t \cot(\theta_2/2) + \left(-\sin\overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)\right) \cot^2(\theta_2/2) + \sin\sigma_2 \sin\overline{\beta}_2\ )$ • as there; $\Phi_0 = 0$ by its (2.4)

$\Phi_1 := \sin\alpha_2 \sin\gamma_2\ \cot(\theta_2/2) + \sin\overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right)\ t \cot^2(\theta_2/2)$ • as there

$$+ \sin \overline{\alpha}_2 \sin(\beta_2 - \overline{\alpha}_2)\ t$$

$$\Phi_2 := -\sin(\beta_2 - \overline{\gamma}_2) \sin\left(\beta_2 - \overline{\delta}_2\right) \cot^2(\theta_2/2) + \sin \overline{\alpha}_2 \sin \overline{\beta}_2$$

- as there

(2.8) $$(-\sin \overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin \overline{\beta}_2 \sin\left(\gamma_2 - \overline{\beta}_2\right))\ \Phi_1 - (\sin \beta_2 \sin \delta_2\, t + (-\sin \overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right) t^2 - \sin \overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)) \cot(\theta_2/2) - \sin \alpha_2 \sin \gamma_2\, t)\ \Phi_2 = -(-\sin(\beta_2 - \overline{\gamma}_2) \sin\left(\beta_2 - \overline{\delta}_2\right) \cot(\theta_2/2) - \sin \overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t)\ \Phi_0$$

☑ expand both sides, with (2.6) of the block of Claim 3: the coefficients of $t^3 \cot^2(\theta_2/2)$, $t^3$, $t^2 \cot^3(\theta_2/2)$, $\cot^3(\theta_2/2)$ agree term by term, and those of $t^2 \cot(\theta_2/2)$, $t \cot^2(\theta_2/2)$, $t$, $\cot(\theta_2/2)$, after taking out the factors $\sin \overline{\alpha}_2 \sin\left(\beta_2 - \overline{\delta}_2\right)$, $\sin(\beta_2 - \overline{\gamma}_2) \sin\left(\beta_2 - \overline{\delta}_2\right)$, $\sin \overline{\alpha}_2 \sin \overline{\beta}_2$, $\sin \overline{\beta}_2 \sin(\beta_2 - \overline{\gamma}_2)$, by its (2.4), (2.3), (2.3) to (2.5), (2.5)

(2.9) $$(-\sin \overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2) t^2 - \sin \overline{\beta}_2 \sin\left(\gamma_2 - \overline{\beta}_2\right)) \cot(\theta_{12'}/2) - \sin \beta_2 \sin \delta_2\, t = (-\sin \overline{\delta}_2 \sin\left(\beta_2 - \overline{\delta}_2\right) t^2 - \sin \overline{\gamma}_2 \sin(\beta_2 - \overline{\gamma}_2)) \cot(\theta_2/2) - \sin \alpha_2 \sin \gamma_2\, t$$

☑ (2.8) divided by $\Phi_2$, which is not 0 by Claim 3 and (2.2) of the block of Claim 2, because $\Phi_1/\Phi_2 = \cot(\theta_{12'}/2)$ by Claim 2 and $\Phi_0 = 0$

(2.10) $$\cot(\theta_{12'}/2) = \left(\varepsilon_2 (t \sqrt{u x_1 \tilde{y}_2 (1 + \tilde{z}_2)(1 + u \tilde{z}_2)} + e_0\ e_2 \sqrt{u \tilde{y}_2 \tilde{z}_2\ D(x_1, t)})\right) \Big/ (\tilde{y}_2\, u\, (\tilde{z}_2 + x_1 t^2))$$

☑ the right sides of (2.7) and (2.9) are equal by (2.3), hence so are their left sides; divide by $-\sin \overline{\alpha}_2 \sin(\gamma_2 - \overline{\alpha}_2)\ t^2 - \sin \overline{\beta}_2 \sin\left(\gamma_2 - \overline{\beta}_2\right)$, which is not 0 by (2.6)

Together, these give Claim 4.

**Section 12: ALL CLAIMS VERIFIED**

# Section 13

In[•]:=
```
(* ============================================================*)
(*PART 1:SETUP*)
(* ============================================================*)
ClearAll[statusChip, smallChip, givenChip, colTest, cols, bullet, idxB, ptB, pt,
  gk, thI, bx, frac, note, firstBase, nrow, prow, hrow, vgap, vgrid, iv, ivp,
  mulR, uPlain, mSym, xPlain, capX, capY, wSym, fSym, zV, uV, powD, hB, hG,
  hpt, hgk, hV, sgD, bAD, bBD, bGD, bDD, trigNames, simpleBoxQ, trigBoxRule,
  dispRule, disp, angAl, angBe, angGa, angDe, allAng, hh, zz, hRuleAll, zeroV,
  nxt, oldV, newV, sigF, ratF, qM, qR, qS, qF, qX, qY, qZ, cDef, cD, dTag, argD,
  famD, etD, etH, sums, normS, pp, qq, tt, ee, pH, qH, tH, eH, rS, sS, rH, sH, Fh,
  wW, XX, YY, k00, k22, k20, k02, k11, cS, zS, xS, uS, dS, normF, ampOf, argOf,
  iSet, tSym, e0D, yS, rPos, pAB, sq, absD, tSq, cotT, wH, wHat, resultsAll];
```

```
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
    Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
    Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
    RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☑", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
givenChip = Style["•", 12, GrayLevel[0.45]];
colTest = RGBColor[0.10, 0.35, 0.75];
cols = {RGBColor[0.10, 0.35, 0.75],
    RGBColor[0.50, 0.20, 0.65], RGBColor[0.00, 0.55, 0.55],
    RGBColor[0.20, 0.50, 0.20], RGBColor[0.75, 0.35, 0.05],
    RGBColor[0.70, 0.15, 0.40], RGBColor[0.30, 0.25, 0.70],
    RGBColor[0.45, 0.35, 0.15], RGBColor[0.35, 0.35, 0.35]};
bullet = Style["■ ", 12, colTest];
idxB[x_] :=
  If[IntegerQ[x], ToString[x], StyleBox[ToString[x], FontSlant → "Italic"]];
ptB[ltr_, i_] := SubscriptBox[StyleBox[ltr, FontSlant → "Italic"], idxB[i]];
pt[ltr_, i_] := DisplayForm[ptB[ltr, i]];
gk[g_, i_] := DisplayForm[SubscriptBox[g, idxB[i]]];
thI[x_] := DisplayForm[SubscriptBox["θ", idxB[x]]];
bx[DisplayForm[b_]] := b; bx[x_] := ToBoxes[x, TraditionalForm];
frac[num_, den_] := DisplayForm[FractionBox[bx[num], bx[den]]];
note[ntxt_] := Style[ntxt, 12, GrayLevel[0.5]];
firstBase[e_] :=
  If[Head[e] === Column, Append[e, BaselinePosition → {1, 1}], e];
nrow[n_, stmt_, mark_, nt_] := {Style["(" <> ToString[n] <> ")", 13],
   Style[firstBase[stmt], 13], mark, note[nt]};
prow[stmt_, mark_, nt_] := {"", Style[firstBase[stmt], 13], mark, note[nt]};
hrow[t_] := {Style[t, 13], SpanFromLeft, SpanFromLeft, SpanFromLeft};
vgap = {"", Spacer[{0, 7}], "", ""};
vgrid[rows_] :=
  Grid[rows, Alignment → {Left, Baseline}, Spacings → {1.2, 0.8}];
mulR[args___] := Row[Riffle[{args}, " "]];
uPlain = DisplayForm[StyleBox["u", FontSlant → "Italic"]];
mSym = DisplayForm[StyleBox["M", FontSlant → "Italic"]];
xPlain = DisplayForm[StyleBox["x", FontSlant → "Italic"]];
capX = DisplayForm[StyleBox["X", FontSlant → "Italic"]];
capY = DisplayForm[StyleBox["Y", FontSlant → "Italic"]];
wSym = DisplayForm[StyleBox["w", FontSlant → "Italic"]];
fSym = DisplayForm[StyleBox["F", FontSlant → "Italic"]];
zV = DisplayForm[StyleBox["Z", FontSlant → "Italic"]];
uV = DisplayForm[StyleBox["U", FontSlant → "Italic"]];
powD[ltr_, n_] := DisplayForm[
   SuperscriptBox[StyleBox[ltr, FontSlant → "Italic"], ToString[n]]];
tSym = DisplayForm[StyleBox["t", FontSlant → "Italic"]];
```

```
e0D = DisplayForm[SubscriptBox[StyleBox["e", FontSlant → "Italic"], "0"]];
iSet[setX_] := DisplayForm[
    RowBox[{StyleBox["I", FontSlant → "Italic"], "(", bx[setX], ")"}]];
rPos = DisplayForm[SubscriptBox["ℝ", RowBox[{">", "0"}]]];
pAB[k_, a_, b_] :=
  DisplayForm[RowBox[{ptB["P", k], "(", bx[a], ", ", bx[b], ")"}]];
sq[radArg_] := DisplayForm[SqrtBox[bx[radArg]]];
absD[e_] := Row[{"|", e, "|"}];
tSq = DisplayForm[SuperscriptBox[StyleBox["t", FontSlant → "Italic"], "2"]];
cotT[k_] := Row[{"cot(", thI[k], "(", tSym, ")/2)"}];
wHat = DisplayForm[OverscriptBox[StyleBox["w", FontSlant → "Italic"], "^"]];
(*a hat marks what is formed after the permutation*)
hB[ltr_] := OverscriptBox[StyleBox[ltr, FontSlant → "Italic"], "^"];
hG[g_] := OverscriptBox[g, "^"];
hpt[ltr_, i_] := DisplayForm[SubscriptBox[hB[ltr], idxB[i]]];
hgk[g_, i_] := DisplayForm[SubscriptBox[hG[g], idxB[i]]];
hV[ltr_] := DisplayForm[hB[ltr]];
(*every symbol that is only displayed is a string,so that no value
 given to a symbol elsewhere in the notebook can enter the display;
the displayed indices are italic strings as well*)
iv = Style["i", Italic]; ivp = Style["i+1", Italic];
sgD[i_] := Subscript["σ", i];
bAD[i_] := Subscript[OverBar["α"], i];
bBD[i_] := Subscript[OverBar["β"], i];
bGD[i_] := Subscript[OverBar["γ"], i];
bDD[i_] := Subscript[OverBar["δ"], i];
trigNames = {"sin", "cos", "tan", "cot", "sec", "csc"};
simpleBoxQ[b_String] := True;
simpleBoxQ[StyleBox[b_, ___]] := simpleBoxQ[b];
simpleBoxQ[(SubscriptBox | SuperscriptBox | OverscriptBox | UnderscriptBox)[
    bs__]] := AllTrue[{bs}, simpleBoxQ];
simpleBoxQ[_] := False;
trigBoxRule =
  {RowBox[{f_String /; MemberQ[trigNames, f], "(", a_, ")"}] /; simpleBoxQ[a] :>
    RowBox[{f, "", a}],
   RowBox[{f_String /; MemberQ[trigNames, f], RowBox[{"(", a_, ")"}]}] /;
     simpleBoxQ[a] :> RowBox[{f, "", a}]};
dispRule = Join[Table[angAl[k] → Subscript["α", k], {k, 4}],
   Table[angBe[k] → Subscript["β", k], {k, 4}],
   Table[angGa[k] → Subscript["γ", k], {k, 4}],
   Table[angDe[k] → Subscript["δ", k], {k, 4}]];
disp[e_] :=
  DisplayForm[ToBoxes[e /. dispRule, TraditionalForm] //. trigBoxRule];
(*a trigonometric identity is decided without
 Simplify:every flat angle is written as twice a half angle,
each trigonometric function of an integer combination of the half
 angles is written through the exponentials of the half angles,
```

```
and the resulting rational function must be 0;
the kernel writes 1/sin as csc,so all six functions are converted*)
allAng = Flatten[Table[{angAl[k], angBe[k], angGa[k], angDe[k]}, {k, 4}]];
hRuleAll = Thread[allAng → Table[2 hh[k], {k, Length[allAng]}]];
zeroV[e_] :=
  Module[{f = e /. hRuleAll, hv, mon}, hv = Union[Cases[f, hh[_], {0, Infinity}]];
   mon[x_] := Product[zz[h]^Coefficient[x, h], {h, hv}];
   f =
    f /. {Sin[x_] :> (mon[x] - 1 / mon[x]) / (2 I), Cos[x_] :> (mon[x] + 1 / mon[x]) / 2,
      Tan[x_] :> (mon[x] - 1 / mon[x]) / (I (mon[x] + 1 / mon[x])),
      Cot[x_] :> I (mon[x] + 1 / mon[x]) / (mon[x] - 1 / mon[x]),
      Sec[x_] :> 2 / (mon[x] + 1 / mon[x]), Csc[x_] :> 2 I / (mon[x] - 1 / mon[x])};
   AllTrue[Flatten[{Together[f]}], # === 0 &]];
(*the numbers of a vertex from its four flat angles,
in the order alpha,beta,gamma,delta;
the vertex i+1 before the permutation is the vertex i after it,
with alpha and gamma exchanged*)
nxt[i_] := Mod[i, 4] + 1;
oldV[k_] := {angAl[k], angBe[k], angGa[k], angDe[k]};
newV[i_] := With[{k = nxt[i]}, {angGa[k], angBe[k], angAl[k], angDe[k]}];
sigF[v_] := Total[v] / 2;
ratF[v_] := With[{sgm = Total[v] / 2}, Sin[v] / Sin[sgm - v]];
qM[v_] := Times @@ ratF[v];
qR[v_] := ratF[v][[1]] ratF[v][[4]];
qS[v_] := ratF[v][[3]] ratF[v][[4]];
qF[v_] := ratF[v][[1]] ratF[v][[3]];
qX[v_] := 1 / (qR[v] - 1);
qY[v_] := 1 / (qS[v] - 1);
qZ[v_] := 1 / (qF[v] - 1);
(*the coefficients c00,c22,c20,c02,c11 of the polynomial of a vertex*)
cDef[v_] := With[{a = v[[1]], b = v[[2]], g = v[[3]], d = v[[4]], sgm = Total[v] / 2},
   {Sin[sgm] Sin[sgm - b], Sin[sgm - d] Sin[sgm - d - b],
    Sin[sgm - a] Sin[sgm - a - b], Sin[sgm - g] Sin[sgm - g - b], -Sin[a] Sin[g]}];
cD[k_String, j_] := DisplayForm[
   SubscriptBox[StyleBox["c", FontSlant → "Italic"], RowBox[{k, idxB[j]}]]];
dTag[argX_, argT_] := Row[{DisplayForm[StyleBox["D", FontSlant → "Italic"]],
    "(", argX, ", ", argT, ")"}];
argD[terms_] :=
  Row[Flatten[Table[{If[k == 1, If[terms[[k, 1]] < 0, "-", ""], If[terms[[k, 1]] < 0,
        " - ", " + "]], terms[[k, 2]]}, {k, Length[terms]}]]];
famD[amp_, terms_] := Row[{amp, "", fSym, "(", argD[terms], ")"}];
etD[k_] := mulR[pt["e", k], pt["t", k]];
etH[k_] := mulR[hpt["e", k], hpt["t", k]];
(*the eight sums of a vertex,
and a sum written with the coefficient+1 of the angle a*)
sums[v_] := Flatten[Table[v[[1]] + s1 v[[2]] + s2 v[[3]] + s3 v[[4]],
    {s1, {1, -1}}, {s2, {1, -1}}, {s3, {1, -1}}]];
```

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```
normS[e_, a_] := If[Coefficient[e, a] < 0, Expand[-e], Expand[e]];
(*the family:the amplitudes pp,qq,the phase shifts tt,
the signs ee and the hatted ones pH,qH,tH,
eH are symbols,Fh is the function F,wW is w*)
normF[e_] := e /. Fh[a_] :> Fh[Expand[a]];
ampOf[e_] := e /. Fh[_] → 1;
argOf[e_] := First[Cases[e, Fh[a_] :> a, {0, Infinity}]];
resultsAll = {};

(* ============================================================*)
(*PART 2:GIVEN PANEL*)
(* ============================================================*)
Print[
  Panel[Column[{Style["Given hypotheses and definitions:", Bold, 14], Style[
      "A 3 x 3 net with, for each i = 1, 2, 3, 4, the indices taken modulo
        4,", Bold, 13],
     Style[Column[{Row[{"the flat angles ", gk["α", "i"], ", ",
         gk["β", "i"], ", ", gk["γ", "i"], ", ", gk["δ", "i"], " ∈ (0, π) at ",
         pt["A", "i"], " and the central dihedral angles ", thI[1],
         ", ", thI[2], ", ", thI[3], ", ", thI[4]}], Spacer[{0, 6}],
       Grid[{{Row[{disp[sgD[iv]], " = ", disp[(Subscript["α", iv] + Subscript[
                 "β", iv] + Subscript["γ", iv] + Subscript["δ", iv]) / 2]}],
         Row[{disp[bAD[iv]], " = ", disp[sgD[iv] - Subscript["α", iv]]}],
         Row[{disp[bBD[iv]], " = ", disp[sgD[iv] - Subscript["β", iv]]}],
         Row[{disp[bGD[iv]], " = ", disp[sgD[iv] - Subscript["γ", iv]]}],
         Row[{disp[bDD[iv]], " = ", disp[sgD[iv] - Subscript["δ", iv]]}]},
        {Row[{pt["a", "i"], " = ", frac[Row[{"sin", gk["α", "i"]}],
            Row[{"sin", disp[bAD[iv]]}]]}], Row[{pt["b", "i"], " = ",
          frac[Row[{"sin", gk["β", "i"]}], Row[{"sin", disp[bBD[iv]]}]]}],
         Row[{pt["c", "i"], " = ", frac[Row[{"sin", gk["γ", "i"]}],
            Row[{"sin", disp[bGD[iv]]}]]}], Row[{pt["d", "i"], " = ",
          frac[Row[{"sin", gk["δ", "i"]}], Row[{"sin", disp[bDD[iv]]}]]}],
         Row[{pt["ε", "i"], " = sign(sin", disp[sgD[iv]], ")"}]},
        {Row[{pt["M", "i"], " = ", mulR[pt["a", "i"],
            pt["b", "i"], pt["c", "i"], pt["d", "i"]]}],
         Row[{pt["r", "i"], " = ", mulR[pt["a", "i"], pt["d", "i"]]}],
         Row[{pt["s", "i"], " = ", mulR[pt["c", "i"], pt["d", "i"]]}],
         Row[{pt["f", "i"], " = ", mulR[pt["a", "i"], pt["c", "i"]]}], ""},
        {Row[{pt["x", "i"], " = ", frac["1", Row[{pt["r", "i"], " - 1"}]]}],
         Row[{pt["y", "i"], " = ", frac["1", Row[{pt["s", "i"], " - 1"}]]}],
         Row[{pt["z", "i"], " = ", frac["1", Row[{pt["f", "i"], " - 1"}]]}],
         Row[{uPlain, " = 1 - ", mSym}], ""}},
       Alignment → Left, Spacings → {2.4, 1.1}], Spacer[{0, 6}],
      Row[{gk["α", "i"], " ± ", gk["β", "i"], " ± ", gk["γ", "i"],
        " ± ", gk["δ", "i"], " ≠ 0 (mod 2π),   ", gk["δ", 1], " + ",
        gk["δ", 2], " + ", gk["δ", 3], " + ", gk["δ", 4], " = 2π"}],
      Row[{pt["M", 1], " = ", pt["M", 2], " = ", pt["M", 3], " = ",
```

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      pt["M", 4], " = ", mSym, " > 0,   ", pt["r", 1], " = ", pt["r", 2],
      ",   ", pt["s", 1], " = ", pt["s", 4], ",   ", pt["r", 3],
      " = ", pt["r", 4], ",   ", pt["s", 2], " = ", pt["s", 3]}],
    Row[{etD[1], " + ", etD[2], " + ", etD[3], " + ", etD[4],
      " ∈ Λ,   ", pt["e", 1], ", ", pt["e", 2], ", ", pt["e", 3],
      ", ", pt["e", 4], " ∈ {-1, 1}"}]}, Spacings → 0.7], 13],
Spacer[{0, 10}], Style["Amplitudes, phase shifts, lattice:",
 Bold, 14],
Style[Column[{Row[{pt["p", "i"], " = ", DisplayForm[
       SqrtBox[RowBox[{ptB["r", "i"], "-", "1"}]]], ",   ", pt["q", "i"],
      " = ", DisplayForm[SqrtBox[RowBox[{ptB["s", "i"], "-", "1"}]]],
      " ∈ ", rPos, " ⋃ ɨ", rPos}],
    Row[{"dn", pt["t", "i"], " = ", DisplayForm[SqrtBox[ptB["f", "i"]]],
      " if ", mSym, " < 1,   dn", pt["t", "i"], " = 1/",
      DisplayForm[SqrtBox[ptB["f", "i"]]], " if ", mSym,
      " > 1,   the interval of ", pt["t", "i"], " fixed by ",
      mulR[pt["p", "i"], pt["q", "i"]], ", ", gk["σ", "i"], ", ", mSym}],
    Row[{fSym, " = sn if ", mSym, " < 1,   ", fSym, " = cn if ",
      mSym, " > 1,   Λ depends only on ", mSym}]}, Spacings → 0.7], 13],
Spacer[{0, 10}], Style["Bricard's equations:", Bold, 14],
Style[
 Column[{Row[{"(", zV, ", ", pt["W", 2], ", ", uV, ", ",
      pt["W", 1], ") = (cot(", thI[1], "/2), cot(", thI[2],
      "/2), cot(", thI[3], "/2), cot(", thI[4], "/2))"}],
    Row[{pAB[1, zV, pt["W", 1]], " = ", pAB[2, zV, pt["W", 2]], " = ", pAB[
       3, uV, pt["W", 2]], " = ", pAB[4, uV, pt["W", 1]], " = 0,   where"}],
    Row[{"          ", pAB["i", capX, capY], " = ", mulR[cD["22", "i"],
       powD["X", 2], powD["Y", 2]], " + ", mulR[cD["20", "i"], powD["X", 2]],
      " + ", mulR[cD["02", "i"], powD["Y", 2]], " + 2",
      mulR[cD["11", "i"], capX, capY], " + ", cD["00", "i"], ","}],
    Row[{"          ", cD["00", "i"], " = sin", disp[sgD[iv]],
      " sin", disp[bBD[iv]], ",   ", cD["22", "i"], " = sin",
      disp[bDD[iv]], " sin(", disp[bDD[iv]], " - ", gk["β", "i"],
      "),   ", cD["20", "i"], " = sin", disp[bAD[iv]],
      " sin(", disp[bAD[iv]], " - ", gk["β", "i"], "),"}],
    Row[{"          ", cD["02", "i"], " = sin", disp[bGD[iv]], " sin(",
      disp[bGD[iv]], " - ", gk["β", "i"], "),   ", cD["11", "i"], " = -sin",
      gk["α", "i"], " sin", gk["γ", "i"]}]}, Spacings → 0.7], 13],
Spacer[{0, 10}], Style["Family and flexion:", Bold, 14],
Style[
 Column[{Row[{"the family of ", pt["e", 1], ", ", pt["e", 2], ", ",
      pt["e", 3], ", ", pt["e", 4], ":   (", zV, ", ", pt["W", 2], ", ",
      uV, ", ", pt["W", 1], ") = (", famD[pt["p", 1], {{1, wSym}}],
      ", ", famD[pt["q", 2], {{1, wSym}, {-1, etD[2]}}], ", ",
      famD[pt["p", 3], {{1, wSym}, {-1, etD[2]}, {-1, etD[3]}}],
      ", ", famD[pt["q", 1], {{1, wSym}, {1, etD[1]}}],
      "),   ", wSym, " ∈ ℂ,   all four finite"}],
```

```
Row[{dTag[xPlain, capX], " = (", mulR[xPlain, powD["X", 2]],
  " - 1)(1 - ", mulR[uPlain, xPlain, powD["X", 2]], "),   ",
  iSet[xPlain], " = {", capX, " ∈ ℝ : ", mulR[xPlain,
   dTag[xPlain, capX]], " > 0},   ", xPlain, ", ", capX, " ∈ ℝ"}],
Row[{"the flexion of ", pt["e", 1], ", ", pt["e", 2], ", ",
  pt["e", 3], ", ", pt["e", 4], ":   for each ", e0D,
  " ∈ {-1, 1} and each admissible ", tSym, ", the point (",
  zV, ", ", pt["W", 2], ", ", uV, ", ", pt["W", 1], ") = (", tSym,
  ", ", cotT[2], ", ", cotT[3], ", ", cotT[4], "), where"}],
With[{x1 = pt["x", 1], x3 = pt["x", 3], y1 = pt["y", 1],
  y2 = pt["y", 2], z1 = pt["z", 1], z2 = pt["z", 2],
  z3 = pt["z", 3], ee2 = mulR[pt["e", 2], pt["e", 3]],
  hh1 = pt["H", 1], hh2 = pt["H", 2], hh3 = pt["H", 3]},
 Column[{Row[{"          ", cotT[2], " = ", frac[Row[{gk["ε", 2], "(",
        mulR[tSym, sq[Row[{mulR[uPlain, x1, y2], "(1 + ", z2, ")(1 + ",
            mulR[uPlain, z2], ")"}]]], " + ", mulR[e0D, pt["e", 2],
         sq[Row[{mulR[uPlain, y2, z2], " ", dTag[x1, tSym]}]]], ")"}],
      Row[{mulR[y2, uPlain], "(", z2, " + ", mulR[x1, tSq], ")"}]]}],
   Row[{"          ", cotT[4], " = ", frac[Row[{gk["ε", 1], "(",
        mulR[tSym, sq[Row[{mulR[uPlain, x1, y1], "(1 + ", z1, ")(1 + ",
            mulR[uPlain, z1], ")"}]]], " - ", mulR[e0D, pt["e", 1],
         sq[Row[{mulR[uPlain, y1, z1], " ", dTag[x1, tSym]}]]], ")"}],
      Row[{mulR[y1, uPlain], "(", z1, " + ", mulR[x1, tSq], ")"}]]}],
   Row[{"          ", cotT[3], " = ", mulR[gk["ε", 2], gk["ε", 3]],
     " sign(", z2, ") ", sq[frac[mulR[z2, z3], mulR[x1, x3]]],
     " · ", frac[Row[{mulR[absD[x1], hh2, hh3, tSym], " + ",
        mulR[e0D, pt["e", 2], hh1, sq[mulR[x1, dTag[x1, tSym]]]]}],
      Row[{mulR[z2, z3], " + ", mulR[uPlain, x1, DisplayForm[
          SuperscriptBox[bx[hh1], "2"]], tSq]}]]}],
   Row[{"          ", hh1, ", ", hh2, ", ", hh3, ":   if ",
     mulR[uPlain, z2, z3], " ≠ 1,"}], Row[{"                    ", hh1,
     " = ", frac[Row[{absD[z2], " ", sq[Row[{z3, "(1 + ", z3, ")(1 + ",
           mulR[uPlain, z3], ")"}]], " + ", ee2, " ", absD[z3], " ",
        sq[Row[{z2, "(1 + ", z2, ")(1 + ", mulR[uPlain, z2], ")"}]]}],
      Row[{mulR[uPlain, z2, z3], " - 1"}]], ",   ", hh2, " = ",
     frac[Row[{absD[mulR[z2, z3]], " ", sq[Row[{"(1 + ", mulR[uPlain,
           z2], ")(1 + ", mulR[uPlain, z3], ")"}]], " + ", ee2, " ",
        sq[Row[{mulR[z2, z3], "(1 + ", z2, ")(1 + ", z3, ")"}]]}],
      Row[{mulR[uPlain, z2, z3], " - 1"}]], ","}],
   Row[{"                  ", hh3, " = ", frac[Row[{uPlain,
        " ", sq[Row[{mulR[z2, z3], "(1 + ", z2, ")(1 + ", z3, ")"}]],
        " sign(", mulR[z2, z3], ") + ", ee2, " ", sq[Row[{"(1 + ",
           mulR[uPlain, z2], ")(1 + ", mulR[uPlain, z3], ")"}]]}],
      Row[{mulR[uPlain, z2, z3], " - 1"}]], ";"}],
   Row[{"                  if ", mulR[uPlain, z2, z3],
     " = 1 and ", pt["e", 2], " = -", pt["e", 3], ","}],
   Row[{"                  ", hh1, " = ", frac[Row[{mulR[z2, z3],
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            "(", z3, " - ", z2, ")"}], Row[{"2 ", sq[Row[{mulR[z2, z3],
               "(1 + ", z2, ")(1 + ", z3, ")"}]]}]], ",   ", hh2, " = ",
         frac[Row[{mulR[z2, z3], "(", z2, " + ", z3, " + 2)"}], Row[{"2 ",
            sq[Row[{mulR[z2, z3], "(1 + ", z2, ")(1 + ", z3, ")"}]]}]],
         ",   ", hh3, " = ", frac[Row[{"(", z2, " + ", z3, " + 2",
            mulR[z2, z3], ") sign(", mulR[z2, z3], ")"}], Row[{"2 ", sq[Row[
              {mulR[z2, z3], "(1 + ", z2, ")(1 + ", z3, ")"}]]}]], ";"}],
       Row[{"          if ", mulR[uPlain, z2, z3], " = 1 and ",
         pt["e", 2], " = ", pt["e", 3], ", the formula for ", cotT[3],
         " is replaced by   ", cotT[3], " = ", mulR[gk["ε", 2], gk["ε", 3]],
         " ", sq[frac[mulR[z2, z3], mulR[x1, x3]]], " · ",
         frac[Row[{"sign(", y2, ")"}], tSym]}]}, Spacings → 0.9]],
   Row[{"admissible ", tSym, ":   ", tSym, " ∈ ", iSet[pt["x", 1]],
     " outside the finitely many values at which a
       denominator of these formulas vanishes"}],
   Row[{"the polyhedra of the flexion with the parameter ",
     tSym, ":   the polyhedra with these flat angles whose (",
     zV, ", ", pt["W", 2], ", ", uV, ", ", pt["W", 1],
     ") is this point for ", e0D, " = 1 or for ", e0D, " = -1"}]},
  Spacings → 0.7], 13], Spacer[{0, 10}],
Style["Cyclic permutation of the indices:",
 Bold,
 14], Style[
 Column[
  {Row[{"by 1:   ", pt["A", "i+1"], ", ", pt["B", "i+1"],
     ", ", pt["C", "i+1"], ", ", thI["i+1"], "  ↦  ",
     pt["A", "i"], ", ", pt["C", "i"], ", ", pt["B", "i"], ", ",
     thI["i"], ";   a hat marks what is formed after it:"}],
   Row[{"          (", hgk["α", "i"], ", ", hgk["β", "i"], ", ",
     hgk["γ", "i"], ", ", hgk["δ", "i"], ") = (", gk["γ", "i+1"],
     ", ", gk["β", "i+1"], ", ", gk["α", "i+1"], ", ",
     gk["δ", "i+1"], "),   ", hpt["e", "i"], " = ", pt["e", "i+1"]}],
   Row[{"by j = 2, 3:   the permutation by 1 applied j times"}]},
  Spacings → 0.7], 13], Spacer[{0, 10}],
Style["Given facts, valid whenever the hypotheses above hold:",
 Bold,
 14],
Style[Column[{Row[{"(a)   ", pt["x", "i"], ", ", pt["y", "i"],
     ", ", pt["z", "i"], " ≠ -1, 0 are well-defined reals,   ",
     mulR[pt["z", "i"], Row[{"(1 + ", pt["z", "i"], ")"}]],
     " > 0,   1 + ", mulR[uPlain, pt["z", "i"]], " > 0"}],
   Row[{"(b)   ", fSym, "(v + λ) = ", fSym, "(v) for v ∈ ℂ, λ ∈ Λ"}],
   Row[{"(c)   every point of the family of ",
     pt["e", 1], ", ", pt["e", 2], ", ", pt["e", 3],
     ", ", pt["e", 4], " solves Bricard's equations"}],
   Row[{"(d)   for admissible ", tSym, ", the points of the family of ",
     pt["e", 1], ", ", pt["e", 2], ", ", pt["e", 3],
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            ", ", pt["e", 4], " with ", zV, " = ", tSym,
            " are the two points of the flexion above, for ",
            e0D, " = 1 and ", e0D, " = -1"}]}, Spacings → 0.7], 13],
      Spacer[{0, 10}], Style[Row[{"Each ", smallChip[True],
         " below certifies that the statement displayed beside it has been
           verified by a symbolic computation at that point; a ",
         smallChip[False], " would mean that this verification failed. A ",
         givenChip,
         " marks a given fact, a definition, a result of an earlier line, or
           an inference stated in words beside it."}], 12, GrayLevel[
        0.35]], "", Style[Row[{Style["We want to test the following claim ",
          Bold, 14], Style["⋆", 14, colTest], Style[" :", Bold, 14]}]],
      Spacer[{0, 6}], Style["Claim.", Bold, 13, colTest],
      Style[
       Row[{"For j = 1, 2, 3 and all but finitely many admissible ",
         tSym, ","}], Bold, 13, colTest], Row[
       {bullet, Style[Row[{"the polyhedra of the flexion with the parameter ",
            tSym, " are polyhedra of the flexion formed after the
              cyclic permutation of the indices by j, with ",
            pt["e", "1+j"], ", …, ", pt["e", "4+j"], " in place of ",
            pt["e", 1], ", …, ", pt["e", 4]}], Bold, 13, colTest]}]},
     Spacings → 1.1], Background → RGBColor[
      0.99,
      0.96,
      0.88],
    FrameMargins → 14]];

(* ============================================================*)
(*PART 3:STEP 1*)
(* ============================================================*)
Module[{col = cols[[9]], oks = {}, chk, all4, c2, c3, c4, c5, c6, rows},
  chk[e_] := Module[{verdict = zeroV[e]}, AppendTo[oks, verdict];
    verdict];
  all4[f_] := And @@ Table[f[i], {i, 4}];
  c2 = all4[Function[i, chk[sigF[newV[i]] - sigF[oldV[nxt[i]]]]]];
  c3 = all4[Function[i, And @@ (chk /@ ((sigF[newV[i]] - newV[i]) -
           (sigF[oldV[nxt[i]]] - oldV[nxt[i]])[[{3, 2, 1, 4}]]))]];
  c4 = all4[Function[i,
     And @@ (chk /@ (ratF[newV[i]] - ratF[oldV[nxt[i]]][[{3, 2, 1, 4}]]))]];
  c5 = all4[Function[i, And @@
      (chk /@ {qM[newV[i]] - qM[oldV[nxt[i]]], qR[newV[i]] - qS[oldV[nxt[i]]],
         qS[newV[i]] - qR[oldV[nxt[i]]], qF[newV[i]] - qF[oldV[nxt[i]]]})]];
  c6 =
   all4[Function[i, And @@ (chk /@ {qX[newV[i]] - qY[oldV[nxt[i]]], qY[newV[i]] -
          qX[oldV[nxt[i]]], qZ[newV[i]] - qZ[oldV[nxt[i]]]})]];
  rows =
   {nrow[1, Row[{"(", hgk["α", "i"], ", ", hgk["β", "i"], ", ", hgk["ɣ", "i"],
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      ", ", hgk["δ", "i"], ") = (", gk["ɣ", "i+1"], ", ", gk["β", "i+1"],
      ", ", gk["α", "i+1"], ", ", gk["δ", "i+1"], ")"}],
   givenChip, "the permutation, see the given panel"],
  nrow[2, Row[{hgk["σ", "i"], " = ", gk["σ", "i+1"]}], smallChip[c2],
   "by (1): the half sum of the same four angles"],
  nrow[3, Row[{"(", hgk["σ", "i"], " - ", hgk["α", "i"], ", ",
      hgk["σ", "i"], " - ", hgk["β", "i"], ", ", hgk["σ", "i"], " - ",
      hgk["ɣ", "i"], ", ", hgk["σ", "i"], " - ", hgk["δ", "i"], ") = (",
      disp[bGD[ivp]], ", ", disp[bBD[ivp]], ", ", disp[bAD[ivp]],
      ", ", disp[bDD[ivp]], ")"}], smallChip[c3], "by (1) and (2)"],
  nrow[4, Row[{"(", hpt["a", "i"], ", ", hpt["b", "i"], ", ",
      hpt["c", "i"], ", ", hpt["d", "i"], ") = (", pt["c", "i+1"], ", ",
      pt["b", "i+1"], ", ", pt["a", "i+1"], ", ", pt["d", "i+1"], ")"}],
   smallChip[c4], "by (1) and (3): each is the sine of an
      angle over the sine of its barred angle"],
  nrow[5, Row[{"(", hpt["M", "i"], ", ", hpt["r", "i"], ", ",
      hpt["s", "i"], ", ", hpt["f", "i"], ") = (", pt["M", "i+1"], ", ",
      pt["s", "i+1"], ", ", pt["r", "i+1"], ", ", pt["f", "i+1"], ")"}],
   smallChip[c5], Row[{"by (4), because ", pt["M", "i"], " = ",
      mulR[pt["a", "i"], pt["b", "i"], pt["c", "i"], pt["d", "i"]],
      ", ", pt["r", "i"], " = ", mulR[pt["a", "i"], pt["d", "i"]],
      ", ", pt["s", "i"], " = ", mulR[pt["c", "i"], pt["d", "i"]],
      ", ", pt["f", "i"], " = ", mulR[pt["a", "i"], pt["c", "i"]]}]],
  nrow[6, Row[{"(", hpt["x", "i"], ", ", hpt["y", "i"], ", ",
      hpt["z", "i"], ") = (", pt["y", "i+1"], ", ", pt["x", "i+1"],
      ", ", pt["z", "i+1"], ")"}], smallChip[c6], "by (5)"],
  nrow[7, Row[{hpt["ε", "i"], " = ", pt["ε", "i+1"]}], givenChip,
   Row[{"by (2), because ", pt["ε", "i"], " = sign(sin", gk["σ", "i"], ")"}]],
  nrow[8, Row[{"(", hpt["p", "i"], ", ", hpt["q", "i"], ") = (", pt["q", "i+1"],
      ", ", pt["p", "i+1"], "),   ", mulR[hpt["p", "i"], hpt["q", "i"]],
      " = ", mulR[pt["p", "i+1"], pt["q", "i+1"]]}], givenChip,
   Row[{"by (5), because ", pt["p", "i"], " = ", DisplayForm[
       SqrtBox[RowBox[{ptB["r", "i"], "-", "1"}]]], " and ", pt["q", "i"],
      " = ", DisplayForm[SqrtBox[RowBox[{ptB["s", "i"], "-", "1"}]]]}]],
  nrow[9, Row[{hpt["t", "i"], " = ", pt["t", "i+1"]}], givenChip,
   Row[{"dn", hpt["t", "i"], " = dn", pt["t", "i+1"], ", because ",
      hpt["f", "i"], " = ", pt["f", "i+1"], " and ", hpt["M", "i"], " = ",
      pt["M", "i+1"], " by (5); the interval is the same, because ",
      mulR[hpt["p", "i"], hpt["q", "i"]], " = ",
      mulR[pt["p", "i+1"], pt["q", "i+1"]], " by (8) and ",
      hgk["σ", "i"], " = ", gk["σ", "i+1"], " by (2)"}]], nrow[10,
   Row[{"(", hV["Z"], ", ", hpt["W", 2], ", ", hV["U"], ", ", hpt["W", 1],
      ") = (", pt["W", 2], ", ", uV, ", ", pt["W", 1], ", ", zV, ")"}],
   givenChip, Row[{thI["i+1"], " is relabeled ", thI["i"]}]]};
AppendTo[resultsAll, And @@ oks];
Print[
 Panel[Column[{Row[{Style["Step 1: the numbers after the permutation", Bold,
```

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        15, col], Spacer[20], statusChip[And @@ oks]}],
      Style["Verification: each line holds for i = 1, 2, 3, 4, and each
        checked line is an identity in the flat angles, checked
        for each of the four values of i.", Bold, 13], vgrid[rows],
      Style["Together, these give the numbers after the permutation.", 13]},
     Spacings → 1], Background → Lighter[col, 0.93],
    FrameMargins → 12]];];

(* ============================================================*)
(*PART 4:STEP 2*)
(* ============================================================*)
Module[{col = cols⟦9⟧, oks = {}, reg, d1, d2, d3, d4, d5, newPairs, oldPairs, rows},
  reg[verdict_] := (AppendTo[oks, verdict];
    verdict);
  d1 = reg[
    And @@ Table[With[{k = nxt[i]}, Sort[normS[#, angAl[k]] & /@ sums[newV[i]]] ===
        Sort[normS[#, angAl[k]] & /@ sums[oldV[k]]]], {i, 4}]];
  d2 = reg[Expand[Sum[newV[i]⟦4⟧, {i, 4}] - Sum[oldV[k]⟦4⟧, {k, 4}]] === 0];
  d3 = reg[Sort[Table[nxt[i], {i, 4}]] === {1, 2, 3, 4}];
  newPairs = {{rH[1], rH[2]}, {sH[1], sH[4]}, {rH[3], rH[4]}, {sH[2], sH[3]}} /.
    {rH[i_] :↦ sS[nxt[i]], sH[i_] :↦ rS[nxt[i]]};
  oldPairs = {{rS[1], rS[2]}, {sS[1], sS[4]}, {rS[3], rS[4]}, {sS[2], sS[3]}};
  d4 = reg[Sort[Sort /@ newPairs] === Sort[Sort /@ oldPairs]];
  d5 = reg[Expand[
      (Sum[eH[i] tH[i], {i, 4}] /. {eH[i_] :↦ ee[nxt[i]], tH[i_] :↦ tt[nxt[i]]}) -
       Sum[ee[k] tt[k], {k, 4}]] === 0];
  rows =
   {nrow[1, Row[{"the eight sums ", hgk["α", "i"], " ± ", hgk["β", "i"], " ± ",
       hgk["ɣ", "i"], " ± ", hgk["δ", "i"], " are, up to sign, the eight sums ",
       gk["α", "i+1"], " ± ", gk["β", "i+1"], " ± ",
       gk["ɣ", "i+1"], " ± ", gk["δ", "i+1"]}], smallChip[d1],
     "for i = 1, 2, 3, 4, by (1) of Step 1; so none of them is 0 modulo 2π"],
    nrow[2, Row[{hgk["α", "i"], ", ", hgk["β", "i"], ", ",
       hgk["ɣ", "i"], ", ", hgk["δ", "i"], " ∈ (0, π)"}],
     givenChip, "by (1) of Step 1: they are flat angles of the net"],
    nrow[3, Row[{hgk["δ", 1], " + ", hgk["δ", 2], " + ", hgk["δ", 3], " + ",
       hgk["δ", 4], " = ", gk["δ", 2], " + ", gk["δ", 3], " + ", gk["δ", 4],
       " + ", gk["δ", 1], " = 2π"}], smallChip[d2], "by (1) of Step 1"],
    nrow[4, Row[{hpt["M", 1], " = ", hpt["M", 2], " = ", hpt["M", 3], " = ",
       hpt["M", 4], " reads ", pt["M", 2], " = ", pt["M", 3], " = ", pt["M", 4],
       " = ", pt["M", 1], ", so that ", hV["M"], " = ", mSym, " > 0"}],
     smallChip[d3], "by (5) of Step 1"], nrow[5, Row[{hpt["r", 1], " = ",
       hpt["r", 2], ", ", hpt["s", 1], " = ", hpt["s", 4], ", ", hpt["r", 3],
       " = ", hpt["r", 4], ", ", hpt["s", 2], " = ", hpt["s", 3], " read ",
       pt["s", 2], " = ", pt["s", 3], ", ", pt["r", 2], " = ", pt["r", 1], ", ",
       pt["s", 4], " = ", pt["s", 1], ", ", pt["r", 3], " = ", pt["r", 4]}],
     smallChip[d4], Row[{"by (5) of Step 1; these are ", pt["r", 1],
```

```
      " = ", pt["r", 2], ", ", pt["s", 1], " = ", pt["s", 4], ", ", pt["r", 3],
      " = ", pt["r", 4], ", ", pt["s", 2], " = ", pt["s", 3]}]], nrow[6,
    Row[{etH[1], " + ", etH[2], " + ", etH[3], " + ", etH[4], " = ", etD[2],
      " + ", etD[3], " + ", etD[4], " + ", etD[1], " ∈ Λ"}], smallChip[d5],
    Row[{"by (9) of Step 1 and ", hpt["e", "i"], " = ", pt["e", "i+1"],
      "; the lattice is the same, because ", hV["M"], " = ", mSym}]]};
 AppendTo[resultsAll, And @@ oks];
 Print[
  Panel[Column[{Row[{Style["Step 2: the hypotheses after the permutation",
       Bold, 15, col], Spacer[20], statusChip[And @@ oks]}],
    Style["Verification: each hypothesis of the given panel,
       written with the hats, is compared with the hypotheses
       written without them.", Bold, 13], vgrid[rows],
    Style[Row[{"Together, these show that the hypotheses of the
         given panel hold after the permutation, with ",
       pt["e", "i+1"], " in place of ", pt["e", "i"], "."}], 13]},
    Spacings → 1], Background → Lighter[col, 0.93], FrameMargins →
    12]];];

(* ============================================================*)
(*PART 5:STEP 3*)
(* ============================================================*)
Module[{col = cols[[9]], oks = {}, reg, famOld, famHat, hatRule,
   step3, exp3, inOld, step5, exp5, sumE, k3, k5, k6, k7, rows},
 reg[verdict_] := (AppendTo[oks, verdict];
   verdict);
 famOld = {pp[1] Fh[wW], qq[2] Fh[wW - ee[2] tt[2]],
   pp[3] Fh[wW - ee[2] tt[2] - ee[3] tt[3]], qq[1] Fh[wW + ee[1] tt[1]]};
 famHat = {pH[1] Fh[wH], qH[2] Fh[wH - eH[2] tH[2]],
   pH[3] Fh[wH - eH[2] tH[2] - eH[3] tH[3]], qH[1] Fh[wH + eH[1] tH[1]]};
 hatRule = {pH[i_] :> qq[nxt[i]],
   qH[i_] :> pp[nxt[i]], tH[i_] :> tt[nxt[i]], eH[i_] :> ee[nxt[i]]};
 (*the hatted family,with (8) and (9) of Step 1,
 is the list of W2,U,W1,Z*)step3 = normF[famHat /. hatRule];
 exp3 = normF[{qq[2] Fh[wH], pp[3] Fh[wH - ee[3] tt[3]],
    qq[4] Fh[wH - ee[3] tt[3] - ee[4] tt[4]], pp[2] Fh[wH + ee[2] tt[2]]}];
 k3 = reg[step3 === exp3];
 inOld = {step3[[4]], step3[[1]], step3[[2]], step3[[3]]};
 step5 =
  normF[(inOld /. {pp[2] → pp[1], qq[4] → qq[1]}) /. wH → wW - ee[2] tt[2]];
 exp5 = normF[{pp[1] Fh[wW],
    qq[2] Fh[wW - ee[2] tt[2]], pp[3] Fh[wW - ee[2] tt[2] - ee[3] tt[3]],
    qq[1] Fh[wW - ee[2] tt[2] - ee[3] tt[3] - ee[4] tt[4]]}];
 k5 = reg[step5 === exp5];
 sumE = Expand[ee[1] tt[1] + ee[2] tt[2] + ee[3] tt[3] + ee[4] tt[4]];
 k6 = reg[Expand[(wW - ee[2] tt[2] - ee[3] tt[3] - ee[4] tt[4]) -
      (wW + ee[1] tt[1]) + sumE] === 0];
```

```
k7 = reg[And @@
    MapThread[(ampOf[#1] === ampOf[#2]) && MemberQ[{0, sumE, Expand[-sumE]},
        Expand[argOf[#1] - argOf[#2]]] &, {step5, normF[famOld]}]];
rows = {nrow[1, Row[{"(", zV, ", ", pt["W", 2], ", ",
      uV, ", ", pt["W", 1], ") = (", famD[pt["p", 1], {{1, wSym}}],
      ", ", famD[pt["q", 2], {{1, wSym}, {-1, etD[2]}}], ", ",
      famD[pt["p", 3], {{1, wSym}, {-1, etD[2]}, {-1, etD[3]}}],
      ", ", famD[pt["q", 1], {{1, wSym}, {1, etD[1]}}], ")"}],
    givenChip, Row[{"the family of ", pt["e", 1], ", ",
      pt["e", 2], ", ", pt["e", 3], ", ", pt["e", 4]}]],
   nrow[2, Row[{"(", hV["Z"], ", ", hpt["W", 2], ", ", hV["U"], ", ",
      hpt["W", 1], ") = (", famD[hpt["p", 1], {{1, wHat}}],
      ", ", famD[hpt["q", 2], {{1, wHat}, {-1, etH[2]}}], ", ",
      famD[hpt["p", 3], {{1, wHat}, {-1, etH[2]}, {-1, etH[3]}}],
      ", ", famD[hpt["q", 1], {{1, wHat}, {1, etH[1]}}], ")"}],
    givenChip, Row[{"the family of ", hpt["e", 1], ", ", hpt["e", 2],
      ", ", hpt["e", 3], ", ", hpt["e", 4], "; its parameter ",
      wHat, " runs over ℂ, as ", wSym, " does in (1); ",
      fSym, " is the same, because ", hV["M"], " = ", mSym}]],
   nrow[3, Row[{"(", pt["W", 2], ", ", uV, ", ", pt["W", 1], ", ",
      zV, ") = (", famD[pt["q", 2], {{1, wHat}}], ", ",
      famD[pt["p", 3], {{1, wHat}, {-1, etD[3]}}], ", ",
      famD[pt["q", 4], {{1, wHat}, {-1, etD[3]}, {-1, etD[4]}}],
      ", ", famD[pt["p", 2], {{1, wHat}, {1, etD[2]}}], ")"}],
    smallChip[k3], Row[{"(2) with (8)-(10) of Step 1 and ",
      hpt["e", "i"], " = ", pt["e", "i+1"]}]], nrow[4,
    Row[{pt["p", 2], " = ", pt["p", 1], ",   ", pt["q", 4], " = ", pt["q", 1]}],
    givenChip, Row[{"by ", pt["r", 2], " = ",
      pt["r", 1], ", ", pt["s", 4], " = ", pt["s", 1]}]],
   nrow[5, Row[{"(", zV, ", ", pt["W", 2], ", ", uV, ", ", pt["W", 1],
      ") = (", famD[pt["p", 1], {{1, wSym}}], ", ",
      famD[pt["q", 2], {{1, wSym}, {-1, etD[2]}}], ", ",
      famD[pt["p", 3], {{1, wSym}, {-1, etD[2]}, {-1, etD[3]}}], ", ",
      famD[pt["q", 1], {{1, wSym}, {-1, etD[2]}, {-1, etD[3]}, {-1, etD[4]}}],
      ")"}], smallChip[k5],
    Row[{"(3) with (4), written at ", wHat, " = ", wSym, " - ", etD[2],
      ": as ", wSym, " runs over ℂ, so does ", wHat, ", so (3) and (5) give
        the same set of points; this shift makes the first entry ",
      famD[pt["p", 1], {{1, wSym}}], ", as in (1)"}]], nrow[6,
    Row[{"(", argD[{{1, wSym}, {-1, etD[2]}, {-1, etD[3]}, {-1, etD[4]}}],
      ") - (", argD[{{1, wSym}, {1, etD[1]}}], ") = -(", etD[1],
      " + ", etD[2], " + ", etD[3], " + ", etD[4], ")"}],
    smallChip[k6], Row[{"the right side is in Λ, because ", etD[1],
      " + ", etD[2], " + ", etD[3], " + ", etD[4], " ∈ Λ"}]],
   nrow[7, Row[{"thus ", famD["", {{1, wSym}, {-1, etD[2]}, {-1, etD[3]},
        {-1, etD[4]}}], " = ", famD["", {{1, wSym}, {1, etD[1]}}]}]],
    givenChip, Row[{"by (6) and (b)"}]], nrow[8,
```

```
      "the formulas (5) are the formulas (1)", smallChip[k7],
      Row[{"by (7): the amplitudes agree, and the arguments of ",
        fSym, " agree or differ by a point of Λ"}]]};
  AppendTo[resultsAll, And @@ oks];
  Print[
   Panel[Column[{Row[{Style["Step 3: the family after the permutation", Bold,
         15, col], Spacer[20], statusChip[And @@ oks]}],
      Style["Verification: the amplitudes, the phase shifts and the signs
         are symbols, and F is a symbol for a function.", Bold, 13],
      vgrid[rows], Style[Row[{"Together, these show that the family of ",
         hpt["e", 1], ", ", hpt["e", 2], ", ", hpt["e", 3], ", ", hpt["e", 4],
         " formed after the permutation is the family of ", pt["e", 1],
         ", ", pt["e", 2], ", ", pt["e", 3], ", ", pt["e", 4], "."}], 13]},
     Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 6:STEP 4*)
(* ============================================================*)
Module[{col = cols[[9]], oks = {}, reg, chk, m1, m2, m3, m4, c11S, pOne,
   rhsD, lhsD, facD, rows}, reg[verdict_] := (AppendTo[oks, verdict];
    verdict);
  chk[e_] := Module[{verdict = zeroV[e]}, AppendTo[oks, verdict];
    verdict];
  m1 = reg[Expand[(k22 XX^2 YY^2 + k20 XX^2 + k02 YY^2 + 2 k11 XX YY + k00) -
       ((k22 XX^2 + k02) YY^2 + 2 k11 XX YY + k20 XX^2 + k00)] === 0];
  m2 = Module[{v = oldV[1], c, x, z, u, dd}, c = cDef[v];
    x = qX[v];
    z = qZ[v];
    u = 1 - qM[v];
    dd = (x XX^2 - 1) (1 - u x XX^2);
    chk[c[[5]]^2 XX^2 -
      (c[[2]] XX^2 + c[[4]]) (c[[3]] XX^2 + c[[1]]) - c[[5]]^2 z dd / (x (1 + z) (1 + u z))]];
  m3 = reg[Together[cS^2 zS dS / (xS (1 + zS) (1 + uS zS)) -
       (cS^2 zS (1 + zS) / (xS^2 (1 + zS)^2 (1 + uS zS))) xS dS] === 0];
  m4 = reg[Expand[(2 k11 XX)^2 - 4 (k22 XX^2 + k02) (k20 XX^2 + k00) -
       4 (k11^2 XX^2 - (k22 XX^2 + k02) (k20 XX^2 + k00))] === 0];
  c11S = DisplayForm[SuperscriptBox[bx[cD["11", 1]], "2"]];
  pOne[y_] := DisplayForm[RowBox[
     {ptB["P", 1], "(", StyleBox["X", FontSlant → "Italic"], ", ", y, ")"}]];
  rhsD = frac[mulR[c11S, pt["z", 1], dTag[pt["x", 1], capX]], Row[{pt["x", 1],
     "(1 + ", pt["z", 1], ")(1 + ", mulR[uPlain, pt["z", 1]], ")"}]];
  lhsD = Row[{mulR[c11S, powD["X", 2]], " - (",
     mulR[cD["22", 1], powD["X", 2]], " + ", cD["02", 1], ")(",
     mulR[cD["20", 1], powD["X", 2]], " + ", cD["00", 1], ")"}];
  facD = frac[mulR[c11S, pt["z", 1], Row[{"(1 + ", pt["z", 1], ")"}]],
    mulR[DisplayForm[SuperscriptBox[bx[pt["x", 1]], "2"]], DisplayForm[
     SuperscriptBox[RowBox[{"(", "1", "+", bx[pt["z", 1]], ")"}], "2"]],
```

```
      Row[{"(1 + ", mulR[uPlain, pt["z", 1]], ")"}]]];
  rows = {nrow[1, Row[{pOne[StyleBox["Y", FontSlant → "Italic"]],
       " = (", mulR[cD["22", 1], powD["X", 2]], " + ", cD["02", 1],
       ") ", powD["Y", 2], " + 2", mulR[cD["11", 1], capX, capY], " + ",
       mulR[cD["20", 1], powD["X", 2]], " + ", cD["00", 1]}], smallChip[m1],
     Row[{"the definition, ordered by the powers of ", capY}]],
    nrow[2, Row[{lhsD, " = ", rhsD}], smallChip[m2],
     Row[{"an identity in the four flat angles at ", pt["A", 1],
       " and ", capX, ", with ", uPlain, " = 1 - ", pt["M", 1]}]], nrow[3,
     Row[{rhsD, " = ", facD, " ", mulR[pt["x", 1], dTag[pt["x", 1], capX]]}],
     smallChip[m3],
     "the same quotient, written with the factor on the right"],
    nrow[4, Row[{facD, " > 0"}], givenChip,
     Row[{"because ", cD["11", 1], " = -sin", gk["α", 1], " sin", gk["ϒ", 1],
       " ≠ 0 by ", gk["α", 1], ", ", gk["ϒ", 1], " ∈ (0, π), and ",
       pt["x", 1], " ≠ 0, ", mulR[pt["z", 1], Row[{"(1 + ", pt["z", 1], ")"}]],
       " > 0, 1 + ", mulR[uPlain, pt["z", 1]], " > 0 by (a)"}]],
    nrow[5, Row[{DisplayForm[SuperscriptBox[
          RowBox[{"(", "2", bx[mulR[cD["11", 1], capX]], ")"}], "2"]],
       " - 4(", mulR[cD["22", 1], powD["X", 2]], " + ", cD["02", 1], ")(",
       mulR[cD["20", 1], powD["X", 2]], " + ", cD["00", 1], ") = 4(", lhsD, ")"}],
     smallChip[m4], Row[{"the left side is the discriminant in ",
       capY, " of ", pOne["·"], ", written as in (1)"}]],
    nrow[6, Row[{"if ", capX, ", ", capY, " ∈ ℝ satisfy ",
       pOne[StyleBox["Y", FontSlant → "Italic"]], " = 0, then ", lhsD, " ≥ 0"}],
     givenChip, Row[{"if ", mulR[cD["22", 1], powD["X", 2]], " + ",
       cD["02", 1], " ≠ 0, then, by (1), ", pOne["·"], " is a quadratic
         polynomial with real coefficients and the real root ",
       capY, ", so its discriminant is ≥ 0, and (5) applies; if ",
       mulR[cD["22", 1], powD["X", 2]], " + ", cD["02", 1],
       " = 0, then the left side is ", mulR[c11S, powD["X", 2]], " ≥ 0"}]],
    nrow[7, Row[{"if ", capX, ", ", capY, " ∈ ℝ satisfy ",
       pOne[StyleBox["Y", FontSlant → "Italic"]], " = 0, then ",
       mulR[pt["x", 1], dTag[pt["x", 1], capX]], " ≥ 0"}],
     givenChip, "by (6), (2), (3) and (4)"]};
  AppendTo[resultsAll, And @@ oks];
  Print[Panel[Column[{Row[{Style[
         "Step 4: the real solutions of the polynomial at the first vertex",
         Bold, 15, col], Spacer[20], statusChip[And @@ oks]}],
      Style["Verification: (2) is an identity in the flat angles and
         X; (1), (3) and (5) are checked with the coefficients
         and the quotients as symbols.", Bold, 13], vgrid[rows],
      Style["Together, these give line (7).", 13]}, Spacings → 1],
    Background → Lighter[col, 0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 7:CLAIM*)
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(* ============================================================*)
Module[{col = cols[[1]], oks = {}, reg, it1, it2, dz, cf, hs, ctD, pD, rows},
  reg[verdict_] := (AppendTo[oks, verdict];
    verdict);
  (*the permutation by j is the permutation by 1 applied j times;
  x1 and the signs after it,written in the numbering before it*)
  it1 = reg[Table[Nest[# /. {xS[k_] :> yS[nxt[k]], yS[k_] :> xS[nxt[k]]} &,
       xS[1], j], {j, 3}] === {yS[2], xS[3], yS[4]}];
  it2 =
   reg[And @@ Table[Nest[# /. ee[k_] :> ee[nxt[k]] &, Table[ee[i], {i, 4}], j] ===
       Table[ee[Mod[i + j - 1, 4] + 1], {i, 4}], {j, 3}]];
  dz = reg[Expand[((xS[1] XX^2 - 1) (1 - uS xS[1] XX^2)) /. XX → 0] === -1];
  cf = reg[
    CoefficientList[k22 XX^2 YY^2 + k20 XX^2 + k02 YY^2 + 2 k11 XX YY + k00, YY] ===
      {k20 XX^2 + k00, 2 k11 XX, k22 XX^2 + k02} &&
     CoefficientList[k22 XX^2 YY^2 + k20 XX^2 + k02 YY^2 + 2 k11 XX YY + k00, XX] ===
      {k02 YY^2 + k00, 2 k11 YY, k22 YY^2 + k20}];
  hs = reg[And @@ Table[Expand[2 sigF[oldV[i]] - Total[oldV[i]]] === 0 &&
       Expand[2 (sigF[oldV[i]] - oldV[i][[2]]) -
           (oldV[i][[1]] - oldV[i][[2]] + oldV[i][[3]] + oldV[i][[4]])] === 0, {i, 4}]];
  ctD = Row[{"cot(", thI["1+j"], "/2)"}];
  pD[k_, a_, b_] :=
   DisplayForm[RowBox[{ptB["P", k], "(", bx[a], ", ", bx[b], ")"}]];
  rows =
   {nrow[1, Row[{hpt["x", 1], " = ", pt["y", 2], ", ", pt["x", 3], ", ", pt["y",
        4], "   and   (", hpt["e", 1], ", ", hpt["e", 2], ", ", hpt["e", 3],
       ", ", hpt["e", 4], ") = (", pt["e", "1+j"], ", ", pt["e", "2+j"],
       ", ", pt["e", "3+j"], ", ", pt["e", "4+j"], ")"}], smallChip[it1 && it2],
     Row[{"for j = 1, 2, 3, respectively: (6) of Step 1 and ",
       hpt["e", "i"], " = ", pt["e", "i+1"], ", applied j times"}]], nrow[2,
     "after the permutation by j, the hypotheses of the given panel hold,
       and the family of the new signs is the family of the old ones",
     givenChip, "by Steps 2 and 3, applied j times; each time the
       hypotheses hold for the numbers they are applied to"], vgap,
    nrow[3, Row[{"(", zV, ", ", pt["W", 2], ", ", uV, ", ", pt["W", 1],
       ") of a polyhedron of the flexion with an admissible parameter ",
       tSym, " is a real point of the family of ", pt["e", 1],
       ", ", pt["e", 2], ", ", pt["e", 3], ", ", pt["e", 4]}],
     givenChip, Row[{"by (d); it is real, because ", thI[1], ", ",
       thI[2], ", ", thI[3], ", ", thI[4], " are real and not 0"}]],
    nrow[4, Row[{"after the permutation by j, this point is a real
         solution of Bricard's equations formed after it, with ",
       hV["Z"], " = ", ctD}], givenChip, Row[
      {"by (2) and by (c) after the permutation, which holds by (2); ",
       hV["Z"], " = ", ctD, " by (10) of Step 1, applied j times"}]],
    nrow[5, Row[{mulR[hpt["x", 1], dTag[hpt["x", 1], ctD]],
       " ≥ 0,   so   ", ctD, " ∈ ", iSet[hpt["x", 1]]],
```

```
    "   unless   ", dTag[hpt["x", 1], ctD], " = 0"}],
  givenChip, Row[{"by (4) and by line (7) of Step 4 after the
      permutation, which holds by (2); ", hpt["x", 1],
    " ≠ 0 by (a) after the permutation, and ", iSet[hpt["x", 1]],
    " = {", capX, " ∈ ℝ : ", mulR[hpt["x", 1], dTag[hpt["x", 1], capX]],
    " > 0}"}]], nrow[6, Row[{"if ", ctD,
    " is admissible after the permutation, then the polyhedron is a
      polyhedron of the flexion formed after it, with the signs ",
    hpt["e", "i"], " = ", pt["e", "i+j"]}], givenChip,
  "by (d) after the permutation, which holds by (2), and by (1)–(4)"],
 vgap, nrow[7, Row[{dTag[xPlain, 0], " = -1"}], smallChip[dz],
  Row[{"so, by (5), a value of ", ctD,
    " that is not admissible after the permutation is one of
      the at most four zeros of the nonzero polynomial ",
    dTag[hpt["x", 1], "·"], " or one of the finitely many excluded
      values; these values form a finite set"}]],
 nrow[8, Row[{"2", gk["σ", "i"], " = ", gk["α", "i"], " + ",
    gk["β", "i"], " + ", gk["γ", "i"], " + ", gk["δ", "i"], "   and   2",
    disp[bBD[iv]], " = ", gk["α", "i"], " - ", gk["β", "i"],
    " + ", gk["γ", "i"], " + ", gk["δ", "i"]}], smallChip[hs],
  Row[{"so ", cD["00", "i"], " = sin", disp[sgD[iv]], " sin", disp[bBD[iv]],
    " ≠ 0, because ", gk["α", "i"], " + ", gk["β", "i"], " + ", gk["γ", "i"],
    " + ", gk["δ", "i"], " ≠ 0 and ", gk["α", "i"], " - ", gk["β", "i"],
    " + ", gk["γ", "i"], " + ", gk["δ", "i"], " ≠ 0 (mod 2π); also ",
    cD["11", "i"], " = -sin", gk["α", "i"], " sin", gk["γ", "i"],
    " ≠ 0, because ", gk["α", "i"], ", ", gk["γ", "i"], " ∈ (0, π)"}]],
 nrow[9, Row[{"the coefficients of ", pD["i", capX, capY], " in ",
    capY, " are ", mulR[cD["20", "i"], powD["X", 2]], " + ",
    cD["00", "i"], ", 2", mulR[cD["11", "i"], capX], ", ",
    mulR[cD["22", "i"], powD["X", 2]], " + ", cD["02", "i"],
    ", and in ", capX, " are ", mulR[cD["02", "i"], powD["Y", 2]],
    " + ", cD["00", "i"], ", 2", mulR[cD["11", "i"], capY], ", ",
    mulR[cD["22", "i"], powD["Y", 2]], " + ", cD["20", "i"]}],
  smallChip[cf], Row[{"so, by (8), for a fixed value of one variable, ",
    pD["i", capX, capY], " is a nonzero polynomial of degree at
      most 2 in the other: its constant coefficient is ",
    cD["00", "i"], " when the fixed value is 0, and its linear
      coefficient is not 0 otherwise"}]],
 nrow[10, Row[{ctD, " is linked to ", tSym, " = ", zV, " by ",
    pD[2, zV, pt["W", 2]], " = 0 for j = 1, by ", pD[2, zV, pt["W", 2]],
    " = ", pD[3, uV, pt["W", 2]], " = 0 for j = 2, and by ",
    pD[1, zV, pt["W", 1]], " = 0 for j = 3"}], givenChip,
  Row[{"by (3) and (c), because ", ctD, " is ", pt["W", 2], ", ", uV,
    ", ", pt["W", 1], " for j = 1, 2, 3, respectively"}]], nrow[11,
  Row[{"each value of ", ctD, " is taken for at most two values of ",
    tSym, " for odd j, and for at most four for j = 2"}],
  givenChip, "by (9) and (10)"], vgap, nrow[12,
```

```
    Row[{"the admissible ", tSym,
      " for which some polyhedron of the flexion with the parameter ",
      tSym, " has ", ctD, " in the finite set of (7) form a finite set"}],
    givenChip, "by (11), applied to each value of this finite set"],
   nrow[13, Row[{"for every other admissible ", tSym,
      ", every polyhedron of the flexion with the parameter ", tSym,
      " is a polyhedron of the flexion formed after the permutation
        by j, with ", pt["e", "1+j"], ", …, ", pt["e", "4+j"],
      " in place of ", pt["e", 1], ", …, ", pt["e", 4]}], givenChip,
    Row[{"by (6), because for such ", tSym, " the value ", ctD,
      " is not in the finite set of (7), so it is
        admissible after the permutation"}]]};
 AppendTo[resultsAll, And @@ oks];
 Print[
  Panel[Column[{Row[{Style["Claim: the flexions after the permutation", Bold,
       15, col], Spacer[20], statusChip[And @@ oks]}],
     Style[Row[{"Verification: in this block, j ∈ {1, 2, 3} is fixed and a
          hat marks what is formed after the permutation by j; ",
       tSym, " is admissible."}], Bold, 13], vgrid[rows],
     Style["Together, these give the claim.", 13]}, Spacings → 1],
   Background → Lighter[col, 0.93], FrameMargins → 12]];];

(* ============================================================*)
(*PART 8:SUMMARY*)
(* ============================================================*)
Print[Framed[Style[If[And @@ resultsAll, "Section 13: ALL CLAIMS VERIFIED",
    "Section 13: SOME CLAIMS FAILED"], Bold, White, 16], Background →
   If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];
```

Given hypotheses and definitions:

A 3 x 3 net with, for each i = 1, 2, 3, 4, the indices taken modulo 4,

the flat angles $\alpha_i, \beta_i, \gamma_i, \delta_i \in (0, \pi)$ at $A_i$ and the central dihedral angles $\theta_1, \theta_2, \theta_3, \theta_4$

$\sigma_i = \frac{1}{2}(\alpha_i + \beta_i + \gamma_i + \delta_i)$ $\quad \overline{\alpha}_i = \sigma_i - \alpha_i$ $\quad \overline{\beta}_i = \sigma_i - \beta_i$ $\quad \overline{\gamma}_i = \sigma_i - \gamma_i$ $\quad \overline{\delta}_i = \sigma_i - \delta_i$

$a_i = \frac{\sin \alpha_i}{\sin \overline{\alpha}_i}$ $\quad b_i = \frac{\sin \beta_i}{\sin \overline{\beta}_i}$ $\quad c_i = \frac{\sin \gamma_i}{\sin \overline{\gamma}_i}$ $\quad d_i = \frac{\sin \delta_i}{\sin \overline{\delta}_i}$ $\quad \varepsilon_i = \text{sign}(\sin \sigma_i)$

$M_i = a_i b_i c_i d_i$ $\quad r_i = a_i d_i$ $\quad s_i = c_i d_i$ $\quad f_i = a_i c_i$

$x_i = \frac{1}{r_i - 1}$ $\quad y_i = \frac{1}{s_i - 1}$ $\quad z_i = \frac{1}{f_i - 1}$ $\quad u = 1 - M$

$\alpha_i \pm \beta_i \pm \gamma_i \pm \delta_i \neq 0 \pmod{2\pi}$, $\quad \delta_1 + \delta_2 + \delta_3 + \delta_4 = 2\pi$

$M_1 = M_2 = M_3 = M_4 = M > 0$, $\quad r_1 = r_2$, $\quad s_1 = s_4$, $\quad r_3 = r_4$, $\quad s_2 = s_3$

$e_1 t_1 + e_2 t_2 + e_3 t_3 + e_4 t_4 \in \Lambda$, $\quad e_1, e_2, e_3, e_4 \in \{-1, 1\}$

Amplitudes, phase shifts, lattice:

$p_i = \sqrt{r_i - 1}$, $q_i = \sqrt{s_i - 1} \in \mathbb{R}_{>0} \bigcup i\mathbb{R}_{>0}$

dn $t_i = \sqrt{f_i}$ if $M < 1$, dn $t_i = 1/\sqrt{f_i}$ if $M > 1$, the interval of $t_i$ fixed by $p_i q_i$, $\sigma_i$, $M$

$F =$ sn if $M < 1$, $F =$ cn if $M > 1$, $\Lambda$ depends only on $M$

Bricard's equations:

$(Z, W_2, U, W_1) = (\cot(\theta_1/2), \cot(\theta_2/2), \cot(\theta_3/2), \cot(\theta_4/2))$

$P_1(Z, W_1) = P_2(Z, W_2) = P_3(U, W_2) = P_4(U, W_1) = 0$, where

$$P_i(X, Y) = c_{22i} X^2 Y^2 + c_{20i} X^2 + c_{02i} Y^2 + 2 c_{11i} XY + c_{00i},$$

$$c_{00i} = \sin\sigma_i \sin\overline{\beta}_i,\quad c_{22i} = \sin\overline{\delta}_i \sin(\overline{\delta}_i - \beta_i),\quad c_{20i} = \sin\overline{\alpha}_i \sin(\overline{\alpha}_i - \beta_i),$$

$$c_{02i} = \sin\overline{\gamma}_i \sin(\overline{\gamma}_i - \beta_i),\quad c_{11i} = -\sin\alpha_i \sin\gamma_i$$

Family and flexion:

the family of $e_1, e_2, e_3, e_4$: $(Z, W_2, U, W_1) = (p_1 F(w), q_2 F(w - e_2 t_2), p_3 F(w - e_2 t_2 - e_3 t_3), q_1 F(w + e_1 t_1))$, $w \in \mathbb{C}$, all four finite

$D(x, X) = (xX^2 - 1)(1 - uxX^2)$, $I(x) = \{X \in \mathbb{R} : x\, D(x, X) > 0\}$, $x, X \in \mathbb{R}$

the flexion of $e_1, e_2, e_3, e_4$: for each $e_0 \in \{-1, 1\}$ and each admissible $t$, the point $(Z, W_2, U, W_1) = (t, \cot(\theta_2(t)/2), \cot(\theta_3(t)/2), \cot(\theta_4(t)/2))$, where

$$\cot(\theta_2(t)/2) = \frac{\varepsilon_2 (t \sqrt{ux_1y_2(1+z_2)(1+uz_2)} + e_0 e_2 \sqrt{uy_2z_2 D(x_1, t)})}{y_2 u (z_2 + x_1 t^2)}$$

$$\cot(\theta_4(t)/2) = \frac{\varepsilon_1 (t \sqrt{ux_1y_1(1+z_1)(1+uz_1)} - e_0 e_1 \sqrt{uy_1z_1 D(x_1, t)})}{y_1 u (z_1 + x_1 t^2)}$$

$$\cot(\theta_3(t)/2) = \varepsilon_2 \varepsilon_3 \operatorname{sign}(z_2) \sqrt{\frac{z_2 z_3}{x_1 x_3}} \cdot \frac{|x_1| H_2 H_3 t + e_0 e_2 H_1 \sqrt{x_1 D(x_1, t)}}{z_2 z_3 + u x_1 H_1^2 t^2}$$

$H_1, H_2, H_3$: if $uz_2z_3 \neq 1$,

$$H_1 = \frac{|z_2| \sqrt{z_3(1+z_3)(1+uz_3)} + e_2 e_3 |z_3| \sqrt{z_2(1+z_2)(1+uz_2)}}{uz_2z_3 - 1},$$

$$H_2 = \frac{|z_2 z_3| \sqrt{(1+uz_2)(1+uz_3)} + e_2 e_3 \sqrt{z_2 z_3 (1+z_2)(1+z_3)}}{uz_2z_3 - 1},$$

$$H_3 = \frac{u \sqrt{z_2 z_3 (1+z_2)(1+z_3)} \operatorname{sign}(z_2 z_3) + e_2 e_3 \sqrt{(1+uz_2)(1+uz_3)}}{uz_2z_3 - 1};$$

if $uz_2z_3 = 1$ and $e_2 = -e_3$,

$$H_1 = \frac{z_2 z_3 (z_3 - z_2)}{2\sqrt{z_2 z_3 (1+z_2)(1+z_3)}},\quad H_2 = \frac{z_2 z_3 (z_2 + z_3 + 2)}{2\sqrt{z_2 z_3 (1+z_2)(1+z_3)}},\quad H_3 = \frac{(z_2 + z_3 + 2 z_2 z_3) \operatorname{sign}(z_2 z_3)}{2\sqrt{z_2 z_3 (1+z_2)(1+z_3)}};$$

if $uz_2z_3 = 1$ and $e_2 = e_3$, the formula for $\cot(\theta_3(t)/2)$ is replaced by $\cot(\theta_3(t)/2) = \varepsilon_2 \varepsilon_3 \sqrt{\frac{z_2 z_3}{x_1 x_3}} \cdot \frac{\operatorname{sign}(y_2)}{t}$

admissible $t$: $t \in I(x_1)$

outside the finitely many values at which a denominator of these formulas vanishes

the polyhedra of the flexion with the parameter $t$: the polyhedra with these flat angles whose $(Z, W_2, U, W_1)$ is this point for $e_0 = 1$ or for $e_0 = -1$

Cyclic permutation of the indices:

by 1: $A_{i+1}, B_{i+1}, C_{i+1}, \theta_{i+1} \mapsto A_i, C_i, B_i, \theta_i$; a hat marks what is formed after it:

$(\hat{\alpha}_i, \hat{\beta}_i, \hat{\gamma}_i, \hat{\delta}_i) = (\gamma_{i+1}, \beta_{i+1}, \alpha_{i+1}, \delta_{i+1}), \quad \hat{e}_i = e_{i+1}$

by j = 2, 3: the permutation by 1 applied j times

Given facts, valid whenever the hypotheses above hold:

(a) $x_i, y_i, z_i \neq -1, 0$ are well −defined reals, $z_i (1 + z_i) > 0$, $1 + u z_i > 0$

(b) $F(v + \lambda) = F(v)$ for $v \in \mathbb{C}, \lambda \in \Lambda$

(c) every point of the family of $e_1, e_2, e_3, e_4$ solves Bricard's equations

(d) for admissible $t$, the points of the family of $e_1, e_2, e_3, e_4$ with $Z = t$ are the two points of the flexion above, for $e_0 = 1$ and $e_0 = -1$

Each ☑ below certifies that the statement displayed beside it has been verified by a symbolic computation at that point; a ✕ would mean that this verification failed. A • marks a given fact, a definition, a result of an earlier line, or an inference stated in words beside it.

We want to test the following claim ★ :

Claim.

For j = 1, 2, 3 and all but finitely many admissible $t$,

- the polyhedra of the flexion with the parameter $t$ are polyhedra of the flexion formed after the cyclic permutation of the indices by j, with $e_{1+j}, \ldots, e_{4+j}$ in place of $e_1, \ldots, e_4$

### Step 1: the numbers after the permutation

Verified

Verification: each line holds for i = 1, 2, 3, 4, and each checked line is an identity in the flat angles, checked for each of the four values of i.

(1) $(\hat{\alpha}_i, \hat{\beta}_i, \hat{\gamma}_i, \hat{\delta}_i) = (\gamma_{i+1}, \beta_{i+1}, \alpha_{i+1}, \delta_{i+1})$ — • the permutation, see the given panel

(2) $\hat{\sigma}_i = \sigma_{i+1}$ — ☑ by (1): the half sum of the same four angles

(3) $(\hat{\sigma}_i - \hat{\alpha}_i, \hat{\sigma}_i - \hat{\beta}_i, \hat{\sigma}_i - \hat{\gamma}_i, \hat{\sigma}_i - \hat{\delta}_i) = (\overline{\gamma}_{i+1}, \overline{\beta}_{i+1}, \overline{\alpha}_{i+1}, \overline{\delta}_{i+1})$ — ☑ by (1) and (2)

(4) $(\hat{a}_i, \hat{b}_i, \hat{c}_i, \hat{d}_i) = (c_{i+1}, b_{i+1}, a_{i+1}, d_{i+1})$ — ☑ by (1) and (3): each is the sine of an angle over the sine of its barred angle

(5) $(\hat{M}_i, \hat{r}_i, \hat{s}_i, \hat{f}_i) = (M_{i+1}, s_{i+1}, r_{i+1}, f_{i+1})$ — ☑ by (4), because $M_i = a_i b_i c_i d_i$, $r_i = a_i d_i$, $s_i = c_i d_i$, $f_i = a_i c_i$

(6) $(\hat{x}_i, \hat{y}_i, \hat{z}_i) = (y_{i+1}, x_{i+1}, z_{i+1})$ — ☑ by (5)

(7) $\hat{\varepsilon}_i = \varepsilon_{i+1}$ — • by (2), because $\varepsilon_i = \text{sign}(\sin \sigma_i)$

(8) $(\hat{p}_i, \hat{q}_i) = (q_{i+1}, p_{i+1})$, $\hat{p}_i \hat{q}_i = p_{i+1} q_{i+1}$ — • by (5), because $p_i = \sqrt{r_i - 1}$ and $q_i = \sqrt{s_i - 1}$

(9) $\hat{t}_i = t_{i+1}$ — • dn $\hat{t}_i$ = dn $t_{i+1}$, because $\hat{f}_i = f_{i+1}$ and $\hat{M}_i = M_{i+1}$ by (5); the interval is the same, because $\hat{p}_i \hat{q}_i = p_{i+1} q_{i+1}$ by (8) and $\hat{\sigma}_i = \sigma_{i+1}$ by (2)

(10) $(\hat{Z}, \hat{W}_2, \hat{U}, \hat{W}_1) = (W_2, U, W_1, Z)$ — • $\theta_{i+1}$ is relabeled $\theta_i$

Together, these give the numbers after the permutation.

### Step 2: the hypotheses after the permutation

Verified

Verification: each hypothesis of the given panel, written with the hats, is compared with the hypotheses written without them.

(1) the eight sums $\hat{\alpha}_i \pm \hat{\beta}_i \pm \hat{\gamma}_i \pm \hat{\delta}_i$ are, up to sign, the eight sums $\alpha_{i+1} \pm \beta_{i+1} \pm \gamma_{i+1} \pm \delta_{i+1}$ — ☑ for i = 1, 2, 3, 4, by (1) of Step 1; so none of them is 0 modulo 2 $\pi$

(2) $\hat{\alpha}_i, \hat{\beta}_i, \hat{\gamma}_i, \hat{\delta}_i \in (0, \pi)$ — • by (1) of Step 1: they are flat angles of the net

(3) $\hat{\delta}_1 + \hat{\delta}_2 + \hat{\delta}_3 + \hat{\delta}_4 = \delta_2 + \delta_3 + \delta_4 + \delta_1 = 2\pi$ — ☑ by (1) of Step 1

(4) $\hat{M}_1 = \hat{M}_2 = \hat{M}_3 = \hat{M}_4$ reads $M_2 = M_3 = M_4 = M_1$, so that $\hat{M} = M > 0$ — ☑ by (5) of Step 1

(5) $\hat{r}_1 = \hat{r}_2, \hat{s}_1 = \hat{s}_4, \hat{r}_3 = \hat{r}_4, \hat{s}_2 = \hat{s}_3$ read $s_2 = s_3, r_2 = r_1, s_4 = s_1, r_3 = r_4$ — ☑ by (5) of Step 1; these are $r_1 = r_2, s_1 = s_4, r_3 = r_4, s_2 = s_3$

(6) $\hat{e}_1 \hat{t}_1 + \hat{e}_2 \hat{t}_2 + \hat{e}_3 \hat{t}_3 + \hat{e}_4 \hat{t}_4 = e_2 t_2 + e_3 t_3 + e_4 t_4 + e_1 t_1 \in \Lambda$ — ☑ by (9) of Step 1 and $\hat{e}_i = e_{i+1}$; the lattice is the same, because $\hat{M} = M$

Together, these show that the hypotheses of the given panel hold after the permutation, with $e_{i+1}$ in place of $e_i$.

### Step 3: the family after the permutation

Verified

Verification: the amplitudes, the phase shifts
and the signs are symbols, and F is a symbol for a function.

(1) $(Z, W_2, U, W_1) = ($
$p_1\ F(w),\ q_2\ F(w - e_2 t_2)$
$,\ p_3\ F(w - e_2 t_2 - e_3 t_3)$
$,\ q_1\ F(w + e_1 t_1))$

- the family of $e_1, e_2, e_3, e_4$

(2) $(\hat{Z}, \hat{W}_2, \hat{U}, \hat{W}_1) = ($
$\hat{p}_1\ F(\hat{w}),\ \hat{q}_2\ F(\hat{w} - \hat{e}_2 \hat{t}_2)$
$,\ \hat{p}_3\ F(\hat{w} - \hat{e}_2 \hat{t}_2 - \hat{e}_3 \hat{t}_3)$
$,\ \hat{q}_1\ F(\hat{w} + \hat{e}_1 \hat{t}_1))$

- the family of $\hat{e}_1, \hat{e}_2, \hat{e}_3, \hat{e}_4$; its parameter $\hat{w}$ runs over $\mathbb{C}$, as $w$ does in (1); $F$ is the same, because $M\hat{} = M$

(3) $(W_2, U, W_1, Z) = ($
$q_2\ F(\hat{w}),\ p_3\ F(\hat{w} - e_3 t_3)$
$,\ q_4\ F(\hat{w} - e_3 t_3 - e_4 t_4)$
$,\ p_2\ F(\hat{w} + e_2 t_2))$

☑ (2) with (8)–(10) of Step 1 and $e\hat{}_i = e_{i+1}$

(4) $p_2 = p_1,\ \ q_4 = q_1$

- by $r_2 = r_1, s_4 = s_1$

(5) $(Z, W_2, U, W_1) = ($
$p_1\ F(w),\ q_2\ F(w - e_2 t_2)$
$,\ p_3\ F(w - e_2 t_2 - e_3 t_3),$
$q_1\ F(w - e_2 t_2 - e_3 t_3 - e_4 t_4))$

☑ (3) with (4), written at $w\hat{} = w - e_2 t_2$: as $w$ runs over $\mathbb{C}$, so does $\hat{w}$, so (3) and (5) give the same set of points; this shift makes the first entry $p_1\ F(w)$, as in (1)

(6) $(w - e_2 t_2 - e_3 t_3 - e_4 t_4) - (w + e_1 t_1) = -(e_1 t_1 + e_2 t_2 + e_3 t_3 + e_4 t_4)$

☑ the right side is in $\Lambda$, because $e_1 t_1 + e_2 t_2 + e_3 t_3 + e_4 t_4 \in \Lambda$

(7) thus $F(w - e_2 t_2 - e_3 t_3 - e_4 t_4) = F(w + e_1 t_1)$

- by (6) and (b)

(8) the formulas (5) are the formulas (1)

☑ by (7): the amplitudes agree, and the arguments of $F$ agree or differ by a point of $\Lambda$

Together, these show that the family of $\hat{e}_1, \hat{e}_2, \hat{e}_3, \hat{e}_4$ formed after the permutation is the family of $e_1, e_2, e_3, e_4$.

### Step 4: the real solutions of the polynomial at the first vertex

Verified

Verification: (2) is an identity in the flat angles and X; (1), (3) and (5) are checked with the coefficients and the quotients as symbols.

(1) $P_1(X, Y) = (c_{221} X^2 + c_{021}) Y^2 + 2 c_{111} X Y + c_{201} X^2 + c_{001}$
☑ the definition, ordered by the powers of $Y$

(2) $c_{111}{}^2 X^2 - (c_{221} X^2 + c_{021})(c_{201} X^2 + c_{001}) = \frac{c_{111}{}^2 z_1 D(x_1, X)}{x_1 (1 + z_1)(1 + u z_1)}$
☑ an identity in the four flat angles at $A_1$ and $X$, with $u = 1 - M_1$

(3) $\frac{c_{111}{}^2 z_1 D(x_1, X)}{x_1 (1 + z_1)(1 + u z_1)} = \frac{c_{111}{}^2 z_1 (1 + z_1)}{x_1{}^2 (1+z_1)^2 (1 + u z_1)} x_1 D(x_1, X)$
☑ the same quotient, written with the factor on the right

(4) $\frac{c_{111}{}^2 z_1 (1 + z_1)}{x_1{}^2 (1+z_1)^2 (1 + u z_1)} > 0$
- because $c_{111} = -\sin \alpha_1 \sin \gamma_1 \neq 0$ by $\alpha_1, \gamma_1 \in (0, \pi)$, and $x_1 \neq 0$, $z_1 (1 + z_1) > 0$, $1 + u z_1 > 0$ by (a)

(5) $(2 c_{111} X)^2 - 4(c_{221} X^2 + c_{021})(c_{201} X^2 + c_{001}) = 4(c_{111}{}^2 X^2 - (c_{221} X^2 + c_{021})(c_{201} X^2 + c_{001}))$
☑ the left side is the discriminant in $Y$ of $P_1(X, \cdot)$, written as in (1)

(6) if $X, Y \in \mathbb{R}$ satisfy $P_1(X, Y) = 0$, then $c_{111}{}^2 X^2 - (c_{221} X^2 + c_{021})(c_{201} X^2 + c_{001}) \geq 0$
- if $c_{221} X^2 + c_{021} \neq 0$, then, by (1), $P_1(X, \cdot)$ is a quadratic polynomial with real coefficients and the real root $Y$, so its discriminant is $\geq 0$, and (5) applies; if $c_{221} X^2 + c_{021} = 0$, then the left side is $c_{111}{}^2 X^2 \geq 0$

(7) if $X, Y \in \mathbb{R}$ satisfy $P_1(X, Y) = 0$, then $x_1 D(x_1, X) \geq 0$
- by (6), (2), (3) and (4)

Together, these give line (7).

### Claim: the flexions after the permutation

Verified

Verification: in this block, j $\in \{1, 2, 3\}$ is fixed and a hat marks what is formed after the permutation by j; $t$ is admissible.

(1) $\hat{x}_1 = y_2, x_3, y_4$ and $(\hat{e}_1, \hat{e}_2, \hat{e}_3, \hat{e}_4) = (e_{1+j}, e_{2+j}, e_{3+j}, e_{4+j})$
☑ for j = 1, 2, 3, respectively: (6) of Step 1 and $\hat{e}_i = e_{i+1}$, applied j times

(2) after the permutation by j, the hypotheses of the given panel hold, and the family of the new signs is the family of the old ones
- by Steps 2 and 3, applied j times; each time the hypotheses hold for the numbers they are applied to

(3) $(Z, W_2, U, W_1)$ of a polyhedron of the flexion with an admissible parameter $t$ is a real point of the family of $e_1, e_2, e_3, e_4$
- by (d); it is real, because $\theta_1, \theta_2, \theta_3, \theta_4$ are real and not 0

(4) after the permutation by j, this point is
- by (2) and by (c) after the permutation ,

a real solution of Bricard's equations formed after it, with $\hat{Z} = \cot(\theta_{1+j}/2)$

which holds by (2); $\hat{Z} = \cot(\theta_{1+j}/2)$ by (10) of Step 1, applied j times

(5) $\hat{x}_1\ D(\hat{x}_1, \cot(\theta_{1+j}/2)) \geq 0$, so $\cot(\theta_{1+j}/2) \in I(\hat{x}_1)$ unless $D(\hat{x}_1, \cot(\theta_{1+j}/2)) = 0$

- by (4) and by line (7) of Step 4 after the permutation, which holds by (2); $\hat{x}_1 \neq 0$ by (a) after the permutation, and $I(\hat{x}_1) = \{X \in \mathbb{R} : \hat{x}_1\ D(\hat{x}_1, X) > 0\}$

(6) if $\cot(\theta_{1+j}/2)$ is admissible after the permutation, then the polyhedron is a polyhedron of the flexion formed after it, with the signs $e\hat{}_i = e_{i+j}$

- by (d) after the permutation, which holds by (2), and by (1)–(4)

(7) $D(x, 0) = -1$

☑ so, by (5), a value of $\cot(\theta_{1+j}/2)$ that is not admissible after the permutation is one of the at most four zeros of the nonzero polynomial $D(\hat{x}_1, \cdot)$ or one of the finitely many excluded values; these values form a finite set

(8) $2\sigma_i = \alpha_i + \beta_i + \gamma_i + \delta_i$ and $2\overline{\beta}_i = \alpha_i - \beta_i + \gamma_i + \delta_i$

☑ so $c_{00i} = \sin\sigma_i \sin\overline{\beta}_i \neq 0$, because $\alpha_i + \beta_i + \gamma_i + \delta_i \neq 0$ and $\alpha_i - \beta_i + \gamma_i + \delta_i \neq 0 \pmod{2\pi}$; also $c_{11i} = -\sin\alpha_i \sin\gamma_i \neq 0$, because $\alpha_i, \gamma_i \in (0, \pi)$

(9) the coefficients of $P_i(X, Y)$ in $Y$ are $c_{20i}X^2 + c_{00i}, 2c_{11i}X, c_{22i}X^2 + c_{02i}$, and in $X$ are $c_{02i}Y^2 + c_{00i}, 2c_{11i}Y, c_{22i}Y^2 + c_{20i}$

☑ so, by (8), for a fixed value of one variable, $P_i(X, Y)$ is a nonzero polynomial of degree at most 2 in the other: its constant coefficient is $c_{00i}$ when the fixed value is 0, and its linear coefficient is not 0 otherwise

(10) $\cot(\theta_{1+j}/2)$ is linked to $t = Z$ by $P_2(Z, W_2) = 0$ for j = 1, by $P_2(Z, W_2) = P_3(U, W_2) = 0$ for j = 2, and by $P_1(Z, W_1) = 0$ for j = 3

- by (3) and (c), because $\cot(\theta_{1+j}/2)$ is $W_2, U, W_1$ for j = 1, 2, 3, respectively

(11) each value of $\cot(\theta_{1+j}/2)$ is taken for at most two values of $t$ for odd j, and for at most four for j = 2

- by (9) and (10)

(12) the admissible $t$ for which some polyhedron of the flexion with the parameter $t$ has $\cot(\theta_{1+j}/2)$ in the finite set of (7) form a finite set

- by (11), applied to each value of this finite set

(13) for every other admissible $t$, every polyhedron of the flexion with the parameter $t$ is a polyhedron of the flexion formed

- by (6), because for such $t$ the value $\cot(\theta_{1+j}/2)$ is not in the finite set of (7), so it is admissible after the permutation

is a polyhedron of the flexion formed
after the permutation by j, with
$e_{1+j}, \ldots, e_{4+j}$ in place of $e_1, \ldots, e_4$

admissible after the permutation

Together, these give the claim.

**Section 13: ALL CLAIMS VERIFIED**